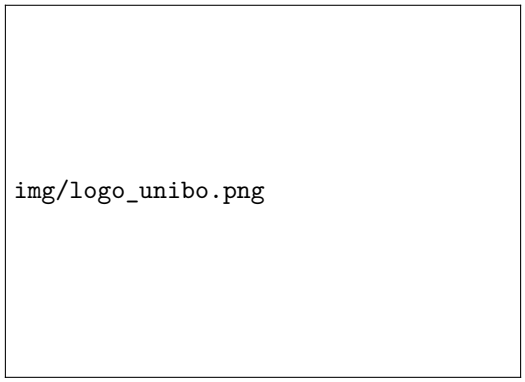


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Quantum Field Theory II

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Introduction

QFT is a fundamental framework in physics that combines classical field theory, special relativity, and quantum mechanics. It provides a powerful language for describing the behavior of particles and fields at the most fundamental level. We are going to assume that the reader has a basic understanding of quantum mechanics and special relativity, as well as some familiarity with classical field theory and quantum field theory of free fields (which will be reviewed anyway in the first chapter).

There exist different motivations for studying QFT: it provides a framework for describing the interactions between particles and fields, and it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. Additionally:

- QFT is the language of the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity); it is used for the quantization of systems with an **infinite number of degrees of freedom**, such as fields; it is essential for understanding the behavior of particles at high energies and small distances, where quantum effects become significant; furthermore it provides a framework for describing the behavior of particles in curved spacetime, which is important for understanding phenomena such as black holes and the early universe.
- It is a successful framework for describing the behavior of particles and fields at the most fundamental level, incorporating both quantum mechanics and special relativity:
  - At high energies we can witness **particles creation and annihilation**, which cannot be described by non-relativistic quantum mechanics; QFT provides a framework for describing these processes, in an infinite-dimensional system, where the number of particles is not fixed.
  - QFT is a relativistic theory, which means that it is consistent with the principles of special relativity, implmenting **causality** and **Lorentz invariance**; this is essential for describing the behavior of particles at high energies and small distances, where relativistic effects become significant. Two events that are spacelike separated cannot influence each other

    $$(x-y)^2 = (x^0-y^0) - \sum_{i=1}^{3}(x^i-y^i)^2 < 0 \quad \Longrightarrow \quad [\mathcal{O}(x), \mathcal{O}(y)] = 0,$$

    thus the events are causally disconnected, ensuring causality.
- Particles are described as excitations of underlying fields, which can be created and annihilated; this allows for a more complete description of particle interactions and the behavior of

particles at high energies. Examples of particles and underlying fields include:

  - **Scalar field**: $\phi(x) \implies$ spin-0 particles (e.g., Higgs boson).
  - **Spinor field**: $\psi(x) \implies$ spin-1/2 particles (e.g., electrons, quarks).
  - **Vector field**: $A_\mu(x) \implies$ spin-1 particles (e.g., photons, W and Z bosons).

- The **Standard Model** of particle physics is a quantum field theory that describes the fundamental particles and their interactions (except for gravity); it has been successful in explaining a wide range of phenomena, from the behavior of subatomic particles to the properties of condensed matter systems. It is a quantum field theory that describes the electromagnetic, weak, and strong interactions, through the exchange of gauge bosons among fields of spin 0, 1/2, and 1 particles.

- The inclusion of gravity in a quantum field theory framework is an open problem in theoretical physics, but can be approached by inserting the metric tensor of general relativity

$$g^{\mu\nu} = \eta^{\mu\nu} + h^{\mu\nu}.$$

  While there are approaches to quantizing gravity, such as string theory and loop quantum gravity, a complete and consistent theory of quantum gravity has not yet been developed. Quanta of the gravitational field are called gravitons, which are hypothetical spin-2 particles that mediate the gravitational interaction, but they have not been observed experimentally. These notes will not cover quantum gravity, but it is an important area of research in theoretical physics which sometimes will be mentioned throughout the notes.

The main goal of these notes is to provide a comprehensive understanding of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level, as well as mathematical methods for making theoretical predictions on interactions among particles. We will cover a wide range of topics, including:

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- The quantization of free fields, which provides the foundation for understanding the behavior of particles and fields in quantum field theory.
- The construction of interacting quantum field theories, which are essential for describing the behavior of particles and fields at the most fundamental level.
- The use of perturbation theory and Feynman diagrams to calculate scattering amplitudes and other physical observables in quantum field theory.
- The renormalization of quantum field theories, which is necessary to make sense of the infinities that arise in perturbation theory and to extract meaningful physical predictions from the theory.
- The application of quantum field theory to the Standard Model of particle physics, which describes the fundamental particles and their interactions (except for gravity).

QFT is one of the most successful and powerful frameworks in physics, and it has been instrumental in our understanding of the fundamental nature of the universe. This theory has been tested and confirmed by a wide range of experiments, some of which tested the predictions of QFT to an extraordinary degree of precision. For example, the anomalous magnetic moment of the electron

has been measured to an accuracy of one part in a trillion ($\sim 10^{-13}$), and the predictions of QFT have been confirmed to this level of precision. Additionally, the discovery of the Higgs boson at the Large Hadron Collider in 2012 provided further confirmation of the predictions of QFT and the Standard Model.

Another essential aspect of these notes is to provide a comprehensive understanding of the importance of symmetries in quantum field theory, which play a crucial role in determining the behavior of particles and fields. We will explore the concept of gauge symmetries, which are local symmetries that arise in quantum field theories and are responsible for the interactions between particles and fields. We will also discuss the role of global symmetries, which are symmetries that do not depend on spacetime coordinates, and their implications for the behavior of particles and fields. In the end we will argue that *a QFT is essentially defined by its fields and its symmetries* (statement that will be made more precise in the end of the notes), and we will see how symmetries can be used to constrain the behavior of particles and fields, and to make predictions about their interactions.

For a more comprehensive introduction to quantum field theory, we recommend the following textbooks:

- **An Introduction to Quantum Field Theory** by Michael E. Peskin and Daniel V. Schroeder: This is a widely used textbook that provides a comprehensive introduction to quantum field theory, covering both the theoretical foundations and practical applications of the subject.
- **Quantum Field Theory** by Mark Srednicki: This textbook offers a clear and concise introduction to quantum field theory, with an emphasis on the physical intuition behind the mathematical formalism.
- **Quantum Field Theory in a Nutshell** by A. Zee: This book provides a more informal and intuitive introduction to quantum field theory, making it accessible to a wider audience while still covering essential topics.
- **The Quantum Theory of Fields** by Steven Weinberg: This three-volume series is a comprehensive and rigorous treatment of quantum field theory, written by one of the pioneers in the field. It is suitable for advanced students and researchers who want a deep understanding of the subject.

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For more information about the notation and conventions used in these notes, please refer to the appendix, where we provide a detailed explanation of the symbols and mathematical conventions that we will use throughout the notes (Appendix A).

Prerequisites: *Familiarity with the Lagrangian and Hamiltonian formalisms of Classical Mechanics, standard Quantum Mechanics, Special Relativity and the basics of Quantum Field Theory (free fields) is required.*

1 | Free QFT Review

A free quantum field theory does not account for interactions between particles. It describes particles that do not interact with each other, and the fields that represent them evolve independently. Free QFTs are often used as a starting point for understanding more complex interacting theories, as they provide a simpler framework to study the properties of quantum fields and particles.

The lagrangian for a free quantum field theory typically consists of kinetic terms and mass terms for the fields, without any interaction terms: it can only contain **quadratic terms** in the fields, without any higher-order interactions. Quanta of the free fields correspond to particles that do not interact with each other, and their dynamics are governed by the **free propagators** of the theory, which describe how particles propagate through spacetime without any interactions: propagators are the Green's functions of the free theory, and they determine how particles propagate from one point to another in spacetime with determined probability amplitudes.

We will review the free quantum field theories for different types of particles, including scalar fields (spin 0), fermionic fields (spin 1/2), and gauge fields (spin 1). We will also discuss the symmetries of these theories and how they lead to conservation laws through Noether's theorem. This review will provide a foundation for understanding more complex interacting quantum field theories, and it will also highlight the differences between free and interacting theories, as well as the role of symmetries in quantum field theory.

1.1 | Spin 0

For a real scalar field $\phi(x)$, the free Lagrangian is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2,$$

where m is the mass of the scalar particle. The equation of motion derived from this Lagrangian is the **Klein-Gordon equation**:

$$(\Box + m^2)\phi = 0,$$

where $\Box = \partial_\mu \partial^\mu$ is the d'Alembertian operator. The solutions to this equation describe free scalar particles with mass m. This is a classical equation of motion, having the same role of Maxwell's equations in classical electromagnetism, but for a scalar field $\phi(x)$ instead of a vector field $A^\mu(x)$.

The idea of quantum field theory is to promote the classical field $\phi(x)$ to a quantum operator $\hat{\phi}(x)$ that acts on a Hilbert space of states. The quantization procedure involves imposing commutation relations on the field operators, which leads to the creation and annihilation of particles. The free scalar field can be expanded in terms of creation and annihilation operators, which allows us to describe the particle content of the theory. Indeed, we can see how decomposing the field into its Fourier modes leads to the interpretation of the quanta of the field as particles with specific momenta and energies:

$$\phi(t,\, \mathbf{x}) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\, e^{i\mathbf{k}\cdot\mathbf{x}} \tilde{\phi}(t,\, \mathbf{k}),$$

where each mode can be computed by using this ansatz in the Klein-Gordon equation in momentum space, leading to the dispersion relation of an harmonic oscillator represented by on-shell particles with energy $E_\mathbf{k} = \omega_\mathbf{k} = \sqrt{\mathbf{k}^2 + m^2}$:

$$\left(\partial_t^2 + \omega_\mathbf{k}^2\right) \tilde{\phi}(t, \mathbf{k}) = 0,$$

showing how the free scalar field can be interpreted as a collection of independent harmonic oscillators, each corresponding to a different momentum mode.

In order to quantize the free scalar field, we have to promote the fields themselves to operators and impose the canonical commutation relations, which leads to the creation and annihilation of particles. Before showing how to do that, it is crucial to note that $\hat{\phi}(x)$ depends on both space and time, enforcing the Heisenberg picture of quantum mechanics, where operators evolve in time while states remain fixed. The canonical commutation relations for the scalar field are given by

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^3(\mathbf{x} - \mathbf{y}),$$

where $\hat{\pi}(x) = \frac{\partial \mathcal{L}}{\partial(\partial_t \hat{\phi}(x))} = \partial_t \hat{\phi}(x)$ is the conjugate momentum operator. This constraint along with reality of the field $\hat{\phi}(x) = \hat{\phi}^\dagger(x)$ allows us to express the field operator in terms of **creation and annihilation operators**:

$$\hat{\phi}(x) = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2\omega_\mathbf{k}}} \left(\hat{a}_\mathbf{k} e^{-ik\cdot x} + \hat{a}_\mathbf{k}^\dagger e^{ik\cdot x}\right),$$

where $k \cdot x = \omega_\mathbf{k} t - \mathbf{k}\cdot\mathbf{x} = k^\mu x_\mu$, and $\omega_\mathbf{k} = k^0$. The creation and annihilation (ladder) operators satisfy the following commutation relations

$$\begin{cases} [\hat{a}_\mathbf{k}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta^3(\mathbf{k} - \mathbf{k}'), \\ [\hat{a}_\mathbf{k}, \hat{a}_{\mathbf{k}'}] = [\hat{a}_\mathbf{k}^\dagger, \hat{a}_{\mathbf{k}'}^\dagger] = 0. \end{cases}$$

Note that while the field operator $\hat{\phi}(x)$ is Hermitian and time dependent, the creation and annihilation operators are not, and they are responsible for creating and destroying particles in the quantum field theory. The **vacuum state** $|0\rangle$ is defined as the state that is annihilated by all annihilation operators:

$$\hat{a}_{\mathbf{k}} |0\rangle = 0 \quad \forall \mathbf{k},$$

while we can create one-particle states by acting with the creation operators on the vacuum, which adds one quanta of the field with momentum $\mathbf{k}$:

$$|\mathbf{k}\rangle = \sqrt{2\omega_{\mathbf{k}}} \hat{a}^{\dagger}_{\mathbf{k}} |0\rangle ,$$

and more generally, we can create multi-particle states by acting with multiple creation operators on the vacuum, which leads to the **Fock space** of the theory. With this definition of single particle states, we can compute the inner product between two single particle states, which is given by

$$\langle \mathbf{k}|\mathbf{k}'\rangle = 2\omega_{\mathbf{k}} \delta^3(\mathbf{k} - \mathbf{k}'),$$

showing that the single particle states are orthogonal and normalized with respect to the relativistic normalization convention. The normalization $\sqrt{2\omega_{\mathbf{k}}}$ is arbitrary, but important for ensuring that the probabilities computed in the theory are consistent with relativistic invariance.

Remark. *To find a Lorentz invariant measure, we can start from the Minkowski space measure* d^4k *and impose the on-shell condition* $k^2 = m^2$ *using a delta function along with a step function to ensure that we only consider positive energy solutions:*

$$\int \mathrm{d}^4k \, \delta(k^2 - m^2)\theta(k^0) = \int \frac{\mathrm{d}^3\mathbf{k}}{2\omega_{\mathbf{k}}}.$$

We have then found the Lorentz invariant measure $\frac{\mathrm{d}^3\mathbf{k}}{2\omega_{\mathbf{k}}}$ *for integrating over the momentum space of on-shell particles, which is crucial for ensuring that the theory is consistent with special relativity. The previous normalization of the single particle states is consistent with this Lorentz invariant measure, since it ensures that the inner product between single particle states is Lorentz invariant, and it also allows us to compute probabilities and cross sections in a way that is consistent with relativistic invariance:*

$$\int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}| = 1,$$

which is the completeness relation for the single particle states in the theory. This ensures that we can express any state in the Fock space as a superposition of single particle states, and it also allows us to compute transition amplitudes and probabilities in a way that is consistent with relativistic invariance.

1.1.1 | Feynman's Propagator

In Quantum Field Theory (QFT), we can compute physical observables in terms of *Green functions*, which are defined as the expectation values of products of field operators evaluated in the vacuum state:

$$\langle 0|T\{\hat{\phi}(x_1) \dots \hat{\phi}(x_n)\}|0\rangle$$

where "T" stands for "time ordering" and defines the time-ordered product. For two bosonic field operators, the time-ordered product is explicitly given by:

$$T\{\hat{O}(x)\hat{O}(y)\} = \Theta(x^0 - y^0)\hat{O}(x)\hat{O}(y) + \Theta(y^0 - x^0)\hat{O}(y)\hat{O}(x)$$

where $\Theta(t)$ is the Heaviside step function, ensuring that operators evaluated at later times always appear to the left.

A very important case is the 2-point function, universally known as the **Feynman Propagator**. For a real scalar field, we define it as:

$$D_F(x,y) = D_F(x-y) \equiv \langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\}|0\rangle$$

Notice that the propagator depends only on the difference $x - y$ due to space-time translation symmetry. Physically, it can be interpreted as the probability amplitude for a "causal" propagation of a particle between the spacetime points x and y.

The exact analytical result for this expectation value (the proof can be found in standard textbooks such as Peskin 2.4 or Schwartz 6.2) is given by the following momentum-space integral:

$$D_F(x,y) = \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik\cdot(x-y)}$$

Comments:

- This result is strictly valid for a *free theory* (non-interacting fields).
- Here, k^0 is an independent integration variable, which means the particle can be "off-shell", i.e., $k^0 \neq \sqrt{\vec{k}^2 + m^2}$.
- The "$i\epsilon$" term is known as the *Feynman prescription*. Mathematically, it shifts the poles of the integrand at $k^2 = m^2$ slightly away from the real axis in the complex k^0-plane, avoiding singularities along the integration contour. This specific shift properly implements the causal time-ordering defined by the Θ functions. It is implied that $\epsilon \to 0^+$, and it is common to write the denominator simply as $k^2 - m^2 + i0^+$ or even leave it implicit.

Furthermore, $D_F(x,y)$ is mathematically a "Green function" of the classical Klein-Gordon equation. This means it acts as the inverse of the Klein-Gordon differential operator:

$$(\Box_x + m^2)D_F(x,y) = -i\delta^{(4)}(x-y)$$

The specific boundary conditions chosen to solve this differential equation are precisely what give rise to the aforementioned "$i\epsilon$" prescription in momentum space.

1.1.2 | Noether's Theorem

Noether's Theorem establishes a profound connection between symmetries and conservation laws: a continuous symmetry of the action implies the existence of a *conserved current* and a corresponding *conserved charge*.

Consider a system described by a Lagrangian density $\mathcal{L} = \mathcal{L}(\phi_k, \partial_\mu \phi_k)$, where ϕ_k can be any type of field. Suppose there is an infinitesimal transformation of the fields ϕ_k and the coordinates x^μ, described by a continuous parameter α. If this transformation leaves the total action $S = \int d^4x\, \mathcal{L}(x)$ invariant, the variation of the Lagrangian density must at most be a total divergence:

$$\delta\mathcal{L} = \alpha \partial_\mu \mathcal{J}^\mu + \mathcal{O}(\alpha^2)$$

for some 4-vector $\mathcal{J}^\mu$. Note that if $\mathcal{L}$ is strictly invariant point-by-point, then $\mathcal{J}^\mu = 0$.

By applying the Euler-Lagrange equations, we can extract the general form of the conserved Noether current J^μ:

$$J^\mu = \sum_k \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_k)} \frac{\partial \phi_k}{\partial \alpha} - \mathcal{J}^\mu$$

which satisfies the continuity equation:

$$\partial_\mu J^\mu = 0$$

This local conservation law implies the existence of a globally conserved charge Q, obtained by integrating the time-component of the current (J^0) over all space:

$$Q = \int d^3x \, J^0 \implies \frac{dQ}{dt} = 0$$

Energy-Momentum Tensor

An exceptionally important symmetry is space-time translation: $x^\nu \to x^\nu + a^\nu$. Because the parameter a^ν has 4 independent components (for $\nu = 0, 1, 2, 3$), this symmetry yields 4 distinct conserved currents. These currents are naturally arranged into a rank-2 tensor known as the **Energy-Momentum Tensor**, $T^{\mu\nu}$, which satisfies $\partial_\mu T^{\mu\nu} = 0$.

Applying the general formula for J^μ to translations (where the variation of the field is $\delta\phi_k = a^\nu \partial_\nu \phi_k$ and $\mathcal{J}^\mu = g^{\mu\nu}\mathcal{L}$), we obtain:

$$T^{\mu\nu} = \sum_k \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_k)} \partial^\nu \phi_k - g^{\mu\nu}\mathcal{L}$$

A crucial component of this tensor is the *energy density* of the field configuration, given by the 00-component:

$$\mathcal{E} = T^{00}$$

For Lagrangians where the canonical momentum is standardly defined as $\pi \equiv \frac{\partial \mathcal{L}}{\partial(\partial_t \phi)}$, this energy density perfectly coincides with the Hamiltonian density: $\mathcal{E} = \mathcal{H}$.

For a real scalar field, after performing canonical quantization (promoting fields to operators and imposing commutation relations), we obtain the full quantum Hamiltonian operator by integrating the Hamiltonian density over space:

$$\hat{H} = \int \frac{d^3k}{(2\pi)^3} E_{\vec{k}} \, \hat{a}^\dagger_{\vec{k}} \hat{a}_{\vec{k}} + \text{const}$$

where $E_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$ is the energy of a single mode. (Deriving this from the expression of T^{00} is a standard and highly recommended exercise).

> **Remark.** *The "+const" term represents the vacuum zero-point energy, which in QFT is technically infinite. We typically eliminate it by defining normal ordering $(: \hat{H} :)$ so that the vacuum state has exactly zero energy.*

1.1.3 | Complex Scalar Field

The complex scalar field generalizes the real scalar field by introducing a field ϕ that takes complex values. The Lagrangian density for a free complex scalar field is given by:

$$\mathcal{L} = |\partial_\mu \phi|^2 - m^2 |\phi|^2 = (\partial_\mu \phi^*)(\partial^\mu \phi) - m^2 \phi^* \phi$$

A complex field ϕ inherently possesses two independent degrees of freedom. We can parameterize it using two real scalar fields, for instance ϕ_1 and ϕ_2, such that $\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$. However, it is mathematically more convenient to treat ϕ and its complex conjugate ϕ^* as the two independent degrees of freedom when applying the Euler-Lagrange equations.

Continuous Symmetries and Noether's Current

The Lagrangian $\mathcal{L}$ is manifestly invariant under a global $U(1)$ phase transformation. If we rotate the field by a continuous phase α, the transformations are:

$$\phi \to e^{i\alpha}\phi, \qquad \phi^* \to e^{-i\alpha}\phi^*$$

For an infinitesimal transformation ($\alpha \ll 1$), the field variations are $\delta\phi = i\alpha\phi$ and $\delta\phi^* = -i\alpha\phi^*$. Because this continuous symmetry leaves the action invariant, Noether's theorem dictates the existence of a conserved current J^μ.

We can derive this current explicitly (solving the suggested exercise) using the standard Noether formula:

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\delta\phi^*$$

Dropping the overall constant parameter α, and substituting the derivatives $\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \partial^\mu\phi^*$ and $\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)} = \partial^\mu\phi$, we obtain:

$$J^\mu = (\partial^\mu\phi^*)(i\phi) + (\partial^\mu\phi)(-i\phi^*) = -i(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi)$$

By virtue of the equations of motion (the Klein-Gordon equation for both ϕ and ϕ^*), this current is strictly conserved, meaning $\partial_\mu J^\mu = 0$.

Quantization and Antiparticles

To quantize the theory, we promote the classical fields ϕ and ϕ^* to operator-valued distributions $\hat{\phi}$ and $\hat{\phi}^\dagger$. Because the field is complex ($\hat{\phi} \neq \hat{\phi}^\dagger$), the Fourier expansion requires two distinct sets of creation and annihilation operators:

$$\hat{\phi}(x) = \int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{k}}}}\left(\hat{a}_{\vec{k}}e^{-ik\cdot x} + \hat{b}^\dagger_{\vec{k}}e^{ik\cdot x}\right)$$

$$\hat{\phi}^\dagger(x) = \int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{k}}}}\left(\hat{a}^\dagger_{\vec{k}}e^{ik\cdot x} + \hat{b}_{\vec{k}}e^{-ik\cdot x}\right)$$

where $E_{\vec{k}} = \sqrt{\vec{k}^2 + m^2}$. The physical interpretation of these operators is fundamental to QFT:

- $\hat{a}_{\vec{k}}$ and $\hat{a}^\dagger_{\vec{k}}$ annihilate and create **particles**, respectively.
- $\hat{b}_{\vec{k}}$ and $\hat{b}^\dagger_{\vec{k}}$ annihilate and create **anti-particles**, respectively.

This implies the existence of two distinct types of particle states. Because they both originate from the same Lagrangian and share the same dispersion relation $E_{\vec{k}}$, particles and anti-particles possess the exact same mass.

The Conserved Charge

The conserved Noether current J^μ leads to a conserved Noether charge $Q = \int d^3x\, J^0$. Upon quantization, promoting the fields to operators and substituting the mode expansions into the integral yields the quantum charge operator $\hat{Q}$:

$$\hat{Q} = \int \frac{d^3k}{(2\pi)^3} \left(\hat{a}^\dagger_{\vec{k}} \hat{a}_{\vec{k}} - \hat{b}^\dagger_{\vec{k}} \hat{b}_{\vec{k}} \right) + \text{const}$$

The additive constant arises from commutation relations during the derivation and represents the vacuum charge. In practice, we apply *normal ordering* $(: \hat{Q} :)$ to subtract this infinite vacuum contribution, setting the charge of the vacuum state to zero.

The resulting operator algebraically "counts" the number of particles minus the number of anti-particles. This demonstrates that while particles and anti-particles have the same mass, they carry **opposite conserved charges** under the $U(1)$ symmetry.

The Feynman Propagator

For the complex scalar field, the causal propagation between two spacetime points is described by the time-ordered expectation value of the field and its conjugate. The Feynman propagator is defined as:

$$D_F(x,y) = \langle 0|T\{\hat{\phi}(x)\hat{\phi}^\dagger(y)\}|0\rangle = \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i0^+} e^{-ik\cdot(x-y)}$$

Physically, this represents the amplitude for a particle to propagate from y to x (if $x^0 > y^0$), or for an anti-particle to propagate from x to y (if $y^0 > x^0$).

Remark. *The expectation values of two identical fields vanish identically:*

$$\langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\}|0\rangle = \langle 0|T\{\hat{\phi}^\dagger(x)\hat{\phi}^\dagger(y)\}|0\rangle = 0$$

This happens because the operator $\hat{\phi}\hat{\phi}$ changes the net $U(1)$ charge by two units, meaning it connects the vacuum (charge 0) to a state with charge +2 or -2, making the inner product with the vacuum state strictly zero.

1.2 | Spin 1/2

1.2.1 | Dirac Equation and Gamma Matrices

Fermions of spin-1/2 are mathematically described by a 4-component complex object known as a Dirac spinor, $\psi(x)$. The classical field equation governing the dynamics of this free spinor is the **Dirac equation**:

$$(i\partial\!\!\!/ - m)\psi = 0$$

where we introduce the Feynman slash notation defined as $\partial\!\!\!/ \equiv \gamma^\mu \partial_\mu$. The fundamental structural elements here are the γ **matrices** (γ^μ, with $\mu = 0, 1, 2, 3$). These are 4×4 matrices that strictly obey the fundamental Clifford algebra anticommutation relation:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{1}_{4\times 4}$$

Remark. *The 4×4 identity matrix $\mathbb{1}_{4\times4}$ on the right-hand side is very often left implicit in QFT literature. Furthermore, by squaring the Dirac equation operator $(i\partial\!\!\!/ - m)(-i\partial\!\!\!/ - m) = \Box + m^2$, one can show that every component of the Dirac spinor individually satisfies the Klein-Gordon equation $(\Box + m^2)\psi = 0$, ensuring the correct relativistic energy-momentum dispersion relation $p^2 = m^2$.*

While the physics is independent of the chosen matrix representation, an explicit and highly convenient choice for relativistic physics is the **Weyl (or Chiral) representation**. Using 2×2 block matrices:

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}$$

where $\sigma^\mu = (\mathbb{1}_{2\times2}, \vec{\sigma})$ and $\bar{\sigma}^\mu = (\mathbb{1}_{2\times2}, -\vec{\sigma})$. The spatial components $\vec{\sigma}$ are the standard 2×2 Pauli matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

1.2.2 | Free Spinor Solutions and Helicity

We search for plane-wave solutions to the Dirac equation of the form:

$$\psi(x) = \begin{cases} u_s(p)e^{-ip\cdot x} & \text{(positive frequency)} \\ v_s(p)e^{+ip\cdot x} & \text{(negative frequency)} \end{cases}$$

where $u_s(p)$ is a spinor representing a particle, and $v_s(p)$ is an anti-spinor representing an anti-particle. The 4-momentum must satisfy $p^2 = m^2$ and $p^0 > 0$. Substituting these into the Dirac equation yields momentum-space algebraic constraints:

$$(p\!\!\!/ - m)u_s = 0, \qquad (p\!\!\!/ + m)v_s = 0$$

For a given momentum p^μ, there are two linearly independent solutions for both u and v, labeled by an internal index $s = \pm$, which corresponds to the spin state.

To find explicit forms, let us take the momentum $\vec{p}$ to be aligned parallel to the z-axis (which is always possible via spatial rotation). The solutions take the form:

$$u_s = \begin{pmatrix} \sqrt{E - p_z}\,\eta_s \\ \sqrt{E + p_z}\,\eta_s \end{pmatrix}, \qquad v_s = \begin{pmatrix} \sqrt{E - p_z}\,\eta_s \\ -\sqrt{E + p_z}\,\eta_s \end{pmatrix}$$

where η_s are two-component basis vectors. We can explicitly choose the basis of **helicity eigenstates**, where helicity is defined as the projection of the spin vector along the direction of the momentum $\vec{p}$:

$$\eta_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{(spin parallel to } z\text{-axis} \rightarrow \text{positive helicity } +)$$

$$\eta_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{(spin anti-parallel to } z\text{-axis} \rightarrow \text{negative helicity } -)$$

Remark. *Alternatively, one could construct solutions by first defining eigenstates of spin along an arbitrary axis in the particle's rest frame ($E = m, \vec{p} = 0$), and then applying a Lorentz boost to reach the desired momentum $\vec{p}$. However, defining spin along an arbitrary fixed axis requires the particle to be massive ($m \neq 0$) so that a rest frame exists. Helicity states, conversely, remain well-defined even for strictly massless particles ($m = 0$). Note also that spin is fundamentally a quantum observable and is properly defined only after quantization.*

1.2.3 | Lorentz Transformation of Spinors and Chirality

Under a Lorentz transformation, a Dirac spinor transforms according to a specific representation of the Lorentz group. The generators of Lorentz transformations for a Dirac spinor are given by the commutators of the gamma matrices:

$$S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$$

In the Weyl representation, these generators are explicitly *block-diagonal*:

$$S^{\mu\nu} = \begin{pmatrix} (\text{non-zero})_{2\times 2} & 0 \\ 0 & (\text{non-zero})_{2\times 2} \end{pmatrix}$$

This block-diagonal structure implies that the 4-component Dirac representation is mathematically *reducible*. It decomposes into two distinct, independent 2-component representations. We decompose the Dirac spinor ψ into two objects:

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

where ψ_L represents a **Left-Chirality** Weyl spinor, and ψ_R represents a **Right-Chirality** Weyl spinor. Under Lorentz transformations, ψ_L and ψ_R transform independently of each other.

To formalize this mathematically without referencing a specific basis, we introduce the fifth gamma matrix, γ^5 (or γ_5):

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$$

In the Weyl representation, it takes a simple diagonal form:

$$\gamma^5 = \begin{pmatrix} -\mathbb{1}_{2\times 2} & 0 \\ 0 & \mathbb{1}_{2\times 2} \end{pmatrix}$$

A critical property of γ^5 is that it exactly anti-commutes with all the other four gamma matrices: $\{\gamma^\mu, \gamma^5\} = 0$.

Using γ^5, we can define Lorentz-invariant **Chirality Projectors**:

$$P_R = \frac{\mathbb{1} + \gamma^5}{2} = \begin{pmatrix} 0 & 0 \\ 0 & \mathbb{1} \end{pmatrix}, \qquad P_L = \frac{\mathbb{1} - \gamma^5}{2} = \begin{pmatrix} \mathbb{1} & 0 \\ 0 & 0 \end{pmatrix}$$

These operators project out the right-handed ($P_R\psi = \psi_R$) and left-handed ($P_L\psi = \psi_L$) chiral components of any generic Dirac spinor.

Remark. *For the special case of strictly massless fermions ($m = 0$), chirality and helicity are intimately connected but fundamentally distinct concepts:*

- *For massless **particles**: Chirality perfectly coincides with Helicity.*
- *For massless **anti-particles**: Chirality is exactly opposite to Helicity.*

Bilinears and Spinor Sums

To construct Lorentz invariant quantities (like Lagrangians) from spinors, we must define the **Dirac adjoint spinor**:

$$\bar{\psi} \equiv \psi^\dagger \gamma^0$$

This is necessary because the naive hermitian inner product $\psi^\dagger\psi$ is *not* Lorentz invariant. However, utilizing the adjoint spinor, we can construct various objects with definite Lorentz transformation properties:

- $\bar{\psi}\psi$ transforms as a **Lorentz Scalar** (invariant).
- $\bar{\psi}\gamma^\mu\psi$ transforms as a **Lorentz Vector**.
- $\bar{\psi}\gamma^\mu\gamma^\nu \dots \psi$ transforms as a **Lorentz Tensor**.

When calculating cross-sections and decay rates later, we will heavily rely on the **spinor sum completeness relations**:

$$\sum_s u_s(p)\bar{u}_s(p) = \not{p} + m$$

$$\sum_s v_s(p)\bar{v}_s(p) = \not{p} - m$$

These relations mathematically sum over the unobserved spin states of external fermions, converting spinor products into trace operations over gamma matrices.

1.2.4 | Quantization of the Dirac Field

The classical Lagrangian density for the free Dirac field is:

$$\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi$$

This Lagrangian is invariant under a global $U(1)$ phase transformation $\psi \to e^{i\alpha}\psi$, $\bar{\psi} \to e^{-i\alpha}\bar{\psi}$. By Noether's theorem, this yields the conserved vector current:

$$J^\mu = \bar{\psi}\gamma^\mu\psi$$

To quantize the theory, we promote the classical field $\psi(x)$ and its adjoint $\bar{\psi}(x)$ to operators. Following the **Spin-Statistics Theorem**, fermions (particles with half-integer spin) must obey Fermi-Dirac statistics, which requires that their corresponding quantum operators satisfy **anti-commutation relations**, unlike bosons (integer spin) which obey commutation relations.

The Fourier expansion of the quantum field operators is:

$$\hat{\psi}(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{k}}}} \sum_s \left(\hat{a}^s_{\vec{k}} u_s(k) e^{-ik\cdot x} + \hat{b}^{s\dagger}_{\vec{k}} v_s(k) e^{ik\cdot x} \right)$$

$$\hat{\bar{\psi}}(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{k}}}} \sum_s \left(\hat{b}^s_{\vec{k}} \bar{v}_s(k) e^{-ik\cdot x} + \hat{a}^{s\dagger}_{\vec{k}} \bar{u}_s(k) e^{ik\cdot x} \right)$$

where the creation and annihilation operators satisfy the fundamental anti-commutation algebra:

$$\{\hat{a}^r_{\vec{k}}, \hat{a}^{s\dagger}_{\vec{k}'}\} = \{\hat{b}^r_{\vec{k}}, \hat{b}^{s\dagger}_{\vec{k}'}\} = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k}')\delta^{rs}$$

All other anti-commutators strictly vanish. The use of anti-commutators naturally enforces the Pauli Exclusion Principle, guaranteeing that two identical fermions cannot occupy the exact same quantum state.

The Dirac Propagator

The causal propagation of a spin-1/2 fermion is described by the Feynman propagator, defined as the time-ordered expectation value of the field operators:

$$S_F(x, y) = S_F(x - y) = \langle 0 | T\{\hat{\psi}(x)\hat{\bar{\psi}}(y)\} | 0 \rangle$$

Evaluating this using the mode expansion and the anti-commutation relations, one obtains the momentum-space representation:

$$S_F(x, y) = \int \frac{d^4k}{(2\pi)^4} \frac{i(\not{k} + m)}{k^2 - m^2 + i0^+} e^{-ik\cdot(x-y)}$$

Remark. *Algebraically, notice the relation* $(\not{k}+m)(\not{k}-m) = k^2 - m^2$*. This allows us to formally rewrite the momentum-space propagator as the exact inverse matrix of the Dirac differential operator:*

$$\frac{i(\not{k} + m)}{k^2 - m^2} = \frac{i}{\not{k} - m}$$

1.3 | Spin 1

1.3.1 | The Massless Vector Field (Electromagnetism)

A rigorous derivation of the Lagrangian for a vector field A^μ from first principles (group theory and Lorentz representations) can be done, but for now, we simply "copy" it from classical electromagnetism. The classical Lagrangian density for a massless vector field coupled to an external source is:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - A_\mu J^\mu$$

where $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor, and $J^\mu = (\rho, \vec{J})$ is the 4-current density (with ρ being the charge density and $\vec{J}$ the current density). In the vacuum, we simply set $J^\mu = 0$. The 4-potential is defined as $A^\mu = (\phi, \vec{A})$, containing the scalar electric potential ϕ and the vector magnetic potential $\vec{A}$.

The classical equations of motion, derived from the Euler-Lagrange equations, are the covariant Maxwell equations:

$$\partial_\mu F^{\mu\nu} = J^\nu \implies \Box A^\nu - \partial^\nu(\partial_\mu A^\mu) = J^\nu$$

Gauge Invariance and Current Conservation

A fundamental property of this Lagrangian is its behavior under a **gauge transformation**:

$$A_\mu \to A_\mu + \partial_\mu \alpha(x)$$

for any arbitrary scalar function $\alpha(x)$. The field strength tensor $F_{\mu\nu}$ is identically invariant under this transformation (since mixed partial derivatives commute: $\partial_\mu\partial_\nu\alpha = \partial_\nu\partial_\mu\alpha$). Consequently, the free equations of motion are also invariant. This means the field A^μ is physically defined only up to a gauge transformation, creating a redundancy that allows us to remove 2 unphysical degrees of freedom from the 4 components of A^μ.

Common gauge choices include:

- **Coulomb Gauge:** $\partial_i A^i = \nabla \cdot \vec{A} = 0$ (useful but breaks manifest Lorentz covariance).
- **Lorenz Gauge:** $\partial_\mu A^\mu = 0$ (manifestly covariant, heavily used in QFT).

Remark. *There is a profound connection between gauge invariance and the conservation of the 4-current. If we apply the transformation $A_\mu \to A_\mu + \partial_\mu\alpha$ to the interaction term of the Lagrangian, the variation is:*

$$\delta\mathcal{L} = -(\partial_\mu\alpha)J^\mu = -\partial_\mu(\alpha J^\mu) + \alpha(\partial_\mu J^\mu)$$

*The first term is a total derivative. In the action $S = \int d^4x\,\mathcal{L}$, total derivatives integrate to zero (assuming fields vanish at infinity). Thus, for the total action to be completely gauge invariant ($\delta S = 0$), the second term must vanish for any $\alpha(x)$, which strictly requires **current conservation**:*

$$\partial_\mu J^\mu = 0 \iff \frac{\partial\rho}{\partial t} + \nabla \cdot \vec{J} = 0$$

1.3.2 | Classical Solutions and Degrees of Freedom

Consider the free theory in the vacuum ($J^\mu = 0$). We impose the Lorenz gauge condition $\partial_\mu A^\mu = 0$. Using a plane-wave ansatz (representing a Fourier component of the field):

$$A^\mu(x) = \varepsilon^\mu e^{-ik\cdot x}$$

Plugging this into the equations of motion in the Lorenz gauge ($\Box A^\mu = 0$) gives:

$$-k^2\varepsilon^\mu e^{-ik\cdot x} = 0 \implies k^2 = 0$$

This implies that the field A^μ describes a **massless** particle (the photon), as its 4-momentum satisfies $E^2 = |\vec{k}|^2$. Furthermore, the Lorenz gauge condition requires:

$$\partial_\mu A^\mu = 0 \implies -ik_\mu\varepsilon^\mu e^{-ik\cdot x} = 0 \implies k_\mu\varepsilon^\mu = 0$$

This orthogonality condition removes 1 degree of freedom from the 4 components of the polarization vector ε^μ.

However, $\partial_\mu A^\mu = 0$ does *not* completely fix the gauge. We can perform a residual gauge transformation $A^\mu \to A^\mu + \partial^\mu\alpha$ provided that $\Box\alpha = 0$. Choosing $\alpha(x) = iCe^{-ik\cdot x}$ yields:

$$A^\mu \to (\varepsilon^\mu + Ck^\mu)e^{-ik\cdot x}$$

This corresponds to a shift of the polarization vector $\varepsilon^\mu \to \varepsilon^\mu + Ck^\mu$. By choosing the constant $C = -\varepsilon^0/k^0$, we can explicitly set the temporal component $\varepsilon^0 = 0$.

If we align the momentum along the z-axis, $\vec{k} = (0, 0, k^0)$, the 4-momentum is $k^\mu = (k^0, 0, 0, k^0)^T$. The condition $k_\mu\varepsilon^\mu = 0$ with $\varepsilon^0 = 0$ forces:

$$0 - k^0\varepsilon^3 = 0 \implies \varepsilon^3 = 0$$

Thus, the physical polarization vector takes the form $\varepsilon^\mu = (0, \varepsilon^1, \varepsilon^2, 0)^T$. There are **only 2 physical degrees of freedom**, representing transverse polarizations.

1.3.3 | Helicity States

As a basis for the transverse polarization vectors, we usually define the circular polarization states:

$$\varepsilon^\mu_\pm = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix}$$

These vectors are eigenvectors of a rotation matrix around the direction of motion (the z-axis). Applying the rotation matrix $R(\theta)$:

$$R(\theta)^\mu{}_\nu\varepsilon^\nu_\pm = e^{\pm i\theta}\varepsilon^\mu_\pm$$

The eigenvalues $e^{ih\theta}$ reveal the **helicity** $h = \pm 1$. After quantization, these become the helicity states (eigenstates of spin parallel to the direction of propagation). A vector field therefore physically corresponds to a particle with **Spin-1**.

TODO: add diagram showing a rotation around the z-axis and the associated right-handed and left-handed helicity vectors

1.3.4 | The Massive Spin-1 Field (Proca Lagrangian)

To describe a massive vector boson, we explicitly add a mass term to the Lagrangian:

$$\mathcal{L} = \mathcal{L}_{\text{massless}} + \frac{1}{2}m^2 A_\mu A^\mu$$

Remark. *Unlike the massless case, this mass term is strictly* ***not gauge invariant****. Gauge symmetry is explicitly broken by the mass of the vector field.*

The equation of motion (for $J^\mu = 0$) becomes the Proca equation:

$$\Box A^\mu - \partial^\mu(\partial_\nu A^\nu) + m^2 A^\mu = 0$$

If we take the 4-divergence (∂_μ) of both sides, the first two terms cancel out ($\partial_\mu \Box A^\mu - \Box \partial_\mu A^\mu = 0$), leaving:

$$m^2 \partial_\mu A^\mu = 0 \implies \partial_\mu A^\mu = 0 \quad (\text{since } m \neq 0)$$

Here, $\partial_\mu A^\mu = 0$ is *not* an arbitrary gauge choice, but a strict dynamical constraint forced by the equations of motion.

For a plane wave solution $A^\mu(x) = \varepsilon^\mu e^{-ik\cdot x}$, we still have $k_\mu \varepsilon^\mu = 0$. However, because $k^2 = m^2$ (not zero), we cannot use residual gauge freedom to eliminate a third degree of freedom. Thus, a massive spin-1 field has **3 independent polarization states**. If $\vec{k} \parallel \hat{z}$, we have $k^\mu = (k^0, 0, 0, k^3)^T$. Besides the two transverse states $\varepsilon^\mu_\pm$, we now have a physical longitudinal polarization state:

$$\varepsilon^\mu_0 = \frac{1}{m}\begin{pmatrix} k^3 \\ 0 \\ 0 \\ k^0 \end{pmatrix}$$

which satisfies $k_\mu \varepsilon^\mu_0 = 0$ and is properly normalized to $\varepsilon_0^2 = -1$.

Remark. *Massive vector bosons present severe theoretical problems at high energies, largely due to the longitudinal polarization state ε^μ_0 whose components grow with energy ($\propto k/m$), leading to unitarity violations in scattering amplitudes. In the Standard Model, the fundamental vector bosons (W and Z) are originally massless (preserving gauge invariance) and acquire mass dynamically via the* ***Higgs Mechanism****, neatly avoiding this issue.*

1.3.5 | Quantization and Propagators

The canonical quantization of the electromagnetic field is notoriously difficult precisely because of gauge invariance: infinitely many field configurations correspond to just one physical state, making the canonical momentum ill-defined. We will rigorously address this later using the Path Integral formalism (via Faddeev-Popov quantization).

For now, by analogy with scalar fields, we write the free field operator expansion as:

$$\hat{A}^\mu(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{k}}}} \sum_s \left(\varepsilon^\mu_s(\vec{k}) \hat{a}_{\vec{k},s} e^{-ik\cdot x} + \varepsilon^{\mu *}_s(\vec{k}) \hat{a}^\dagger_{\vec{k},s} e^{ik\cdot x} \right)$$

The Feynman propagator is defined as the time-ordered expectation value:

$$\langle 0 | T\{\hat{A}^\mu(x) \hat{A}^\nu(y)\} | 0 \rangle = \int \frac{d^4k}{(2\pi)^4} e^{-ik\cdot(x-y)} \Pi^{\mu\nu}(k)$$

where $\Pi^{\mu\nu}(k)$ is the propagator in momentum space.

- **For a massive field ($m \neq 0$)**, the propagator is uniquely determined as the Green function of the Proca equation:

$$\Pi^{\mu\nu}(k) = \frac{i}{k^2 - m^2 + i0^+}\left(-g^{\mu\nu} + \frac{k^\mu k^\nu}{m^2}\right)$$

- **For a massless field ($m = 0$)**, the operator $(\Box g^{\mu\nu} - \partial^\mu\partial^\nu)$ is not invertible due to gauge invariance. We must explicitly "fix the gauge". The propagator takes the general form:

$$\Pi^{\mu\nu}(k) = \frac{-ig^{\mu\nu}}{k^2 + i0^+} + \text{gauge-dependent terms}$$

  A convenient and widely used choice is the **Feynman gauge**, where the gauge-dependent terms exactly vanish, leaving just $\Pi^{\mu\nu}(k) = -ig^{\mu\nu}/(k^2 + i0^+)$. While intermediate calculations depend on this choice, all physical observables (like scattering cross-sections) remain strictly gauge-independent.

2 | Interacting Fields

Free theories are a good starting point to understand the basic concepts and techniques of quantum field theory, but to capture the full complexity of the physical world, we need to introduce interactions between fields. Interactions allow us to describe processes such as scattering, particle decays, and the formation of bound states, which are essential for understanding the behavior of matter and energy at a fundamental level. In a free theory, the fields do not interact with each other, and the equations of motion are linear: they can exhibit *at most quadratic terms in the fields* (in the lagrangian $\mathcal{L}$). We can then build interacting theories by adding non-linear terms to the Lagrangian, which will lead to non-linear equations of motion. These non-linearities are what give rise to the rich phenomenology of interacting quantum field theories.

The most important tools in the developing of interacting quantum field theories are:

- **Dimensional analysis**, which allows us to determine the relevance of different interaction terms at different energy scales, and to classify interactions as relevant, irrelevant, or marginal.
- **Renormalization**, which is a systematic procedure to deal with the infinities that arise in perturbative calculations of interacting theories, and to extract meaningful physical predictions from them.
- **Symmetry principles**, which play a crucial role in constraining the form of the interactions and in determining the properties of the particles and fields in the theory. For example, gauge symmetries lead to the introduction of gauge fields, which mediate the fundamental forces of nature.

Moreover **locality** is a fundamental principle in quantum field theory, which states that interactions should occur at the same point in spacetime. This principle ensures that the theory respects causality and allows us to construct a consistent framework for describing the dynamics of fields and particles:

$$\mathcal{L}(x) = \mathcal{L}\left(\phi(x), \partial_\mu \phi(x)\right), \quad \mathcal{L}(x) \neq \mathcal{L}\left(\phi(x+y), \partial_\mu \phi(x+y)\right),$$

and the interaction terms we will introduce will be local, meaning that they will involve products of fields and their derivatives evaluated at the same spacetime point x; this ensures that the theory is consistent with the principles of special relativity and quantum mechanics, and it will guide us in constructing the interaction terms in the Lagrangian.

In the next section we will start by performing a dimensional analysis of the fields and the interaction terms, which will help us to understand the behavior of the theory at different energy scales and to identify the most relevant interactions for describing physical phenomena. This will set the

stage for our subsequent discussions on renormalization and symmetry principles in interacting quantum field theories.

2.1 | Dimensional Analysis

In physics, and particularly in High Energy Physics (HEP), we heavily rely on the system of Natural Units where we set the reduced Planck constant and the speed of light to unity: $\hbar = c = 1$. Consequently, mass, energy, momentum, inverse length, and inverse time all share the exact same dimension. There is only one independent dimensional unit left, which we conventionally identify with the dimension of mass.

We define the **mass dimension** of a generic quantity X as $[X]$. By definition:

$$[\text{mass}] = 1$$

From the uncertainty principle or the De Broglie relations ($x \sim \hbar/p$, $t \sim \hbar/E$), coordinates in space and time have an inverse mass dimension:

$$[x] = [t] = [\text{length}] = [\text{mass}^{-1}] = -1$$

Because $\hbar = 1$ is completely dimensionless, the physical action S must also be strictly dimensionless:

$$[S] = 0$$

Since the action is the integral of the Lagrangian density over 4-dimensional spacetime ($S = \int d^4x\, \mathcal{L}$), and the integration measure $[d^4x] = [x^0x^1x^2x^3] = -4$, we immediately deduce the fundamental mass dimension of the Lagrangian density:

$$[\mathcal{L}] = +4$$

From this requirement, combined with the fact that $[m] = [\partial_\mu] = 1$, we can systematically derive the mass dimension of any quantum field simply by inspecting its free kinetic Lagrangian.

- **Scalar (Spin-0) and Vector (Spin-1) fields:** The kinetic terms are proportional to $(\partial\phi)^2$ and $(\partial A)^2$. Therefore, $2 + 2[\phi] = 4 \implies [\phi] = [A^\mu] = 1$.
- **Dirac Spinors (Spin-1/2):** The kinetic term involves only one derivative, $\bar{\psi}\partial\!\!\!/\psi$. Therefore, $1 + 2[\psi] = 4 \implies [\psi] = 3/2$.

2.1.1 | Examples of Interaction Terms

The total Lagrangian for an interacting quantum field theory takes the general form:

$$\mathcal{L} = \mathcal{L}_{\text{FREE}} + \mathcal{L}_{\text{INT}}$$

where $\mathcal{L}_{\text{FREE}}$ is the sum of all free, quadratic Lagrangians for the fields involved in the theory. The interaction part, $\mathcal{L}_{\text{INT}}$, can be written as a sum over various interaction operators:

$$\mathcal{L}_{\text{INT}} = \sum_j g_j \mathcal{O}_j(\phi_k, \partial_\mu \phi_k)$$

Here, $\mathcal{O}_j$ is a composite **operator** (a product of fields and possibly their derivatives). Crucially, these operators must be *Lorentz scalars* to preserve the Lorentz invariance of the action. The coefficients g_j are the **coupling constants**, which dictate the strength of the interaction.

Because the total Lagrangian must have dimension 4, we have a strict constraint on the dimensions of the couplings and the operators:

$$4 = [\mathcal{L}] = [g_j] + [\mathcal{O}_j] \implies [g_j] = 4 - [\mathcal{O}_j]$$

This relation implies that the higher the mass dimension of the interaction operator $\mathcal{O}_j$, the lower (more negative) the mass dimension of its associated coupling constant g_j.

Remark. *Theories containing at least one coupling constant with a negative mass dimension ($[g_j] < 0$, which is completely equivalent to having an operator with $[\mathcal{O}_j] > 4$) are classified as* ***non-renormalizable****. These theories become un-predictive at very high energy scales because they require an infinite number of counterterms to absorb ultraviolet divergences. However, they can still be incredibly useful as Effective Field Theories (EFTs), yielding highly accurate physical predictions at lower energies.*

The Symmetries Dictate the Interactions

A fundamental question arises: which specific operators $\mathcal{O}_j$ should be included in $\mathcal{L}_{\text{INT}}$? The golden rule of modern QFT is:

One must include ALL operators that respect the underlying symmetries of the theory, up to a chosen maximal mass dimension $[\mathcal{O}_j] \leq d_{max}$.

Usually, for a fundamental renormalizable theory, we set $d_{\max} = 4$. For effective theories, $d_{\max}$ is an integer strictly greater than 4. The profound reason behind this requirement is that quantum corrections (loop diagrams) will inevitably generate every possible interaction term allowed by the symmetries, even if we manually set them to zero at the classical tree-level. This is why we state that *symmetries uniquely define the theory.*

ϕ^3 Theory

For pedagogical purposes, we often consider a simplified model known as ϕ^3 theory, which describes a single real scalar field self-interacting via a cubic potential.

$$\mathcal{L} = \mathcal{L}_{\text{K.G.}} + \mathcal{L}_{\text{INT}} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{3!}\phi^3$$

In this specific theory:

- ϕ^3 is the cubic interaction operator. Its mass dimension is $[\phi^3] = 3 \times 1 = 3$.
- λ is the coupling constant. Its mass dimension is $[\lambda] = 4 - 3 = 1$.
- The factor $1/3!$ is a conventional symmetry prefactor that elegantly cancels combinatorial factors when calculating Feynman diagrams.

Despite its mathematical simplicity, **this theory is fundamentally unphysical**. To see why, we compute the classical Hamiltonian density (the energy density). The conjugate momentum is $\pi \equiv \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}$.

$$\mathcal{H} = \pi\dot{\phi} - \mathcal{L} = \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{3!}\phi^3$$

The potential energy density is $V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{3!}\phi^3$.

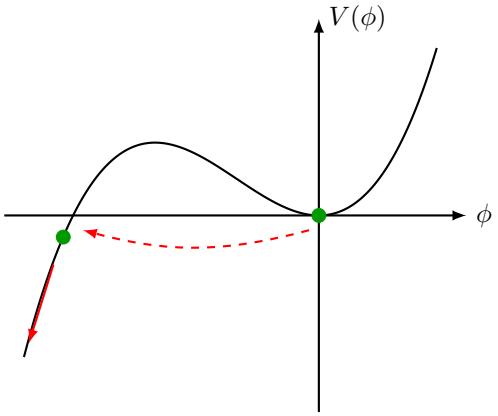


Figure 2.1: Plot of the scalar potential $V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{3!}\phi^3$. The origin $\phi = 0$ represents a metastable **local minimum**, which serves as the starting point for standard perturbation theory. However, the presence of the cubic term implies that **the potential is not bounded from below** for $\phi \to -\infty$. Consequently, quantum mechanics allows for a **tunnel effect into the unstable configuration** (indicated by the dashed red arrow). Once the field tunnels through the classical potential barrier, it inexorably rolls down towards configurations with arbitrarily negative energies (solid red arrow). This global instability implies that such a model cannot support a stable universe.

Because of the cubic term, **the potential is not bounded from below**. As $\phi \to -\infty$ (assuming $\lambda > 0$), $V(\phi) \to -\infty$. This means there exist field configurations with arbitrarily negative energies. While $\phi = 0$ is a local minimum (a metastable vacuum), quantum mechanics allows the field to undergo the *tunnel effect* into the completely unstable configuration. Such a theory cannot support a stable universe.

Nevertheless, we still extensively use ϕ^3 theory as a mathematical **toy model**. The reason is that standard perturbation theory is an expansion purely around the local minimum at $\phi = 0$ and is completely "blind" to the global instability of the potential. It serves as an excellent, simplified arena to learn Feynman diagrams and renormalization without the algebraic clutter of realistic models with spin.

ϕ^4 Theory

The ϕ^4 theory is often considered the simplest "realistic" model of an interacting scalar field. Unlike the ϕ^3 theory, its potential is bounded from below, making the vacuum state stable. The Lagrangian is constructed by adding a quartic interaction term to the free Klein-Gordon Lagrangian:

$$\mathcal{L} = \mathcal{L}_{\text{K.G.}} + \mathcal{L}_{\text{INT}} = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$$

From dimensional analysis, since $[\phi] = 1$ and the interaction term must have dimension 4, the coupling constant λ is **dimensionless** ($[\lambda] = 0$). This implies the theory is perfectly renormalizable in 4 spacetime dimensions.

By applying the Euler-Lagrange equations, we obtain the equation of motion for the interacting field:

$$(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3$$

Remark. *Notice that the equation of motion is* ***no longer linear*** *in ϕ. In a free theory (linear equation), we can solve for the field by superposing independent Fourier modes (plane waves). Here, the non-linearity implies that we cannot solve for the Fourier modes independently; the*

λ term dynamically couples them together. Physically, this coupling of Fourier modes is the mathematical manifestation of ***interactions (scattering) between the quanta*** *of the field.*

ϕ^n General Theory

We can generalize the self-interacting scalar field to an arbitrary power n:

$$\mathcal{L} = \mathcal{L}_{\text{K.G.}} - \frac{\lambda}{n!}\phi^n$$

Using dimensional analysis, the mass dimension of the coupling constant is given by:

$$[\lambda] = 4 - n$$

For any $n > 4$, we find $[\lambda] < 0$. As discussed previously, this means that scalar theories with interaction powers higher than 4 are **non-renormalizable** in 4 spacetime dimensions, losing predictive power at arbitrarily high energy scales.

Complex Scalar Field Interactions

For a complex scalar field ϕ, the free Lagrangian is $\mathcal{L}_{\text{FREE}} = |\partial_\mu \phi|^2 - m^2|\phi|^2$. A typical interaction term is the quartic self-interaction:

$$\mathcal{L}_{\text{INT}} = -\frac{\lambda}{4}|\phi|^4 = -\frac{\lambda}{4}(\phi^*\phi)^2$$

Remark. *The free Lagrangian possesses a global $U(1)$ continuous symmetry: it is invariant under the phase transformations $\phi \to e^{i\alpha}\phi$ and $\phi^* \to e^{-i\alpha}\phi^*$. If we demand that the interacting theory preserves this fundamental symmetry, it imposes severe constraints on the allowed interaction terms $\mathcal{L}_{INT}$. Specifically, operators that have an unequal number of ϕ and ϕ^* fields (such as ϕ^n or $(\phi^*)^n$) will transform with a net phase $e^{in\alpha}$ and are strictly* ***not allowed****.*

Quantum Electrodynamics (QED)

Quantum Electrodynamics (QED) is one of the most successful and experimentally verified QFTs in physics. It describes the interaction between a spin-1/2 Dirac fermion (like the electron) and a massless spin-1 vector field (the photon). The total Lagrangian is the sum of the free Maxwell, free Dirac, and interaction Lagrangians:

$$\mathcal{L} = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\not{\partial} - m)\psi}_{\mathcal{L}_{\text{FREE}}} + \mathcal{L}_{\text{INT}}$$

The interaction term couples the electromagnetic 4-potential A_μ to the conserved Noether current J^μ of the Dirac field:

$$\mathcal{L}_{\text{INT}} = A_\mu J^\mu \qquad \text{with} \quad J^\mu = -e\bar{\psi}\gamma^\mu\psi$$

Here, the coupling constant "$-e$" represents the fundamental electric charge. The current J^μ is exactly the conserved current arising from the global $U(1)$ symmetry of the free Dirac Lagrangian ($\psi \to e^{i\alpha}\psi$).

We can elegantly rewrite the full Lagrangian by combining the derivative in the Dirac term with the interaction term. We introduce the **Covariant Derivative** $D_\mu \equiv \partial_\mu + ieA_\mu$. Replacing $\partial_\mu \to D_\mu$ in the free Dirac Lagrangian yields the full QED Lagrangian:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\not{D} - m)\psi$$

Remark. *This theoretical construction is profound: it promotes the global $U(1)$ symmetry to a* ***Local Gauge Symmetry****. The QED Lagrangian is perfectly invariant under local phase transformations where the parameter α depends on spacetime coordinates $\alpha(x)$:*

$$\begin{aligned} A_\mu(x) &\to A_\mu(x) - \frac{1}{e}\partial_\mu \alpha(x) \\ \psi(x) &\to e^{i\alpha(x)}\psi(x) \\ \bar{\psi}(x) &\to e^{-i\alpha(x)}\bar{\psi}(x) \end{aligned}$$

This invariance is easily verified by noting that the covariant derivative of the field transforms exactly like the field itself: $D_\mu\psi \to e^{i\alpha(x)}(D_\mu\psi)$, allowing the phase to trivially cancel with the one from $\bar{\psi}$.

Yukawa Theory

The Yukawa theory describes the interaction between a Dirac fermion field ψ and a real scalar field ϕ. The total Lagrangian is:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m_\phi^2\phi^2 + \bar{\psi}(i\not{\partial} - m_f)\psi + \mathcal{L}_{\text{INT}}$$

The interaction term is defined as:

$$\mathcal{L}_{\text{INT}} = -g\bar{\psi}\psi\phi$$

Dimensional analysis confirms that $[\bar{\psi}\psi\phi] = \frac{3}{2}+\frac{3}{2}+1 = 4$, so the Yukawa coupling g is dimensionless ($[g] = 0$), making the theory renormalizable. The physics is conceptually similar to QED, but the fundamental force is mediated by a scalar particle instead of a vector particle. Crucially, the couplings between the Higgs boson and fermions in the Standard Model take exactly this Yukawa form!

Remark. *Following the strict theoretical criterion that an interacting Lagrangian must include all terms allowed by its symmetries up to dimension 4, the naive Yukawa interaction written above is mathematically incomplete. Since there is no symmetry preventing scalar self-interactions, we must rigorously add two more terms to $\mathcal{L}_{INT}$ for the theory to be fully consistent and renormalizable under loop corrections:*

$$\mathcal{L}_{INT} = -g\bar{\psi}\psi\phi - \frac{\lambda}{4!}\phi^4 - \frac{\lambda'}{3!}\phi^3$$

(Note: A linear term proportional to ϕ is also technically allowed, but it can always be eliminated by performing a constant field shift $\phi \to \phi + const$, which simply re-defines the masses and coupling constants of the other terms).

Scalar QED (sQED)

Scalar QED describes the gauge interaction between the electromagnetic field and a charged (complex) scalar field. The Lagrangian is obtained via the "minimal coupling" prescription, replacing the standard derivative with the covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ in the free complex scalar Lagrangian:

$$\mathcal{L} = \mathcal{L}_{\text{MAXWELL}} + \mathcal{L}_{\text{COMPLEX SCALAR}}\big|_{\partial_\mu \to D_\mu} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + |D_\mu\phi|^2 - m^2|\phi|^2$$

Expanding the kinetic term $|D_\mu \phi|^2$ reveals both a 3-point interaction vertex ($A_\mu \phi^* \partial^\mu \phi$) and a 4-point "seagull" interaction vertex ($A_\mu A^\mu |\phi|^2$).

Similarly to standard QED, this Lagrangian is fully invariant under local $U(1)$ gauge transformations.

Remark. *Once again, applying the criterion of including all symmetry-allowed terms up to mass dimension 4, we must explicitly add a scalar self-interaction term "* $-\frac{\lambda}{4}|\phi|^4$*" to the sQED Lagrangian, as it is perfectly invariant under local* $U(1)$ *transformations and is required for the renormalizability of the theory.*

2.1.2 | Master Plan

In the next sections, we will introduce interactions in the theory by adding interaction terms to the Lagrangian. The simplest way to do this is to add a term that is a polynomial in the field ϕ, such as $\lambda\phi^4$ or $g\phi^3$. These interaction terms will lead to non-linear equations of motion, which will make the theory more complicated to solve. However, they will also allow us to describe a wider range of physical phenomena, such as scattering processes and particle decays.

More precisely, we are going to:

- Define useful concepts and observables in the theory, such as **cross sections** and **decay rates**.
- Express these observables in terms of the **S-matrix**, which encodes the transition amplitudes between initial and final states in scattering processes: the matrix elements of a generic operator represent the probability amplitude of measuring a specific eigenvalue, corresponding to a transition from an initial state $|i\rangle$ to a final state $|f\rangle$.
- We will use theorems and techniques developed in a QFT framework, such as the **LSZ reduction formula**, to relate the S-matrix elements to correlation functions of the fields or their Fourier transforms, which are the fundamental objects of study in QFT. For example, correlation functions of the form

  $$\langle 0| T\{\phi(x_1)\phi(x_2)\cdots\phi(x_n)\} |0\rangle ,$$

  will show how the fields interact at different points in spacetime, and can be used to compute scattering amplitudes and other physical observables.
- We will develop in the end procedures and techniques to compute these correlation functions from an interacting lagrangian in **perturbation theory**; using **Feynman diagrams** and **Feynman rules** we will be able to systematically calculate the contributions of different interaction terms to the physical observables.

2.2 | Scattering Processes

Scattering processes are currently the most important tool to probe the fundamental interactions of nature, and thus to test our theories of particle physics. They let us infer and understand the properties of particles and their interactions at a microscopic level by analyzing the outcomes of experiments and collisions at macroscopic distances.

We are forced to cite some of the most important results of scattering theory, as well as experiments, which is a vast and complex subject, in order to understand its importance in particle physics. Some of the most important results and experiments in scattering theory include:

- **Rutherford scattering**, which was the first experiment to probe the structure of the atom and led to the discovery of the nucleus. Scattering alpha particles off a thin gold foil, Rutherford observed that most of the particles passed through the foil with little deflection, while a small fraction were scattered at large angles, leading to the conclusion that the atom has a small, dense nucleus at its center. This experiment lead to our modern understanding of the atomic structure of matter at a subatomic level, and it was a crucial step in the development of quantum mechanics and quantum field theory.
- **Lowrence Berkeley National Laboratory**, which was the first particle accelerator and allowed for the discovery of many new particles and interactions: in 1931 Lowrence was the first to accelerate particles to relativistic energies, firstly only around 80 keV, and then with Livingston he built the first cyclotron, which was able to accelerate protons to energies of about 1.22 MeV, leading to the discovery of the neutron in 1932 by James Chadwick.
- **Large Hadron Collider (LHC)**, which is the largest and most powerful particle accelerator in the world, located at CERN, and has been instrumental in the discovery of the Higgs boson in 2012, as well as in probing the properties of the Standard Model and searching for new physics beyond it. It has a circumference of about 27 km and can accelerate protons to energies of up to 7 TeV per beam, allowing for collisions at unprecedented energies and luminosities (actually around 13 TeV, with future upgrades planning further increases).
- **Future colliders**, such as the proposed Future Circular Collider (FCC) at CERN, which aims to reach even higher energies and luminosities than the LHC, and could potentially discover new particles and interactions that are beyond the reach of current experiments. Basically all we know about particle physics and the fundamental interactions of nature has been inferred from the observations and measurements of scattering processes, either in natural phenomena (such as cosmic rays) or in controlled experiments.

Such great energies and luminosities allow us to probe the fundamental interactions of nature at a microscopic level, and to test our theories of particle physics with high precision: the **indetermination principle** allows us to probe smaller and smaller distances by increasing the energy of the particles, thus allowing us to explore the structure of matter and the fundamental forces at a deeper level. By analyzing the outcomes of scattering experiments, we can infer the properties of particles and their interactions, and test our theoretical models against experimental data.

Also the **decay rate** of unstable particles is an important observable that can be measured in experiments and computed in theory, providing insights into the properties of the particles and their interactions. In these decay processes, an unstable particle decays into other smaller and lighter particles, and the decay rate quantifies the probability per unit time that this decay will occur.

By studying the decay rates of particles, we can gain insights into their properties, such as their masses, lifetimes, and interaction strengths, and test our theoretical models against experimental data.

2.2.1 | S-Matrix

We prepare a state $|i,\, t_i\rangle$ at an initial time t_i which will transition after some interaction (or decay, scatter...) into a state $|f,\, t_f\rangle$, measured at a final time t_f with a probability given by the square of the amplitude $|\langle f,\, t_f | i,\, t_i\rangle|^2$. Here we can make the following observations:

- The initial and final states are defined in the limit of infinite past and future, where the particles are free and non-interacting. This is a common assumption in scattering theory, as it allows us to define **asymptotic states** that can be used to compute scattering amplitudes and cross sections.
- The process happens on microscopic time and length scales, while being measured on macroscopic scales, thus we can treat the interaction as an instantaneous event, letting us ignore the detailed time dependencies of the process and focus on the transition between the initial and final states.
- The transition from the initial to the final state can be described by a unitary operator called the **S-matrix**, which encodes the **transition amplitudes** between the asymptotic states.

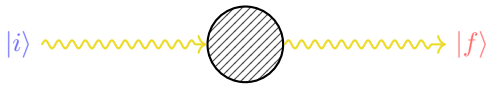


Figure 2.2: Schematic representation of a generic process, where an initial state $|i\rangle$ evolves into a final state $|f\rangle$ through some interaction described by the S-matrix. We are considering asymptotic states, which are defined in the limit of infinite past and future, where the particles are free and non-interacting. The S-matrix encodes the transition amplitudes between these asymptotic states describing the scattering process happening on microscopic time scales.

We tipically use **Heisenberg picture**, so if we want to compute the amplitude of a determined process, we can write it as

$$\langle f|\, \hat{S}\, |i\rangle = \langle f, \infty | i, -\infty\rangle\,,$$

where $|i\rangle$ and $|f\rangle$ are the asymptotic states of the system (time independent and directly related to the free particle states), respectively. *The S-matrix element* $\langle f|\, \hat{S}\, |i\rangle$ *gives us the amplitude for the transition* from the initial state $|i\rangle$ to the final state $|f\rangle$ due to the interaction described by the S-matrix itself. The S-matrix is a powerful tool in quantum field theory, as it allows us to compute **scattering amplitudes** and **cross sections** for a wide range of processes, and to make predictions that can be tested against experimental data. By analyzing the S-matrix elements for different processes, we can gain insights into the properties of particles and their interactions, testing our theoretical models against experimental observations.

2.3 | Cross Sections

If we analyze a scattering process, such as the Rutherford scattering, we can see how a beam of particles is scattered by a target; in the case of Rutherford scattering, it were alpha particles being scattered by a thin gold foil, where the alpha particles are much smaller than the nucleus of the gold atoms, thus we can treat the nucleus as a still sphere and the alpha particles as point-like particles. We will make the same assumption for the following scattering processes, thus treating the particles as point-like and the interactions as instantaneous events.

In this case, we can define the **cross section** of the target, which quantifies the probability of a scattering event occurring when a particle from the beam interacts with the target. The cross section is an effective area that represents the likelihood of a scattering event, and it can be computed using the S-matrix elements for the process.

The **geometric cross section** of the target, which encodes the effective area of the target for scattering, is given by

$$\sigma \equiv \pi R^2, \tag{2.3.1}$$

where R is the radius of the target. One can compute the number of particles that are scattered by the target, which is given by

$$N = n_B \cdot \sigma, \quad n_B = \frac{N_B}{A},$$

where n_B is the number density of the beam, N_B is the number of particles in the beam, and A is the area of the target. The cross section σ can be interpreted as an effective area that quantifies the likelihood of a scattering event occurring when a particle from the beam interacts with the target. The power of this approach resides in its simplicity: with few assumptions it allows us to compute the probability of scattering events for a wide range of processes, while linking macroscopical observables, such as the number of scattered particles, to the microscopic dynamics of the interactions encoded in the S-matrix elements.

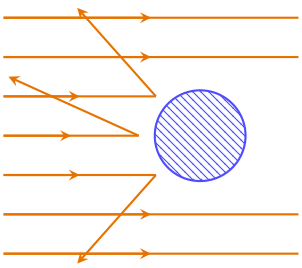


Figure 2.3: Schematic representation of a scattering process, where a beam of particles (orange arrows) is scattered by a target (blue circle). The geometric cross section of the target is given by $\sigma = \pi R^2$, with R being the radius of the target, while the number of scattered particles depends on the interaction probability. We are going to generalize this model to quantum field theory, computing the probability of scattering processes and expressing them in terms of the S-matrix elements.

Now we want to take this model and generalize it to quantum field theory, where we can compute the probability of scattering processes; we want to extend the model

- for any type of scattering process, or interaction;
- for a probability of scattering, not just a number of particles.

2.3.1 | Impact Parameter

If we assume a beam of particles along the z-axis, we can define the **impact parameter b**, which represents the perpendicular distance among the trajectory of the incoming particle and the center of the target:

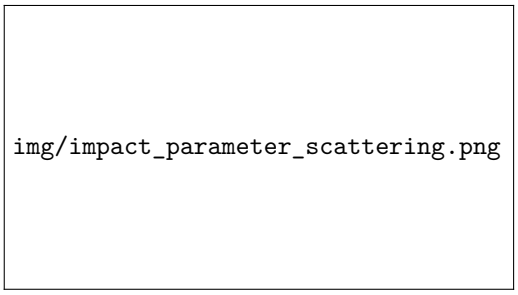
$$N = \int \mathrm{d}^2\mathbf{b}\, n_B\, P(\mathbf{b}) = n_B \int \mathrm{d}^2\mathbf{b}\, P(\mathbf{b}),$$

thus we can now define the **cross section** as

$$\sigma = \int \mathrm{d}^2\mathbf{b}\, P(\mathbf{b}), \tag{2.3.2}$$

where $P(\mathbf{b})$ is the probability of scattering for a given impact parameter $\mathbf{b}$, generalizing the concept of a geometric cross section. This expression allows us to compute the cross section for a scattering process by integrating the probability of scattering over all possible impact parameters, thus providing a more general and flexible framework for analyzing scattering processes in quantum field theory.

Figure 2.4: Schematic representation of a scattering process in terms of the impact parameter $\mathbf{b}$: the probability of scattering can be expressed as a function of the impact parameter, allowing us to compute the cross section by integrating the probability of scattering over all possible impact parameters.



We can also express it in its **differential form**, which ideally counts only a subset of events, with respect to the **solid angle** Ω as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \int \mathrm{d}^2\mathbf{b}\, \frac{P(\mathbf{b})}{\mathrm{d}\Omega}, \tag{2.3.3}$$

where we are considering only the events that are scattered in a specific direction defined by the solid angle $\mathrm{d}\Omega$, thus providing a more detailed description of the scattering process. Alternatively, we can express it with respect to the measure of the space of **final states** $|f\rangle$ as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}f} = \int \mathrm{d}^2\mathbf{b}\, \frac{P(\mathbf{b})}{\mathrm{d}f}. \tag{2.3.4}$$

The consequent step towards the computation of the cross section in QFT are:

- Express the probability of scattering $P(\mathbf{b})$ in terms of the S-matrix elements for the process

  $$\langle f|\, \hat{S}\, |i\rangle = \langle f, \infty | i, -\infty \rangle\,,$$

  which encode the transition amplitudes between the initial and final states.

- Integrate the probability of scattering over all possible impact parameters $\mathbf{b}$ to compute the cross section for the scattering process, using the expressions in equations (2.3.2), (2.3.3), or (2.3.4); then express the result in terms of observables that can be measured in experiments, such as the scattering angle or the energy or momenta of the scattered particles.

2.3.2 | Wave Packets and Regularization

Now one of the natural questions that arises is: what are the initial and final states $|i\rangle$ and $|f\rangle$ that we should use to compute the S-matrix elements for the scattering process? In typical scattering processes, the quantum dispersion of the particle momenta Δp is smaller than the experimental resolution, thus it makes sense to use **eigenstates of the momentum operator**, such as plane wave states. However, there are some issues:

- If the quantum dispersion of the particle momenta Δp is comparable to zero, for the Heisenberg uncertainty principle, the particle is completely delocalized in space

  $$\Delta p \Delta x \geq \frac{\hbar}{2}, \quad \Delta p = 0 \implies \Delta x \to \infty,$$

  thus it can interact with the target at any point in space, leading to divergences in the scattering amplitudes.

- We will show that S-matrix elements for plane wave states are proportional to delta functions, which leads to divergences when we compute the probability of scattering by taking the square of the amplitude: the probability of scattering is given by the square of the amplitude $\langle f | \hat{S} | i \rangle \sim \delta^{(4)}(p_i - p_f)$, but then

  $$P \sim |\langle f | \hat{S} | i \rangle|^2 \sim |\delta^{(4)}(p_i - p_f)|^2 = \delta^{(4)}(p_i - p_f)\delta^{(4)}(0).$$

  The term $\delta^{(4)}(0)$ is divergent, and it arises from the fact that we are using plane wave states, which are not normalizable and lead to divergences in the scattering amplitudes. This is a common issue in quantum field theory, and it requires regularization techniques to handle these divergences and obtain finite results for physical observables.

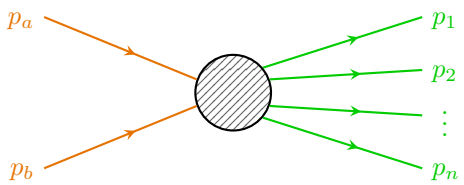
One way to overcome the last problem is to consider a finite volume V and a finite time T, so that the delta function can be regularized and then take only at the end the limit $V \to \infty$ and $T \to \infty$. This is a common technique in quantum field theory to handle divergences and make sense of scattering amplitudes. By working in a finite volume and time, we can ensure that the calculations are well-defined and then take the appropriate limits to recover physical results. However, this approach can be obscure and cumbersome, and it can lead to complications in the calculations.

TODO: Add reference to book by schwarz.

We will instead follow the Peskin and Schroeder approach, which is to consider the states as **wave packets** in the momentum space, which are **localized in both position and momentum space**, thus avoiding the issues with the delta function. This approach allows us to compute scattering amplitudes and cross sections without encountering the divergences associated with plane wave states. By using wave packets, we can ensure that the initial and final states are well-defined and that the scattering probabilities are finite and physically meaningful.

TODO: add reference to Peskin and Schroeder.

We can consider a general process in which the initial state is a two particle state, with momenta p_a and p_b, while the final state is a N particle state:



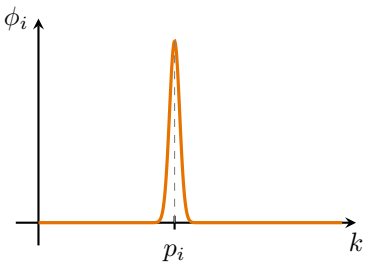
For incoming states we replace $|p_i\rangle$ with $|\phi_i\rangle$, which is a **wave packet** centered around the momentum p_i:

$$|p_i\rangle \to |\phi_i\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{k}}} \phi_i(\mathbf{k}) \, |\mathbf{k}\rangle \,,$$

with

$$\langle \phi_i | \phi_i \rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} |\phi_i(\mathbf{k})|^2 = 1,$$

remembering that $\langle \mathbf{k} | \mathbf{k}' \rangle = 2E_\mathbf{k} \, (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{k}')$; all of this indicates that $\phi_i(\mathbf{k})$ is a very narrow normalized function centered around p_i.



Example (LHC collider). We can assume, very conservatively, that the LHC collider works with $\Delta p \sim 1\,\mathrm{MeV}$, which is unrealistic (since it would correspond to a resolution of about 10^{-6}, not realistic) and leads to a delocalization of the particles of about

$$\Delta x \sim \frac{\hbar}{2\Delta p} \sim \frac{1}{2\,\mathrm{MeV}} \sim 100\,\mathrm{fm} = 10^{-13}\,\mathrm{m},$$

which, compared to the size of the particles bunches in the LHC, which is of the order of $1\,\mu\mathrm{m} = 10^{-6}\,\mathrm{m}$, is negligible, thus we can safely treat the particles as localized in space, and thus we can use wave packets to compute the scattering amplitudes and cross sections without encountering divergences.

This is a common assumption in high-energy physics, with typically well-localized particles and small quantum dispersion of their momenta compared to the experimental resolution; this allows us to use wave packets to describe the initial and final states in scattering processes.

Now we are left with the task of computing the S-matrix elements for the scattering process using wave packets, and then integrating the probability of scattering over all possible impact parameters to compute the cross section for the scattering process. But how are we going to model the wave packets? We can consider different types of wave packets, such as **Gaussian wave packets**, which are commonly used in quantum mechanics and quantum field theory due to their nice mathematical properties and their ability to represent *localized states in both position and momentum space*. By choosing appropriate wave packet profiles, we can ensure that the initial and final states are well-defined and that the scattering probabilities are finite and physically meaningful.

Example (Gaussian wave packet). We can consider a Gaussian wave packet centered around p_i with a width Δp:

$$\phi(\mathbf{k}) = N \exp\left[-\frac{1}{4} \left(\frac{(\mathbf{k} - \mathbf{p})^2}{\Delta p} \right)^2 - i\mathbf{k} \cdot \langle \mathbf{x} \rangle \right], \tag{2.3.5}$$

where N is a normalization constant, Δp is the width of the wave packet in momentum space, and $\langle \mathbf{x} \rangle$ is the average position of the wave packet in real space: using $\hat{x}^\mu = i \frac{\partial}{\partial p_\mu}$, one can check that the uncertainty principle is satisfied with $\Delta x \Delta p = \frac{1}{2}$.

Thus the wave packet is normalized and has a well-defined momentum distribution, with the properties of a momentum eigenstate while being relatively well localized in space for finite Δp. By using Gaussian wave packets, we can compute the S-matrix elements for the scattering process without encountering divergences, and we can ensure that the initial and final states are well-defined and physically meaningful.

Initial states. Sticking to the Gaussian wave packet example, we can write the initial states as the following superposition of momentum eigenstates:

$$|p_a\rangle \to |\phi_a\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \phi_a(\mathbf{k}) \, |\mathbf{k}\rangle \,,$$
$$|p_b\rangle \to |\phi_b\rangle = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{k}}}} \phi_b(\mathbf{k}) \, e^{-i\mathbf{k}\cdot\mathbf{b}} \, |\mathbf{k}\rangle \,,$$

where we shifted the second wave packet by the impact parameter $\mathbf{b}$, defined as

$$\mathbf{b} = \langle \phi_b | \, \hat{\mathbf{x}}^{\perp} \, | \phi_b \rangle \,,$$

with $\hat{\mathbf{x}}^{\perp}$ being the position operator in the plane perpendicular to the beam axis. Looking at the expression for the gaussian wave packet in (2.3.5), we can think of having took out of ϕ_b the exponential factor $e^{-i\mathbf{k}\cdot\langle\hat{\mathbf{x}}^{\perp}\rangle}$, whereas in ϕ_a it can be set to zero by choosing the average position of the wave packet to be at the origin. In this way we can model the initial states as wave packets that are localized in both position and momentum space, with the second wave packet shifted by the impact parameter $\mathbf{b}$ with respect to the first one, to account for the spatial separation of the incoming particles in the scattering process.

Outgoing states. For the outgoing states we can consider just the momentum eigenstates, since the scattering has already happened, thus we can write

$$|\phi_i\rangle = |p_i\rangle \,, \quad \forall i = 1, \ldots, N.$$

Since the outgoing particles are detected at macroscopic distances, we can treat them as free particles and thus we can use momentum eigenstates to describe them: the scattering already happened, thus we can treat the outgoing particles as free particles.

Now we can compute the **S-matrix elements** for the scattering process using the initial and final states defined above. We can write the S-matrix elements for momentum eigenstates as

$$\langle f, \infty | i, -\infty \rangle = \langle f | \, \hat{S} \, | i \rangle = \langle p_1, \ldots, p_N | \, \hat{S} \, | p_a, p_b \rangle \,,$$

where we are considering the initial state as a two particle state with momenta p_a and p_b, and the final state as a N particle state with momenta $p_1, \ldots, p_N$. The S-matrix can be expressed as

$$\hat{S} = \mathbb{I} + i\hat{T},$$

where the identity term corresponds to the case where no scattering occurs, while the $i\hat{T}$ term corresponds to the case some interactions among the fields, i.e. the scattering occurs. Thus we can write[1]

$$\langle p_1, \ldots, p_N | \, i\hat{T} \, | p_a, p_b \rangle = (2\pi)^4 \delta^{(4)}(p_a + p_b - \sum_{i=1}^{N} p_i) i\mathcal{M},$$

where the delta function ensures the conservation of energy and momentum, and $\mathcal{M}$ is the **invariant matrix element** or **scattering amplitude** for the process, which encodes the dynamics

[1] Note that we evaluate the transition amplitude here using exact momentum eigenstates $|p_a, p_b\rangle$ rather than the initial wave packets $|\phi_a, \phi_b\rangle$. This is done to formally define the invariant scattering amplitude $\mathcal{M}$ and cleanly factor out the energy-momentum conserving Dirac delta $\delta^{(4)}$. The physical wave packets enter the calculation in the subsequent step: exploiting the linearity of the $\hat{T}$ operator, the realistic transition amplitude is obtained by integrating these exact-momentum matrix elements over the momentum distributions $\phi_a(\mathbf{k})$ and $\phi_b(\mathbf{k})$ of the incoming wave packets.

of the interaction (we omitted its dependencies for simplicity, but it relies on the initial and final states, thus $\mathcal{M} = \mathcal{M}(p_a, p_b \to p_1, \dots, p_N)$). The invariant amplitude can be computed using Feynman diagrams and the rules of quantum field theory, and it is a key quantity in determining the cross section for the scattering process.

Thus we can now compute the **differential cross section** in the space of final configurations as in (2.3.4):

$$\frac{\mathrm{d}\sigma}{\mathrm{d}f} = \int \mathrm{d}^2\mathbf{b}\, \frac{\mathrm{d}P}{\mathrm{d}f},$$

where the differential probability of scattering in the space of final states is given by

$$\frac{\mathrm{d}P}{\mathrm{d}f} = |\, \langle p_1, \dots, p_N, +\infty |\, i\hat{T}\, | \phi_a, \phi_b, -\infty \rangle\, |^2,$$

and the measure of the space of final states for N particles is normalized as[2]

$$\mathrm{d}f = \prod_{i=1}^{N} \frac{\mathrm{d}^3\mathbf{p}_i}{(2\pi)^3} \frac{1}{2E_{p_i}}.$$

TODO: write it.

Now we can put everything back together and after some algebra (full computation in Appendix B) we can write the final expression for the differential cross section

$$\begin{aligned}
\mathrm{d}\sigma &= \frac{1}{\Phi} |\mathcal{M}(p_a, p_b \to p_1, \dots, p_N)|^2 \mathrm{d}\Pi_N, \\
\mathrm{d}\Pi_N &= \left[\prod_{i=1}^{N} \frac{\mathrm{d}^3\mathbf{p}_i}{(2\pi)^3 2E_{p_i}} \right] (2\pi)^4 \delta^{(4)}(p_a + p_b - \sum_{i=1}^{N} p_i), \\
\Phi &= 2E_a 2E_b |v_a - v_b| = 4|E_b p_a^z - E_a p_b^z|,
\end{aligned}$$

where:

- $\mathrm{d}\Pi_N$ is the Lorentz-invariant phase space measure for N final state particles, which accounts for the density of final states and the conservation of energy and momentum through the delta function.
- Φ is the flux factor, which accounts for the incoming particles and their relative velocity, ensuring that the cross section is properly normalized with respect to the initial state of the scattering process.
- $\mathcal{M}$ is again the invariant matrix element or scattering amplitude, which encodes the dynamics of the interaction and can be computed using Feynman diagrams and the rules of quantum field theory.

Φ is the only non Lorentz invariant term: we have indeed

$$\Phi = 4|E_b p_a^z - E_a p_b^z|,$$

where z is the beam axis, thus it is not invariant. But we can recognize its invariance under:

[2] We can convince ourself with the resolution of the identity for 1 particle states, which reads

$$\mathbb{I} = \int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{2E_{\mathbf{k}}} |\mathbf{k}\rangle \langle \mathbf{k}|,$$

where $\mathrm{d}f = \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3} \frac{1}{2E_{\mathbf{k}}}$ is the measure of the space of final states for 1 particle.

1. boosts along the beam axis, since E_a and p_a^z transform in the same way, and E_b and p_b^z also transform in the same way, thus the combination $E_b p_a^z - E_a p_b^z$ is invariant under boosts along the beam axis.

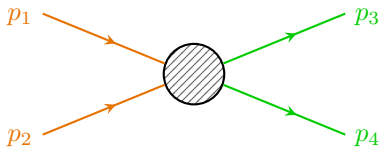
2. rotations around the beam axis (in the perpendicular plane), since E_a, E_b, p_a^z and p_b^z are scalars under rotations around the beam axis, thus the combination $E_b p_a^z - E_a p_b^z$ is invariant under rotations around the beam axis.

Thus the flux factor Φ transforms as the inverse of an area in the plane perpendicular to the beam axis, behaving as a cross section (which is consistent with the interpretation of the cross section as an effective area that quantifies the likelihood of a scattering event occurring when a particle from the beam interacts with the target).

Even if we focus on the case of a beam of particles scattering on a target, the computed expression for the differential cross section remains symmetric for exchange of beam and target, thus it can be applied to any scattering process, regardless of the specific roles of the particles involved. We can also imagine to use it for collisions of two beams of particles, as in e^+e^- collisions in LEP, $p\bar{p}$ collisions in the Tevatron, or more recently pp collisions in the LHC.

We are basically performing a change of reference frame in order to reverse the problem: we can consider the target as the beam and the beam as the target, it all depends on the symmetries and the choice of the reference frame. This is a common technique in scattering theory, where we can choose a reference frame that simplifies the calculations (even center of mass, it depends) and makes it easier to analyze the scattering process. By considering different reference frames, we can gain insights into the dynamics of the interaction and compute cross sections more efficiently.

Example ($2 \to 2$ scattering in center of mass frame). We can consider a simple scattering process where two particles collide and produce two particles in the final state. In the center of mass frame, the initial momenta of the particles are equal and opposite, and we can analyze the scattering process using the invariant amplitude $\mathcal{M}$ and the phase space factor Φ to compute the differential cross section for this process.



The incoming particles have momenta p_1 and p_2, and the outgoing particles have momenta p_3 and p_4, and their spatial parts is conserved: $\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{p}_3 + \mathbf{p}_4 = 0$ (since we are in the center of mass frame). The energy of the particles is given by the energy momentum relation $E_i = \sqrt{\mathbf{p}_i^2 + m_i^2}$, where m_i is the mass of the particle. We can write

$$\mathrm{d}\Pi_2 = \frac{\mathrm{d}^3\mathbf{p}_3}{(2\pi)^3 2E_3} \frac{\mathrm{d}^3\mathbf{p}_4}{(2\pi)^3 2E_4} (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4),$$

which, after integrating in $\mathrm{d}^3\mathbf{p}_4$ using the delta function, gives $\mathbf{p}_4 = -\mathbf{p}_3$; using then polar coordinates for $\mathrm{d}^3\mathbf{p}_3 = p^2 \, \mathrm{d}p \, \mathrm{d}\Omega$, we can write

$$E_3 = \sqrt{p^2 + m_3^2}, \quad E_4 = \sqrt{p^2 + m_4^2}, \quad p = |\mathbf{p}_3| = |\mathbf{p}_4|,$$

and thus

$$\int \mathrm{d}p\, \delta(E_1 + E_2 - E_3 - E_4) = \left| \frac{p}{E_3} + \frac{p}{E_4} \right|^{-1} = \frac{E_3 E_4}{p(E_3 + E_4)} = \frac{E_3 E_4}{pE},$$

where the last energy E is the total energy of the system in the center of mass frame, which is given by $E = E_1 + E_2 = E_3 + E_4$. Thus we can write

$$\mathrm{d}\Pi_2 = \mathrm{d}\Omega \frac{p}{16\pi^2 E},$$

with a flux $\Phi = 2E_1 2E_2 |v_1 - v_2|$, with now

$$|v_1 - v_2| = \left| \frac{|\mathbf{p}_1|}{E_1} + \frac{|\mathbf{p}_2|}{E_2} \right| = p_{in} \frac{E}{E_1\, E_2},$$

where p_{in} is the magnitude of the incoming momenta (which are equal in the center of mass frame) $p_{in} = |\mathbf{p}_1| = |\mathbf{p}_2|$. Thus we can write the final expression for the differential cross section as

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{64\pi^2 E^2} \frac{p}{p_{in}} |\mathcal{M}|^2,$$

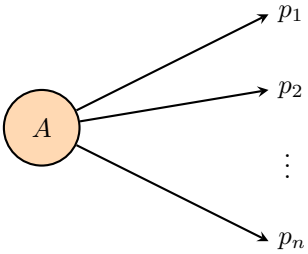
where we used the fact that $E_1 E_2 = \frac{E^2}{4}$ in the center of mass frame. This expression allows us to compute the *differential cross section for the 2 → 2 scattering process only in the center of mass frame*, given the invariant amplitude $\mathcal{M}$ and the kinematic variables of the process.

Remark. *Most of the time this problem is independent of the azimuthal angle ϕ, thus we can integrate over it and write $\mathrm{d}\Omega = 2\pi \mathrm{d}\cos\theta$, thus simplifying the expression for the differential cross section.*

Furthermore $\mathrm{d}\sigma$ is invariant under boosts along the beam axis, but the solid angle $\mathrm{d}\Omega$ is not, thus the expression for the differential cross section $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ is not invariant.

2.3.3 | Decay Rates

We can similarly deal with **unstable particles**, which after a certain lifetime can decay into other lighter particles. Systems composed by unstable particles can be analyzed using the S-matrix formalism in a similar way to how we compute scattering amplitudes and cross sections for stable particles. By analyzing the transition amplitudes for the decay process, we can determine the decay rate and the branching ratios for different decay channels, which provide insights into the properties of the unstable particle and its interactions with other particles.



The **decay rate** Γ of an unstable particle can be computed as

$$\Gamma = \frac{\text{Number of decays per unit time}}{\text{Number of particles}},$$

and we can define the **lifetime** τ of the particle as the inverse of the decay rate:

$$\tau = \frac{1}{\Gamma}.$$

Decay rates and lifetimes are defined in the rest frame of the particle, where the particle is at rest and its energy is equal to its mass $E = m$ and its momentum is zero $\mathbf{p} = 0$.

The decay rate can be computed using the S-matrix element for the transition from the initial state (the unstable particle) to the final state (the decay products), and it is related to the **invariant**

amplitude $\mathcal{M}$ for the decay process. We cannot directly apply the limit for $t_i \to -\infty$, since the particle is unstable and it would have decayed before, it cannot exist for an infinite amount of time. We therefore need to consider a finite time interval for the decay process, but we can make a simplification: if we assume the particle to have a lifetime way longer than the time scale of the interaction (always true for most practical cases), we can still apply the S-matrix formalism and compute the decay rate using the invariant amplitude for the decay process, and we will have to take into account the instability of the particle only in one step of the calculation.[3]

TODO: check footnote.

Defining the **differential decay rate** as

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}f} = \frac{1}{T}\mathrm{d}P, \tag{2.3.6}$$

where T is the time interval during which the decay process occurs $T = t_f - t_i$, and $\mathrm{d}P$ is the probability of decaying from the initial state to the final state within that time interval; we can compute the total decay rate by integrating over the phase space of the final states; let us first write the expression for the delta function in the limit of finite and large T:

$$\delta(0) = \int_{t_i}^{t_f} \mathrm{d}t\, e^{i(E_i - E_f)t}|_{t_f\to\infty, t_i\to-\infty},$$

which is the critical step in the calculation, since it allows us to regularize the delta function and make sense of the transition amplitude for the decay process; the complete calculation is shown in Appendix B. Having solved the subtleties of defining the asymptotic states for unstable particles, the final expression for the decay rate after integrating over the phase space of the final states is

TODO: write it.

$$\mathrm{d}\Gamma = \frac{1}{2m}|\mathcal{M}_{p_i\to p_f}|^2 \mathrm{d}\Pi_N, \tag{2.3.7}$$

where m is the mass of the unstable particle, $\mathcal{M}_{p_i\to p_f}$ is the invariant amplitude for the decay process, and $\mathrm{d}\Pi_N$ is the phase space factor for the N-particle final state, which accounts for the kinematics of the decay products and ensures that energy and momentum are conserved in the decay process. It is very similar to the expression for the differential cross section, with some differences concerning the flux factor term $\Phi = 2E_a 2E_b|v_a - v_b|$:

- $2E_a \to 2m$ since we are in the rest frame of the unstable particle, thus $E_a = m$;
- $2E_b|v_a - v_b|$ is not present since we are not dealing with a scattering process, but rather with a decay process, thus there is no incoming particle b and no relative velocity to consider.

This expression allows us to compute the decay rate for an unstable particle given the invariant amplitude and the kinematic variables of the decay process.

Identical particles. When dealing with identical particles among the decay products, the previous formulae are still valid, with the appropriate modifications to take into account the symmetrization (for bosons) or antisymmetrization (for fermions) of the wave function. This means that the scattering amplitude must be modified to include the appropriate symmetry factors, which can affect the calculation of cross sections and decay rates in order to consider the spin statistics theorem

$$|p_3,\, p_4\rangle = \pm\,|p_4,\, p_3\rangle\,.$$

[3] We will consider a finite time interval for the decay process and then take the limit of large time intervals to recover the usual expression for the decay rate.

This can be accomplished in two ways: one can restrict the integration over the phase space to a specific region that accounts for the indistinguishability of the particles (thus over all final inequivalent states), or one can include a symmetry factor in the calculation of the amplitude (such as the permutation number $\frac{1}{N!}$) to account for the number of identical particles in the final state. Both approaches ensure that the physical predictions are consistent with the principles of quantum mechanics and the statistics of identical particles.

2.4 | LSZ reduction and Green's functions

In this section we aim to show how the S-matrix elements can be computed from the Green's functions of the interacting theory. This is done by the LSZ reduction formula, which relates the S-matrix elements to the time-ordered correlation functions of the fields.

The Green's functions of the interacting theory are defined as the time-ordered correlation functions of the field operators in the interacting theory, evaluated in the vacuum state of the interacting theory:

$$\langle f|\,\hat{S}\,|i\rangle \to \langle\Omega|\,T\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle \tag{2.4.1}$$

where

- $\hat{\phi}$ is the field operator of the interacting theory,
- $|\Omega\rangle$ is the vacuum state of the interacting theory,

where we highlighted the difference among the vacuum state of the interacting theory and the free theory one: interactions can modify the structure of the vacuum and the Green's functions of the interacting theory encode all the information about its dynamics; they are the building blocks for computing physical observables such as scattering amplitudes and cross sections.

We now start from the structure analysis of the two point functions $\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle$ in an interacting theory, investigating its physical content and how it relates to the S-matrix elements $\langle f|\,\hat{S}\,|i\rangle$, finally relating it to the LSZ reduction formula for connecting the scattering matrix elements to the n-point Green's functions. We will start from a **real scalar theory** before generalizing to more complicated theories.

2.4.1 | Two Point Functions

TODO: Add reference to Peskin 7.1

To analyze the structure of the two point function, we start from the definition of the time-ordered two point function in the interacting theory:

$$\langle\Omega|\,T\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle\,,$$

the idea is now to insert a complete set of physical states $|\lambda\rangle$[4] as a resolution of the identity $\mathbb{I} = \sum_\lambda |\lambda\rangle\langle\lambda|$ between the two field operators; but we have to be more precise, since the vacuum state $|\Omega\rangle$ is a physical state of the theory, we have to separate it from the other states in the spectrum of the theory, so that we can write the resolution of the identity as

$$\mathbb{I} = |\Omega\rangle\langle\Omega| + \sum_{\lambda\neq\Omega} |\lambda\rangle\langle\lambda|\,,$$

where $|\lambda\rangle$ are seen as eigenstates of the momentum operator $\hat{\mathbf{P}}$ plus other quantum numbers, so that we can write them as $|\lambda_{\mathbf{p}}\rangle$ to make the momentum dependence explicit. They are normalized according to the relativistic normalization convention, so that we have

$$\langle\lambda_{\mathbf{p}}|\lambda'_{\mathbf{p}'}\rangle = (2\pi)^3 2E_{\mathbf{p}}(\lambda)\delta^{(3)}(\mathbf{p}-\mathbf{p}')\delta_{\lambda,\lambda'},$$

[4] Here we can interpret λ as the set of quantum numbers characterizing the different states of our theory.

where $E_{\mathbf{p}}(\lambda) = \sqrt{\mathbf{p}^2 + m_\lambda^2}$ is the energy of the state $|\lambda_{\mathbf{p}}\rangle$ with momentum $\mathbf{p}$ and mass m_λ. The sum over λ includes all the physical states of the theory, such as the one particle states, the two particle states, the bound states, etc.

We cited the mass m_λ of the state $|\lambda_{\mathbf{p}}\rangle$ as a label for the state, but it is important to note that the mass m_λ is not a free parameter, but it is determined by the dynamics of the theory and it is related to the eigenvalue of the momentum operator $\hat{P}^\mu$ on the state $|\lambda_{\mathbf{p}}\rangle$ by the relativistic dispersion relation:

$$\hat{P}^\mu |\lambda_{\mathbf{p}}\rangle = p^\mu |\lambda_{\mathbf{p}}\rangle , \quad p^2 = p_\mu p^\mu = m_\lambda^2,$$

so that the mass m_λ is determined by the eigenvalue of $\hat{P}^2$ on the state $|\lambda_{\mathbf{p}}\rangle$:

$$\hat{P}^2 |\lambda_{\mathbf{p}}\rangle = m_\lambda^2 |\lambda_{\mathbf{p}}\rangle .$$

Putting everything together, we can write the resolution of the identity as

$$\mathbb{I} = |\Omega\rangle \langle\Omega| + \sum_{\lambda \neq \Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| ,$$

with the factor of $1/(2E_{\mathbf{p}}(\lambda))$ coming from the relativistic normalization of the states $|\lambda_{\mathbf{p}}\rangle$.

If we now assume $x^0 > y^0$, the time-ordering operator T does not change the order of the fields, and we can neglect it in order to write the two point function as:

$$\begin{aligned}
\langle\Omega| \hat{\phi}(x)\hat{\phi}(y) |\Omega\rangle &= \langle\Omega| \hat{\phi}(x) \left(|\Omega\rangle \langle\Omega| + \sum_{\lambda \neq \Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \right) \hat{\phi}(y) |\Omega\rangle \\
&= \langle\Omega| \hat{\phi}(x) |\Omega\rangle \langle\Omega| \hat{\phi}(y) |\Omega\rangle + \sum_{\lambda \neq \Omega} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}(\lambda)} \langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle \langle\lambda_{\mathbf{p}}| \hat{\phi}(y) |\Omega\rangle ,
\end{aligned}$$

and if we now relate $\hat{\phi}(x)$ to $\hat{\phi}(0)$ by a spacetime translation, using the unitary operator $\hat{U}_{\mathbf{p}} = e^{i\hat{P}\cdot x}$ that implements the translation, so that

$$\hat{U}_{\mathbf{p}} = e^{i\hat{P}\cdot x} \implies \hat{\phi}(x) = e^{i\hat{P}\cdot x} \hat{\phi}(0) e^{-i\hat{P}\cdot x} = \hat{U}_{\mathbf{p}} \hat{\phi}(0) \hat{U}_{\mathbf{p}}^\dagger,$$

we can compute the terms by assuming that the vacuum is *translationally invariant*, and since we said that $|\lambda_{\mathbf{p}}\rangle$ are eigenstates of the momentum operator $\hat{P}$ with eigenvalue p, we can write:

$$\begin{aligned}
\langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle &= \langle\Omega| e^{i\hat{P}\cdot x} \hat{\phi}(0) e^{-i\hat{P}\cdot x} |\lambda_{\mathbf{p}}\rangle \\
&= e^{-ip\cdot x} \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{p}}\rangle ,
\end{aligned}$$

from vacuum invariance and momentum eigenstate properties. We can now assume a Lorentz transformation Λ and the corresponding unitary operator $\hat{U}(\Lambda)$ to exist such that $\Lambda \cdot p = (m_\lambda, \mathbf{0})$ and $\hat{U}(\Lambda) |\lambda_{\mathbf{p}}\rangle = |\lambda_{\mathbf{0}}\rangle$. Hence, inserting the identity operator $\hat{U}(\Lambda)^\dagger \hat{U}(\Lambda) = \mathbb{I}$ two times, we can write

$$\langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{p}}\rangle = \left(\langle\Omega| \hat{U}(\Lambda)^\dagger \right) \left(\hat{U}(\Lambda) \hat{\phi}(0) \hat{U}(\Lambda)^\dagger \right) \left(\hat{U}(\Lambda) |\lambda_{\mathbf{p}}\rangle \right) = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle ,$$

since the vacuum is invariant under Lorentz transformations, and the field operator transforms trivially: $\hat{\phi}(\Lambda 0) = \hat{\phi}(0)$. Thus we can write

$$\langle\Omega| \hat{\phi}(x) |\lambda_{\mathbf{p}}\rangle = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle e^{-ip\cdot x} = e^{-ip\cdot x} F(\lambda),$$

where we have defined the function $F(\lambda) = \langle\Omega| \hat{\phi}(0) |\lambda_{\mathbf{0}}\rangle$ which encodes the information about the coupling of the field operator to the state $|\lambda\rangle$. In the particular case of $|\lambda_{\mathbf{p}}\rangle = |\Omega\rangle$, we get

$$\langle\Omega| \hat{\phi}(x) |\Omega\rangle = v = \text{constant},$$

which is a consequence of the translational invariance of the vacuum state and represents the vacuum expectation value of the field operator. We can then always redefine the field operator by subtracting the vacuum expectation value

$$\phi(x) = v + \tilde{\phi}(x), \quad \langle\Omega|\,\tilde{\phi}(x)\,|\Omega\rangle = 0,$$

since the vacuum expectation value of the field operator does not affect the dynamics of the theory, and it can be absorbed into a redefinition of the field operator (we can always build a field operator such that its vacuum expectation value is zero, simplifying the analysis of correlation functions). Thus the two point function can be written (after the redefinition $\phi \to \tilde{\phi}$) as

$$\langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle = \sum_\lambda |F(\lambda)|^2 \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2E_\mathbf{p}(\lambda)}e^{-ip\cdot(x-y)},$$

valid for $x^0 > y^0$ and where the last term

$$D(x,\,y,\,m_\lambda) = \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{2E_\mathbf{p}(\lambda)}e^{-ip\cdot(x-y)} = \langle 0|\,\hat{\phi}_{0,\lambda}(x)\hat{\phi}_{0,\lambda}(y)\,|0\rangle\,, \tag{2.4.2}$$

is the **free propagator** of a particle with mass m_λ, with $\hat{\phi}_{0,\lambda}$ the free field operator of a particle with mass m_λ: it is the amplitude for a free scalar particle to propagate from point y to point x. We can now reintroduce the time-ordering to write the two point function in a more general form

$$\begin{aligned}\langle\Omega|\,T\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle &= \Theta(x^0-y^0)\,\langle\Omega|\,\hat{\phi}(x)\hat{\phi}(y)\,|\Omega\rangle + \Theta(y^0-x^0)\,\langle\Omega|\,\hat{\phi}(y)\hat{\phi}(x)\,|\Omega\rangle \\ &= \Theta(x^0-y^0)\sum_\lambda |F(\lambda)|^2 D(x,\,y,\,m_\lambda) + \Theta(y^0-x^0)\sum_\lambda |F(\lambda)|^2 D(y,\,x,\,m_\lambda).\end{aligned}$$

In this last expression we can recognize the **Feynman propagator** of a particle with mass m_λ:

$$\begin{aligned}D_F(x-y,\,m_\lambda) &= \Theta(x^0-y^0)D(x,\,y,\,m_\lambda) + \Theta(y^0-x^0)D(y,\,x,\,m_\lambda) \\ &= \langle 0|\,\hat{T}\{\hat{\phi}_{0,\lambda}(x)\hat{\phi}_{0,\lambda}(y)\}\,|0\rangle = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\frac{i}{p^2-m_\lambda^2+i\epsilon}e^{-ip\cdot(x-y)},\end{aligned} \tag{2.4.3}$$

which is the amplitude for a free scalar particle to propagate from point y to point x in a time-ordered manner, and it is given by the Fourier transform of the free propagator with the appropriate $i\epsilon = i0^+$ prescription to ensure causality.

With this last definition, we can now write the final expression, the **Källén-Lehmann spectral representation** of the two point function, as:

$$\langle\Omega|\,T\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle = \int \frac{\mathrm{d}M^2}{2\pi}g(M^2)D_F(x,\,y,\,M), \tag{2.4.4}$$

where we have defined the **spectral density** function $g(M^2)$ as

$$g(M^2) = \sum_\lambda (2\pi)|F(\lambda)|^2\delta(M^2-m_\lambda^2). \tag{2.4.5}$$

The spectral density function encodes the information about the spectrum of the theory,[5] and it is a positive function since it is defined as a sum of positive terms. The Källén-Lehmann representation of the two point function shows that it can be written as a superposition of free propagators with different masses, weighted by the spectral density function.

[5]The spectral density function is a measure of the number of states with a given mass: the spectrum of the theory is the set of all possible masses of physical particles.

This spectral representation is very useful for analyzing the properties of the theory, and it will be crucial for deriving the LSZ reduction formula in the next section: it will allow us to relate the S-matrix elements to the poles of the two point function (which correspond to the physical particles in the spectrum of the theory) and to the residues of these poles (which correspond to the field strength renormalization constants of these particles).

Let us make some observations about the Källén-Lehmann representation to understand better the look and physical meaning of the spectral density function $g(M^2)$:

- If $M^2 = m^2$ with m the physical mass of a **one particle state** (i.e. m^2 eigenvalue of $\hat{P}^2$), then the spectral density function has a delta function contribution

$$g(M^2)\Big|_{M^2=m^2} = Z(2\pi)\delta(M^2 - m^2),$$

  with $Z = |F(\lambda = 1\text{PS})|^2$ which represents the *field strength renormalization constant* of the one particle state. This means that the two point function has a pole at $p^2 = m^2$ with residue Z, which is a consequence of the presence of a one particle state in the spectrum of the theory.

- For a **two particle bound state** we have two cases:

  - If $M^2 \leq (2m)^2$ then the spectral density function has a delta function contribution

$$g(M^2)\Big|_{M^2\leq(2m)^2} = \sum_{\lambda\in 2\text{PBS}} Z_B(2\pi)\delta(M^2 - m_\lambda^2),$$

    with $Z_B = |F(\lambda = 2\text{PBS})|^2$ which represents the *field strength renormalization constant* of the two particle bound state. This means that the two point function has a pole at $p^2 = M_B^2$ with residue Z_B, which is a consequence of the presence of a two particle bound state in the spectrum of the theory.

  - If $M^2 > (2m)^2$ then the spectral density function has a *continuous* contribution given by the two particle continuum states, which represent the scattering states of two particles with mass m

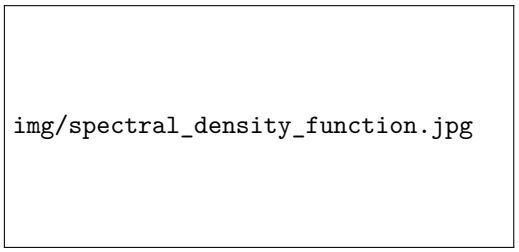
$$g(M^2)\Big|_{M^2>(2m)^2} = Z_C \sum_{\lambda\in 2\text{PCS}} (2\pi)\delta(M^2 - m_\lambda^2),$$

    with $Z_C = |F(\lambda = 2\text{PCS})|^2$ which represents the *field strength renormalization constant* of the two particle continuum states. This means that the two point function has a branch cut starting at $p^2 = (2m)^2$, which is a consequence of the presence of the two particle continuum states in the spectrum of the theory.

- The spectral density function $g(M^2)$ is a positive function of the **bare mass** M, since it is defined as a sum of positive terms. This means that the two point function is a positive function, which is a consequence of the unitarity of the theory.

- The **physical mass** m, whose square m^2 is the exact eigenvalue of the invariant operator $\hat{P}^2$ for a single-particle state, represents the true, measurable mass found in the spectrum of the theory (the physical pole of the full two-point function). This must be carefully distinguished from the **bare mass** M^2 which appears as a parameter in the original, unrenormalized Lagrangian. Because the particle constantly interacts with vacuum fluctuations (getting "dressed" by self-energy contributions), the physical mass gets shifted from the bare mass. The two quantities are ultimately connected through a *renormalization procedure*.

Thus we can now understand a typical look of the spectral density function $g(M^2)$ for a theory with a one particle state, some two particle bound states and a two particle continuum states:



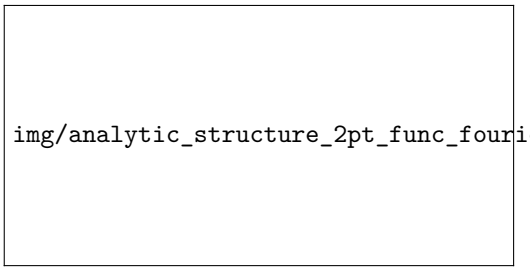


Figure 2.5: The left image displays the spectral density function for a theory featuring a one-particle state alongside two-particle bound and continuum states, illustrating both the delta-function contributions and the continuous spectrum. The right image shows the analytic structure of the two-point function's Fourier transform in the complex p^2-plane, highlighting the single-particle pole at $p^2 = m^2$, the two-particle bound state poles at $p^2 = M_B^2 \leq (2m)^2$, and the branch cut starting at $p^2 = (2m)^2$ for the two-particle continuum states.

We can now study again the two point function in momentum space, by performing a Fourier transform of the Källén-Lehmann representation:

$$\begin{aligned}\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega| T\{\hat{\phi}(x)\hat{\phi}(0)\} |\Omega\rangle &= \int \mathrm{d}^4x\, e^{ip\cdot x} \int \frac{\mathrm{d}M^2}{2\pi} g(M^2) D_F(x,\, 0,\, M) \\ &= \frac{iZ}{p^2 - m^2 + i\epsilon} + \int_{\sim(2m)^2}^{\infty} \frac{\mathrm{d}M^2}{2\pi} \frac{ig(M^2)}{p^2 - M^2 + i\epsilon},\end{aligned}$$

where we have divided the integral over M^2 into the contribution from the one particle state, which gives a pole at $p^2 = m^2$ with residue iZ, and the contribution from the two particle states, which gives a branch cut starting at $p^2 = (2m)^2$ (these are regular terms for $p^2 \sim m^2$). The contribution from the two particle bound states, if present, would give poles at $p^2 = M_B^2 \leq (2m)^2$ with residue iZ_B (as shown in the above figure), but we have not explicitly written them in the last expression for simplicity. To sum up, in momentum space the two point function has the following analytic structure in the complex p^2-plane:

1. A branch cut starting at $p^2 = (2m)^2$ due to the presence of the two particle continuum states in the spectrum of the theory, which represent the scattering states of two particles with mass m.
2. A pole at $p^2 = m^2$ with residue iZ, which is a consequence of the presence of a one particle state in the spectrum of the theory.
3. Poles at $p^2 \leq (2m)^2$ for each two particle bound state in the spectrum of the theory, with residue iZ_B.

For a free theory we would have only the one particle state in the spectrum of the theory, so that the Källén-Lehmann representation of the two point function would reduce to

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle 0| T\{\hat{\phi}_0(x)\hat{\phi}_0(0)\} |0\rangle = \frac{i}{p^2 - m^2 + i0^+},$$

with $Z = 1$ since the field operator creates a one particle state with unit probability, and the physical mass m^2 coincides with the bare mass M^2 since there are no interactions in the theory.

For an interacting theory we will have corrections to the physical mass and to the field strength renormalization constant, so that we can write

$$\begin{cases} m = m_\lambda + O(g), \\ Z = 1 + O(g), \end{cases}$$

where g is the coupling constant of the theory, so that the physical mass m^2 and the field strength renormalization constant Z receive corrections from the interactions in the theory.

Example (Dirac Field). In the end we will give rules for the generalization of the LSZ reduction formula to more complicated theories, such as theories with fermions and gauge fields, but let us first analyze the structure of the two point function for a Dirac field. If we compute these last results for a Dirac field, we would find

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega|\, \hat{T}\{\hat{\psi}(x)\hat{\bar{\psi}}(0)\}\, |\Omega\rangle = \frac{iZ(\not{p}+m)}{p^2-m^2+i\epsilon},$$

such that

$$\langle\Omega|\, \hat{\psi}(0)\, |\lambda_{\mathbf{p},s}\rangle \Big|_{\lambda=1\mathrm{PS}} = u_s(\mathbf{p})F(\lambda=1\mathrm{PS}) = u_s(\mathbf{p})\sqrt{Z},$$

where $u_s(\mathbf{p})$ is the Dirac spinor for a particle with momentum $\mathbf{p}$ and spin s, with

$$(\not{p}+m) = \sum_s u_s(\mathbf{p})\overline{u}_s(\mathbf{p}),$$

which is the completeness relation for the Dirac spinors. The spectral density function for the Dirac field has a delta function contribution at $M^2 = m^2$ with residue Z, which corresponds to the one particle state in the spectrum of the theory. However, because the operator $\hat{\psi}$ carries half-integer spin and changes the fermion number by one, its matrix elements with any two-particle state (which necessarily has integer spin and fermion number 0 or ± 2) are strictly zero. Therefore, the spectral density of the fundamental Dirac field does not contain branch cuts for two-particle continuum states, nor poles for two-particle bound states. Such multi-particle features can only appear in the spectral density of composite operators (e.g., $\hat{\bar{\psi}}\gamma^\mu\hat{\psi}$) in interacting theories.

Example (Unstable Particle State). If we are considering unstable particles, i.e particles with a decay rate $\Gamma \neq 0$, then the last results are not valid anymore, since we adopted eigenvalues of the momentum operator $\hat{P}^\mu$, and in particular of $\hat{H}$, which cannot be defined for unstable particles:

$$\hat{H}\,|\psi\rangle = E\,|\psi\rangle\,, \quad \hat{S}(t,\,t_0)\,|\psi\rangle = e^{i\hat{H}(t-t_0)}\,|\psi\rangle = e^{iE(t-t_0)}\,|\psi\rangle\,,$$

since an eigenstate by itself can only get a phase and does not decay, while an unstable particle state has to decay in order to be unstable.

Thus a rigorous definition of the physical mass for an unstable particle is rather non-trivial; however the decay rate Γ of the unstable particle state is usually very small compared to its lifetime, so that we can consider those as approximated eigenstates of the Hamiltonian.

We have to take into account the next facts:

- If $\Gamma \neq 0$ the pole of the two point function is not anymore on the real axis, but it is shifted in the complex plane, so that it acquires an imaginary part proportional to the decay rate Γ of the unstable particle state.

- If $\Gamma \ll m$, we may still define the physical mass m of the unstable particle state as the real part of the location of the pole of the two point function, since the imaginary part is small compared to the real part, and it represents a small correction to the physical mass of the unstable particle state.

Thus in the end we can define the physical mass m of the unstable particle state as

$$m^2 = \mathrm{Re}\left[\text{propagator pole}\right],$$

which gives us a well-defined physical mass for the unstable particle state, and we can write the two point function in momentum space as

$$\int \mathrm{d}^4x\, e^{ip\cdot x} \langle\Omega| T\{\hat{\phi}(x)\hat{\phi}(0)\} |\Omega\rangle = \frac{iZ}{p^2 - m^2 + im\Gamma} + \Lambda(p^2),$$

where $\Lambda(p^2)$ is a regular function for $p^2 \sim m^2$ which encodes the regular contributions from the other states in the spectrum of the theory.

2.4.2 | LSZ Reduction Formula

TODO: Add reference to Schwartz 6.1

Let us start from the definition of the S-matrix elements in terms of the asymptotic in and out states:

$$\langle f| \hat{S} |i\rangle,$$

where $|i\rangle$ and $|f\rangle$ are the in and out states of the theory, which are defined as the asymptotic states of the theory at $t \to -\infty$ and $t \to +\infty$ respectively. Asymptotically, the interacting theory reduces to a free theory, so that the in and out states can be approximated by the free particle states of the free theory. This means that the field operators of the interacting theory can be approximated by the field operators of the free theory in this limit, in which they act as creation and annihilation operators for the free particle states (or 1-particle):

$$\hat{\phi}(x)\Big|_{t\to\pm\infty} \sim \sqrt{Z} \int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3} \frac{1}{\sqrt{2E_\mathbf{p}}} \left(\hat{a}_\mathbf{p} e^{-ip\cdot x} + \hat{a}^\dagger_\mathbf{p} e^{ip\cdot x}\right),$$

where Z is the field strength renormalization constant of the one particle state in the spectrum of the theory, and $\hat{a}_\mathbf{p}$ and $\hat{a}^\dagger_\mathbf{p}$ are the annihilation and creation operators for the free particle states of the free theory. The energy appearing in the denominator is the physical energy of the one particle state, which is related to the physical mass m by the relativistic dispersion relation $E_\mathbf{p} = \sqrt{\mathbf{p}^2 + m^2} = p^0$.

Remark. *Note that the annihilation and creation operators $\hat{a}_\mathbf{p}$ and $\hat{a}^\dagger_\mathbf{p}$ are time independent in a free theory. We will assume that they are also time independent in the interacting theory, when considering the asymptotic limit $t \to \pm\infty$, even if in general*

$$\hat{a}_\mathbf{p}(t \to \infty) \neq \hat{a}_\mathbf{p}(t \to -\infty),$$

since the interactions can change the number of particles in the theory, so that the annihilation and creation operators at $t \to +\infty$ can be different from those at $t \to -\infty$ (the free-like field configurations may be different after the scattering process).

If we consider a scattering process with 2 incoming particles and n outgoing particles, we can write the in and out states as

$$|i\rangle = \sqrt{2E_a}\sqrt{2E_b}\hat{a}^\dagger_{\mathbf{p}_a}\, \hat{a}^\dagger_{\mathbf{p}_b} |\Omega\rangle,$$
$$|f\rangle = \sqrt{2E_1}\sqrt{2E_2}\cdots\sqrt{2E_n}\, \hat{a}^\dagger_{\mathbf{p}_1}\hat{a}^\dagger_{\mathbf{p}_2}\cdots\hat{a}^\dagger_{\mathbf{p}_n} |\Omega\rangle,$$

where we have properly normalized the states by the factors of $\sqrt{2E}$ according to the relativistic normalization convention. It is now time to compute the S-matrix element $\langle f|\,\hat{S}\,|i\rangle$ by inserting the expressions for the in and out states and recalling that $\hat{S} = \mathbb{I} + i\hat{T}$, such that we can write

$$\langle f|\,\hat{S}\,|i\rangle = \langle f|\,\mathbb{I} + i\hat{T}\,|i\rangle = \langle f|\,i\hat{T}\,|i\rangle = \sqrt{2E_a}\sqrt{2E_b}\sqrt{2E_1}\sqrt{2E_2}\cdots\sqrt{2E_n}\times$$
$$\times\,\langle\Omega|\,\hat{a}_{\mathbf{p}_1}(+\infty)\hat{a}_{\mathbf{p}_2}(+\infty)\cdots\hat{a}_{\mathbf{p}_n}(+\infty)\left(i\hat{T}\right)\hat{a}^\dagger_{\mathbf{p}_a}(-\infty)\,\hat{a}^\dagger_{\mathbf{p}_b}(-\infty)\,|\Omega\rangle\,.$$

The idea is now to rewrite everyting in terms of Green's functions of the theory, so let us first prove the following identity:

$$\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_x^2 + m^2)\hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\ldots,y_n)\}$$
$$= \sqrt{2E_{\mathbf{p}}}\left(\hat{a}_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\ldots,y_n) - \hat{T}\hat{O}(y_1,\ldots,y_n)\hat{a}_{\mathbf{p}}(-\infty)\right),$$

where $\hat{O}(y_1,\ldots,y_n)$ is an arbitrary product of operators of the theory, and we have used the fact that the field operator $\hat{\phi}(x)$ can be approximated by the free field operator in the asymptotic limit.

In order to prove this identity, we now define $\hat{\mathcal{O}}(x) = \hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\ldots,y_n)\}$ as the time ordered product of the field operator and the arbitrary product of operators, so that we can write

$$\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_x^2 + m^2)\hat{\mathcal{O}}(x) = \frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_t^2 - \partial_i^2 + m^2)\hat{\mathcal{O}}(x),$$

where we have explicitly written the derivatives in terms of the time and spatial derivatives. We can now integrate by parts the spatial derivatives, so that it changes sign and start acting on the left $\overrightarrow{\partial}_i \to \overleftarrow{\partial}_i$,[6] getting to

$$\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x\, e^{ip\cdot x}(\partial_t^2 + \mathbf{p}^2 + m^2)\hat{\mathcal{O}}(x),$$

where we can recognize the relativistic dispersion relation $E_{\mathbf{p}}^2 = \mathbf{p}^2 + m^2$ for a particle with mass m and momentum $\mathbf{p}$, so that we can write

$$\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\, i\, e^{ip\cdot x}(\partial_t^2 + E_{\mathbf{p}}^2)\hat{\mathcal{O}}(x) = \frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\, \partial_t\left(e^{ip\cdot x}(i\partial_t + E_{\mathbf{p}})\hat{\mathcal{O}}(x)\right).$$

Now we can split the integral in a spatial and a time integral, so that

$$\frac{1}{\sqrt{Z}}\int \mathrm{d}t\, \partial_t e^{iE_{\mathbf{p}}t}\int \mathrm{d}^3\mathbf{x}\, e^{-i\mathbf{p}\cdot\mathbf{x}}(i\partial_t + E_{\mathbf{p}})\hat{\mathcal{O}}(x),$$

which in this form is a total time derivative (the first integral), so that we can evaluate it at the boundaries $t = \pm\infty$, where we know the field operators to behave as free field operators, and the ladder operators to be time independent; so we can now work on the second integral and substitute the expression for $\hat{\mathcal{O}}(x)$ and the asymptotic behavior of the field operator, getting to

$$\int \frac{\mathrm{d}^3\mathbf{k}}{(2\pi)^3}\int \mathrm{d}^3\mathbf{x}\left[\frac{E_{\mathbf{p}} + E_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}}e^{-iE_{\mathbf{k}}t}e^{-i(\mathbf{p}-\mathbf{k})\cdot\mathbf{x}}\hat{T}\{\hat{a}_{\mathbf{k}}\hat{O}\} + \frac{E_{\mathbf{p}} - E_{\mathbf{k}}}{\sqrt{2E_{\mathbf{k}}}}e^{iE_{\mathbf{k}}t}e^{i(\mathbf{p}-\mathbf{k})\cdot\mathbf{x}}\hat{T}\{\hat{a}^\dagger_{\mathbf{k}}\hat{O}\}\right],$$

in which a spatial integration gives us a delta function $\delta^{(3)}(\mathbf{p}-\mathbf{k})$ which allows us to perform the integral over $\mathbf{k}$, getting to

$$\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\, i\, e^{ip\cdot x}(\partial_t^2 + E_{\mathbf{p}}^2)\hat{\mathcal{O}}(x) = \int \mathrm{d}t\, \partial_t\left[e^{iE_{\mathbf{p}}t}\sqrt{2E_{\mathbf{p}}}\left(e^{-iE_{\mathbf{p}}t}\hat{T}\{\hat{a}_{\mathbf{p}}(t)\hat{O}\}\right)\right]$$
$$= \sqrt{2E_{\mathbf{p}}}\left[\hat{a}_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\ldots,y_n) - \hat{T}\hat{O}(y_1,\ldots,y_n)\hat{a}_{\mathbf{p}}(-\infty)\right],$$

[6] We are assuming that the fields vanish at spatial infinity, so that the boundary terms from the integration by parts vanish: then the spatial derivatives will change sign and start acting on the left.

yielding the desired result. We can obtain a similar relation holding for its hermitian conjugate, which is

$$\begin{aligned}&\frac{1}{\sqrt{Z}}\int \mathrm{d}^4x\,(-i)\,e^{-ip\cdot x}(\partial_x^2+m^2)\hat{T}\{\hat{\phi}(x)\hat{O}(y_1,\dots,y_n)\}\\ &=\sqrt{2E_{\mathbf{p}}}\left[\hat{a}^\dagger_{\mathbf{p}}(+\infty)\hat{T}\hat{O}(y_1,\dots,y_n)-\hat{T}\hat{O}(y_1,\dots,y_n)\hat{a}^\dagger_{\mathbf{p}}(-\infty)\right].\end{aligned}$$

Now we can apply the derived result to the expression for the S-matrix element $\langle f|\,\hat{S}\,|i\rangle$ that we wrote above, so that we can write

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle &= \sqrt{2E_a 2E_b 2E_1 2E_2\cdots 2E_n}\\ &\quad\times\langle\Omega|\,\hat{T}\hat{a}_{\mathbf{p}_1}(+\infty)\hat{a}_{\mathbf{p}_2}(+\infty)\cdots\hat{a}_{\mathbf{p}_n}(+\infty)\left(i\hat{T}\right)\hat{a}^\dagger_{\mathbf{p}_a}(-\infty)\hat{a}^\dagger_{\mathbf{p}_b}(-\infty)\,|\Omega\rangle\,,\end{aligned}$$

where we inserted a time ordering operator T to use the derived result (the time ordering does not change the result since the ladder operators are already time-ordered). We can now apply the derived result for each of the ladder operators in the above expression: while we consider $\hat{a}_{\mathbf{p}_1}$ all the other ladder operators can be grouped in the arbitrary product of operators $\hat{O}$ in the derived result, so that we can apply it to each of the ladder operators in the above expression, getting to the famous **LSZ reduction formula**, which is essentially

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle &= \left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_a\,e^{-ip_a\cdot x_a}(\partial^2_{x_a}+m^2)\phi(x_a)\right]\left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_b\,e^{-ip_b\cdot x_b}(\partial^2_{x_b}+m^2)\phi(x_b)\right]\\ &\quad\times\left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_1\,e^{ip_1\cdot x_1}(\partial^2_{x_1}+m^2)\phi(x_1)\right]\cdots\left[\frac{i}{\sqrt{Z}}\int \mathrm{d}^4x_n\,e^{ip_n\cdot x_n}(\partial^2_{x_n}+m^2)\phi(x_n)\right]\\ &\quad\times\langle\Omega|\,T\{\phi(x_1)\phi(x_2)\cdots\phi(x_n)\phi(x_a)\phi(x_b)\}\,|\Omega\rangle\,,\end{aligned}\tag{2.4.6}$$

which is a way to relate the S-matrix elements to the time-ordered correlation functions of the fields, which are the Green's functions of the theory. The LSZ reduction formula is a powerful tool for computing the S-matrix elements from the Green's functions, and it is widely used in quantum field theory to compute scattering amplitudes and cross sections for various processes.

The LSZ reduction formula can be interpreted as a way to amputate the external legs of the Green's functions, which correspond to the free propagators of the theory, and to put them on shell by applying the operator (∂^2+m^2) to them, which gives zero when acting on the free propagators. The factors of $1/\sqrt{Z}$ in the formula are necessary to cancel the field strength renormalization constants that appear in the residues of the poles of the Green's functions, so that we can obtain finite results for the S-matrix elements.

Substantially, it is a bunch of Fourier transforms of the field operators, where the only difference is in the phase signs: incoming particles have a negative phase sign, while outgoing particles have a positive phase sign.

If we rewrite it in the momentum space, integrating by parts the derivatives so that they act on the exponentials on their left (which corresponds to the substitution $i(\partial_i^2+m^2)\to -i(p_i^2-m^2)$) we can write then the LSZ reduction formula as

$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle &= \prod_{i=a,b,1,\dots,n}\left[\frac{i}{\sqrt{Z}}(p_i^2-m^2)\right]\\ &\quad\times\int \mathrm{d}^4x_a\,\mathrm{d}^4x_b\,\mathrm{d}^4x_1\,\cdots\,\mathrm{d}^4x_n\,e^{i(p_1\cdot x_1+\cdots+p_n\cdot x_n-p_a\cdot x_a-p_b\cdot x_b)}\\ &\quad\times\langle\Omega|\,T\{\phi(x_1)\cdots\phi(x_n)\phi(x_a)\phi(x_b)\}\,|\Omega\rangle\,.\end{aligned}\tag{2.4.7}$$

We have assumed asymptotic particles as the final state, with on shell momenta $p_i^2 = m^2$, so that the factors of $(\partial^2 + m^2)$ in the LSZ reduction formula will give zero when acting on the external legs of the Green's functions, which are the free propagators of the theory. This means that the only contribution to the S-matrix elements will come from the poles of the Green's functions, which correspond to the physical particles in the spectrum of the theory:

$$\frac{iZ}{p_i^2 - m^2}.$$

The residues of these poles will give the field strength renormalization constants Z for each particle, which will cancel with the factors of $1/\sqrt{Z}$ in the LSZ reduction formula, giving a finite result for the S-matrix elements. Indeed, as $p_i \to \pm m$, the factors of $(p_i^2 - m^2)$ in the LSZ reduction formula will cancel with the poles of the Green's functions, giving a finite result for the S-matrix elements, since namely its just a series of regular terms for $p_i^2 \sim m^2$ multiplied by $\sqrt{Z}$.

2.4.3 | Generalization to Other Fields

If we repeat the same analysis for different types of fields, such as Dirac fields or gauge fields, we will find similar results for the two point functions and the LSZ reduction formula, with some modifications due to the different spin and statistics of the fields:

- For a **complex scalar field**, we would have two types of particles, the particle and the antiparticle, with different creation and annihilation operators, and the LSZ reduction formula would involve both types of operators for the in and out states:
  - $\phi^\dagger$ for *incoming particles* and *outgoing antiparticles*, with a negative phase sign in the Fourier transform.
  - ϕ for *outgoing particles* and *incoming antiparticles*, with a positive phase sign in the Fourier transform.

  This would lead to a more complicated LSZ reduction formula, but the general structure would be the same as for the real scalar field, and with this prescription we could also compute the results for Dirac theory ($\phi \to \psi$ and $\phi^\dagger \to \bar{\psi}$).

- If we are considering a **theory of several fields**, then the Green function will contain all the different fields along with the different masses of the quanta, and the LSZ reduction formula will involve the corresponding field operators for each type of particle in the in and out states, with the appropriate phase signs in the Fourier transforms.

- When considering **particle with different spins**, we have to take into account the different transformation properties of the fields under Lorentz transformations, which will affect the structure of the two point functions and the LSZ reduction formula. For example, for a Dirac field we would have the following field operators for the in and out states:
  - $\overline{u}_{s,\alpha}$ for *outgoing particles*;
  - $u_{s,\alpha}$ for *incoming particles*;
  - $v_{s,\alpha}$ for *outgoing antiparticles*;
  - $\overline{v}_{s,\alpha}$ for *incoming antiparticles*.

If we instead consider a spin-1 gauge field, we would have the following field operators for the in and out states:

- $\epsilon_\mu(p,\,\lambda)$ for *outgoing particles*;
- $\epsilon^*_\mu(p,\,\lambda)$ for *incoming particles*.

The spinorial/vectorial indices (α, μ etc.) of the field operators will be contracted with the corresponding indices of the Green's functions in the LSZ reduction formula, and the final expression for the S-matrix elements will involve the appropriate spinor/vector structures for each type of particle in the in and out states:

$$\langle\Omega|\, T\{\dots A^\mu(x)\dots\psi_\alpha(y)\dots\}\,|\Omega\rangle\,.$$

- For a Dirac fermion the factor $(\partial^2 - m^2)$ in the LSZ reduction formula will be replaced by the Dirac operator $(\not{\partial} - m)$, which gives zero when acting on the free propagator of the Dirac field, and the field strength renormalization constant Z will be replaced by the corresponding field strength renormalization constant for the Dirac field. Furthermore external factors $\not{p} - m$ will cancel out the poles of the Green's functions, leaving a factor of $\sqrt{Z}$ as $p_i \to$ *on-shell*.

We have to make one final observation: we did not make any assumption on the behaviour of the field operators of the interacting theory, apart from the fact that they can be approximated by the field operators of the free theory in the asymptotic limit (as ladder operators), which is roughly equivalent to the statement

$$\langle\Omega|\,\hat{\phi}(x)\,|\lambda_{\mathbf{p}}\rangle = \sqrt{Z}e^{-ip\cdot x},$$

whose validity was argued when we derived the Källén-Lehmann representation of the two point function. This means that the LSZ reduction formula is valid for any interacting theory, as long as the field operators of the interacting theory can be approximated by the field operators of the free theory in the asymptotic limit, which is a very general condition that is satisfied by most of the theories we are interested in.

Hence, our derivation is still valid when replacing the one particle state $|\mathbf{p}\rangle$ to any state $|\lambda_{\mathbf{p}}\rangle$ and the field operator $\hat{\phi}(x)$ to any operator $\hat{O}(x)$ of the theory, as long as the following condition is satisfied:

$$\langle\Omega|\,\hat{O}(x)\,|\lambda_{\mathbf{p}}\rangle \sim e^{-ip\cdot x},$$

which is valid as long as $\langle\Omega|\,\hat{O}(x)\,|\lambda_{\mathbf{p}}\rangle \neq 0$, so that the operator $\hat{O}(x)$ creates a one particle state from the vacuum with non-zero probability, and the LSZ reduction formula can be applied to any such operator. The LSZ reduction formula is a very general result that can be applied to a wide range of theories and operators, and it is a powerful tool for computing the S-matrix elements from the Green's functions of the theory.

Finally, inserting any operator $\hat{O}(x)$ among asymptotic states in the LSZ reduction formula is allowed in the form of the following substitutions:

$$\begin{cases}\langle f|\,\hat{S}\,|i\rangle = \langle f, t=+\infty|i, t=-\infty\rangle \\ \langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\}\,|\Omega\rangle\end{cases} \quad\Longrightarrow\quad \begin{cases}\langle f, t=+\infty|\,\hat{O}(x)\,|i, t=-\infty\rangle\,, \\ \langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\hat{O}(x)\}\,|\Omega\rangle\,,\end{cases}$$

respectively for the l.h.s and the r.h.s of the LSZ reduction formula. This means that we can compute matrix elements of any operator $\hat{O}(x)$ between asymptotic states by using the LSZ reduction formula, which is a very powerful result that allows us to compute a wide range of physical observables.

3 | Feynman Diagrams

In the previous chapters we have seen how to calculate Green's functions in QFT and, as we have notice, the number of terms in the expansion of the Green's function grows very quickly as we increase the order of the correction, making it very difficult to keep track of all the terms and their contributions. This is where Feynman diagrams come in handy, as they provide a **visual representation of the interactions** and allow us to easily keep track of the different terms in the expansion.

In this chapter we will see how to derive a set of practical rules, known as Feynman rules, that allow us to compute Green's functions and scattering amplitudes in a systematic way for an *interacting theory in perturbation theory*, without having to calculate all the contractions explicitly. This method is based on Tylor expansions in the coupling constants g_j, assuming a weak enough interaction, and it is a very powerful tool for calculating physical quantities in QFT, as it allows us to write down the contribution of each diagram without having to calculate all the contractions explicitly, which is a very powerful tool for calculating physical quantities in QFT.

We will start by reviewing the Hamiltonian formalism and the interaction picture, which are essential for understanding how to calculate Green's functions in an interacting theory, and then we will see how to derive the Feynman rules for a simple interacting theory, and how to use them to calculate physical quantities. Through some examples, we will see how to apply the Feynman rules in practice, and then we will see how to generalize them to more complicated theories and to momentum space, which is the most common way to use Feynman diagrams in QFT.

3.1 | Hamiltonian Formalism and Interaction Picture

In QFT is often more convenient to work in the Hamiltonian formalism, where we express the dynamics of the system in terms of a Hamiltonian operator. The Hamiltonian can be split into a free part, which describes non-interacting particles, and an interaction part, which describes the interactions between particles.

The **Heisenberg picture** is often used in QFT, where operators evolve in time while states remain constant. However, for perturbative calculations, it is more convenient to use the **interaction picture**, where both operators and states evolve in time. Let us start from the Heisenberg equation of motion for an operator $\hat{O}(x)$:

$$i\partial_t \hat{O}(x) = [\hat{O}(x), \hat{H}(x)],$$

where $\hat{H}(x)$ is the Hamiltonian operator. We can write the solution to this equation through the time evolution operator $\hat{S}(t, t_0)$ as

$$\hat{O}(\mathbf{x}, t) = \hat{S}^\dagger(t, t_0)\hat{O}(\mathbf{x}, t_0)\hat{S}(t, t_0),$$

which can be inserted back into the Heisenberg equation of motion to get an equation for the time evolution operator:

$$i\partial_t \hat{S}(t, t_0) = \hat{H}(t)\hat{S}(t, t_0).$$

Now let us split the Hamiltonian into a free part and an interaction part:

$$\hat{H}(t) = \hat{H}_0 + \hat{V}(t),$$

and the Lagrangian into a free part and an interaction part:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}.$$

In practice, $\hat{H}_0$ will be the Hamiltonian of a free theory, which describes non-interacting particles whose dynamics can be solved exactly, while $\hat{V}(t)$ will be the interaction Hamiltonian, which will be treated as a perturbation and encodes all the interaction terms. The free part of the Lagrangian $\mathcal{L}_0$ will describe the dynamics of the free fields, while the interaction part $\mathcal{L}_{\text{int}}$ will describe the interactions between the fields, indeed we can write the Legendre transform of the Lagrangian to get the Hamiltonian as

$$H = \int \mathrm{d}^3\mathbf{x}\, \mathcal{H}(\mathbf{x}), \quad \begin{cases} \mathcal{H} = \pi\dot{\phi} - \mathcal{L}, \\ \pi = \partial\mathcal{L}/\partial\dot{\phi}, \end{cases}$$

and if $\mathcal{L}_{\text{int}}$ does not contain derivatives, then we simply have $\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}}$, and thus we can write the interaction Hamiltonian as

$$V = -\int \mathrm{d}^3\mathbf{x}\, \mathcal{L}_{\text{int}}.$$

If $\mathcal{L}_{\text{int}}$ does indeed contain derivatives, then similar results can be obtained, but it will be easier to derive them in the path integral formalism, which is more convenient for theories with derivative interactions, and we will see how to do that in the next chapters.

3.1.1 | Interaction Picture

We are now moving from the Heisenberg picture to the interaction picture (also called the Dirac picture), where both operators and states evolve in time. In the interaction picture, the *fields evolve*

according to the free Hamiltonian H_0, while the *states evolve according to the total Hamiltonian* $H = H_0 + V$: the interaction is encoded in the evolution of the states, while the fields evolve as if they were free fields. This is a very important point, as it allows us to use the free fields to calculate the interactions, which is much easier than dealing with the full interacting fields.

Then of course the *matrix elements* we are interested in computing *will be independent from the chosen picture*, since they are physical quantities, and thus we can choose the picture that is more convenient for our calculations. Schrödinger, Heisenberg and Dirac picture are just different ways of describing the same physics, and they are identically equivalent equivalent at a conventional reference time $t = t_0$.

Thus we can start from the Schrödinger picture, where the fields are time-independent, and then we can move to the interaction and Heisenberg picture, where the fields evolve in time according to the free Hamiltonian and the total Hamiltonian respectively. We can write the fields in the different pictures and see how they are related to each other as

$$\begin{cases} \hat{\phi}_0 = \hat{\phi}_I = e^{i\hat{H}_0(t-t_0)}\hat{\phi}_S e^{-i\hat{H}_0(t-t_0)}, \quad \hat{\phi}_S = \hat{\phi}_I(t=t_0) \\ \hat{\phi} = \hat{\phi}_H = \hat{S}^\dagger(t,t_0)\hat{\phi}_S\hat{S}(t,t_0) = \hat{U}^\dagger(t,t_0)\hat{\phi}_I\hat{U}(t,t_0), \end{cases}$$

where

$$\hat{U}(t,t_0) = e^{i\hat{H}_0(t-t_0)}\hat{S}(t,t_0),$$

is the time evolution operator connecting the interaction picture to the Heisenberg picture, while $\hat{S}(t,t_0)$ is the time evolution operator connecting the Schrödinger picture to the Heisenberg picture.

Now having $\hat{H}_H = \hat{H}_S$ and $\hat{H}_0 = \hat{H}_{0,S}$ (i.e. the Heisenberg Hamiltonian is the same as the Schrödinger Hamiltonian, and the free Hamiltonian is the same in both pictures) we can write the equation of motion for $\hat{U}$ by inserting the fields in the Heisenberg equation of motion, which gives

$$i\partial_t\hat{U}(t,t_0) = \hat{V}_I(t,t_0)\hat{U}(t,t_0), \tag{3.1.1}$$

with

$$\hat{V}_I = e^{i\hat{H}_0(t-t_0)}\hat{V}_S e^{-i\hat{H}_0(t-t_0)},$$

which is the **interaction potential** (potential in the interaction picture). In practice, since $\hat{V}$ is given by the integral of the interaction Lagrangian, we can write it as

$$\hat{V}_I = \hat{V}_S(\hat{\phi})\big|_{\hat{\phi}\to\hat{\phi}_I} = \hat{V}(\hat{\phi}_0),$$

which means that the potential in the interaction picture is the same as the potential in the Schrödinger picture, but with the fields replaced by their interaction picture counterparts (which is the same as the free fields). This is a very important result, as it allows us to use the free fields to calculate the interactions, which is much easier than dealing with the full interacting fields.

Dyson Series

The solution to the evolution equation for $\hat{U}$ (3.1.1) can be written as the following well-known time-ordered exponential:

$$\hat{U}(t,t_0) = \hat{T}\exp\left(-i\int_{t_0}^{t}\mathrm{d}t'\hat{V}_I(t')\right),$$

which can be expanded in a power series as

$$\hat{U}(t,t_0) = \mathbb{I} + (-i)\int_{t_0}^{t}\mathrm{d}t'\hat{V}_I(t') + \frac{(-i)^2}{2!}\int_{t_0}^{t}\mathrm{d}t'\int_{t_0}^{t}\mathrm{d}t''\hat{T}\{\hat{V}_I(t')\hat{V}_I(t'')\} + \dots$$

where if we now partition the integration domain of the n-th term of the series in a set of $n!$ subdomains, with the idea to apply the time-ordering operator: up to renaming the integration variables, we can reorder the integration limits as

$$t' > t'' > \cdots > t^{(n)} > t_0,$$

making the integral of the n-th term of the series as

$$\hat{U}(t,t_0) = \mathbb{I} + (-i)\int_{t_0}^{t} \mathrm{d}t' \hat{V}_I(t') + (-i)^2 \int_{t_0}^{t} \mathrm{d}t' \int_{t_0}^{t'} \mathrm{d}t'' \hat{V}_I(t')\hat{V}_I(t'') + \dots$$

which is known as the **Dyson series**, and it is the solution to the evolution equation for $\hat{U}$. Thus we can write the Dyson series in such a way that the time-ordering operator is not needed, since the integration limits ensure that the operators are ordered correctly in time. In the end we have

$$\hat{U}(t,t_0) = \mathbb{I} - i\int_{t_0}^{t} \mathrm{d}t' \hat{V}_I(t')\hat{U}(t',t_0),$$

which is an integral equation for $\hat{U}$ that can be solved iteratively, and we can see that the Dyson series is the solution to this integral equation. This is a very important result, as it allows us to calculate the time evolution operator in the interaction picture through

$$i\partial_t \hat{U}(t,t_0) = \hat{V}_I(t)\hat{U}(t,t_0),$$

which is essential for calculating scattering amplitudes and other physical quantities in QFT, even if the computation remains quite complicated.

3.1.2 | Computing Green's Functions

Consider a Green's function of the form

$$\langle\Omega|\, \hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\}\, |\Omega\rangle\,,$$

where $|\Omega\rangle$ is the vacuum state of the interacting theory, and $\hat{\phi}(x) = \hat{\phi}_H(x)$ are the interacting fields in the Heisenberg picture.

To evaluate this, we need to relate the interacting vacuum $|\Omega\rangle$ to the free theory vacuum $|0\rangle$. We assume that at the infinite past ($t = -\infty$), the interacting vacuum behaves like the free vacuum, meaning it is annihilated by the Heisenberg annihilation operator $a_{\mathbf{p}}(t)$:

$$a_{\mathbf{p}}(t)\,|\Omega\rangle = 0 \quad \text{at } t = -\infty.$$

Using the time evolution operator $S(t,t_0)$, we can rewrite the annihilation operator as $a_{\mathbf{p}}(t) = S^\dagger(t,t_0)a_{\mathbf{p}}(t_0)S(t,t_0)$. This implies that $a_{\mathbf{p}}(t_0)$ annihilates the state $S(t,t_0)\,|\Omega\rangle$ at $t = -\infty$.

At the reference time $t = t_0$, the interaction and Heisenberg pictures coincide. Thus, $a_{\mathbf{p}}(t_0)$ is identical to the annihilation operator of the free theory, which naturally annihilates the free vacuum: $a_{\mathbf{p}}(t_0)\,|0\rangle = 0$. Furthermore, the free vacuum is an eigenstate of the free Hamiltonian $\hat{H}_0$. Assuming the vacuum energy is zero ($E_0 = 0$), we have $e^{i\hat{H}_0(t-t_0)}\,|0\rangle = |0\rangle$, which means:

$$a_{\mathbf{p}}(t_0)e^{i\hat{H}_0(t-t_0)}\,|0\rangle = 0.$$

Because both $S(t,t_0)\,|\Omega\rangle\,\big|_{t=-\infty}$ and $e^{i\hat{H}_0(t-t_0)}\,|0\rangle$ are annihilated by the exact same operator $a_{\mathbf{p}}(t_0)$, the two states must be proportional. We can express this proportionality using the standard interaction picture time-evolution operator $U(t_f,t_i)$:

$$|\Omega\rangle = N_i U(0,-\infty)\,|0\rangle\,,$$

for some normalization constant N_i. By an analogous argument for the bra vector at $t = +\infty$, we find:

$$\langle\Omega| = N_f \langle 0| U(+\infty, 0).$$

We can now evaluate the Green's function. Let us assume, up to a relabelling of the coordinates $x_i = (t_i, \mathbf{x}_i)$, that the times are strictly ordered: $t_1 > t_2 > \cdots > t_n$. Because of this explicit ordering, we can temporarily drop the time-ordering operator $\hat{T}$. We also express the Heisenberg fields in terms of the free interaction-picture fields $\hat{\phi}_0$ using $\hat{\phi}(x_i) = U(0, t_i)\hat{\phi}_0(x_i)U(t_i, 0)$. Substituting these elements yields:

$$\langle\Omega| \hat{T}\{\hat{\phi}(x_1) \dots \hat{\phi}(x_n)\} |\Omega\rangle = \langle\Omega| \hat{\phi}(x_1) \dots \hat{\phi}(x_n) |\Omega\rangle$$
$$= N_i N_f \langle 0| U(+\infty, 0) \Big(U(0, t_1)\hat{\phi}_0(x_1)U(t_1, 0)\Big) \dots \Big(U(0, t_n)\hat{\phi}_0(x_n)U(t_n, 0)\Big) U(0, -\infty) |0\rangle .$$

Using the composition property of the evolution operator ($U(t_a, t_b)U(t_b, t_c) = U(t_a, t_c)$), all the intermediate adjacent operators combine gracefully:

$$= N_i N_f \langle 0| U(+\infty, t_1)\hat{\phi}_0(x_1)U(t_1, t_2)\hat{\phi}_0(x_2) \dots \hat{\phi}_0(x_n)U(t_n, -\infty) |0\rangle .$$

At this point, every operator inside the expectation value $\langle 0| \dots |0\rangle$ is time-ordered by construction (since $U(t_a, t_b)$ already contains time-ordered products of the interaction potential $\hat{V}_I(t)$ with $t_b \leq t \leq t_a$). We can therefore safely put the global time-ordering operator $\hat{T}$ back in. Since operators commute inside $\hat{T}$, we can group all the independent evolution operators together into one:

$$\begin{aligned}\langle\Omega| \hat{T}\{\hat{\phi}(x_1) \dots \hat{\phi}(x_n)\} |\Omega\rangle &= N_i N_f \langle 0| \hat{T}\{\hat{\phi}_0(x_1) \dots \hat{\phi}_0(x_n)U(+\infty, t_1)U(t_1, t_2) \dots U(t_n, -\infty)\} |0\rangle \\ &= N_i N_f \langle 0| \hat{T}\{\hat{\phi}_0(x_1) \dots \hat{\phi}_0(x_n)U(+\infty, -\infty)\} |0\rangle .\end{aligned}$$

Finally, we fix the overall constant $N_i N_f$ by enforcing the standard normalization condition for the vacuum, $\langle\Omega|\Omega\rangle = \langle 0|0\rangle = 1$. Setting $n = 0$ (no fields) in our derivation above gives $\langle\Omega|\Omega\rangle = N_i N_f \langle 0| \hat{T}\{U(+\infty, -\infty)\} |0\rangle$, which implies:

$$N_i N_f = \frac{1}{\langle 0| \hat{T}\{U(+\infty, -\infty)\} |0\rangle}.$$

Recalling that $U(+\infty, -\infty) = \exp\left[-i \int_{-\infty}^{+\infty} \mathrm{d}t\, \hat{V}_I(t)\right]$, we obtain the final formula (also known as the *Gell-Mann and Low* theorem):

$$\langle\Omega| \hat{T}\{\hat{\phi}(x_1) \dots \hat{\phi}(x_n)\} |\Omega\rangle = \frac{\langle 0| \hat{T}\{\hat{\phi}_0(x_1) \dots \hat{\phi}_0(x_n) \exp\left[-i \int_{-\infty}^{\infty} \mathrm{d}t\, \hat{V}_I(t)\right]\} |0\rangle}{\langle 0| \hat{T}\{\exp\left[-i \int_{-\infty}^{\infty} \mathrm{d}t\, \hat{V}_I(t)\right]\} |0\rangle}.$$

The right-hand side is now entirely expressed in terms of free fields $\hat{\phi}_0$ and the free vacuum $|0\rangle$, objects we know how to deal with using Wick's theorem and standard perturbation theory.

We can expand the computation considering that the interaction Hamiltonian is given by the integral of the interaction Lagrangian, which is the case for a large class of theories, and thus we can write

$$\mathcal{H}_{int} = -\mathcal{L}_{int}, \quad \int \mathrm{d}t\, V_I(t) = -\int \mathrm{d}^4x\, \mathcal{L}_{int}(x),$$

so that the Green's function can be written as

$$\langle\Omega| \hat{T}\{\hat{\phi}(x_1) \dots \hat{\phi}(x_n)\} |\Omega\rangle = \frac{\langle 0| \hat{T}\{\hat{\phi}(x_1) \dots \hat{\phi}(x_n) \exp\left[i \int \mathrm{d}^4x\, \hat{\mathcal{L}}_{int}\right]\} |0\rangle}{\langle 0| \hat{T}\{\exp\left[i \int \mathrm{d}^4x\, \hat{\mathcal{L}}_{int}\right]\} |0\rangle},$$

which is a very important result, as it allows us to calculate Green's functions in the interaction picture, which is much easier than doing the same in the Heisenberg picture, since we can use the free fields to calculate the interactions. Furthermore the right hand side is now manifestly Lorentz invariant. This is the basis for perturbation theory in QFT, where we can expand the exponential of the interaction Lagrangian and calculate the Green's functions order by order in perturbation theory.

3.1.3 | Wick's Theorem

Wick's theorem is a fundamental result in quantum field theory that allows us to express the time-ordered product of field operators as a sum of normal-ordered products and contractions. It is particularly useful for calculating Green's functions and scattering amplitudes in perturbation theory. Let us consider a time-ordered Green's function of the form

$$\langle 0|\, \hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\}\,|0\rangle\,,$$

where $\hat{\phi}(x)$ is a free scalar field operator (we are dropping the subscript 0 for simplicity), thus behaving as a ladder operator. We can manipulate the fields as

$$\hat{\phi}=\hat{\phi}^+ + \hat{\phi}^-, \quad \begin{cases}\hat{\phi}^+|0\rangle = 0,\\ \langle 0|\,\hat{\phi}^- = 0,\end{cases}, \quad \begin{cases}\hat{\phi}^+ = \displaystyle\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}}\hat{a}_{\mathbf{p}}e^{-ip\cdot x},\\ \hat{\phi}^- = \displaystyle\int \frac{\mathrm{d}^3\mathbf{p}}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}}\hat{a}^\dagger_{\mathbf{p}}e^{ip\cdot x},\end{cases}$$

where $\hat{\phi}^+$ is the annihilation part of the field, and $\hat{\phi}^-$ is the creation part of the field. We can then use the normal ordering of the product of $\hat{a}$ and $\hat{a}^\dagger$ to simplify the expression: it puts all the creation operators to the left of the annihilation operators, thus ensuring that the vacuum expectation value of any normal-ordered product is zero. We will use the notation $:\hat{O}:$ to denote the normal ordering of an operator (or a general product of operators) $\hat{O}$.

Example. Let us consider the product of four ladder operators:

$$\hat{a}(p_1)\hat{a}^\dagger(p_2)\hat{a}(p_3)\hat{a}^\dagger(p_4),$$

then we can normal order this product as

$$:\hat{a}(p_1)\hat{a}^\dagger(p_2)\hat{a}(p_3)\hat{a}^\dagger(p_4):=\hat{a}^\dagger(p_2)\hat{a}^\dagger(p_4)\hat{a}(p_1)\hat{a}(p_3),$$

and we can see that the vacuum expectation value of this normal-ordered product is zero:

$$\langle 0|:\hat{a}(p_1)\hat{a}^\dagger(p_2)\hat{a}(p_3)\hat{a}^\dagger(p_4):|0\rangle = 0.$$

If we now consider the time-ordered product of two fields, a **2-point T-product**, assuming $x^0 > y^0$, we can write it as

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}_{x^0>y^0} = \hat{\phi}^+(x)\hat{\phi}^+(y)+\hat{\phi}^+(x)\hat{\phi}^-(y)+\hat{\phi}^-(x)\hat{\phi}^+(y)+\hat{\phi}^-(x)\hat{\phi}^-(y),$$

where the only term that is not normal-ordered is the second term. We can then write the normal-ordered product of the two fields as

$$:\hat{\phi}(x)\hat{\phi}(y):=\hat{\phi}^-(x)\hat{\phi}^-(y)+\hat{\phi}^-(x)\hat{\phi}^+(y)+\hat{\phi}^-(y)\hat{\phi}^+(x)+\hat{\phi}^+(x)\hat{\phi}^+(y),$$

which enforces

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}_{x^0>y^0} =: \hat{\phi}(x)\hat{\phi}(y) : +[\hat{\phi}^+(x), \hat{\phi}^-(y)].$$

Since a symmetric expression exists for the case $x^0 < y^0$, we can write in general that the time-ordered product of two fields can be written as:

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) : +\theta(x^0 - y^0)[\hat{\phi}^+(x), \hat{\phi}^-(y)] + \theta(y^0 - x^0)[\hat{\phi}^+(y), \hat{\phi}^-(x)].$$

If we now take the vacuum expectation value of this time-ordered product (the commutator is not an operator, it is just a function multiplied by the identity), we can see that the normal-ordered part will vanish, since it contains only properly ordered creation and annihilation operators, and thus we are left with[1]

$$\langle 0|\, \hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\, |0\rangle = \Theta(x^0 - y^0)[\hat{\phi}^+(x), \hat{\phi}^-(y)] + \Theta(y^0 - x^0)[\hat{\phi}^+(y), \hat{\phi}^-(x)] = D_F(x, y),$$

where $D_F(x, y)$ is the Feynman propagator, describing the propagation of particles between two points in spacetime:

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) : + D_F(x, y).$$

With these results in hand, we can now define the contraction of two fields, which is a very important concept in Wick's theorem and in the calculation of Green's functions in perturbation theory.

Definition 3.1 (Contraction of two fields)**.** The **contraction of two fields** is defined as *the vacuum expectation value of the time-ordered product* of those two fields, which is the Feynman propagator

$$\hat{\phi}_1\hat{\phi}_2 = \langle 0|\, \hat{T}\{\hat{\phi}_1\hat{\phi}_2\}\, |0\rangle = D_F(x_1, x_2).$$

This time-ordered product of two fields can be rewritten in a more convenient form as the normal-ordered product of the same two fields summed to their contraction:

$$\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\} =: \hat{\phi}(x)\hat{\phi}(y) + \hat{\phi}(x)\hat{\phi}(y) :$$

where the contraction is denoted by a line connecting the two fields, and it represents the Feynman propagator between those two fields.

This is the simplest form of Wick's theorem, which allows us to express the time-ordered product of two fields as a sum of a normal-ordered product and a contraction; we already proven it in the previous steps, since it is the basis for the more general form of Wick's theorem, which we will see next.

This last result can be generalized by induction to the time-ordered product of n fields, which is the most general form of Wick's theorem:

Theorem 3.1 (Wick's theorem)**.** *The time-ordered product of n fields can be written as the normal-ordered product of those n fields plus the sum of all normal-ordered terms containing one, two, or any number of simultaneous pairwise contractions between those fields:*

$$\hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\} =: \hat{\phi}(x_1)\dots\hat{\phi}(x_n) + \text{ all possible contractions } :$$

[1] Here we simplified the product $\langle 0|0\rangle = 1$, since the commutator is just a function multiplied by the identity operator; we could have recognized the Feynman propagator even without taking the vacuum expectation value, but it is important to see how the normal-ordered part vanishes, since it is a crucial step in the proof of Wick's theorem.

where a contraction between two fields is defined as the vacuum expectation value of the time-ordered product of those two fields, which is the Feynman propagator between those two fields:

$$\hat{\phi}_1\hat{\phi}_2 = \langle 0|\,\hat{T}\{\hat{\phi}_1\hat{\phi}_2\}\,|0\rangle = D_F(x_1, x_2).$$

Example (4-point function). Let us consider the following 4-point function

$$\langle 0|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\}\,|0\rangle\,,$$

and let us apply Wick's theorem to it. We can write the time-ordered product of the four fields as the normal-ordered product of those four fields plus all possible contractions between those fields:

$$\begin{aligned}\hat{T}\{\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4\} =& :\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4: \\ &+ \hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4 + \hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4 + \dots \\ &+ \hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4 + \hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4 + \hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4 :\end{aligned}$$

where we denoted $\hat{\phi}(x_i) \equiv \hat{\phi}_i$. Note that when contracting two fields in a product of more than two fields, we do not need to worry about having adjacent fields, since the contraction is well defined as

$$:\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4: = :\hat{\phi}_2\hat{\phi}_4 : D_F(x_1, x_3),$$

and thus we can contract any two fields in the product, regardless of their position, and we will always get a well-defined result.

Using this last observation, we can notice how all the contractions of only two fields will give us a normal-ordered product of the remaining two fields, which will have zero vacuum expectation value, thus the only terms that will contribute to the vacuum expectation value of the 4-point function are the terms with two contractions, which will give us a product of two Feynman propagators: taking all the possible contractions of two pairs of fields, we get

$$\begin{aligned}\langle 0|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\}\,|0\rangle &= D_F(x_1, x_2)D_F(x_3, x_4) \\ &+ D_F(x_1, x_3)D_F(x_2, x_4) + D_F(x_1, x_4)D_F(x_2, x_3),\end{aligned}$$

which shows the extent to which the number of terms grows as we increase the number of fields, since we have to sum over all possible contractions.

Proof. Wick's theorem can be easily proven by induction. Starting from the case of two fields, which we already proved, we can assume that the theorem holds for $n-1$ fields, and then we can write the time-ordered product of n fields, assuming without loss of generality that $x_1^0 > x_j^0$ for $j > 1$, as

$$\hat{T}\{\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n\} = \hat{\phi}_1\hat{T}\{\hat{\phi}_2\cdots\hat{\phi}_n\} = (\hat{\phi}_1^+ + \hat{\phi}_1^-) : \hat{\phi}_2\cdots\hat{\phi}_n + (\,{}^{\text{all contractions of}}_{\hat{\phi}_2\cdots\hat{\phi}_n}\,) :,$$

where we note how one can put $\hat{\phi}_1^-$ right back in the normal ordering, since it is a creation operator and thus it does not change the normal ordering of the remaining fields, while $\hat{\phi}_1^+$ is an annihilation operator and thus it has to be commuted with the remaining fields to be put in normal order, and thus it will give us a contraction with each of the remaining fields. Thus we can write

$$\begin{aligned}\hat{T}\{\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n\} =& :\hat{\phi}_1\hat{\phi}_2\cdots\hat{\phi}_n + \hat{\phi}_1 \times (\,{}^{\text{all contractions of}}_{\hat{\phi}_2\cdots\hat{\phi}_n}\,) : \\ &+ \left[\hat{\phi}_1^+\,, :\hat{\phi}_2\cdots\hat{\phi}_n + (\,{}^{\text{all contractions of}}_{\hat{\phi}_2\cdots\hat{\phi}_n}\,) :\right],\end{aligned}$$

where in order to write the expression for the normal ordered product of the n fields, we had to sum the commutator of $\hat{\phi}_1^+$ with the normal-ordered product and contractions of the remaining

fields, exactely as done in the case of two fields. We can now develop the commutator, which in general behaves as

$$[A, B_1 \cdots B_n] = [A, B_1]B_2 \cdots B_n + B_1[A, B_2] \cdots B_n + \ldots + B_1 \cdots B_{n-1}[A, B_n],$$

where the sum of commutators between $\hat{\phi}_1^+$ and the remaining fields generates all the possible contractions of $\hat{\phi}_1$ with those fields; combining these terms with the normal-ordered product gives us all possible contractions of the n fields, and thus we have proven that the time-ordered product of n fields can be written as the normal-ordered product of those n fields plus all possible contractions between those fields

$$\hat{T}\{\hat{\phi}_1\hat{\phi}_2 \cdots \hat{\phi}_n\} =: \hat{\phi}_1\hat{\phi}_2 \cdots \hat{\phi}_n + \text{ all possible contractions } :,$$

which is exactly what we wanted to prove: □

Let us now see some important consequences of Wick's theorem, which will be crucial for the calculation of Green's functions in perturbation theory.

Corollary 3.2. *As we said before, the vacuum expectation value of any normal-ordered product is zero, thus we have*

$$\langle 0| : \hat{\phi}(x_1) \ldots \hat{\phi}(x_n) : |0\rangle = 0,$$

and thus we can write the vacuum expectation value of the time-ordered product of n fields as

$$\langle 0| \hat{T}\{\hat{\phi}(x_1) \ldots \hat{\phi}(x_n)\} |0\rangle = \textit{all possible } \textbf{\textit{fully contracted}} \textit{ terms},$$

since non-fully contracted terms will be multiplied by a normal ordered product (of the non-contracted fields) which vacuum expectation value is zero.

This is a very important result, as it allows us to calculate Green's functions in terms of Feynman propagators, which are much easier to calculate than the full time-ordered product of fields.

Corollary 3.3. *If in a free theory we have an odd number of fields, then there are no possible fully contracted terms, since we cannot fully contract an odd number of fields, thus we have*

$$\langle 0| \hat{T}\{\hat{\phi}(x_1) \ldots \hat{\phi}(x_{2n+1})\} |0\rangle = 0.$$

This is very important, as it tells us that the vacuum expectation value of the time-ordered product of an odd number of fields is always zero (there will always be at least one field that is not contracted, and thus there will remain a normal-ordered product, whose vacuum expectation value is zero). Fundamentally, this is a consequence of the exact $\mathbb{Z}_2$ symmetry ($\hat{\phi} \to -\hat{\phi}$) of the free Lagrangian, which forces any odd-point correlation function to vanish.

For an interacting theory, this is not necessarily true, but it depends strictly on the symmetries of the interaction. If the interacting Lagrangian preserves the $\mathbb{Z}_2$ symmetry (like in a ϕ^4 theory), the vacuum expectation values of an odd number of fields will still remain zero. However, if the interaction explicitly breaks this symmetry (e.g., a ϕ^3 interaction) or if the symmetry is spontaneously broken (as in the Higgs mechanism, where $\langle \Omega| \hat{\phi} |\Omega\rangle \neq 0$), then odd-point Green's functions can acquire non-zero values.

3.2 | Feynman Diagrams and Rules

Let us start to put all the pieces together with an example. We will first derive the Feynman rules for a simple interacting theory in the coordinate space, and then we will see how to generalize them to more complicated theories and to momentum space.

3.2.1 | Coordinates Space

Example (ϕ^3 theory). Let us study a ϕ^3 theory, which is a simple interacting theory of a scalar field with a cubic interaction term in the Lagrangian:

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{3!}\phi^3,$$

where λ is the coupling constant that controls the strength of the interaction. We want to calculate the 2-point function in this theory, which is given by

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\}\,|\Omega\rangle\,,$$

where $|\Omega\rangle$ is the vacuum state of the interacting theory and $\hat{\phi}$ are the field operators. Using the formula we derived before for the Green's function in the interaction picture, we can write

$$\frac{\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\exp\big[i\int \mathrm{d}^4x\,\mathcal{L}_{\text{int}}\big]\}\,|0\rangle}{\langle 0|\,\hat{T}\{\exp\big[i\int \mathrm{d}^4x\,\mathcal{L}_{\text{int}}\big]\}\,|0\rangle},$$

with the free field operators $\hat{\phi}_0$; now the idea is to expand the exponential in a power series in λ, assuming it small. Let us start from the numerator, which we can write as

$$\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\}\,|0\rangle + (-i\frac{\lambda}{3!})\,\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\int \mathrm{d}^4x\,\hat{\phi}_0(x)^3\}\,|0\rangle + \\ +(-i\frac{\lambda}{3!})^2\,\langle 0|\,\hat{T}\{\hat{\phi}_0(x_1)\hat{\phi}_0(x_2)\int \mathrm{d}^4x\,\hat{\phi}_0(x)^3\int \mathrm{d}^4y\,\hat{\phi}_0(y)^3\}\,|0\rangle + \dots$$

where we have expanded the exponential of the interaction Lagrangian in a power series, and we have also used the fact that the fields in the interaction picture are the same as the free fields, which we have denoted with a subscript 0. With $\lambda = 0$ we get the free theory back, with the usual Feynman propagator as we expected.

The first term in the expansion (order $\mathcal{O}(\lambda^0)$) corresponds to the free theory, which is just a single line connecting the two external points x_1 and x_2, representing the propagation of a free particle. The second term (order $\mathcal{O}(\lambda)$) strictly vanishes by Wick's theorem, since it contains an odd number of fields (five in total: two external and three from the interaction vertex), making it impossible to fully contract them. The third term (order $\mathcal{O}(\lambda^2)$) is the first non-vanishing correction, containing eight fields. Its connected part corresponds to a **diagram with a single loop**, where the particle propagates from x_1 to an interaction vertex x, splits into two virtual particles that propagate to another vertex y (forming the loop), and then recombines to propagate to x_2.

If λ remains small, we can *truncate the expansion* at a certain order and get an approximation for the 2-point function. This is the basis for perturbation theory in QFT, where we can calculate physical quantities as a power series in the coupling constant, and each term in the series corresponds to a Feynman diagram that represents a particular process or interaction.

As we have highlighted, due to Wick's theorem, all corrections of odd order in λ vanish for the 2-point function in ϕ^3 theory. The physical corrections start at $\mathcal{O}(\lambda^2)$ with a one-loop diagram, and higher-order terms will progressively include more complex multi-loop topologies.

Shortcut Notation

Let us introduce a simpler **shortcut notation** for the following computations, since we will have to deal with a large number of fields and propagators, and it will be easier to keep track of them in this way. We will denote the Feynman propagator between two external points x_i and x_j as D_{ij}, the Feynman propagator between an external point x_i and an internal point x as D_{ix}, and the free field operator at a point x_i as ϕ_i. Thus we have

$$\begin{cases} D_{ij} = D_F(x_i, x_j), \\ D_{xi} = D_F(x, x_i), \\ \phi_i = \phi_0(x_i), \end{cases}$$

so that we can write the second order correction to the free theory as

$$(-\frac{i\lambda}{3!})^2 \frac{1}{2!} \int \mathrm{d}^4x \, \mathrm{d}^4y \, \langle 0| \hat{T}\{\hat{\phi}_{x_1}\hat{\phi}_{x_2}\hat{\phi}_x\hat{\phi}_x\hat{\phi}_x\hat{\phi}_y\hat{\phi}_y\hat{\phi}_y\} |0\rangle \,,$$

where now we can use Wick's theorem to calculate the vacuum expectation value of this time-ordered product, which will give us a sum of all possible contractions between the fields:

$$(-\frac{i\lambda}{3!})^2 \frac{1}{2!} \int \mathrm{d}^4x \, \mathrm{d}^4y \, (D_{1x}D_{2y}D_{xy}^2 + \dots),$$

where we have to note that the number of contractions grows very quickly and different contractions may give us the same contribution, since they may be equivalent under permutations of the fields.

But maybe there is a way in which we can **list all the possible contractions** without having to write them all down, since as we can see, the number of contractions grows very quickly as we increase the order of the correction, and they may also be equivalent. This is where **Feynman diagrams** come in handy, as they provide a visual representation of the interactions and allow us to easily keep track of the different terms in the expansion.

Feynman Rules

Now we can start to derive the Feynman rules for this theory, which will allow us to write down the contribution of each diagram without having to calculate all the contractions explicitly. The Feynman graphical representation can be derived from the previous example, where we had the following elements:

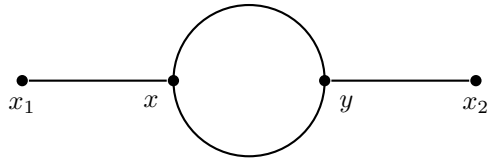


Figure 3.1: Feynman diagram for the second order correction to the 2-point function in a ϕ^3 theory: each internal point corresponds to an interaction vertex, connected to three lines, each corresponding to a propagator.

- x_1, x_2 external points, represented as dots.
- x, y internal points, represented as vertices.
- D_{x1} and D_{y2} propagators, represented as lines connecting the external points to the internal points.
- D_{xy}^2 propagators, represented as two lines connecting the internal points x and y.

More generally, we can write the **Feynman rules** for an n-point function as

- N external points $x_1, x_2, \ldots, x_n$, each connected to only one propagator (line) in every theory.
- Any number of vertexes, each corresponding to an interaction term in the Lagrangian

  $$i \int \mathrm{d}^4 z \, \mathcal{L}_{\text{int}}(z),$$

  where in particular for our ϕ^3 theory, we had

  $$i \int \mathrm{d}^4 z \, (-\frac{\lambda}{3!}\phi^3(z)) = -i\frac{\lambda}{3!} \int \mathrm{d}^4 z \, \phi^3(z),$$

  which means that each vertex has to be connected to three lines, since the interaction term is cubic in the field.
- Every vertex is of order $O(\lambda)$, since it comes from the interaction Lagrangian, and thus the order of the correction is given by the number of vertexes in the diagram.
- The propagators as lines connects ALL the points, since the contractions has to be complete, since we cannot have uncontracted fields in the vacuum expectation value.

These rules telling us how to draw the Feynman diagrams for a given process, and they also allow us to write down the mathematical expression corresponding to each diagram without having to calculate all the contractions explicitly, which is a very powerful tool for calculating physical quantities in QFT.

The procedure of calculating a physical quantity using Feynman diagrams can be summarized as follows:

1. Draw all the possible Feynman diagrams for the process you want to calculate, following the Feynman rules.
2. For each vertex, we have an expression to write down the corresponding factor from the interaction Lagrangian $\rightarrow -i\alpha \int \mathrm{d}^4 z$.
3. For each line, write down the corresponding Feynman propagator $D_F(x, y)$.

Diagram Prefactors

We have to understand how to put the correct prefactors α in front of the integrals. Let us start from the interaction Lagrangian, which is given by

$$\mathcal{L}_{\text{int}} = \frac{\lambda^N}{N!}\phi^N,$$

then a diagram with M vertexes (obtained considering corrections up to order $O(\lambda^M)$) and any number of external points will have

- $M!$ ways of permuting the vertexes, since they are indistinguishable.
- $N!$ ways of permuting the fields in each vertex, since they are also indistinguishable, thus a total of $(N!)^M$ ways of permuting the fields in all the vertexes.

Putting all these factors together, we get a factor of $(N!)^M \times M!$ only from the counting of the permutations of the vertexes and fields, and thus we need to divide by this factor to avoid overcounting, since we are summing over all the possible contractions, which already takes into account all the possible permutations of the vertexes and fields.

An example of naive counting of the factors for a diagram with M vertexes and N fields in each vertex would give us a factor of

$$(N!)^M \times M! \times (\frac{\lambda}{N!})^M \times \frac{1}{M!},$$

where the first factor comes from the permutations of fields in each vertex, the second factor comes from the permutations of the vertexes, the third factor comes from the interaction Lagrangian, and the last factor comes from the expansion of the exponential of the interaction Lagrangian. Is it possible that all these factors cancel out and we are left with a factor of λ^M for a diagram with M vertexes?

No, this counting is not correct, since we have to take into account that some diagrams may be equivalent under permutations of the vertexes and fields, and thus we need to divide by the number of symmetries of the diagram: *the number of permutations of the vertexes and fields that leave the diagram unchanged.* This is known as the **symmetry factor** of the diagram, and it can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram.

Thus the correct factor for a diagram with M vertexes and N fields in each vertex is given by the inverse of the symmetry factor of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram:

Definition 3.2 (Symmetry factor)**.** The symmetry factor of a Feynman diagram is the **number of independent transformations** (permutations of the vertexes and fields in the diagram) **mapping the diagram onto itself** (with external points fixed), and it is denoted by S. The contribution of a diagram to a physical quantity is given by the inverse of the symmetry factor, which is $1/S$. The identity has to be counted as a symmetry, while swapping external points is not a symmetry, since they are fixed. One can sum the symmetries of the drawing by hand, or one can multiply the symmetry prefactors of the subdiagrams (D_{xy}, D^2_{xx}...).

Example (Back to the cubic interaction). Let us go back to the second order correction to the 2-point function in the ϕ^3 theory, which corresponds to the following diagram:

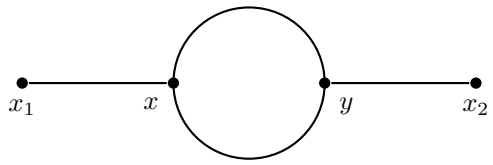


Figure 3.2: Feynman diagram for the second order correction to the 2-point function in a ϕ^3 theory: each internal point corresponds to an interaction vertex, connected to three lines, each corresponding to a propagator.

This diagram has two vertexes, each connected to three lines, and thus it corresponds to a correction of order $O(\lambda^2)$. It can be obtained from 18 different contractions:

- 3 choices of ϕ_x in $\underbrace{\phi_1}\phi_x$;
- 3 choices of ϕ_y in $\underbrace{\phi_2}\phi_y$;
- 2 choices of the remaining ϕ_y to be contracted with the first remaining ϕ_x.

These 18 contractions have to be multiplied by 2 for the swap of the two vertexes $x \leftrightarrow y$. Thus we get a total of 36 contractions which has to be divided by $(3!)^2 \times 2! = 72$, which is the

number of permutations of the fields in the vertexes and the number of permutations of the vertexes, thus we get a symmetry factor of $S = 2$ for this diagram, and thus its contribution to the 2-point function is given by

$$\frac{1}{2} \times (-i\lambda)^2 \int \mathrm{d}^4 z_1 \, \mathrm{d}^4 z_2 \, D_{x_1 x} D_{x_2 y} D_{xy}^2,$$

where the factor of $1/2$ comes from the symmetry factor of the diagram, and the rest of the factors come from the interaction Lagrangian and the Feynman propagators.

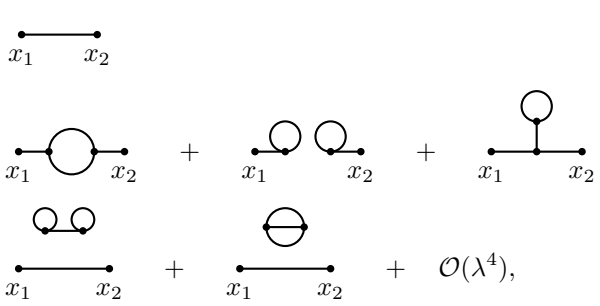
In order to understand the symmetry factor of this diagram, we can also look at it from a different perspective; if we rewrite the contraction, we can see that permuting the propagators between the internal points x and y does not change the diagram, since they are indistinguishable:

$$\phi_1\phi_2\phi_x\phi_x\phi_x\phi_y\phi_y\phi_y \quad = \quad \phi_1\phi_2\phi_x\phi_x\phi_x\phi_y\phi_y\phi_y,$$

and if we compare it to the graphical representation of the diagram, we can see that this contraction symmetry corresponds to the invariance of the diagram under the **vertical flip symmetry**, which is a symmetry of the diagram that leaves the external points fixed and swaps the internal points x and y. This symmetry is the only non-trivial symmetry of the diagram and thus we have a total of 2 symmetries (do not forget the identity), which gives us a symmetry factor of $S = 2$, and thus a contribution of $1/2$ for this diagram.

It can even happen to have an internal subdiagram that is disconnected from the external points, which is called a *vacuum bubble*, and it is possible to show that the contribution of a vacuum bubble cancels out with the denominator of the Green's function. For example, if we were to compute the next order corrections to the 2-point function, we would get the following diagrams:

$\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle\,\Big|_{\text{num}} =$



where the fifth and sixth diagrams have vacuum bubble. The complete expression associated to these diagrams is given by

$$\begin{aligned}\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle\,\Big|_{\text{num}} = D_{12} + (-i\lambda)^2 \int \mathrm{d}^4x \, \mathrm{d}^4y \, \big(&\tfrac{1}{2} D_{1x} D_{2y} D_{xy}^2 + \tfrac{1}{4} D_{1x} D_{xx}^2 D_{2y} D_{yy}^2 \\ &+ \tfrac{1}{2} D_{1x} D_{2x} D_{xy} D_{yy} + D_{12}\big(\tfrac{1}{8} D_{xx} D_{xy} D_{yy} + \tfrac{1}{12} D_{xy}^3\big)\big) + \mathcal{O}(\lambda^4),\end{aligned}$$

where the contribution of the vacuum bubbles can be canceled out with the denominator of the Green's function, which is given by

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle\,\Big|_{\text{den}} = \frac{1}{\langle 0|\,\hat{T}\{\exp\big[i\int \mathrm{d}^4 z\, \mathcal{L}_{int}\big]\}\,|0\rangle}.$$

Thus the denominator is exactly a Green's function with no external points, and thus it contains only **vacuum diagrams**, which are disconnected from the external points, and thus they do not contribute to the physical quantity we are calculating, and thus they can be canceled out with the denominator. This is a very important result, as it tells us that we can ignore the vacuum bubbles

when calculating physical quantities in QFT: they do not contribute to the final result, and thus we can focus only on the connected diagrams, which are the ones that contribute to the physical quantity we are calculating. Indeed we have:

- Numerator: $(\sum \text{connected diagrams})\,(1 + \sum \text{vacuum diagrams})$.
- Denominator: $(1 + \sum \text{vacuum diagrams})$.

Thus in the end, the contribution of the vacuum diagrams cancels out, and we are left with only the connected diagrams, which are the ones that contribute to the physical quantity we are calculating:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle = \sum \text{connected diagrams}.$$

Example (Still on the cubic interaction). If we were to show the previous result explicitly for the ϕ^3 theory, we would have to compute the green's function up to order $\mathcal{O}(\lambda^4)$, which would give us the following diagrams:

$\langle\Omega|\,\hat{T}\{\hat{\phi}(x)\hat{\phi}(y)\}\,|\Omega\rangle$

$$= \frac{\left(\underset{x_1}{\bullet}\!\!-\!\!\underset{x_2}{\bullet} \;+\; \underset{x_1}{\bullet}\!\!-\!\!\bigcirc\!\!-\!\!\underset{x_2}{\bullet} \;+\; \underset{x_1}{\bullet}\!\!\bigcirc\!\!\bigcirc\!\!\underset{x_2}{\bullet} \;+\; \underset{x_1}{\bullet}\!\!-\!\!\bigcirc\!\!-\!\!\underset{x_2}{\bullet} \right)\left(1 + \bigcirc\!\!-\!\!\bigcirc + \bigcirc\!\!\!\bigcirc\right)}{\left(1 + \bigcirc\!\!-\!\!\bigcirc + \bigcirc\!\!\!\bigcirc\right)} + \mathcal{O}(\lambda^4)$$

$$= \underset{x_1}{\bullet}\!\!-\!\!\underset{x_2}{\bullet} \;+\; \underset{x_1}{\bullet}\!\!-\!\!\bigcirc\!\!-\!\!\underset{x_2}{\bullet} \;+\; \underset{x_1}{\bullet}\!\!\bigcirc\!\!\bigcirc\!\!\underset{x_2}{\bullet} \;+\; \underset{x_1}{\bullet}\!\!-\!\!\bigcirc\!\!-\!\!\underset{x_2}{\bullet} + \mathcal{O}(\lambda^4),$$

which shows explicitly that the contribution of the vacuum bubbles cancels out with the denominator, and thus we are left with only the connected diagrams, which are the ones that contribute to the physical quantity we are calculating.

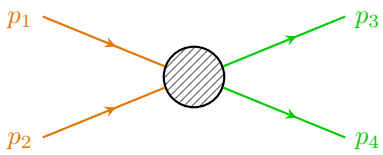
Summary of Feynman Rules in Coordinate Space

Let us group and summarize all the Feynman rules we have derived so far for a scalar field theory in position space:

- **External point**: x_i, represented as a dot, and it is connected to only one propagator.
- **Vertex**: $i\int \mathrm{d}^4 z\, \mathcal{L}_{\text{int}}(z)$, represented as a dot and connected to a number of lines equal to the number of fields in the interaction term. It is specific to the theory we are considering, since it depends on the interaction Lagrangian, and thus it can be different for different theories.
- **Propagator**: $D_F(x, y)$, represented as a line connecting two points, and corresponding to the Feynman free propagator between those two points.
- **Symmetry factor**: $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram. It contains the identity as a symmetry, while swapping external points is not a symmetry, since they are fixed.
- $\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle = \sum \text{connected diagrams}$, since the contribution of the vacuum diagrams cancels out with the denominator of the Green's function, and we can focus only on the connected diagrams, which are the ones contributing to the physical quantity we are calculating.

3.2.2 | Momentum Space

Let us study the same ϕ^3 theory $\mathcal{L}_{int} = -\frac{\lambda}{3!}\phi^3$ for a $2 \to 2$ scattering process.

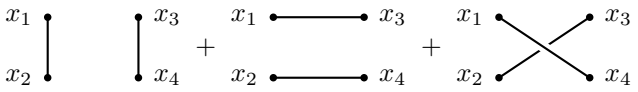


We work up to the **leading order** (LO) correction, which means up to the first non zero matrix element. It is not a general a priori order, such as $\mathcal{O}(\lambda^0)$ or $\mathcal{O}(\lambda^2)$; the leading order is determined by the structure of the interaction term, and is given by the lowest power of the coupling constant λ which gives a non zero matrix element.

The Green's function we are considering can be computed as

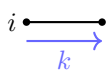
$$\begin{aligned}\langle f|\,\hat{S}\,|i\rangle = \langle p_3, p_4|\,\hat{S}\,|p_1, p_2\rangle = \prod_{j=1}^{4}\left(-\frac{i}{\sqrt{Z}}(p_j^2 - m^2)\right) \\ \times \int \mathrm{d}^4x_1 \ldots \mathrm{d}^4x_4 e^{i(p_3x_3 + p_4x_4 - p_1x_1 - p_2x_2)} \\ \times \langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\hat{\phi}(x_3)\hat{\phi}(x_4)\}\,|\Omega\rangle\,,\end{aligned}$$

where we have used the LSZ reduction formula to relate the S-matrix element to the Green's function. Now for a free theory $Z = 1$, while for an interacting theory $Z = 1 + O(\lambda)$, thus at the leading order we can set $Z = 1$ (since we have lower order terms). Now we can use the Feynman rules in position space to write down the contribution of the leading order diagram, given by

$\mathcal{O}(\lambda^0) =$ 

$$= D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23},$$

where we have three possible contractions between the four fields, and thus we have three possible diagrams. Now we can write down the contribution of each diagram in momentum space by using the Fourier transform of the Feynman propagator, which is given assigning a momentum k to the internal line (with definite direction $i \to j$), and thus we have

$$D_{ij} = \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{e^{-ik(x_i - x_j)}}{k^2 - m^2 + i\epsilon} \quad = \quad$$ 

Following this procedure for the first diagram, we get

$$\begin{aligned}\text{Diag I: } \prod_{j=1}^{4}\left(-i(p_j^2 - m^2)\right) \times \int \mathrm{d}^4x_1 \ldots \mathrm{d}^4x_4 \int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\frac{\mathrm{d}^4k_2}{(2\pi)^4} \\ \times e^{-ix_1(p_1 - k_1)} e^{-ix_2(p_2 + k_1)} e^{ix_3(p_3 - k_2)} e^{ix_4(p_4 + k_2)} \times \frac{i}{k_1^2 - m^2}\frac{i}{k_2^2 - m^2},\end{aligned}$$

where we can recognize the Fourier transform of the propagators, and thus we can perform the integrals over the external points x_1, x_2, x_3, x_4, which gives us delta functions that set the momenta of the internal lines to be equal to the momenta of the external lines, enforcing momentum

conservation at each vertex. Thus simplifing a factor of $(2\pi)^4$ for each delta function, we get a term of the form

$$\delta^{(4)}(p_1-k_1)\delta^{(4)}(p_2+k_1)\delta^{(4)}(p_3-k_2)\delta^{(4)}(p_4+k_2) = \delta^{(4)}(p_1+p_2)\delta^{(4)}(p_3+p_4)\delta^{(4)}(k_1-p_1)\delta^{(4)}(k_2-p_3),$$

where the first two delta functions enforce momentum conservation at the vertices, while the last two delta functions set the internal momenta to be equal to the external momenta. Thus we can perform the integrals over the internal momenta k_1 and k_2, enforcing $k_1 = p_1$ and $k_2 = p_3$, which leads us to

$$\text{Diag I: } \prod_{j=1}^{4}\left(-i(p_j^2-m^2)\right) \times \delta^{(4)}(p_1+p_2)\delta^{(4)}(p_3+p_4) \times \frac{i}{p_1^2-m^2}\frac{i}{p_3^2-m^2} = 0,$$

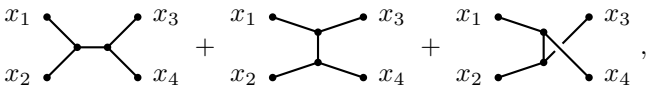
since the delta functions enforce a condition that makes the product vanish: $p_1 + p_2 \neq 0$ as well as $p_3 + p_4 \neq 0$, since they are both positive-energy momenta.

For the second diagram we could repeat the same procedure, and we would get

$$\text{Diag II: } \prod_{j=1}^{4}\left(-i(p_j^2-m^2)\right) \times \delta^{(4)}(p_1-p_3)\delta^{(4)}(p_2-p_4) \times \frac{i}{p_1^2-m^2}\frac{i}{p_2^2-m^2},$$

where the delta functions set $p_3 = p_1$ and $p_4 = p_2$ this time, describing the process where the particles p_1 and p_2 do not scatter, leading to a final state that is the same as the initial state (it is the identity process of $\hat{S} = \mathbb{I} + i\hat{T}$). The third diagram gives us the same contribution as the second diagram, since it is just a permutation of the external points, and we can ignore these two processes.

We are interested in the scattering process where the particles p_1 and p_2 interact and scatter into the final state p_3 and p_4 such that $|f\rangle \neq |i\rangle$. The first non trivial diagrams come from the second order correction $\mathcal{O}(\lambda^2)$, which is given by

$\mathcal{O}(\lambda^2) =$ 

,

where we have three possible diagrams, corresponding to the three possible channels of the scattering process: the s-channel, the t-channel and the u-channel (the origin of the names will be cleared in the following sections).

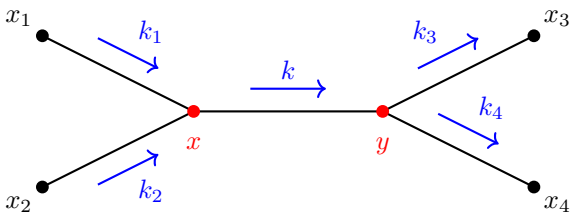


Figure 3.3: s-channel with indication of the exchanged four-momenta at vertices x and y.

Let us focus on the s-channel diagram, the first one, where as done before we assign a momentum k to the internal lines, getting from each vertex and momenta a term of the form $e^{\pm ik\cdot x}$, with a positive sign for outgoing momenta and a negative sign for incoming momenta.

Thus we get the following Fourier transform for the contribution of the first diagram:

$$\text{Diag I: } \prod_{j=1}^{4}\left(-i(p_j^2-m^2)\right) \times \int \mathrm{d}^4x_1 \ldots \mathrm{d}^4x_4\, e^{i(p_3x_3+p_4x_4-p_1x_1-p_2x_2)}$$
$$\times (-i\lambda)^2 \int \mathrm{d}^4x\, \mathrm{d}^4y\, D_{1x}D_{2x}D_{xy}D_{y3}D_{y4}$$

where we have used the Feynman rules in position space to write down the contribution of the diagram, and now we can use the Fourier transform of the propagators to write down the contribution of the diagram in momentum space, which is given by

$$\begin{aligned}\text{Diag I: } \prod(\cdot) \times \int \mathrm{d}^4x_1 \ \ldots \ \mathrm{d}^4x_4 \, \mathrm{d}^4x \, \mathrm{d}^4y \ \times (-i\lambda)^2 \int \frac{\mathrm{d}^4k_1}{(2\pi)^4} \cdots \frac{\mathrm{d}^4k_4}{(2\pi)^4} \, \frac{\mathrm{d}^4k}{(2\pi)^4} \\ \times e^{-ix_1(p_1-k_1)} e^{-ix_2(p_2-k_2)} e^{-ix_3(p_3-k_3)} e^{-ix_4(p_4-k_4)} \\ \times e^{ix(k-k_1-k_2)} e^{iy(k_3+k_4-k)} \times \prod_{q\in\{k_1,\ldots,k_4,k\}} \left(\frac{i}{q^2-m^2}\right).\end{aligned}$$

Paying again attention to factors, we can perform the integrals over the external points x_1, x_2, x_3, x_4, obtaining delta functions that set the momenta of the internal lines to be equal to the momenta of the external lines, enforcing momentum conservation at each vertex. Thus simplifying a factor of $(2\pi)^4$ for each delta function, we get a term of the form

$$\delta^{(4)}(p_1 - k_1) \cdots \delta^{(4)}(p_4 - k_4) \delta^{(4)}(k - k_1 - k_2) \delta^{(4)}(k - k_3 - k_4),$$

where the first four delta functions set the internal momenta to be equal to the external momenta, while the last two delta functions enforce momentum conservation at the vertices. We can rewrite the previous expression as

$$\delta^{(4)}(p_1 - k_1) \cdots \delta^{(4)}(p_4 - k_4) \delta^{(4)}(k - p_1 - p_2) \delta^{(4)}(p_1 + p_2 - p_3 - p_4),$$

in order to make the momentum conservation more explicit, setting $k = p_{in}$ and momentum conservation throughout the whole diagram. To recap:

- We rewrote every **propagator in position space as a Fourier transform**, which introduced an integral over the internal momenta $\int \frac{\mathrm{d}^4k_j}{(2\pi)^4}$ for each internal line (assingning to each of them a momentum vector), and a factor of $e^{\pm ik_j \cdot x}$ for each vertex, with a positive sign for outgoing momenta and a negative sign for incoming momenta.

- Each of the six integrals over internal and external points $\int \mathrm{d}^4x_j$ from the LSZ reduction formula give us delta functions that set the momenta $k_j = p_j$.

- The integrals over the internal points $\int \mathrm{d}^4x$ and $\int \mathrm{d}^4y$ give us delta functions that set the momenta of the internal lines to be equal to the momenta of the external lines, **enforcing momentum conservation at each vertex**.

- The combination of the delta functions from the external points and the internal points give us a global delta function that enforces momentum conservation for the whole scattering process, which is given by $(2\pi)^4\delta^{(4)}(p_{in} - p_{out})$, where p_{in} and p_{out} are the total incoming and outgoing momenta, respectively.

If we conclude the calculation for the first diagram, we note how each of the factors $\frac{i}{q^2-m^2}$ cancels with a factor of $-i(p_j^2 - m^2)$ from the LSZ reduction formula, since the external momenta are on-shell, thus we are left with only the contribution of the delta enforcing $\delta^{(4)}(k - p_1 - p_2) \implies k = p_1 + p_2$, leaving us with a final contribution of the form

$$\text{Diag I: } \frac{i}{(p_1 + p_2)^2 - m^2}.$$

The delta functions do not always cancel out all the integrals over momenta, and one can show indeed that if the diagram contains a loop, then we are left with an integral over the loop momentum, which is not fixed by the delta functions, and thus we need to integrate over it. For example, if we consider the following diagram, which is a one-loop correction to the 2-point function:

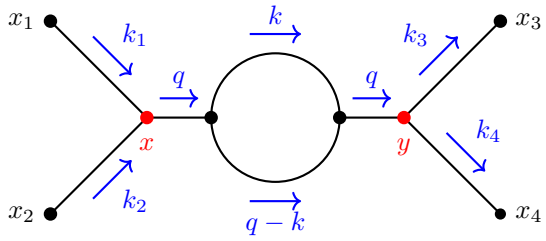


Figure 3.4: 1-loop correction of the s-channel diagram of the $2 \to 2$ scattering process.

we get always the deltas enforcing internal momenta to be equal to the external momenta, but we are left with an integral over the loop momentum k, since among the four internal lines (giving rise to four delta functions, one for each line), only three of the assigned momenta are fixed by the delta functions, while the fourth one remains unfixed for the presence of the global $\delta^{(4)}(p_{in} - p_{out})$, which is always present up to rewriting the delta's arguments: we are left with

$$\int \frac{\mathrm{d}^4 k}{(2\pi)^4}, \quad k \equiv \text{loop momentum},$$

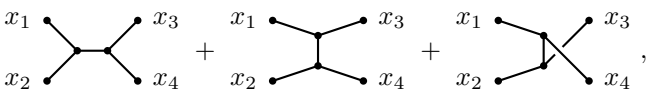
which is not fixed by the delta functions, and thus we need to integrate over it. This is a general feature of loop diagrams, and it is one of the main reasons why loop diagrams are more difficult to calculate than tree-level diagrams, since they involve integrals over loop momenta that can be divergent, and thus we need to use regularization and renormalization techniques to make sense of them.

Thus remembering that $\hat{S} = \mathbb{I} + i\hat{T}$, we found that up to $\mathcal{O}(\lambda^0)$ we have the identity process of the scattering, while the first non trivial contribution to the scattering amplitude comes from $\mathcal{O}(\lambda^2)$ where, after performing the integrals over the internal momenta k_1, k_2, k_3, k_4 (enforcing $k_j = p_j$ for $j = 1, 2, 3, 4$ as we said), we are left with a contribution of the form

$$\langle f | \, i\hat{T} \, | i \rangle = i\mathcal{M}(2\pi)^4 \delta^{(4)}(p_{in} - p_{out}),$$

where $\mathcal{M}$ is the scattering amplitude, which is a function of the external momenta, and it encodes all the information about the scattering process; this is the quantity that we are interested in calculating, since it encodes the probability amplitude for the scattering process, and we can obtain the cross section and the decay rate from it, which are the physical observables that we can measure in experiments. The Feynman rules in momentum space allow us to calculate $\mathcal{M}$ for any given scattering process, by summing over all the fully connected diagrams that contribute to the process, and applying the rules we have just described.

Finishing the computation for the leading order correction, we had the three channel diagrams:

$\mathcal{O}(\lambda^2) =$ 

,

where the contribution of the deltas in the internal lines (internal momenta) is respectively given by

- $p_1 + p_2$ for the s-channel diagram;

- $p_1 - p_3$ for the t-channel diagram;
- $p_1 - p_4$ for the u-channel diagram.

Thus in the end we get the following contribution for the leading order correction:

$$i\mathcal{M} = (-i\lambda)^2 \left[\frac{i}{(p_1 + p_2)^2 - m^2} + \frac{i}{(p_1 - p_3)^2 - m^2} + \frac{i}{(p_1 - p_4)^2 - m^2} \right],$$

where we have a contribution from each of the three channels, and we can see that the scattering amplitude is a function of the external momenta, encoding all the information about the scattering process. The $(-i\lambda)^2$ factor comes from the two vertexes, common for all the three diagrams, while inside the parenthesis we have the three propagator contributions of each diagram, which is given by the product of the propagators corresponding to the internal lines, and the delta functions that enforce momentum conservation at each vertex have already been integrated out: only applying the rules we got the result which would have implied a lot of calculations otherwise.

Preliminary Summary of Rules in Momentum Space

Let us summarize the Feynman rules in momentum space for a scalar field theory; for leading order (LO) corrections, this will be plenty enough to understand the scattering process and calculate the scattering amplitude, while for higher order corrections, we will need to take into account also the renormalization of the theory, which is a more advanced topic that we will have to cover in the next chapter. The preliminary Feynman rules in momentum space are the following:

- $i\mathcal{M} = \sum$ fully connected diagrams, since we have seen that only these diagrams contribute to the scattering process , while the diagrams with bubbles are canceled out by the denominator of the green function; $\mathcal{M}$ is the scattering amplitude, which encodes all the information about the scattering process.
- We assign a momentum k to each internal line, with the convention that the momentum flows from the initial state to the final state (thus the direction matters, even if it is a four-vector).
- The external lines are on-shell, thus they satisfy the mass-shell condition $p_j^2 = m^2$ for $j = 1, 2, 3, 4$ and each of them contributes with a factor of $-i(p_j^2 - m^2)$ from the LSZ reduction formula; we have seen that these factors cancel with the propagators of the internal lines, thus external lines contribute with a factor of 1.
- Each of the internal lines corresponds to a propagator, given by $\frac{i}{k^2 - m^2}$ in the momentum space, where k is the momentum flowing through the internal line.
- The vertexes correspond to factors from the interaction Lagrangian, which in momentum space are given by $-i\lambda$ for a ϕ^3 theory, and more generally by the Fourier transform of the interaction term in the Lagrangian. They depend on the theory under consideration, and they also depend on the momenta flowing through the vertex.
- **Momentum conservation at each vertex**: the sum of the incoming momenta must equal the sum of the outgoing momenta, which is enforced by delta functions in the mathematical expression corresponding to each diagram.
- Each loop in the diagram corresponds to an unresolved integral over the loop momentum, which is not fixed by the delta functions, and thus we need to integrate over it. The measure of the integral is given by $\int \frac{\mathrm{d}^4 k}{(2\pi)^4}$ for each unresolved loop momentum k.

- Prefactor: $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram. Same as in position space.

When saying **fully connected diagrams**, we mean that the diagram is connected and that it *does not contain any disconnected subdiagram*, i.e. we do not include in this definition diagrams which are formed by 2 or more connected components, which are not connected to each other, and thus they do not describe a global interaction of the incoming particles scattering into the outgoing particles, but rather they describe separate processes that are not related to each other.

Example (Fully connectedness in $2 \to 4$ scattering). In the following $2 \to 4$ diagram, where we have two incoming particles and four outgoing particles, we have two possible diagrams, one of them is fully connected, while the other one is not fully connected:

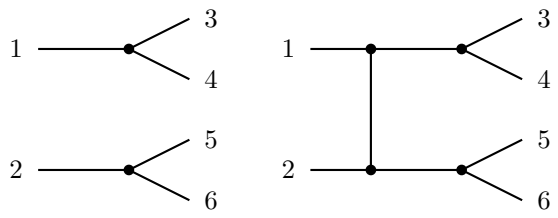


Figure 3.5: In the calculation of scattering amplitudes and fully interacting correlation functions, disconnected sub-diagrams (left) are factored out and excluded. Only fully connected topologies (right) contribute to the non-trivial S-matrix elements.

The non fully connected diagram describes to separate processes more than a global interaction of 2 incoming particles scattering into 4 outgoing particles, and thus it does not contribute to the scattering amplitude of the process we are interested in, while the fully connected diagram describes exactely the process we are interested in.

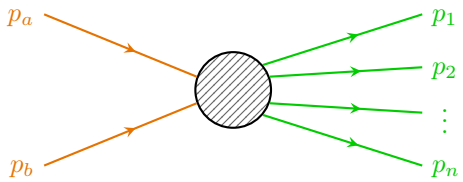
Remark. *At leading order we have set $Z = 1$, and we confused the lagrangian mass with the physical one: in the next chapter we will see that at higher order corrections we need to take into account the renormalization of the theory, which will lead us to distinguish between the lagrangian mass and the physical mass, and to introduce a renormalization factor Z for the fields:*

$$\begin{cases} Z = 1 + \mathcal{O}(\lambda), \\ m_{phys} = m_{\mathcal{L}} + \mathcal{O}(\lambda). \end{cases}$$

this will modify the Feynman rules we have just described, but for now we can ignore these complications and use the rules we have just derived. In our Feynman rules we are using the lagrangian mass $m_{\mathcal{L}}$, but in future it will be convenient to make the substitution $m_{\mathcal{L}} \to m_{phys} + \delta m$, where δm represents additional terms added to the lagrangian.

Back to LSZ Reduction Formula

Now consider a $2 \to n$ scattering process, where we have two incoming particles and n outgoing particles.



If we want to calculate the scattering amplitude for this process, we can use the LSZ reduction formula, which relates the S-matrix element to the Green's function. The Green's function we need to calculate is given by

$$\langle \Omega | \hat{T}\{\hat{\phi}(x_1)\dots\hat{\phi}(x_n)\hat{\phi}(x_a)\hat{\phi}(x_b)\} | \Omega \rangle .$$

We have seen thay at leading order (LO) we have $Z = 1$ and in the LSZ reduction formula we have a factor of $\prod_{j=1}^{n+2}\left(-i(p_j^2 - m^2)\right)$, where p_j are the external momenta, which are on-shell, thus they satisfy the mass-shell condition $p_j^2 = m^2$ for $j = 1, 2, \dots, n+2$. This means that each of the external lines contributing with a factor of $-i(p_j^2 - m^2)$ cancels with a factor of $\frac{i}{p_j^2 - m^2}$ from the propagator of the internal line.

But what happens at higher order corrections? We can schematically represent the contribution of the diagrams to the scattering amplitude as follows:

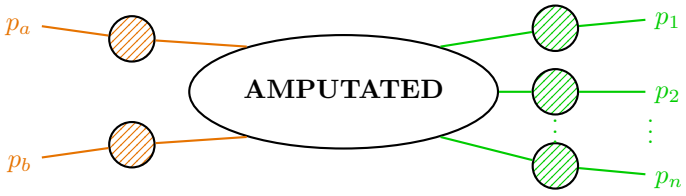
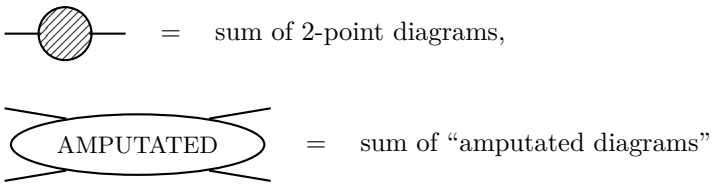


Figure 3.6: Structure of a fully interacting correlation function. To obtain the S-matrix element, the external legs containing the full propagator corrections (hatched circles) are amputated, leaving only the amputated connected Green's function in the center.

Where:



Definition 3.3 (Amputated diagram)**.** An amputated diagram is a Feynman diagram where we cannot disconnect any external line by cutting a single internal line. In other words, an amputated diagram is a diagram that does not contain any disconnected subdiagram, and thus it is fully connected.

The contribution of an amputated diagram to the scattering amplitude is given by the *product of the propagators corresponding to the internal lines*, and the *factors from the vertexes*, without including the external propagators, since they are on-shell and thus they do not contribute to the scattering amplitude.

Example. In the following diagram, we have a cut procedure used to amputate a 1-loop self-energy correction from an incoming external leg. Thanks to the LSZ reduction formalism, we know that the external legs of the Green's function contain full propagator corrections, which are given by the sum of all the 2-point diagrams, and thus we can amputate them in order to

extract more efficiently the scattering amplitude: we will see how we are ignoring self-deleting terms, since in the computation of the green's function in the LSZ reduction formula these contributions cancel out.

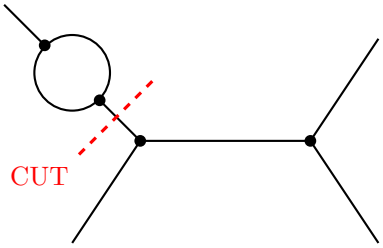
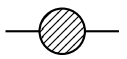
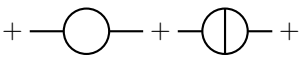


Figure 3.7: Example of a cutting procedure used to amputate a 1-loop self-energy correction from an incoming external leg. In the LSZ reduction formalism, isolating and removing these external full-propagator corrections is required to extract the pure amputated scattering amplitude.

Let us show that the contribution of the external subdiagrams we are amputating is negligible for the scattering amplitude. Each of the external lines in the LSZ reduction contributes with a factor

$$-\frac{i}{\sqrt{Z}}\left(p_j^2 - m^2\right),$$

and we know these momenta to be on-shell, thus they satisfy the mass-shell condition $p_j^2 = m^2$ for $j = 1, 2, \ldots, n+2$ (m is the physical mass). Thus each of the external subdiagrams we are truncating is the sum of two point functions:

$p_j \to$  $=$  $+ \ldots$

$$= \text{ Fourier transform of } \langle \Omega | \, T\hat{\phi}(x_1)\hat{\phi}(x_2) \, | \Omega \rangle$$

$$= \frac{iZ}{p_j^2 - m^2} + \text{ regular terms at } p_j^2 \sim m^2,$$

which implies that each of the $\frac{i}{p_j^2 - m^2}$ terms cancel with the $-i(p_j^2 - m^2)$ factor from the LSZ reduction formula, leaving us with a factor of $\sqrt{Z}$ for each external line, which is the pole residue of the full propagator.

Summary of Rules in Momentum Space

- $i\mathcal{M} = \sum$ fully connected amputated diagrams.
- Assign momentum to internal lines, with the convention that the momentum flows from the initial state to the final state, and it is conserved at each vertex.
- For each unresolved momentum, i.e. for each loop, we need to integrate over it with the measure $\int \frac{\mathrm{d}^4 k}{(2\pi)^4}$.
- External lines contribute with a factor of 1 each, since they are on-shell and thus they do not contribute to the scattering amplitude.
- Internal lines contribute with a factor of $\frac{i}{p^2 - m^2}$ each, where p is the momentum flowing through the internal line: propagator $\equiv$ internal line.
- Vertexes contribute with a factor of $-i\lambda$ for a ϕ^3 theory, and more generally by the Fourier transform of the interaction term in the Lagrangian, which depends on the theory under consideration and on the momenta flowing through the vertex.

- The symmetry factor of the diagram is given by $1/S$, where S is the number of symmetries of the diagram, which can be calculated by counting the number of ways to permute the vertexes and fields without changing the diagram.
- For each external particle, we need to multiply by a factor of $\sqrt{Z}$, which comes from the LSZ reduction formula: Z is the pole residue of the full propagator, and it is equal to 1 at the leading order, but it can receive corrections at higher orders.

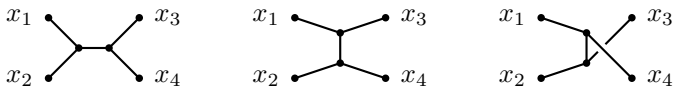
We derived these feynman rules from a real scalar theory with a cubic interaction, but they can be easily generalized to other theories: the only theory specific point is the vertex factor, which depends on the interaction term in the Lagrangian, while the other rules are general and can be applied to any theory.

3.3 | Applications of Feynman Diagrams

Let us now apply the Feynman rules to compute scattering amplitudes for various processes in different quantum field theories. We will start with the simplest case of a scalar field theory and then move on to more complex theories involving fermions and gauge bosons.

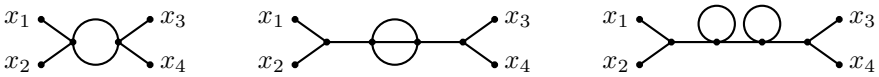
We will divide diagrams in two categories:

1. **The tree-level diagrams**: These are the diagrams that do not contain any loops. They represent the leading order contribution to the scattering amplitude and are often the simplest to compute.



2. **The loop diagrams**: These are the diagrams that contain one or more loops. They represent higher-order corrections to the scattering amplitude and can be more challenging to compute due to the need for regularization and renormalization techniques.

TODO: improve diagrams.



Thus in the loop diagrams, one or more internal lines form a closed loop, which can lead to unresolved integrals like $\int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2-m^2}$, which is divergent and requires regularization and renormalization to obtain a finite result.

Usually the first type of diagrams (tree-level) gives the dominant contribution to the scattering amplitude, while the second type (loop diagrams) provides corrections that can be important for precision calculations. For the first ones, usually some algebra is enough to compute the amplitude, since we have no integrations left, while for the second ones, one needs to use more advanced techniques to handle the divergences and obtain meaningful results. Loop diagrams can also give rise to interesting phenomena such as virtual particles and vacuum polarization, which can have important implications for the behavior of quantum fields and the structure of spacetime, but they are in general more difficult to solve.

Remark. *We said that usually at LO only the tree diagrams appears, but this is nothing general: there exist cases in which we have loops appearing alredy at the leading order, which are more complicated.*

Example ($2 \to 2$ scattering in ϕ^3 theory). The idea is to compute the **differential cross section** $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ for the $2 \to 2$ scattering process of identical particles in ϕ^3 theory at tree level, in the center of mass frame, as a function of the total center of mass energy E and the scattering angle θ.

At the leading order, we have already computed the matrix element for the $2 \to 2$ scattering process in ϕ^3 theory: the Feynman rules told us that the relevant diagrams at tree level include the s-channel, t-channel, and u-channel diagrams:

$$i\mathcal{M} = \quad + \quad + \quad + \mathcal{O}(\lambda^4)$$

$$= (-i\lambda)^2 \left[\frac{1}{(p_1+p_2)^2 - m^2} + \frac{1}{(p_1-p_3)^2 - m^2} + \frac{1}{(p_1-p_4)^2 - m^2} \right] + O(\lambda^4).$$

We can define the **scattering angle** θ as the angle between the incoming particle with momentum p_1 and the outgoing particle with momentum p_3:

$$\cos\theta = \frac{\mathbf{p}_1 \cdot \mathbf{p}_3}{|\mathbf{p}_1||\mathbf{p}_3|},$$

and this will allow us to express the scattering amplitude in terms of the Mandelstam variables, which are functions of E and θ. Before proceeding with the calculation, we have to study in more detail the kinematics of the process, which is the subject of the next section.

3.3.1 | 4-Point Kinematics

Consider a more general $2 \to 2$ scattering process, involving two incoming particles with momenta p_1 and p_2, and two outgoing particles with momenta p_3 and p_4. We are not interested in the specific theory describing the interaction or the type of particles involved, but we are requesting that all particles are on-shell, so that

$$p_i^2 = m^2, \quad i = 1, 2, 3, 4.$$

The scattering amplitude is lorentz invariant, so that it is useful to express it in terms of the **Mandelstam variables** s, t, and u.

Definition 3.4 (Mandelstam variables)**.** The **Mandelstam variables** s, t, and u, are defined as follows:

$$\begin{aligned} s &= (p_1 + p_2)^2 = (p_3 + p_4)^2, \\ t &= (p_1 - p_3)^2 = (p_4 - p_2)^2, \\ u &= (p_1 - p_4)^2 = (p_2 - p_3)^2. \end{aligned}$$

These variables are not independent, since they satisfy the relation

$$s + t + u = \sum_j m_j^2,$$

as we will later prove. These variables are defined in such a way that they are invariant under Lorentz transformations, and provide a convenient way to express the scattering amplitude in terms of the kinematic variables of the process.

Now the idea is to express the scattering amplitude in terms of the Mandelstam variables, which will allow us to compute the differential cross section as a function of the scattering angle θ and the total center of mass energy E.

Starting from their definitions, we can express

$$s = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = m_1^2 + m_2^2 + 2p_1 \cdot p_2,$$

so that it becomes clear that the Mandelstam variables are related to the Lorentz invariants $p_i \cdot p_j$. In particular, we have

$$\begin{cases} p_1 \cdot p_2 = \dfrac{1}{2}\left(s - m_1^2 - m_2^2\right), \\ p_1 \cdot p_3 = \dfrac{1}{2}\left(t - m_1^2 - m_3^2\right), \\ p_2 \cdot p_3 = \dfrac{1}{2}\left(u - m_2^2 - m_3^2\right), \\ p_4 = p_1 + p_2 - p_3, \end{cases}$$

so that every Lorentz invariant $p_i \cdot p_j$ can be expressed in terms of the Mandelstam variables, since p_4 can be always expressed in terms of the other three momenta using momentum conservation. Thus we are going to square the last relation, and we will obtain

$$\begin{aligned} m_4^2 = p_4^2 = (p_1 + p_2 - p_3)^2 &= m_1^2 + m_2^2 + m_3^2 + 2p_1 \cdot p_2 - 2p_1 \cdot p_3 - 2p_2 \cdot p_3 \\ &= m_1^2 + m_2^2 + m_3^2 + 2\left(\tfrac{1}{2}\right)\left(s - m_1^2 - m_2^2\right) \\ &- 2\left(-\tfrac{1}{2}\right)\left(t - m_1^2 - m_3^2\right) - 2\left(-\tfrac{1}{2}\right)\left(u - m_2^2 - m_3^2\right) \\ &= s + t + u - m_1^2 - m_2^2 - m_3^2. \end{aligned}$$

Thus we have

$$s + t + u = \sum_j m_j^2,$$

which is the relation we wanted to prove. For an N-particle scattering process, we can define the Mandelstam variables as ($N > 4$):

1. $4N$ momentum components p_i^μ for $i = 1, \ldots, N$.
2. N on-shell conditions $p_i^2 = m_i^2$.
3. 4 momentum conservation conditions $\sum_{i=1}^N p_i^\mu = 0$ (from translational invariance).
4. 6 independent Lorentz invariants $p_i \cdot p_j$ for $i < j$, i.e. 6 lorentz transformations leaving the amplitude invariant.

Thus in the end we have

$$4N - N - 4 - 6 = 3N - 10 \textbf{ independent invariants},$$

which can be chosen as

$$s_{ij} = (p_i + p_j)^2, \quad i < j,$$

these represent the generalization of the Mandelstam variables for N-particle scattering processes.

Applications

Now for a $2 \to 2$ general process, in the center of mass frame, we have two incoming particles parallel along the z-axis, and two outgoing particles with momenta $\mathbf{p}_3$ and $\mathbf{p}_4$ forming an angle θ with the z-axis

$$p_1 = \begin{pmatrix} E_1 \\ 0 \\ 0 \\ p \end{pmatrix}, \quad p_2 = \begin{pmatrix} E_2 \\ 0 \\ 0 \\ -p \end{pmatrix},$$

where $p = |\mathbf{p}_1| = |\mathbf{p}_2|$ since $\mathbf{p}_1 + \mathbf{p}_2 = 0$ in the center of mass frame. Now considering the outgoing particles, we can rotate the coordinate system so that they lie in the z-y-plane, with momenta

$$\mathbf{p}_4 = \mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_3 = -\mathbf{p}_3, \quad p' = |\mathbf{p}_3| = |\mathbf{p}_4|,$$

$$\mathbf{p}_3 = \begin{pmatrix} E_3 \\ 0 \\ p' \sin\theta \\ p' \cos\theta \end{pmatrix}, \quad \mathbf{p}_4 = \begin{pmatrix} E_4 \\ 0 \\ -p' \sin\theta \\ -p' \cos\theta \end{pmatrix},$$

since we are still in the center of mass frame; θ is the scattering angle defined previously. Note that the scattering amplitude is invariant under rotations, so that we can always choose the coordinate system in such a way that the outgoing particles lie in the z-y-plane, and this will not affect the final result for the scattering amplitude.

Now let us compute the center of mass energy E and the Mandelstam variables in terms of the scattering angle θ. We have

$$E \equiv E_{cm} = E_1 + E_2 = E_3 + E_4,$$

with the first Mandelstam variable defined as

$$s = (p_1 + p_2)^2 = (E_1 + E_2)^2 = E^2,$$

since the spatial momenta of the incoming particles cancel out. Now

$$(p_1 + p_2) \cdot p_1 = E\, E_1,$$

and we can also express it as

$$(p_1 + p_2) \cdot p_1 = m_1^2 + p_2 \cdot p_1 = m_1^2 + \frac{1}{2}(s - m_1^2 - m_2^2) = \frac{1}{2}(E^2 + m_1^2 - m_2^2).$$

Thus we have

$$\begin{cases} E_1 = \dfrac{E^2 + m_1^2 - m_2^2}{2E}, \\ E_2 = \dfrac{E^2 + m_2^2 - m_1^2}{2E}, \end{cases}$$

and now

$$m_1^2 = p_1^2 = E_1 - |\mathbf{p}_1|^2, \quad p = \frac{\sqrt{[E^2 - (m_1 + m_2)^2]\,[E^2 - (m_1 - m_2)^2]}}{2E},$$

obtaining the expression for the momentum of the incoming particles in the center of mass frame. The problem is symmetric in the incoming and outgoing particles, so that we can do the same calculation for the outgoing particles, and we will obtain

$$\begin{cases} E_3 = \dfrac{E^2 + m_3^2 - m_4^2}{2E}, \\ E_4 = \dfrac{E^2 + m_4^2 - m_3^2}{2E}, \end{cases} \quad p' = \frac{\sqrt{[E^2 - (m_3 + m_4)^2]\,[E^2 - (m_3 - m_4)^2]}}{2E},$$

where the same calculation enforces the substitutions $1 \to 3$, $2 \to 4$ and $p \to p'$.

For the second Mandelstam variable, we have similarly

$$t = (p_1 - p_3)^2 = m_1^2 + m_3^2 - 2p_1 \cdot p_3 = m_1^2 + m_2^2 - 2(E_1 E_3 - pp' \cos\theta) = t(E, \theta),$$

showing that t is a function of the center of mass energy E and the scattering angle θ. For the third Mandelstam variable, we have

$$u = (p_1 - p_4)^2 = m_1^2 + m_4^2 - 2p_1 \cdot p_4 = m_1^2 + m_4^2 - 2(E_1 E_4 + pp' \cos\theta) = u(E, \theta),$$

which is also a function of the invariants E and θ. Thus we have expressed the Mandelstam variables in terms of the center of mass energy E and the scattering angle θ, which will allow us to compute the scattering amplitude and the differential cross section as a function of these variables.

Special cases

- $(m_1 = m_2)$: In this case, we have
$$E_1 = E_2 = \frac{E}{2}, \quad p = \frac{\sqrt{E^2 - 4m^2}}{2}.$$
- $(m_3 = m_4)$: In this case, we have
$$E_3 = E_4 = \frac{E}{2}, \quad p' = \frac{\sqrt{E^2 - 4m^2}}{2}.$$
- **Elastic scattering**: In this case, we have $m_1 = m_3$ and $m_2 = m_4$, so that
$$E_1 = E_3, \quad E_2 = E_4, \quad p = p'.$$
- $(m_1 = m_2 = m_3 = m_4)$: In this case, we have
$$E_i = \frac{E}{2}, \quad p = p' = \frac{\sqrt{E^2 - 4m^2}}{2}.$$
This is the combination of the previous two cases, and it is the most common one in practice, since it describes the scattering of identical particles. The mandelstam variables in this case are given by
$$\begin{cases} s = E^2, \\ t = -\frac{1}{2}(E^2 - 4m^2)(1 - \cos\theta), \\ u = -\frac{1}{2}(E^2 - 4m^2)(1 + \cos\theta), \end{cases}$$
with the usual relation $s + t + u = 4m^2$.

Back to The Example

Now we have what we need to compute the scattering amplitude for the $2 \to 2$ scattering process in ϕ^3 theory at tree level, which is given by the sum of the s-channel, t-channel, and u-channel diagrams. Let us come back to this example.

Example ($2 \to 2$ scattering in ϕ^3 theory). Remember that at the leading order, the matrix element for the $2 \to 2$ scattering process in ϕ^3 theory can be computed using the Feynman rules: the relevant Feynman diagrams at tree level include the s-channel, t-channel, and u-channel diagrams, which have the following momentum dependence:

$i\mathcal{M} =$  $+ O(\lambda^4),$

which contribute to the scattering amplitude as follows:

$$i\mathcal{M} = (-i\lambda)^2 \left[\frac{1}{s - m^2} + \frac{1}{t - m^2} + \frac{1}{u - m^2} \right] + O(\lambda^4),$$

from which it is easy to understand why they are called s-channel, t-channel, and u-channel diagrams; remember that in this case the Mandelstam variables are given by

$$\begin{cases} s = E^2, \\ t = -\frac{1}{2}(E^2 - 4m^2)(1 - \cos\theta), \\ u = -\frac{1}{2}(E^2 - 4m^2)(1 + \cos\theta), \end{cases}$$

since we are considering the scattering of identical particles, so that $m_1 = m_2 = m_3 = m_4 = m$.

Now we can express the scattering amplitude in terms of the Mandelstam variables (expressing them in terms of E and θ as we found before), which will allow us to compute the differential cross section as a function of the scattering angle θ and the total center of mass energy E.

In particular, we have the differential cross section found at the beginning of the section, which in the center of mass frame can be expressed as

$$\begin{aligned}\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} &= \frac{1}{64\pi^2 E^2}\frac{p'}{p}|\mathcal{M}|^2 = \frac{1}{64\pi^2 E^2}\frac{\sqrt{E^2-4m^2}}{\sqrt{E^2-4m^2}}|\mathcal{M}|^2 \\ &= \frac{1}{64\pi^2 s}|\mathcal{M}|^2 = \frac{\lambda^4}{64\pi^2 s}\left|\frac{1}{s-m^2}+\frac{1}{t-m^2}+\frac{1}{u-m^2}\right|^2 + O(\lambda^6).\end{aligned}$$

This is the final result for the differential cross section for the $2 \to 2$ scattering process in ϕ^3 theory at tree level, as a function of the total center of mass energy E and the scattering angle θ.

3.3.2 | Feynman Rules for Other Theories

TODO: check below

We can generalize some of the Feynman rules we have found for the real scalar field to other theories, such as for different types of fields (complex scalar fields, Dirac fermion fields, vector boson fields) and for different interactions (Yukawa interactions, gauge interactions).

For example, the generalization to a ϕ^n theory is basically trivial:

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{n!}\phi^n,$$

we just have to take into account the fact that each interaction vertex will now involve n lines instead of 3, and the Feynman rule for the vertex will be given by $-i\lambda$ times a combinatorial factor that takes into account the number of ways to connect the lines to the vertex.

Remark. *Remember that the Feynman rules for vertices are rigorously derived from the action integral $\int d^4x\, i\mathcal{L}_{int}(x)$ after performing all possible Wick contractions between the fields in the interaction Lagrangian and the external states (or internal propagators).*

For example, for a $2 \to 2$ process in a ϕ^4 theory, since the vertexes come from the integrals $\int \mathrm{d}^4x\mathcal{L}_{\text{int}}$ after contracting all the fields, at the tree level we have only one diagram:

$$i\mathcal{M} = \quad \begin{matrix} x_1 & & x_3 \\ & \times & \\ x_2 & & x_4 \end{matrix} \quad + \mathcal{O}(\lambda^2) = -i\lambda + \mathcal{O}(\lambda^2),$$

which is the one with a single vertex, and the Feynman rule for the vertex will be given by $-i\lambda$.

Interactions Involving Derivatives

A physically richer and mathematically more subtle case arises when the interaction Lagrangian contains derivatives of the fields. To illustrate this, let us consider a toy model with the following interaction term:

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{2!}\phi(\partial_\mu\phi)(\partial^\mu\phi)$$

Note the symmetry factor $\frac{1}{2!}$: permuting the two identical $(\partial_\mu\phi)$ terms leaves the operator unchanged, but the plain field ϕ without derivatives plays a distinct mathematical role.

To derive the momentum-space Feynman rule for this vertex, we recall that the spacetime integrals $\int d^4x$ yield the overall momentum conservation delta functions. When we evaluate the Wick contractions, the fields $\phi(x)$ are replaced by their Fourier modes. The spacetime derivative ∂_μ acting on a plane wave $e^{\pm ip\cdot x}$ pulls down a factor of momentum.

Specifically, the general prescription for translating spacetime derivatives into momentum space is:

$$\partial_\mu \to \begin{cases} +ip_\mu & \text{for OUTGOING momenta} \\ -ip_\mu & \text{for INCOMING momenta} \end{cases}$$

Remark. *In this simplified derivation, we are conveniently ignoring subtle issues regarding the strict time-ordering (T-product) of derivative operators inside the path integral or Dyson series. While a completely rigorous derivation requires more care (often involving the Hamiltonian formalism or path integral subtleties), the resulting momentum-space Feynman rule obtained via this simple substitution is indeed correct.*

Let us apply this to our toy model to find the vertex rule for a $1 \to 2$ process, with one incoming momentum p_1 and two outgoing momenta p_2, p_3.

TODO: add diagram showing a 3-point vertex with one incoming line p1 and two outgoing lines p2, p3

The fully contracted term in the amplitude involves the integral:

$$\approx i \int d^4x\, \mathcal{L}_{\text{int}} = \frac{-i\lambda}{2!} \int d^4x\, e^{-ip_1\cdot x} \left(\partial_\mu e^{+ip_2\cdot x}\right) \left(\partial^\mu e^{+ip_3\cdot x}\right) + \text{permutations}$$

When the derivatives act on the outgoing plane waves, they pull down factors of $(+ip_2)$ and $(+ip_3)$. We must also sum over all possible ways to contract the external momenta (p_1, p_2, p_3) with the specific fields in the Lagrangian $(\phi, \partial_\mu\phi, \partial^\mu\phi)$.

Evaluating the sum over permutations:

$$\begin{aligned} &\frac{-i\lambda}{2!}\Big[(ip_{2\mu})(ip_3^\mu) \times 2 \quad \text{(from permuting } p_2 \leftrightarrow p_3 \text{ on the derivative fields)} \\ &+ (-ip_{1\mu})(ip_2^\mu) \times 2 \quad \text{(from contracting } p_1 \text{ and } p_2 \text{ with the derivatives)} \\ &+ (-ip_{1\mu})(ip_3^\mu) \times 2 \quad \text{(from contracting } p_1 \text{ and } p_3 \text{ with the derivatives)}\Big] \times (2\pi)^4 \delta^{(4)}(p_1 - p_2 - p_3) \end{aligned}$$

Simplifying the algebra, the explicit Feynman rule for this derivative vertex is:

$$\text{Vertex}(p_1, p_2, p_3) = (-i\lambda)\big[-(p_2 \cdot p_3) + (p_1 \cdot p_2) + (p_1 \cdot p_3)\big]$$

Crossing Symmetry

Notice that in the momentum space rules, swapping an incoming particle with an outgoing particle is mathematically equivalent to flipping the overall sign of its momentum vector ($p \to -p$). This is a specific manifestation of a very deep and important principle in QFT known as **Crossing Symmetry**:

> An incoming particle with momentum p and polarization s is mathematically equivalent to an outgoing anti-particle with momentum $-p$ and polarization $-s$ in the evaluation of scattering matrix elements.

3.4 | General Approach for Feynman Rules

As a more general and robust approach for finding Feynman rules, we should analyze the direct effect of the interaction operator $i\mathcal{L}_{\text{int}}$ on the quantum states. This approach is powerful because it applies equally well to virtual particles where $p^2 \neq m^2$.

Let us explicitly recall the operator action for a complex scalar field:

- $\hat{\phi} = \hat{\phi}(\hat{a}, \hat{b}^\dagger) \implies$ DESTROYS incoming particles OR CREATES outgoing anti-particles.
- $\hat{\phi}^\dagger = \hat{\phi}(\hat{a}^\dagger, \hat{b}) \implies$ DESTROYS incoming anti-particles OR CREATES outgoing particles.

(For a real scalar field, particles and anti-particles coincide, so $\hat{\phi} = \hat{\phi}^\dagger$).

We can formally define contractions directly between the field operators in the Lagrangian and the physical external states:

$$\overbrace{\hat{\phi}|p^{\phi}\rangle}, \quad \langle p^{\bar{\phi}}|\hat{\phi}, \qquad \overbrace{\hat{\phi}^\dagger|p^{\bar{\phi}}\rangle}, \quad \langle p^{\phi}|\hat{\phi}^\dagger,$$

Here, $|p^{\phi}\rangle$ denotes a state containing a particle, and $|p^{\bar{\phi}}\rangle$ denotes a state containing an anti-particle. Importantly, when applying this technique to derive vertex rules, these states do not strictly need to be on-shell; we are merely extracting the algebraic effect of the operators $\hat{\phi}, \hat{\phi}^\dagger$ inside the Feynman diagrams.

To find the vertex rule, one computes the matrix element of the interaction Hamiltonian density between the desired external states and sums over all possible allowed non-zero Wick contractions.

Returning to our previous derivative toy model $\mathcal{L}_{\text{int}} = -\frac{\lambda}{2!}\phi(\partial_\mu\phi)(\partial^\mu\phi)$ (where ϕ is real, so $\phi = \phi^\dagger$), we evaluate the matrix element:

$$\langle p_2, p_3|\frac{-i\lambda}{2!}\hat{\phi}(\partial_\mu\hat{\phi})(\partial^\mu\hat{\phi})|p_1\rangle.$$

Applying Wick's theorem, we contract the operators with the states. For example, one specific contraction pairing is:

$$\frac{-i\lambda}{2!}\langle p_2, \overbrace{p_3|\hat{\phi}(\partial_\mu\hat{\phi})(\partial^\mu\hat{\phi})}|p_1\rangle.$$

This specific contraction yields:

$$= \left(\frac{-i\lambda}{2!}\right)(ip_{2\mu})(ip_3^\mu) \times 2!$$

The 2! symmetry factor arises because permuting the two final states $p_2 \leftrightarrow p_3$ into the two indistinguishable derivative operators $(\partial_\mu\phi)(\partial^\mu\phi)$ gives an identical mathematical result. Summing this result with all other valid permutations of the momenta into the different field operators reproduces the exact same vertex rule derived previously.

3.4.1 | Complex Scalar Field

For a complex scalar field, the total Lagrangian is given by $\mathcal{L} = \mathcal{L}_{\text{free}} + \mathcal{L}_{\text{int}}$. Unlike the real scalar field, the field operator ϕ and its conjugate ϕ^* (or $\phi^\dagger$) are distinct. We recall their effects on quantum states:

- $\phi = \phi(a, b^\dagger)$ destroys incoming particles or creates outgoing anti-particles.
- $\phi^* = \phi(a^\dagger, b)$ destroys incoming anti-particles or creates outgoing particles.

The Feynman propagator is defined as the time-ordered product:

$$D_F(x_1, x_2) = \langle 0|T\{\phi^*(x_1)\phi(x_2)\}|0\rangle$$

Because x_1 and x_2 are not equivalent in this expectation value, we must place an **arrow** on the propagator line from x_1 to x_2.

TODO: add diagram: solid line from x1 to x2 with an arrow pointing towards x2

By writing ϕ and ϕ^* in terms of creation and annihilation operators $(a, a^\dagger, b, b^\dagger)$, one can physically interpret this propagator as either a particle propagating from x_1 to x_2, or an anti-particle propagating backward from x_2 to x_1.

> **Remark.** *This leads to a fundamental convention in Feynman diagrams (consistently with crossing symmetry): An **INCOMING ARROW** towards a vertex can represent either an incoming particle or an outgoing anti-particle. Conversely, an **OUTGOING ARROW** from a vertex represents either an outgoing particle or an incoming anti-particle.*

Consider the quartic interaction for a complex scalar field:

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4}|\phi|^4 = -\lambda\left(\frac{\phi^2}{2!}\frac{\phi^{*2}}{2!}\right)$$

Note that there is no 4! factorial in the denominator, but rather a factor of 4 (which can be split into $2! \times 2!$ to account for the symmetries of ϕ and ϕ^* independently). The corresponding Feynman rule is simply $-i\lambda$ for a vertex with exactly 2 incoming arrows and 2 outgoing arrows.

TODO: add diagram: a 4-point vertex with 2 arrows pointing inwards and 2 arrows pointing outwards, labeled with -i lambda

Because every vertex must have an equal number of incoming and outgoing arrows, it strictly follows that the charge $Q \equiv N_{\text{particles}} - N_{\text{anti-particles}}$ is conserved in this theory.

Notation Note: In theories containing both scalars and Dirac fermions, we use different line styles to distinguish them:

- **Real scalar:** Plain dashed line.
- **Complex scalar:** Dashed line with an arrow.
- **Dirac fermion:** Solid line with an arrow.

3.4.2 | Dirac Fermion Field

For particles with spin, the LSZ reduction formula dictates that the amplitude must be multiplied by specific polarization states (spinors) for external particles. This introduces non-trivial Feynman rules for the external legs of Dirac fermions:

- **Incoming fermion (f):** $u_s(p)_\alpha$.
- **Outgoing fermion (f):** $\bar{u}_s(p)_\alpha$.
- **Incoming anti-fermion ($\bar{f}$):** $\bar{v}_s(p)_\alpha$.
- **Outgoing anti-fermion ($\bar{f}$):** $v_s(p)_\alpha$.

TODO: add diagram: solid line with arrow pointing towards the vertex (arrow opposes momentum)

Note: The spinor index α is often left implicit in standard calculations, adopting a matrix multiplication ordering.

The Feynman propagator for a Dirac fermion in momentum space is given by:

$$\frac{i(\not{p}+m)}{p^2-m^2} \quad \text{or with explicit spinor indices:} \quad \frac{i(\not{p}+m)_{\alpha\beta}}{p^2-m^2}$$

TODO: add diagram: solid line with an arrow in the direction of momentum p

Additional Feynman Rules for Fermions

Because fermions obey **anti-commutation relations** (due to the spin-statistics theorem), there are crucial additional rules when evaluating Feynman diagrams:

1. **Relative Minus Sign for Exchange:** Diagrams that can be mapped into each other by explicitly exchanging two external identical fermions (such as in t-channel vs u-channel scattering) must have a **relative minus sign**.

2. **Fermion Loops:** Any closed fermion loop inside a Feynman diagram must be multiplied by an overall factor of (-1). (For a detailed theoretical justification, one can refer to Schwartz 13.1.1).

TODO: add diagram: showing two interaction diagrams mapping to each other via exchanging two fermion lines, with a "multiply this by (-1)" note

TODO: add diagram: a circular solid line with an arrow representing a closed fermion loop, labeled "multiplied by (-1)"

Example (Yukawa theory). The interaction Lagrangian for the Yukawa theory is given by:

$$\mathcal{L}_{\text{int}} = -g\phi\bar{\psi}\psi$$

The total Lagrangian is $\mathcal{L} = \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{Klein-Gordon}} + \mathcal{L}_{\text{int}}$. The Feynman rule for this interaction vertex is simply:

$$\text{Vertex} = -ig$$

Note: Strictly speaking, this rule should be multiplied by the 4×4 identity matrix in Dirac space, as the vertex connects two spinors and a scalar field.

TODO: add diagram: a 3-point vertex with one dashed scalar line and two solid fermion lines, labeled -ig

Example (Fermion-Fermion Scattering in Yukawa Theory). Let us compute the tree-level amplitude for the scattering process of a fermion and an anti-fermion: $f(p_1)\bar{f}(p_2) \to f(p_3)\bar{f}(p_4)$.

TODO: add diagram: s-channel and t-channel diagrams for fermion-antifermion scattering mediated by a scalar

At leading order $\mathcal{O}(g^2)$, there are two Feynman diagrams that contribute: an s-channel annihilation diagram and a t-channel scattering diagram.

Applying the Feynman rules (including spinors for external legs and the scalar propagator $i/(q^2-m_\phi^2)$):

$$i\mathcal{M} = (-ig)^2\left[\frac{i}{s-m_\phi^2}(\bar{v}_2 u_1)(\bar{u}_3 v_4) + \frac{i}{t-m_\phi^2}(\bar{u}_3 u_1)(\bar{v}_2 v_4)\right] + \mathcal{O}(g^4)$$

where we used the shorthand notation $u_i \equiv u_{s_i}(p_i)$ and $v_i \equiv v_{s_i}(p_i)$. (Note: a relative minus sign between the diagrams arises from exchanging the identical external fermions, which we will formally justify later).

3.4.3 | Vector Bosons

In Feynman diagrams, vector bosons (spin-1 particles like photons, W, or Z bosons) are conventionally drawn as **wavy lines**.

TODO: add diagram: a wavy line representing a vector boson

Because vector fields A^μ carry a Lorentz index, we must assign explicit Lorentz indices to each end of a vector boson line in a diagram. Similar to fermions, external vector bosons require specific polarization states:

- **Incoming vector boson:** $\varepsilon^s_\mu(p)$
- **Outgoing vector boson:** $\varepsilon^{s*}_\mu(p)$

TODO: add diagram: a wavy line with an incoming arrow, labeled with index mu

TODO: add diagram: a wavy line with an outgoing arrow, labeled with index mu

The momentum-space propagators for vector bosons depend on whether they are massive or massless:

- **Massive vector boson ($m \neq 0$):**

$$\Pi^{\mu\nu}(p) = \frac{i}{p^2 - m^2 + i0^+}\left(-g^{\mu\nu} + \frac{p^\mu p^\nu}{m^2}\right)$$

- **Massless vector boson ($m = 0$, e.g., the photon):**

$$\Pi^{\mu\nu}(p) = \frac{-ig^{\mu\nu}}{p^2 + i0^+} + \text{"gauge-dependent terms"}$$

  By choosing the **Feynman gauge**, the gauge-dependent terms are exactly zero, leaving only the simple $-ig^{\mu\nu}$ numerator.

TODO: add diagram: a wavy line with momentum p and indices mu, nu at the ends

3.4.4 | Quantum Electrodynamics (QED)

The interaction Lagrangian for QED couples the Dirac fermion current to the electromagnetic field:

$$\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu$$

The fundamental QED vertex involves two fermions and one photon. The Feynman rule is:

$$\text{Vertex}^\mu = -ie\gamma^\mu$$

TODO: add diagram: a 3-point vertex with two solid fermion lines and one wavy photon line with index mu, labeled -ie gamma

Remark. *For particles with a generic electric charge Q (e.g., quarks where $Q = -1/3$ or $+2/3$), the rule generalizes to $iQe\gamma^\mu$. With explicit spinor indices α, β, the vertex is written as $-ie(\gamma^\mu)_{\alpha\beta}$, emphasizing that it acts as a matrix in Dirac space connecting the incoming and outgoing fermion lines.*

3.4.5 | Scalar Quantum Electrodynamics (sQED)

Scalar QED describes the electromagnetic interactions of a charged complex scalar field. The full Lagrangian is generated by substituting the standard derivative with the covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ in the free complex scalar Lagrangian:

$$\mathcal{L} = \mathcal{L}_{\text{Maxwell}} + \mathcal{L}_{\text{Complex Scalar}}\Big|_{\partial_\mu \to D_\mu}$$

Expanding the $|D_\mu\phi|^2$ term reveals two distinct interaction vertices (an exercise left for the reader, using the derivative prescription derived earlier):

1. **The 3-point vertex** (photon-scalar-scalar):

$$\text{Vertex}^{\mu} = -ie(p_1^{\mu} + p_2^{\mu})$$

where p_1 and p_2 are the momenta of the scalar fields, directed according to the arrow convention.

TODO: add diagram: a 3-point vertex with one wavy photon line and two dashed scalar lines with momenta p1, p2

2. **The 4-point "seagull" vertex** (two photons-two scalars):

$$\text{Vertex}^{\mu\nu} = 2ie^2 g^{\mu\nu}$$

TODO: add diagram: a 4-point vertex with two wavy photon lines and two dashed scalar lines

Example (Two different types of particles in sQED). Consider an extension of sQED containing two distinct, minimally coupled complex scalar fields, ϕ_A and ϕ_B, with masses m_A and m_B respectively.

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + |D_\mu \phi_A|^2 - m_A^2|\phi_A|^2 + |D_\mu \phi_B|^2 - m_B^2|\phi_B|^2$$

Let us compute the tree-level amplitude for the annihilation process $\phi_A(p_1)\bar{\phi}_A(p_2) \to \phi_B(p_3)\bar{\phi}_B(p_4)$. Because ϕ_A and ϕ_B are distinct fields, they only interact through the exchange of a photon. Thus, there is only one s-channel diagram contributing at tree-level.

TODO: add diagram: s-channel diagram with incoming A scalars, exchanging a photon, and outgoing B scalars

We assign Lorentz indices μ and ν to the ends of the internal photon propagator. Applying the sQED 3-point vertex rules:

$$i\mathcal{M} = [-ie(p_1^{\mu} - p_2^{\mu})] \frac{-ig_{\mu\nu}}{(p_1 + p_2)^2} [-ie(p_3^{\nu} - p_4^{\nu})]$$

Remark. *Notice the crucial* ***minus sign*** *in the momentum factor $(-p_2^{\mu})$ for the incoming anti-particle. This is a direct application of the* ***Crossing Rule****: the rule for the vertex is $-ie(p_{in} + p_{out})$. Since p_2 is an incoming anti-particle, we must treat it as an outgoing particle with momentum $-p_2$, hence the minus sign. The same logic applies to the outgoing anti-particle p_4, which yields $(-p_4^{\nu})$.*

Contracting the indices with the metric tensor and using Mandelstam variables ($s = (p_1+p_2)^2$):

$$i\mathcal{M} = \frac{ie^2}{s}(p_1 \cdot p_3 - p_1 \cdot p_4 - p_2 \cdot p_3 + p_2 \cdot p_4)$$

Using kinematic identities (e.g., $p_1 \cdot p_3 = p_2 \cdot p_4 = -\frac{1}{2}(t - m_A^2 - m_B^2)$ and $p_1 \cdot p_4 = p_2 \cdot p_3 = -\frac{1}{2}(u - m_A^2 - m_B^2)$), this beautifully simplifies to:

$$i\mathcal{M} = ie^2 \frac{u - t}{s} \qquad (\text{with } s + t + u = 2m_A^2 + 2m_B^2)$$

Kinematic Note: Momentum conservation implies that the center-of-mass energy $\sqrt{s} = E_{\text{c.m.}} = E_3 + E_4 \geq 2m_B$. Therefore, if $m_A < m_B$, this reaction is only physically possible if the initial particles $\phi_A, \bar{\phi}_A$ are accelerated to a sufficiently high center-of-mass energy ($\sqrt{s} \geq 2m_B$) to overcome the mass threshold of the heavier B particles.

3.4.6 | Observations on Diagrams

Feynman diagrams are fundamental mathematical tools used to make perturbative predictions in Quantum Field Theory (QFT). Each diagram represents a specific mathematical contribution to a Green function or an element of the scattering matrix (S-matrix).

It is crucial to understand that Feynman diagrams are **not physical objects**. You cannot measure or experimentally observe a single Feynman diagram; only their coherent sum corresponds to a measurable physical probability. Furthermore, a single diagram generally does **not** respect the fundamental symmetries of the underlying theory (such as gauge invariance or Bose/Fermi symmetry). Only the complete sum of all diagrams at a given perturbative order perfectly restores these physical symmetries.

Despite these strict mathematical realities, it remains extremely common and useful to assign a heuristic physical meaning to the components of these diagrams to guide our physical intuition (provided one keeps the aforementioned caveats in mind). We can classify the components into two distinct categories:

- **External Lines (Real Particles):** These represent the actual external particles associated with the incoming and outgoing asymptotic states.
  - Their momenta are strictly **on-shell**, meaning they satisfy the relativistic energy-momentum relation $p_i^2 = m_i^2$.
  - They are considered to live for a *long time* (macroscopic distances) compared to the microscopic time scale of the quantum interaction.
- **Internal Lines (Virtual Particles):** These mathematically correspond to the propagators bridging the interaction vertices.
  - They represent intermediate particles that live for a very *short time*.
  - Their momenta are strictly **off-shell**, meaning $p_i^2 \neq m_i^2$.
  - In tree-level diagrams, the 4-momentum of a virtual particle is entirely fixed by momentum conservation at the connected vertices.
  - In loop diagrams, the internal momenta are not uniquely fixed by conservation laws, meaning we must "sum" (integrate) over all possible momentum states via $\int d^4k$.
  - Virtual particles, regardless of whether they appear in trees or loops, only "exist" as intermediate transition states and are **never directly observed**.

As an illustrative example, consider the annihilation process $\phi_A\bar{\phi}_A \to \phi_B\bar{\phi}_B$ in scalar Quantum Electrodynamics (sQED).

In this specific diagram, the initial ϕ_A states are the "prepared incoming states" and the final ϕ_B states are the "observed outgoing states". The internal wavy line is the mediating **virtual photon**. Because it is short-lived and off-shell ($p^2 \neq m_\gamma^2 = 0$), it is inherently unobservable.

TODO: add diagram: s-channel diagram showing prepared incoming states phiA, bar phiA, an internal wavy line representing the virtual photon, and observed outgoing states phiB, bar phiB

Remark. *A profound consequence of quantum mechanics is that virtual particles can appear in Feynman diagrams—and thus indirectly, but measurably, affect our observations of a process—even if the system does not possess enough center-of-mass energy to produce them as actual, real particles.*

3.5 | Crossing Symmetry

We have seen that Feynman rules obey a fundamental property known as **Crossing Symmetry**. Mathematically, this principle states an equivalence in the evaluation of scattering matrix elements:

$$\text{INCOMING } \phi_s(p) \iff \text{OUTGOING } \bar{\phi}_{-s}(-p)$$

where $\phi_s(p)$ represents a particle with momentum p and polarization state s, and $\bar{\phi}$ represents its corresponding anti-particle.

This powerful symmetry allows us to write just **one** Feynman rule per interaction term, rather than having to define separate rules for every possible combination of incoming and outgoing particles. For example, in scalar QED, the fundamental 3-point vertex rule is $-ie(p_1^\mu + p_2^\mu)$ for two incoming scalars. By applying crossing symmetry, this single rule automatically implies the rule for a process with one incoming particle (p_1) and one outgoing particle (p_2), which simply becomes $-ie(p_1^\mu - p_2^\mu)$, and so on. We only need to write one generic combination, and all others are implicitly derived.

TODO: add diagram: showing the basic sQED 3-point vertex with two incoming scalars p1, p2, and a second diagram showing the crossed version with p1 incoming and p2 outgoing

Because full scattering amplitudes $\mathcal{M}$ are constructed purely from these symmetric Feynman rules, it logically follows that the scattering amplitudes themselves strictly obey crossing symmetry.

> **Remark.** *This statement is also a direct corollary of the LSZ reduction formula. When applied to a complete amplitude, crossing symmetry must be understood mathematically as an **Analytic Continuation**. Physical scattering processes only make sense for external particles with strictly positive energy ($p^0 > 0$). However, because the amplitudes are analytic functions of the external momenta, we can mathematically continue these functions into the unphysical region where $p^0 < 0$. This analytic continuation is precisely what validates the transformation $p \to -p$ used in crossing.*

Thanks to crossing symmetry, once we have computed the analytic expression of an amplitude for a specific process, we can obtain the amplitudes for all related crossed processes essentially "for free".

> **Example (Crossing in sQED).** Suppose we have already computed the amplitude for the annihilation process $\phi_A\bar{\phi}_A \to \phi_B\bar{\phi}_B$, which we found to be $\mathcal{M} = ie^2\frac{u-t}{s}$.
>
> We now want to compute the amplitude for the scattering process $\phi_A(p_1)\phi_B(p_2) \to \phi_A(p_3)\phi_B(p_4)$. We can define the crossed momenta k_i for this new process in terms of the original annihilation momenta:
>
> $$k_1 = p_1, \quad k_2 = -p_3, \quad k_3 = p_4, \quad k_4 = -p_2$$
>
> Substituting these crossed momenta into the Mandelstam variables of the original process:
>
> $$\begin{aligned} s(k_i) &= (k_1 + k_2)^2 = (p_1 - p_3)^2 = t \\ t(k_i) &= (k_3 - k_1)^2 = (p_4 - p_1)^2 = u \\ u(k_i) &= (k_1 - k_4)^2 = (p_1 + p_2)^2 = s \end{aligned}$$
>
> Therefore, the crossing transformation elegantly maps the Mandelstam variables as $s \to t$, $t \to u$, and $u \to s$. Applying this mapping to the original amplitude gives us the new amplitude immediately:
>
> $$i\mathcal{M}(\phi_A\phi_B \to \phi_A\phi_B) = i\mathcal{M}(\phi_A\bar{\phi}_A \to \phi_B\bar{\phi}_B)\big|_{\substack{s\to t \\ t\to u \\ u\to s}} = ie^2\frac{s-u}{t}$$

This is the exact same result one would obtain by manually computing the corresponding t-channel Feynman diagram for the scattering process.

TODO: add diagram: t-channel Feynman diagram for phiA phiB scattering

As a theoretical extreme, one could use crossing symmetry to compute a completely unphysical amplitude where all particles are in the outgoing state, e.g., $0 \to \phi_A \bar{\phi}_A \phi_B \bar{\phi}_B$ (vacuum decaying into particles). While this amplitude describes a physically impossible process (due to energy conservation), valid physical amplitudes can be effortlessly extracted from it by crossing two of the final state particles back into the initial state.

3.5.1 | Crossing Symmetry and Identical Fermions

We previously established a critical rule: diagrams that differ by the exchange of two external identical fermions must have a relative (-1) sign. We can now rigorously justify this using crossing symmetry.

Consider Bhabha scattering in QED ($e^- e^+ \to e^- e^+$). At tree level, the amplitude is the sum of an s-channel (annihilation) and a t-channel (scattering) diagram.

TODO: add diagram: s-channel and t-channel diagrams for Bhabha scattering e- e+ -> e- e+

$$i\mathcal{M}(e^- e^+ \to e^- e^+) = \text{Diagram}_s + \text{Diagram}_t + \mathcal{O}(e^4)$$

These two diagrams must have a relative (-1) sign. To see this clearly without tracking complex spinor indices, we use crossing symmetry to move all particles into the outgoing state, evaluating the fictitious process $0 \to e^+ e^- e^- e^+$.

TODO: add diagram: two diagrams for the vacuum decay 0 -> e+ e- e- e+, showing the exchange of the two e- lines

$$i\mathcal{M}(0 \to e^+ e^- e^- e^+) = \text{Diagram}_A + \text{Diagram}_B + \mathcal{O}(e^4)$$

In this completely crossed configuration, it becomes immediately visually clear that Diagram A and Diagram B differ only by the explicit exchange of the two identical outgoing electrons (e^-), labeled 2 and 3 in the diagrams. Because fermions are identical particles obeying Fermi-Dirac statistics, their final state wavefunction must be totally antisymmetric under particle exchange. This antisymmetry forces the relative minus sign between the two diagrams.

Remark. *More generally, to derive the exact signs for complex diagrams, one would need to carefully count the anti-commutations of the fermion field operators during the Wick contractions. While instructive to do by hand once, today this tedious combinatorial information is easily and reliably provided by automated computer programs (e.g., FeynArts or QGRAF).*

4 | Quantum Electrodynamics

[...]

4.1 | Scattering Processes in QED

Consider Quantum Electrodynamics (QED) with two flavors of Dirac fermions, namely the electron (e^-) and the muon (μ^-). The Lagrangian for this theory is given by:

$$\mathcal{L} = -\frac{1}{4}F_{\nu\rho}F^{\nu\rho} + \bar{\psi}_e(i\partial\!\!\!/ - m_e)\psi_e + \bar{\psi}_\mu(i\partial\!\!\!/ - m_\mu)\psi_\mu + Q_e e\bar{\psi}_e\gamma^\nu\psi_e A_\nu + Q_\mu e\bar{\psi}_\mu\gamma^\nu\psi_\mu A_\nu,$$

Remark. *Note that in the subscripts, μ stands for the muon flavor, and is not a Lorentz index.*

Here, Q_i (with $i = e, \mu$) represents the electric charge of the respective fermion in units of the elementary charge e. In this theory, we can study various scattering processes, such as electron-muon scattering, Compton scattering, and more. In the following calculations, we will set $Q_e = Q_\mu = -1$, as this reflects the physical nature of these particles. However, it is worth noting that QED is mathematically consistent for any arbitrary value of Q_i.

> **Remark.** *This is a highly non-trivial theoretical statement. As you will see in more advanced courses, assigning arbitrary charges is perfectly valid in Abelian gauge theories (like QED), but it is strictly not allowed in non-Abelian gauge theories (like QCD or the electroweak sector), where anomaly cancellation imposes rigid constraints on the charge assignments.*

4.1.1 | Electron - Muon Scattering ($e^-e^+ \to \mu^-\mu^+$)

Let us first consider the scattering process $e^-e^+ \to \mu^-\mu^+$ at tree-level. At this leading order in perturbation theory, there is only one Feynman diagram contributing to the process: an s-channel exchange of a virtual photon.

TODO: add the Feynman diagram here: an s-channel diagram with incoming e- (p1) and e+ (p2), an internal wavy photon line with Lorentz indices mu and nu, and outgoing mu- (p3) and mu+ (p4). Assign Lorentz indexes to photon lines.

To compute the amplitude $i\mathcal{M}$, we generally avoid writing explicit Dirac spinor indices ($\alpha, \beta, \ldots$). Instead, we follow a simple and equivalent golden rule:

Apply the Feynman rules for Dirac fermions while following the fermion lines backwards.

By doing so, we construct the amplitude piece by piece. We get:

$$i\mathcal{M} = \underbrace{\bar{v}_{s_2}(p_2)(-ie\gamma^\nu)u_{s_1}(p_1)}_{\text{1st fermion line}} \frac{-ig_{\nu\mu}}{(p_1+p_2)^2} \underbrace{\bar{u}_{s_3}(p_3)(-ie\gamma^\mu)v_{s_4}(p_4)}_{\text{2nd fermion line}},$$

where the first bracketed term is the electron line (traced backwards from the outgoing positron to the incoming electron), the middle term is the photon propagator, and the second bracketed term is the muon line. The indices s_i represent the polarization (spin) states of the specific particles.

To simplify the notation significantly, we define $u_i \equiv u_{s_i}(p_i)$ and $v_i \equiv v_{s_i}(p_i)$. With this compact notation, the amplitude simplifies to:

$$i\mathcal{M} = \frac{ie^2}{s}\left(\bar{v}_2\gamma^\nu u_1\right)\left(\bar{u}_3\gamma_\nu v_4\right),$$

where $s = (p_1+p_2)^2$ is the standard Mandelstam variable.

To compute the physical cross-section, we need the squared amplitude $|\mathcal{M}|^2$, which inherently requires computing the complex conjugate $\mathcal{M}^*$. Recalling the Hermitian conjugation property for Dirac bilinears:

$$\left(\bar{\psi}_1\gamma^{\mu_1}\cdots\gamma^{\mu_n}\psi_2\right)^* = \bar{\psi}_2\gamma^{\mu_n}\cdots\gamma^{\mu_1}\psi_1,$$

the complex conjugated matrix element can be effortlessly written as:

$$i\mathcal{M}^* = \frac{ie^2}{s} \left(\bar{u}_1 \gamma^\nu v_2\right) \left(\bar{v}_4 \gamma_\nu u_3\right).$$

TODO: add the Feynman diagram for the conjugate process i M* here. It should look like the original iM diagram but FLIPPED horizontally, with mu- (p3) and mu+ (p4) on the left going into the vertex, and e- (p1) and e+ (p2) on the right coming out.

Unpolarized Cross Section. At this point, we could specify the explicit polarization states for the initial and final-state fermions. However, in many high-energy experiments, the particles in the initial state are **unpolarized**, meaning they are not prepared in any specific spin state (their spin is essentially random). Moreover, detectors often do not measure the specific spin of the outgoing particles in the final state.

Because of this, we typically compute an **unpolarized cross section**. This requires us to:

- **Average** over all possible initial state polarizations.
- **Sum** over all possible final state polarizations.

The unpolarized squared scattering amplitude is thus:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{1}{2}\sum_{s_1}\frac{1}{2}\sum_{s_2}\sum_{s_3}\sum_{s_4} |\mathcal{M}_{s_1\, s_2 \to s_3\, s_4}|^2 = \frac{1}{4}\sum_{s_i} |\mathcal{M}|^2.$$

To compute these extensive sums efficiently without evaluating each spin state manually, we use the completeness relations for Dirac spinors:

$$\sum_s u_s(p)_\alpha \bar{u}_s(p)_\beta = (\not{p} + m)_{\alpha\beta}, \quad \sum_s v_s(p)_\alpha \bar{v}_s(p)_\beta = (\not{p} - m)_{\alpha\beta}.$$

Using these relations, the spin sums elegantly turn into **traces of γ matrices**.

Let us demonstrate this for a single fermion line. Taking the product of a current from $\mathcal{M}^*$ and $\mathcal{M}$, we explicitly write the spinor indices:

$$\sum_{s_1,s_2} \underbrace{\left(\bar{u}_{1\,\alpha} \gamma^\nu_{\alpha\beta} v_{2\,\beta}\right)}_{\text{from } \mathcal{M}^*} \underbrace{\left(\bar{v}_{2\,\alpha'} \gamma^\mu_{\alpha'\beta'} u_{1\,\beta'}\right)}_{\text{from } \mathcal{M}}.$$

Because these components are merely commuting complex numbers, we can rearrange them to bring the spinors associated with the same momentum next to each other:

$$= \sum_{s_1,s_2} u_{1\,\beta'} \bar{u}_{1\,\alpha} \gamma^\nu_{\alpha\beta} v_{2\,\beta} \bar{v}_{2\,\alpha'} \gamma^\mu_{\alpha'\beta'}.$$

Now we clearly recognize the completeness relations in the sums over s_1 and s_2. Substituting them directly closes the index loop ($\beta' \to \alpha \to \beta \to \alpha' \to \beta'$), which is the precise mathematical definition of a matrix trace:

$$= \text{Tr}\left[(\not{p}_1 + m)\gamma^\nu (\not{p}_2 - m)\gamma^\mu\right].$$

TODO: Insert diagram of the squared amplitude here. It should show the M* diagram and the M diagram connected at the external legs, forming closed fermion loops joined by the wavy photon lines. Label

Thus, the square of a fermion line, when summed over all polarizations, always yields a trace. In future calculations, we will skip the explicit indices and write the trace directly. The summarized algorithm is:

- Follow the fermion lines backward.
- Replace the "cut" lines connecting $\mathcal{M}$ and $\mathcal{M}^*$ with a sum over polarizations ($\sum u\bar{u} \to \not{p} + m$, $\sum v\bar{v} \to \not{p} - m$), which naturally forms a continuous trace over the Dirac matrices.

Applying this powerful machinery to our full $e^-e^+ \to \mu^-\mu^+$ process (incorporating both the electron and muon lines, the two photon propagators, and the 1/4 initial spin-averaging factor), we obtain:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{e^4}{4s^2} \operatorname{Tr}\left[(\not{p}_1 + m_e)\gamma^\nu(\not{p}_2 - m_e)\gamma^\mu\right] \operatorname{Tr}\left[(\not{p}_3 + m_\mu)\gamma_\mu(\not{p}_4 - m_\mu)\gamma_\nu\right].$$

Before we can extract physical predictions from this equation, we must learn how to systematically compute traces of Dirac γ matrices, which is the focus of the next section.

4.1.2 | Dirac Algebra

We list some important identities regarding the Dirac matrices and their proofs. All these properties mathematically stem from the fundamental anticommutation relations of the Clifford algebra:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{I}_{4\times4}, \qquad \{\gamma^5, \gamma^\mu\} = 0, \qquad (\gamma^5)^2 = \mathbb{I}_{4\times4}$$

(Note that the 4×4 identity matrix $\mathbb{I}_{4\times4}$ on the right-hand side is often left understood). The first relation implies that we can always swap two adjacent gamma matrices at the cost of a minus sign and a metric tensor term:

$$\gamma^\mu\gamma^\nu = -\gamma^\nu\gamma^\mu + 2g^{\mu\nu}$$

A very useful preliminary identity is the contraction of a gamma matrix with itself:

$$\gamma^\mu\gamma_\mu = 4\mathbb{I}_{4\times4}$$

Proof. We can split the product into its symmetric and antisymmetric parts:

$$\gamma^\mu\gamma_\mu = \frac{1}{2}\{\gamma^\mu, \gamma_\mu\} + \frac{1}{2}[\gamma^\mu, \gamma_\mu]$$

The commutator vanishes identically because we are contracting the exact same index, while the anticommutator yields $2g^\mu_\mu$. Since the trace of the metric in 4 dimensions is $g^\mu_\mu = 4$, we get $\frac{1}{2}(2 \cdot 4) = 4\mathbb{I}$. □

When computing unpolarized cross sections, remember that we are tracing with respect to the spinor indices; thus we are evaluating the trace over 4×4 matrices. The following fundamental identities hold for these traces:

$$\operatorname{Tr}[\mathbb{I}_4] = 4$$

$$\operatorname{Tr}[\gamma^{\mu_1} \dots \gamma^{\mu_N}] = 0 \quad \text{for } N \text{ odd}$$

Proof. Let us prove it for a generic odd number N of gamma matrices. We insert the identity in the form of $(\gamma^5)^2 = \mathbb{I}$ inside the trace:

$$\operatorname{Tr}[\gamma^{\mu_1} \dots \gamma^{\mu_N}] = \operatorname{Tr}\left[\gamma^{\mu_1} \dots \gamma^{\mu_N}\gamma^5\gamma^5\right]$$

Using the cyclic property of the trace, we move the last γ^5 to the front:

$$= \mathrm{Tr}\big[\gamma^5\gamma^{\mu_1}\dots\gamma^{\mu_N}\gamma^5\big]$$

Now we systematically anti-commute the first γ^5 to the right, passing it through all the N gamma matrices. Since $\{\gamma^5,\gamma^\mu\}=0$, each swap generates a minus sign. After N swaps, it recombines with the other γ^5:

$$= (-1)^N\,\mathrm{Tr}\big[\gamma^{\mu_1}\dots\gamma^{\mu_N}(\gamma^5)^2\big] = (-1)^N\,\mathrm{Tr}[\gamma^{\mu_1}\dots\gamma^{\mu_N}]$$

If N is odd, $(-1)^N=-1$. A mathematical object that is equal to minus itself must be identically zero, hence the second identity is proven. □

Remark. *How do we compute* $\mathrm{Tr}[\gamma^{\mu_1}\dots\gamma^{\mu_N}]$ *for* N *even? The general algorithm is:*

1. *Use the cyclic property of the trace:* $\mathrm{Tr}[\gamma^{\mu_1}\dots\gamma^{\mu_N}]=\mathrm{Tr}[\gamma^{\mu_2}\dots\gamma^{\mu_N}\gamma^{\mu_1}]$.
2. *Anti-commute* μ_1 *to the left* $N-1$ *times. Each anti-commutation yields a* $g^{\mu_1\mu_k}\mathbb{I}$ *term and therefore leaves a trace with* $N-2$ *gamma matrices.*
3. *The final term will be* $(-1)^{N-1}\,\mathrm{Tr}[\gamma^{\mu_1}\dots\gamma^{\mu_N}]$*, which can be moved to the left-hand side to solve for the original trace.*

Let us apply this algorithm for $N=2$:

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 4g^{\mu\nu}$$

Proof. Using the cyclic property of the trace, we can write:

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu] = \mathrm{Tr}[\gamma^\nu\gamma^\mu]$$

Now we anti-commute the two matrices on the right-hand side using $\gamma^\nu\gamma^\mu=-\gamma^\mu\gamma^\nu+2g^{\mu\nu}$:

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu] = -\,\mathrm{Tr}[\gamma^\mu\gamma^\nu] + 2g^{\mu\nu}\,\mathrm{Tr}[\mathbb{I}_4]$$

Bringing the trace to the left side and recalling that $\mathrm{Tr}[\mathbb{I}_4]=4$, we get:

$$2\,\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 8g^{\mu\nu} \implies \mathrm{Tr}[\gamma^\mu\gamma^\nu] = 4g^{\mu\nu}$$

□

For $N=4$, let us use a simplified notation in which $\mathrm{Tr}[\mu\nu\rho\sigma]\equiv\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma]$. Then we can prove the following highly utilized identity:

$$\mathrm{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = 4\left(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}\right)$$

Proof. We start by using the cyclic property to move μ to the right end:

$$\mathrm{Tr}[\mu\nu\rho\sigma] = \mathrm{Tr}[\nu\rho\sigma\mu]$$

Now, we systematically anti-commute μ back to the left, step by step. First, we swap μ with σ:

$$= -\,\mathrm{Tr}[\nu\rho\mu\sigma] + 2g^{\mu\sigma}\,\mathrm{Tr}[\nu\rho]$$

Next, we swap μ with ρ inside the first trace:

$$= +\,\mathrm{Tr}[\nu\mu\rho\sigma] - 2g^{\mu\rho}\,\mathrm{Tr}[\nu\sigma] + 2g^{\mu\sigma}\,\mathrm{Tr}[\nu\rho]$$

Finally, we swap μ with ν:

$$= -\operatorname{Tr}[\mu\nu\rho\sigma] + 2g^{\mu\nu}\operatorname{Tr}[\rho\sigma] - 2g^{\mu\rho}\operatorname{Tr}[\nu\sigma] + 2g^{\mu\sigma}\operatorname{Tr}[\nu\rho]$$

Notice that the first term on the right-hand side is exactly minus our original trace. Bringing it to the left side gives:

$$2\operatorname{Tr}[\mu\nu\rho\sigma] = 2g^{\mu\nu}\operatorname{Tr}[\rho\sigma] - 2g^{\mu\rho}\operatorname{Tr}[\nu\sigma] + 2g^{\mu\sigma}\operatorname{Tr}[\nu\rho]$$

Dividing by 2 and substituting the known 2-gamma traces (e.g., $2g^{\mu\sigma}\operatorname{Tr}[\nu\rho] = 8g^{\mu\sigma}g^{\nu\rho}$) yields the final result. □

We are not going to prove the following traces, but they are very useful in computations involving polarized cross-sections or chiral theories. Recalling the definition $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, we have the following identities:

$$\begin{cases} \operatorname{Tr}\left[\gamma^5\right] = 0 \\ \operatorname{Tr}\left[\gamma^{\mu_1}\cdots\gamma^{\mu_N}\gamma^5\right] = 0 \quad \text{for } N \text{ odd} \\ \operatorname{Tr}\left[\gamma^\mu\gamma^\nu\gamma^5\right] = 0 \\ \operatorname{Tr}\left[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^5\right] = -4i\epsilon^{\mu\nu\rho\sigma} \end{cases}$$

where $\epsilon^{\mu\nu\rho\sigma}$ is the totally antisymmetric Levi-Civita symbol. The proofs for these are left as a standard exercise.

Example (Exercise). Using the anticommutation relations repeatedly, one can show that contracting a string of gamma matrices with the same index reverses their order on the right-hand side. For instance, show that:

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = -2\gamma^\sigma\gamma^\rho\gamma^\nu$$

4.1.3 | Back to Electron - Muon Scattering

Let us explicitly compute the unpolarized squared amplitude for the electron-muon annihilation process ($e^-e^+ \to \mu^-\mu^+$). We previously found:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{e^4}{4s^2}\operatorname{Tr}\left[(\not{p}_1 + m_e)\gamma^\nu(\not{p}_2 - m_e)\gamma^\mu\right]\operatorname{Tr}\left[(\not{p}_3 + m_\mu)\gamma_\mu(\not{p}_4 - m_\mu)\gamma_\nu\right]$$

Let us compute the first trace (the electron line). We expand the product inside the trace and use the property that the trace of an odd number of γ matrices strictly vanishes. This leaves us with terms containing either 4 or 2 gamma matrices:

$$\operatorname{Tr}\left[(\not{p}_1 + m_e)\gamma^\nu(\not{p}_2 - m_e)\gamma^\mu\right] = \operatorname{Tr}\left[\not{p}_1\gamma^\nu\not{p}_2\gamma^\mu\right] - m_e^2\operatorname{Tr}[\gamma^\nu\gamma^\mu]$$

Because the four-momenta p_1 and p_2 are just sets of commuting numbers, we can pull them out of the trace using the definition of the Feynman slash ($\not{p} = p_\rho\gamma^\rho$):

$$\operatorname{Tr}\left[\not{p}_1\gamma^\nu\not{p}_2\gamma^\mu\right] = p_{1\rho}p_{2\sigma}\operatorname{Tr}[\gamma^\rho\gamma^\nu\gamma^\sigma\gamma^\mu]$$

Applying our previously derived 4-gamma trace identity, this evaluates to:

$$= p_{1\rho}p_{2\sigma}\cdot 4(g^{\rho\nu}g^{\sigma\mu} - g^{\rho\sigma}g^{\nu\mu} + g^{\rho\mu}g^{\nu\sigma}) = 4(p_1^\nu p_2^\mu - (p_1\cdot p_2)g^{\nu\mu} + p_1^\mu p_2^\nu)$$

For the mass term, we use the 2-gamma trace identity $\text{Tr}[\gamma^\nu\gamma^\mu] = 4g^{\nu\mu}$. Putting it all together, the first trace is:

$$= 4\left(p_1^\nu p_2^\mu + p_1^\mu p_2^\nu - (p_1 \cdot p_2 + m_e^2)g^{\mu\nu}\right)$$

The second trace (the muon line) is structurally identical. We simply replace the electron labels with the muon labels ($1 \to 3$, $2 \to 4$, and $m_e \to m_\mu$) and lower the Lorentz indices to match:

$$\text{Tr}\left[(\not{p}_3 + m_\mu)\gamma_\mu(\not{p}_4 - m_\mu)\gamma_\nu\right] = 4\left(p_{3\mu}p_{4\nu} + p_{3\nu}p_{4\mu} - (p_3 \cdot p_4 + m_\mu^2)g_{\mu\nu}\right)$$

High-Energy Approximation ($m_e \to 0$). To significantly simplify the algebra, we can look at the physical mass scales. For real electrons and muons, $m_e \simeq 0.5$ MeV and $m_\mu \simeq 106$ MeV. For this annihilation process to even occur, the center-of-mass energy must satisfy $E_{\text{c.m.}} \geq 2m_\mu \simeq 212$ MeV. Because the ratio $m_e^2/m_\mu^2 \simeq 2 \cdot 10^{-5}$ is exceptionally small, we can safely neglect the electron mass $m_e \to 0$ in our calculations. (To require precision where m_e terms matter, one would first need to compute much higher-order loop corrections in perturbation theory!).

Setting $m_e = 0$, we contract the two evaluated traces. Note that $g^{\mu\nu}g_{\mu\nu} = 4$ and $p_1^\mu p_{3\mu} = (p_1 \cdot p_3)$:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{16e^4}{4s^2}\left[2(p_1 \cdot p_3)(p_2 \cdot p_4) + 2(p_1 \cdot p_4)(p_2 \cdot p_3)\right] \text{ (after cancelling cross-terms)}$$

Using the massless electron approximation $p_1^2 = p_2^2 = 0$, the Mandelstam variables simplify to:

$$\begin{aligned}
s &= (p_1 + p_2)^2 = 2p_1 \cdot p_2 \\
t &= (p_1 - p_3)^2 = m_\mu^2 - 2p_1 \cdot p_3 \implies (p_1 \cdot p_3) = (p_2 \cdot p_4) = -\frac{1}{2}(t - m_\mu^2) \\
u &= (p_1 - p_4)^2 = m_\mu^2 - 2p_1 \cdot p_4 \implies (p_1 \cdot p_4) = (p_2 \cdot p_3) = -\frac{1}{2}(u - m_\mu^2)
\end{aligned}$$

Substituting these kinematic relations into our amplitude:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{8e^4}{s^2}\left[\left(\frac{t - m_\mu^2}{-2}\right)^2 + \left(\frac{u - m_\mu^2}{-2}\right)^2 + m_\mu^2\frac{s}{2}\right]$$

Simplifying the fractions, we arrive at the beautiful, manifestly Lorentz-invariant result:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{2e^4}{s^2}\left[(t - m_\mu^2)^2 + (u - m_\mu^2)^2 + 2m_\mu^2 s\right]$$

This expression is extremely powerful as it holds in any inertial reference frame.

Center of Mass Frame

To extract a physical cross-section, it is most practical to evaluate these invariants in the Center of Mass (CM) frame. In this frame, the total spatial momentum is strictly zero. Let E_e be the energy of the incoming electrons.

$$E_1 = E_2 = E_3 = E_4 = \frac{E_{\text{c.m.}}}{2} \equiv E_e \geq m_\mu$$

We define the scattering angle θ as the angle between the incoming electron and outgoing muon. The 4-momenta are explicitly parameterized as:

$$p_1 = \begin{pmatrix} E_e \\ 0 \\ 0 \\ E_e \end{pmatrix}, \quad p_2 = \begin{pmatrix} E_e \\ 0 \\ 0 \\ -E_e \end{pmatrix}, \quad p_3 = \begin{pmatrix} E_e \\ 0 \\ p' \sin\theta \\ p' \cos\theta \end{pmatrix}, \quad p_4 = \begin{pmatrix} E_e \\ 0 \\ -p' \sin\theta \\ -p' \cos\theta \end{pmatrix}$$

where the outgoing muon 3-momentum magnitude is fixed by the on-shell condition: $p' = \sqrt{E_e^2 - m_\mu^2}$.

Evaluating the Mandelstam variables in this specific frame:

$$\begin{cases} s = (p_1 + p_2)^2 = (2E_e)^2 = 4E_e^2 \\ t = (p_3 - p_1)^2 = m_\mu^2 - 2E_e^2 + 2E_e p' \cos\theta \\ u = (p_3 - p_2)^2 = m_\mu^2 - 2E_e^2 - 2E_e p' \cos\theta \end{cases}$$

Substituting these into our invariant amplitude formula and carefully expanding the squares:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{e^4}{8E_e^4}\left[4(E_e^2 - E_e p' \cos\theta)^2 + 4(E_e^2 + E_e p' \cos\theta)^2 + 8E_e^2 m_\mu^2\right]$$

The linear terms in $\cos\theta$ cleanly cancel out. Grouping the remaining terms and utilizing the identity $(p')^2 = E_e^2 - m_\mu^2$:

$$|\mathcal{M}|^2_{\text{unpol}} = e^4\left[(1 + \cos^2\theta) + \frac{m_\mu^2}{E_e^2}(1 - \cos^2\theta)\right]$$

With the squared amplitude fully evaluated, we plug it into the standard phase-space master formula for the differential cross-section of a $2 \to 2$ scattering process:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{64\pi^2 E_{\text{c.m.}}^2}\frac{|\vec{p}_f|}{|\vec{p}_i|}|\mathcal{M}|^2_{\text{unpol}}$$

Here, $E_{\text{c.m.}}^2 = s = 4E_e^2$, and the momentum ratio is given by $|\vec{p}_f|/|\vec{p}_i| = p'/E_e = \sqrt{1 - m_\mu^2/E_e^2}$. Substituting these components and introducing the fundamental fine-structure constant $\alpha \equiv e^2/(4\pi) \approx 1/137$, we obtain the final differential cross-section:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\alpha^2}{16E_e^2}\sqrt{1 - \frac{m_\mu^2}{E_e^2}}\left[(1 + \cos^2\theta) + \frac{m_\mu^2}{E_e^2}(1 - \cos^2\theta)\right]$$

Because the physics is azimuthally symmetric (independent of the azimuthal angle φ), we can trivially integrate $\int \mathrm{d}\varphi = 2\pi$ to obtain the distribution over the polar angle:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\cos\theta} = 2\pi\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$$

Integrating this over all possible scattering angles $\cos\theta \in [-1, 1]$ yields the total physical cross-section:

$$\sigma = \frac{\pi\alpha^2}{3E_e^2}\sqrt{1 - \frac{m_\mu^2}{E_e^2}}\left(1 + \frac{m_\mu^2}{2E_e^2}\right)$$

Example (Exercise). Use crossing symmetry to compute the unpolarized squared amplitude for electron-muon scattering ($e^-\mu^- \to e^-\mu^-$), again taking the massless electron limit ($m_e \to 0$). **Result:**

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{2e^4}{t^2}\left[(u - m_\mu^2)^2 + (s - m_\mu^2)^2 + 2m_\mu^2 t\right]$$

Remark (Note on Crossing Symmetry for Dirac Fermions). *When directly applying crossing symmetry to an unpolarized squared amplitude ($\sum|\mathcal{M}|^2$), one must be exceptionally careful regarding an inherent sign flip arising from the mathematical phase conventions of Dirac spinors. Consider crossing an incoming particle into an outgoing anti-particle. The completeness relation transforms as:*

$$\sum_s u_s\bar{u}_s = \not{p} + m \xrightarrow{\text{crossing } p \to -p} -\not{p} + m = -(\not{p} - m) = -\sum_s v_s\bar{v}_s$$

Because of this algebraic sign flip in the trace, you must strictly ***multiply the overall result by*** (-1) ***each time you cross a fermion*** *to maintain a positive definite squared amplitude.*

4.1.4 | Polarized Matrix Elements

There are several formalisms to compute polarized matrix elements, which are the scattering amplitudes for specific, defined spin states of the particles involved (rather than averaging over them). Let us review the procedures for the most common cases:

- **Massless Case (Helicity States):** For massless fermions, we can elegantly use helicity states u_L and u_R. We exploit the chirality projectors defined as:

$$P_L = \frac{1-\gamma^5}{2}, \quad P_R = \frac{1+\gamma^5}{2}$$

  These operators perfectly project massless spinors onto their left-handed and right-handed components:

$$\begin{cases} P_L u_L = u_L, & P_L v_R = v_R \\ P_R u_R = u_R, & P_R v_L = v_L \\ P_L u_R = P_R u_L = P_L v_L = P_R v_R = 0 \end{cases}$$

  Let us consider calculating $|\mathcal{M}|^2$ for a process involving a strictly left-handed incoming fermion. We need to evaluate the spinor product $u_L(p)\bar{u}_L(p)$. We can rewrite this using the projectors:

$$u_L(p)\bar{u}_L(p) = \sum_s P_L u_s \bar{u}_s = P_L(\not{p} + \underbrace{m}_{=0}) = \frac{1-\gamma^5}{2}\not{p}$$

  Remark. *This is a very powerful result: we can still use the standard trace technology to compute polarized matrix elements! We simply insert the chirality projectors P_L or P_R directly into the Dirac traces. The computations will be slightly more involved because we must now use the trace identities involving γ^5 (which yield Levi-Civita tensors $\epsilon^{\mu\nu\rho\sigma}$). Similar approaches apply for other helicity states and anti-particles.*

- **Massive Case (Spin Projectors):** In the massive case, chirality and helicity do not coincide, so we must use explicit spin states u_s and v_s. We report the following result without formal proof:

$$\begin{cases} u_s(p)\bar{u}_s(p) = P_S(\not{p} + m) \\ v_s(p)\bar{v}_s(p) = P_S(\not{p} - m) \end{cases}$$

  where P_S is the **spin projector** operator defined as:

$$P_S = \frac{1}{2}\left(1 + \gamma^5 \not{S}\right)$$

  Here, S^μ is the spin 4-vector. In the rest frame of the particle, it takes the purely spatial form $S^\mu = (0, \vec{s})$, where $\vec{s}$ is the physical 3-dimensional direction of the spin (with $|\vec{s}|^2 = 1$). To evaluate the matrix element in any other reference frame, one simply obtains the correct expression for S^μ by applying a standard Lorentz boost to the rest-frame vector.

- **Alternative Formalisms:** The **spinor-helicity formalism** offers a highly sophisticated alternative way of handling spin states without relying on squaring the amplitude and taking traces. It computes the amplitude $\mathcal{M}$ directly as a complex number. While exceptionally powerful for multi-leg processes, it is not covered in this introductory course.

4.1.5 | External Photons and Compton Scattering

TODO: add the Feynman diagram showing a generic amplitude blob with an external wavy photon line attached, labeled with momentum p and index mu

Consider a generic scattering amplitude involving an external physical photon with momentum p^μ (such as in Compton scattering).

Using the Feynman rules, we can factor out the polarization vector of this specific external photon:

$$i\mathcal{M} = i\epsilon_\mu(p)\mathcal{M}^\mu$$

where $\mathcal{M}^\mu$ mathematically represents the rest of the Feynman diagram (the "blob") evaluated with an open Lorentz index μ.

Assume we are working in the Lorenz gauge, which enforces the transversality condition:

$$p_\mu \epsilon^\mu(p) = 0$$

Up to a spatial rotation, let us align the photon's momentum along the z-axis, so its 4-momentum is $p^\mu = (E, 0, 0, E)$. We have previously seen that the condition $p_\mu\epsilon^\mu = 0$ does not completely fix the gauge. We can perform an additional residual gauge transformation of the form:

$$\epsilon^\mu(p) \to \epsilon^\mu(p) + \alpha p^\mu$$

for any constant α. We can use this residual freedom to explicitly set the temporal component $\epsilon^0 = 0$. This leaves us with a basis of two physical, transverse polarization states (e.g., circular polarizations):

$$\epsilon^\mu_\pm(p) = (0, 1, \pm i, 0)$$

The Ward Identity

However, the choice to shift ϵ^μ by αp^μ is entirely arbitrary. For our physical scattering amplitude $i\mathcal{M} = i\epsilon_\mu\mathcal{M}^\mu$ to be unambiguously defined and **gauge invariant**, its computed value must not change if we perform this shift:

$$i(\epsilon_\mu + \alpha p_\mu)\mathcal{M}^\mu = i\epsilon_\mu\mathcal{M}^\mu \implies \alpha p_\mu \mathcal{M}^\mu = 0$$

Since α is arbitrary, gauge invariance strictly demands that:

$$p_\mu \mathcal{M}^\mu = 0$$

This profound equation is known as the **Ward Identity**. In practical terms, it states that if you take any valid QED amplitude and explicitly replace the polarization vector of an external photon with its own 4-momentum ($\epsilon^\mu(p) \to p^\mu$), the entire amplitude must identically vanish. We will rigorously prove that this holds in QED later using the path integral formalism.

> **Remark.** *We can interpret the Ward Identity physically: the term $\mathcal{M}^\mu$ (the "blob" the photon attaches to) must act as a **conserved current** that couples to the photon. Recall that the classical continuity equation for a conserved current, $\partial_\mu J^\mu = 0$, transforms in momentum space precisely to $p_\mu J^\mu = 0$. The Ward Identity is the quantum mechanical manifestation of current conservation.*

Polarization Sums

To compute the unpolarized squared amplitude for a process involving external photons, we must square the matrix element $|\mathcal{M}|^2 = (\epsilon_s^\mu \mathcal{M}_\mu)(\epsilon_s^{*\nu} \mathcal{M}_\nu^*)$ and sum over the physical polarization states of the photon:

$$\sum_{s=\pm} \epsilon_s^\mu \epsilon_s^{*\nu}.$$

Assuming the photon momentum p is aligned along the z-axis, the physical transverse polarization vectors are $\epsilon_\pm^\mu = (0, 1, \pm i, 0)^T$. The sum over these states gives a diagonal matrix:

$$\sum_{s=\pm} \epsilon_s^\mu \epsilon_s^{*\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

We can rewrite this expression in a manifestly "covariant-like" form:

$$\sum_{s=\pm} \epsilon_s^\mu \epsilon_s^{*\nu} = -g^{\mu\nu} + \frac{1}{p \cdot \eta}(p^\mu \eta^\nu + p^\nu \eta^\mu),$$

where η^μ is an arbitrary massless four-vector chosen such that $\epsilon_s \cdot \eta = 0$. For a photon moving along the z-axis, choosing $\eta^\mu = (1, 0, 0, -1)$ reproduces the exact matrix above.

What if we choose a generic η such that initially $\epsilon_s \cdot \eta \neq 0$? We can always perform a residual gauge transformation $\epsilon_s^\mu \to \epsilon_s'^\mu = \epsilon_s^\mu + \alpha p^\mu$. By setting $\alpha = -(\epsilon_s \cdot \eta)/(p \cdot \eta)$, we guarantee that the new polarization vector satisfies $\epsilon_s' \cdot \eta = 0$. Therefore, the choice of the vector η is fundamentally just a **choice of gauge** (perfectly compatible with the Lorenz condition $p_\mu \epsilon^\mu = 0$).

Plugging this covariant expression back into the squared amplitude, we distribute the contraction over the two terms:

$$|\mathcal{M}|^2 = -\mathcal{M}_\mu \mathcal{M}_\nu^* g^{\mu\nu} + \frac{1}{p \cdot \eta}\Big[(p^\mu \mathcal{M}_\mu)(\eta^\nu \mathcal{M}_\nu^*) + (\eta^\mu \mathcal{M}_\mu)(p^\nu \mathcal{M}_\nu^*)\Big].$$

At this point, the **Ward Identity** comes to our rescue. It dictates that replacing the polarization vector of an external photon with its four-momentum exactly annihilates the QED amplitude:

$$p^\mu \mathcal{M}_\mu = 0 \quad \text{and} \quad p^\nu \mathcal{M}_\nu^* = 0.$$

Because of this, both terms inside the square brackets strictly vanish, completely removing the dependence on the unphysical gauge-fixing vector η. Therefore, in practical calculations, whenever we sum over the polarizations of an external photon contracted with a conserved amplitude, we are mathematically allowed to make the effective replacement:

$$\sum_s \epsilon_s^\mu \epsilon_s^{*\nu} \longrightarrow -g^{\mu\nu}.$$

While this derivation applies to a single external photon, a stronger version of the Ward identity holds in QED. If an amplitude $\mathcal{M}^{\mu\,\nu_1 \cdots \nu_N}$ involves $N+1$ external photons (with their polarization vectors removed), replacing any one polarization vector with its momentum yields zero:

$$p_\mu \mathcal{M}^{\mu\,\nu_1 \cdots \nu_N} = 0.$$

This powerful result justifies replacing the polarization sum with $-g^{\mu\nu}$ for **any number of external photons** in QED. (Note: This simple substitution rule is true in Abelian gauge theories like QED, but it fails in non-Abelian theories like QCD due to the self-interactions of the gauge bosons, requiring the introduction of Faddeev-Popov ghosts).

Compton Scattering

Let us explicitly verify the Ward identity for Compton scattering: $e^-(p_1)+\gamma(p_2) \to e^-(p_3)+\gamma(p_4)$. We will set the electron mass to zero ($m_e = 0$) for simplicity.

TODO: add Feynman diagrams for Compton scattering: an s-channel diagram and a u-channel diagram, with incoming momenta p1, p2 and outgoing p3, p4. Label the incoming photon index mu and outgoing photon index nu

At tree-level, the amplitude is the sum of an s-channel and a u-channel diagram.

Applying the Feynman rules, we write the full amplitude as $i\mathcal{M} = i\mathcal{M}^{\mu\nu}\epsilon_{2\mu}\epsilon^*_{4\nu}$, where the core tensor is:

$$i\mathcal{M}^{\mu\nu} = -(ie)^2 \left[\overline{u}_3\gamma^\nu \frac{i(\not{p}_1 + \not{p}_2)}{(p_1+p_2)^2}\gamma^\mu u_1 + \overline{u}_3\gamma^\mu \frac{i(\not{p}_3 - \not{p}_2)}{(p_2-p_3)^2}\gamma^\nu u_1\right].$$

To verify the Ward identity for the incoming photon, we substitute its polarization vector with its momentum: $\epsilon_{2\mu} \to p_{2\mu}$, and contract it with the amplitude:

$$p_{2\mu}(i\mathcal{M}^{\mu\nu}) = -ie^2 \left[\overline{u}_3\gamma^\nu \frac{\not{p}_1 + \not{p}_2}{(p_1+p_2)^2}\not{p}_2 u_1 + \overline{u}_3\not{p}_2 \frac{\not{p}_3 - \not{p}_2}{(p_2-p_3)^2}\gamma^\nu u_1\right].$$

We simplify the numerators using the anti-commutation relations of the Dirac algebra and the massless Dirac equations for the external on-shell fermions ($\not{p}_1 u_1 = 0$ and $\overline{u}_3\not{p}_3 = 0$). Also note that $p_2^2 = 0 \implies \not{p}_2\not{p}_2 = 0$.

For the first term (s-channel):

$$(\not{p}_1 + \not{p}_2)\not{p}_2 u_1 = \not{p}_1\not{p}_2 u_1 = (-\not{p}_2\not{p}_1 + 2p_1\cdot p_2)u_1 = 2(p_1\cdot p_2)u_1.$$

For the second term (u-channel):

$$\overline{u}_3\not{p}_2(\not{p}_3 - \not{p}_2) = \overline{u}_3\not{p}_2\not{p}_3 = \overline{u}_3(-\not{p}_3\not{p}_2 + 2p_2\cdot p_3) = \overline{u}_3(2p_2\cdot p_3).$$

Substituting these simplified numerators back into the contracted amplitude, and noting that the denominators are $(p_1+p_2)^2 = 2p_1\cdot p_2$ and $(p_2-p_3)^2 = -2p_2\cdot p_3$, we obtain:

$$p_{2\mu}(i\mathcal{M}^{\mu\nu}) = -ie^2\left[\overline{u}_3\gamma^\nu \frac{2p_1\cdot p_2}{2p_1\cdot p_2}u_1 + \overline{u}_3\frac{2p_2\cdot p_3}{-2p_2\cdot p_3}\gamma^\nu u_1\right] = -ie^2\left[\overline{u}_3\gamma^\nu u_1 - \overline{u}_3\gamma^\nu u_1\right] = 0.$$

Remark. *The Ward identity is strictly valid for the total amplitude $i\mathcal{M}$, but **not** for the individual diagrams! As shown above, the s-channel and u-channel diagrams are not independently gauge invariant; it is their exact cancellation that ensures gauge invariance. Furthermore, this cancellation holds identically for any choice of the outgoing polarization vector ϵ_4^ν, even if it represents an unphysical state.*

Example (Compton Scattering in QED with $m_e = 0$). Let us compute the unpolarized squared amplitude $|\mathcal{M}|^2_{\text{unpol}}$ and the differential cross-section $\frac{d\sigma}{d\Omega}$ in the center of mass (CM) frame for Compton scattering at tree-level, in the massless limit $m_e = 0$. The Mandelstam variables satisfy $s+t+u = 0$.

The amplitude is the sum of two diagrams: $\mathcal{M} = \mathcal{M}_s + \mathcal{M}_u$. The unpolarized squared amplitude expands into three distinct pieces to consider:

$$|\mathcal{M}|^2_{\text{unpol}} = \frac{1}{4}\sum_{\text{spins}}|\mathcal{M}_s|^2 + \frac{1}{4}\sum_{\text{spins}}|\mathcal{M}_u|^2 + \frac{1}{2}\text{Re}\sum_{\text{spins}}(\mathcal{M}_s\mathcal{M}_u^*).$$

TODO: add diagram showing the interference term Ms Mu*, drawing the s-channel diagram connected to the complex conjugated u-channel diagram forming closed loops

1. The s-channel contribution: Using the Feynman rules and replacing the photon polarizations with $-g^{\mu\nu}$ via the Ward identity, the squared amplitude for the s-channel becomes a trace:

$$\frac{1}{4}\sum|\mathcal{M}_s|^2 = \frac{e^4}{4s^2}\,\text{Tr}\left[\not{p}_3\gamma^\nu(\not{p}_1+\not{p}_2)\gamma^\mu\not{p}_1\gamma_\mu(\not{p}_1+\not{p}_2)\gamma_\nu\right].$$

Applying the contraction identities $\gamma^\mu \not{\phi} \gamma_\mu = -2\not{\phi}$, we simplify the trace from the inside out. First, $\gamma^\mu \not{p}_1 \gamma_\mu = -2\not{p}_1$. The central block becomes $(\not{p}_1 + \not{p}_2)(-2\not{p}_1)(\not{p}_1 + \not{p}_2) = -2s\not{p}_2$. Finally, contracting the outer indices gives $\gamma^\nu(-2s\not{p}_2)\gamma_\nu = 4s\not{p}_2$. The trace simplifies to:

$$\frac{1}{4}\sum |\mathcal{M}_s|^2 = \frac{e^4}{4s^2}\,\mathrm{Tr}\left[\not{p}_3(4s\not{p}_2)\right] = \frac{e^4}{s}4(p_2\cdot p_3) = \frac{e^4}{s}4\left(-\frac{u}{2}\right) = -2e^4\frac{u}{s}.$$

2. The u-channel and interference: By crossing symmetry ($s \leftrightarrow u$), the u-channel contribution is immediately:

$$\frac{1}{4}\sum |\mathcal{M}_u|^2 = -2e^4\frac{s}{u}.$$

The interference term identically vanishes in the strict massless limit because the resulting trace is proportional to $(p_1 + p_2)\cdot(p_1 - p_4) = (s + u + t)/2 = 0$. The total unpolarized amplitude is therefore:

$$|\mathcal{M}|^2_{\text{unpol}} = 2e^4\left(-\frac{u}{s} - \frac{s}{u}\right).$$

3. Differential cross-section in the CM frame: In the center of mass frame, the kinematics yield $t = -s(1-\cos\theta)/2$ and $u = -s(1+\cos\theta)/2$, where θ is the scattering angle. Substituting u into our amplitude:

$$|\mathcal{M}|^2_{\text{unpol}} = 2e^4\left(\frac{1+\cos\theta}{2} + \frac{2}{1+\cos\theta}\right).$$

Using the standard massless 2-body cross-section formula $\frac{d\sigma}{d\Omega} = \frac{|\mathcal{M}|^2}{64\pi^2 s}$ and defining $\alpha = \frac{e^2}{4\pi}$, we obtain the differential cross section:

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{2s}\left(\frac{1+\cos\theta}{2} + \frac{2}{1+\cos\theta}\right).$$

Example (Ward Identity in Scalar QED Compton Scattering). Let us verify the Ward identity explicitly for Compton scattering in scalar QED (sQED) at tree-level. In sQED, there are three tree-level diagrams: the s-channel, the u-channel, and a 4-point contact (seagull) vertex arising from the $A_\mu A^\mu|\phi|^2$ term in the Lagrangian.

We write the sub-amplitudes stripped of their polarization vectors and factor out the common coupling components. We want to show that contracting the total amplitude with the incoming photon momentum $p_{2\mu}$ yields zero. The three contributions $i\mathcal{M}^{\mu\nu}$ are:

$$\begin{aligned}
i\mathcal{M}_s^{\mu\nu} &= (-ie)^2\frac{i}{(p_1+p_2)^2 - m^2}(2p_1+p_2)^\mu(p_1+p_2+p_3)^\nu \\
i\mathcal{M}_u^{\mu\nu} &= (-ie)^2\frac{i}{(p_1-p_4)^2 - m^2}(2p_1-p_4)^\nu(p_1-p_4+p_3)^\mu \\
i\mathcal{M}_c^{\mu\nu} &= 2ie^2 g^{\mu\nu}
\end{aligned}$$

Now, we contract each term with $p_{2\mu}$:

- **s-channel:** The contracted numerator is $p_2\cdot(2p_1+p_2) = 2p_1\cdot p_2 + p_2^2$. Since the photon is massless ($p_2^2 = 0$), this is $2p_1\cdot p_2$, which exactly matches the on-shell sQED denominator $(p_1+p_2)^2 - m^2$. Thus, they cancel out:

$$p_{2\mu}(i\mathcal{M}_s^{\mu\nu}) = -ie^2(p_1+p_2+p_3)^\nu.$$

- **u-channel:** To simplify the contraction, we use momentum conservation $p_2 = p_3 - (p_1 - p_4)$. The contracted numerator becomes $p_2\cdot(p_3+p_1-p_4) = p_3^2 - (p_1-p_4)^2 = m^2 - (p_1-p_4)^2$.

This is exactly the negative of the u-channel denominator. Canceling them introduces a minus sign:

$$p_{2\mu}(i\mathcal{M}_u^{\mu\nu}) = +ie^2(2p_1 - p_4)^\nu.$$

- **Contact channel:** The metric tensor simply lowers the index of the momentum:

$$p_{2\mu}(i\mathcal{M}_c^{\mu\nu}) = 2ie^2 p_2^\nu.$$

Summing the three contracted pieces together, we can factor out ie^2:

$$p_{2\mu}(i\mathcal{M}_{\text{tot}}^{\mu\nu}) = ie^2\left[-(p_1 + p_2 + p_3)^\nu + (2p_1 - p_4)^\nu + 2p_2^\nu\right].$$

Combining the momentum terms inside the bracket yields:

$$-p_1 - p_2 - p_3 + 2p_1 - p_4 + 2p_2 = p_1 + p_2 - p_3 - p_4.$$

By global energy-momentum conservation ($p_1 + p_2 = p_3 + p_4$), this expression is identically zero. Therefore, $p_{2\mu}\mathcal{M}^{\mu\nu} = 0$, explicitly verifying the Ward identity for tree-level scalar QED.

4.1.6 | A Note on Convergence (or lack thereof)

Throughout our study of Quantum Electrodynamics (QED), we compute amplitudes and Green's functions using perturbative expansions (the Dyson series). These observables are expanded as power series in the fine-structure constant $\alpha = \frac{e^2}{4\pi}$. A fundamental mathematical question arises: *are these series expansions actually convergent, and if so, what is their radius of convergence?*

Surprisingly, Freeman Dyson famously showed in 1952 that the perturbative series in QED is strictly **divergent for any non-zero value of** α.

His beautiful physical argument works as follows:

- Let us assume that an observable $\mathcal{O}(\alpha) = \sum_{n=0}^{\infty} \mathcal{O}_n \alpha^n$ is a convergent series for some physical, positive value of $\alpha \neq 0$.
- By the standard theorems of complex analysis, if a power series converges for some α, it must have a finite, non-zero radius of convergence R. Consequently, the series must converge for any value of α within the interval $(-R, R)$.
- This implies the series must also converge for some small negative value $\alpha < 0$ such that $-R < \alpha < 0$. Let us consider the physical meaning of a "QED with negative α".
- Since $\alpha \propto e^2$, making α negative effectively flips the overall sign of the Coulomb force. In this hypothetical universe, **like charges attract and opposite charges repel**.
- In such a universe, the vacuum would be catastrophically unstable. A spontaneous quantum fluctuation creating an e^-e^+ pair would not annihilate back into the vacuum. Instead, the electron and positron would repel each other. Furthermore, electrons would attract other electrons, and positrons would attract other positrons, forming infinitely large, separate clusters of pure negative and pure positive charge to lower the energy of the system.
- Because the vacuum state is entirely unstable for $\alpha < 0$, the physical observables cannot be analytically continued to negative couplings. Therefore, the radius of convergence must be exactly zero ($R = 0$), meaning the series **cannot be convergent** for any $\alpha \neq 0$.

If the series diverges, why does perturbation theory work so incredibly well in QED? The answer is that the perturbative expansion is an **asymptotic series**.

Asymptotic convergence is a weaker form of convergence. It means that if we truncate the series at a fixed order N,

$$\mathcal{O}(\alpha, N) = \sum_{n=0}^{N} \mathcal{O}_n \alpha^n,$$

this partial sum provides an approximation of the true observable $\mathcal{O}$ that is valid up to $\mathcal{O}(\alpha^{N+1})$ errors *strictly in the mathematical limit* as $\alpha \to 0$.

However, for our universe where α is fixed to a finite value ($\approx 1/137$), the behavior is different. Adding the first few "M" terms of the series yields a better and better approximation of the true physical result. But eventually, there is an optimal truncation order (roughly $N \sim 1/\alpha \sim 137$ for QED). If you continue adding more terms beyond this point, the coefficients $\mathcal{O}_n$ grow so rapidly (typically factorially, $\sim n!$) that the series starts to violently diverge.

In practice, this is completely fine. The computational complexity of Feynman diagrams grows so fast that we are typically only able to compute a very small number of terms in the series (usually up to 2, 4, or 5 loops at most, depending on the observable). At these low orders, the series behaves beautifully and yields spectacularly precise predictions. The divergence only becomes a practical issue in specific theoretical frameworks where certain classes of diagrams must be considered at all orders simultaneously, requiring dedicated mathematical techniques known as **resummation**.

4.1.7 | Ward Identity and Lorentz Transformations

TODO: check

We have seen that if the amplitude for a process involving an external photon of momentum p^μ is written as $i\mathcal{M} = \epsilon_\mu(p) i\mathcal{M}^\mu$, the **Ward Identity** requires that $p_\mu \mathcal{M}^\mu = 0$.

Another profound way to understand the necessity of this identity is by analyzing how the photon's polarization states transform under the Lorentz group. Specifically, we can look at the *Little Group* of the photon: the subgroup of Lorentz transformations that leaves the photon's momentum vector invariant.

Let us construct such a transformation, T, as a sequence of three operations: a boost along the y-axis, a rotation around the x-axis, and a compensating boost along the z-axis:

$$T = \Lambda_z R_x \Lambda_y.$$

Without loss of generality, choose the photon momentum along the z-axis: $p^\mu = (E, 0, 0, E)$.

Step 1: Boost along the y-axis
We define the boost Λ_y with velocity parameter β (where $0 \leq \beta < 1$ and $\gamma = 1/\sqrt{1-\beta^2}$):

$$\Lambda_y = \begin{pmatrix} \gamma & 0 & -\beta\gamma & 0 \\ 0 & 1 & 0 & 0 \\ -\beta\gamma & 0 & \gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Applying this to the momentum, we get a vector that has acquired a transverse component:

$$\Lambda_y \cdot p = (E\gamma, 0, -\beta E\gamma, E).$$

Step 2: Rotation around the x-axis
We want to remove the new transverse momentum. We define a rotation R_x such that the resulting momentum is once again parallel to the z-axis:

$$R_x = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\theta & -\sin\theta \\ 0 & 0 & \sin\theta & \cos\theta \end{pmatrix},$$

by carefully choosing the angle such that $\cos\theta = 1/\gamma$ and $\sin\theta = -\beta$. Applying this to our boosted momentum eliminates the y-component and boosts the overall energy/z-momentum:

$$R_x \cdot \Lambda_y \cdot p = (E\gamma, 0, 0, E\gamma).$$

Step 3: Boost along the z-axis
Finally, we ask for a boost Λ_z that scales the momentum back down to its original value, requiring $\Lambda_z \cdot R_x \cdot \Lambda_y \cdot p = p = (E, 0, 0, E)$. This requires:

$$\Lambda_z = \begin{pmatrix} \gamma' & 0 & 0 & -\beta'\gamma' \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\beta'\gamma' & 0 & 0 & \gamma' \end{pmatrix} \quad \text{with} \quad \beta' = \frac{\beta^2}{2-\beta^2} \quad \text{and} \quad \gamma' = \frac{1}{\sqrt{1-\beta'^2}}.$$

By construction, one can check that the combined transformation T leaves the momentum strictly invariant:

$$T^\mu{}_\nu p^\nu = p^\mu = (E, 0, 0, E).$$

The Effect on Polarization
However, if we apply this exact same transformation T to the physical circular polarization vectors $\epsilon^\mu_\pm = \frac{1}{\sqrt{2}}(0, 1, \pm i, 0)$, one finds that they do *not* remain invariant. Instead, a direct calculation shows they transform as:

$$T^\mu{}_\nu \epsilon^\nu_\pm = \epsilon^\mu_\pm \mp \frac{i\beta}{E} p^\mu.$$

This reveals a striking property of massless spin-1 particles: **after a Lorentz transformation that leaves p^μ invariant, the polarization vector is shifted by a term proportional to the momentum p^μ.**

This shifted polarization vector no longer belongs strictly to our standard Hilbert space of independent, purely transverse polarizations. We can mathematically "fix" this by performing a gauge transformation ($\epsilon^\mu \to \epsilon^\mu + \alpha p^\mu$), which we know removes the longitudinal part.

However, nature does not care about our coordinate choices. If the scattering amplitude $i\mathcal{M}$ is to be genuinely **Lorentz invariant**, and if the space of physical polarization states must remain closed under Lorentz transformations, the physical observables must be completely insensitive to this momentum shift. This strictly implies that the amplitude dotted into the momentum must vanish:

$$p_\mu \mathcal{M}^\mu = 0.$$

Thus, the Ward identity is deeply rooted in the fundamental requirement of Lorentz invariance for massless vector fields. (This identity will be formally and systematically proven when we study symmetries in QFT through the path integral and Noether's theorem).

4.2 | QED as an Abelian Gauge Theory

Up to this point, we have never actually derived the Lagrangian of Quantum Electrodynamics (QED) from fundamental quantum principles. Instead, we borrowed the interaction structure from classical electromagnetism (letting Maxwell do the heavy lifting) and only verified *a posteriori* that the resulting theory was invariant under the gauge transformations:

$$\begin{aligned}\psi &\to e^{i\alpha(x)}\psi \\ \bar{\psi} &\to \bar{\psi}e^{-i\alpha(x)} \\ A_\mu &\to A_\mu - \frac{1}{e}\partial_\mu\alpha(x)\end{aligned}$$

Let us take a step back, pretend we know absolutely nothing about Maxwell's equations, and see if we can derive QED entirely from a fundamental symmetry principle.

4.2.1 | Promoting a Global Symmetry to a Local Symmetry

Consider the free Dirac Lagrangian:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\not{\partial} - m)\psi$$

It is immediately obvious that this Lagrangian is invariant under a **global** $U(1)$ **transformation**, where we shift the phase of the fields by a constant amount α: $\psi \to e^{i\alpha}\psi$ and $\bar{\psi} \to \bar{\psi}e^{-i\alpha}$.

However, in special relativity, Lagrangians and equations of motion are strictly local. It is philosophically unsatisfying to demand that the phase of a field must be rotated by the exact same amount simultaneously everywhere in the universe. It makes sense to ask if this symmetry can also be made **local**. In other words, can we promote the constant α to a spacetime-dependent function $\alpha(x)$?

Let us test the Dirac Lagrangian under the local transformation $\psi(x) \to e^{i\alpha(x)}\psi(x)$. The mass term remains perfectly invariant:

$$-m\bar{\psi}\psi \to -m\left(\bar{\psi}e^{-i\alpha(x)}\right)\left(e^{i\alpha(x)}\psi\right) = -m\bar{\psi}\psi.$$

The derivative term, however, breaks the invariance because the derivative acts on the new local phase:

$$\begin{aligned}i\bar{\psi}\not{\partial}\psi &\to i\bar{\psi}e^{-i\alpha(x)}\gamma^\mu\partial_\mu\left(e^{i\alpha(x)}\psi\right) \\ &= i\bar{\psi}e^{-i\alpha(x)}\gamma^\mu\left[e^{i\alpha(x)}(\partial_\mu\psi) + i(\partial_\mu\alpha(x))e^{i\alpha(x)}\psi\right] \\ &= i\bar{\psi}\not{\partial}\psi - (\partial_\mu\alpha(x))\bar{\psi}\gamma^\mu\psi.\end{aligned}$$

This failure is not surprising. A standard derivative is defined by comparing the field at two infinitesimally close, but distinct, points:

$$v^\mu\partial_\mu\psi = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[\psi(x+\epsilon v) - \psi(x)].$$

Under a local transformation, $\psi(x+\epsilon v)$ and $\psi(x)$ pick up completely different phase shifts, rendering their direct subtraction physically meaningless (akin to subtracting vectors from different tangent spaces in General Relativity).

4.2.2 | The Covariant Derivative and the Gauge Field

To fix this, we must replace the standard derivative ∂_μ with a **covariant derivative** D_μ that possesses "better" transformation properties. We need a way to transport the phase transformation from x to $x + \epsilon v$.

We define the covariant derivative using a comparator $U(y, x)$, which is a scalar function defined such that if $\psi(x) \to e^{i\alpha(x)}\psi(x)$, then $U(y, x)\psi(x)$ transforms precisely like $\psi(y)$:

$$U(y, x)\psi(x) \to e^{i\alpha(y)}U(y, x)\psi(x).$$

This implies the comparator itself must transform as:

$$U(y, x) \to e^{i\alpha(y)}U(y, x)e^{-i\alpha(x)}.$$

We also require $U(x, x) = 1$. Since a $U(1)$ transformation is purely a phase redefinition, we can parameterize $U(y, x)$ as a phase: $U(y, x) = \exp(i\phi(y, x))$, with $\phi(x, x) = 0$. We can now define a new vector field $A_\mu(x)$ based on the infinitesimal behavior of this phase. Taking the derivative with respect to y^μ and evaluating at $x = y$:

$$\frac{\partial}{\partial y^\mu}\phi(y, x)\bigg|_{x=y} \equiv gA_\mu(x),$$

where g is a constant added for later convenience (the coupling strength). For points infinitesimally close together, we can expand the comparator:

$$U(x + \epsilon v, x) = 1 + ig\epsilon v^\mu A_\mu(x) + \mathcal{O}(\epsilon^2).$$

Let us see how this new field A_μ must transform to keep $U(y, x)$ consistent with our rules. Applying the local $U(1)$ transformation to the expansion:

$$1 + ig\epsilon v^\mu A_\mu \to e^{i\alpha(x+\epsilon v)}(1 + ig\epsilon v^\mu A_\mu)e^{-i\alpha(x)}.$$

Expanding $e^{i\alpha(x+\epsilon v)} \approx e^{i\alpha(x)}(1 + i\epsilon v^\mu \partial_\mu \alpha)$ and keeping terms up to $\mathcal{O}(\epsilon)$, we find:

$$ig\epsilon v^\mu A_\mu \to ig\epsilon v^\mu A_\mu + i\epsilon v^\mu \partial_\mu \alpha \quad \implies \quad A_\mu(x) \xrightarrow{\text{local U(1)}} A_\mu(x) + \frac{1}{g}\partial_\mu \alpha(x).$$

Using this newly constructed field, the modified limit definition of the derivative yields the explicit form of the covariant derivative:

$$D_\mu = \partial_\mu - igA_\mu$$

By construction, $D_\mu\psi$ transforms neatly: $D_\mu\psi \to e^{i\alpha(x)}D_\mu\psi$. Consequently, if we replace ∂_μ with D_μ in the Dirac Lagrangian, the entire Lagrangian becomes invariant under local $U(1)$ transformations:

$$\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi = \bar{\psi}(i\not{\partial} - m)\psi + gA_\mu\bar{\psi}\gamma^\mu\psi.$$

We have just uniquely derived the interaction term of QED!

4.2.3 | The Field Strength Tensor

In order to define D_μ, we were forced to introduce a new dynamical vector field A_μ. For this field to be physical, it must have its own kinetic term in the Lagrangian, and this kinetic term must also be locally gauge invariant.

We know that $D_\mu\psi$ transforms exactly like ψ. Therefore, applying the covariant derivative twice, $D_\mu D_\nu\psi$, also transforms like ψ. The commutator $[D_\mu, D_\nu]\psi$ must do the same. However, $[D_\mu, D_\nu]$ is fundamentally different: it is *not* a derivative operator, but merely a multiplicative function. Let us check this explicitly:

$$\begin{aligned}
[D_\mu, D_\nu]\psi &= [\partial_\mu - igA_\mu, \partial_\nu - igA_\nu]\psi \\
&= \underbrace{[\partial_\mu, \partial_\nu]\psi}_{=0} - \underbrace{g^2[A_\mu, A_\nu]\psi}_{=0} - ig([\partial_\mu, A_\nu] + [A_\mu, \partial_\nu])\psi \\
&= -ig\left(\partial_\mu(A_\nu\psi) - A_\nu\partial_\mu\psi + A_\mu\partial_\nu\psi - \partial_\nu(A_\mu\psi)\right) \\
&= -ig\left((\partial_\mu A_\nu)\psi + A_\nu\partial_\mu\psi - A_\nu\partial_\mu\psi + A_\mu\partial_\nu\psi - (\partial_\nu A_\mu)\psi - A_\mu\partial_\nu\psi\right) \\
&= -ig(\partial_\mu A_\nu - \partial_\nu A_\mu)\psi.
\end{aligned}$$

Notice that the first commutator vanishes because partial derivatives commute on smooth functions ($[\partial_\mu, \partial_\nu] = 0$), and the second commutator vanishes because the gauge fields A_μ are simply commuting functions in an Abelian theory ($[A_\mu, A_\nu] = 0$). The rest of the calculation is a straightforward application of the Leibniz product rule (e.g., $\partial_\mu(A_\nu\psi) = (\partial_\mu A_\nu)\psi + A_\nu\partial_\mu\psi$), which causes all the terms where the derivative acts directly on ψ to perfectly cancel out.

We define the final term in the parentheses as the field strength tensor:

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

Since $[D_\mu, D_\nu]\psi = -igF_{\mu\nu}\psi$ must transform as $e^{i\alpha(x)}(-igF_{\mu\nu}\psi)$, and ψ already absorbs the phase factor, the function $F_{\mu\nu}$ itself must be strictly **gauge invariant**.

To build a Lorentz invariant kinetic term, we simply square this tensor. Adding it to the Lagrangian with the standard normalization factor $-1/4$ yields:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\not{D} - m)\psi$$

If we identify the arbitrary coupling constant as the electron charge ($g = -e$), this is precisely the full QED Lagrangian. We derived QED without ever looking at Maxwell's equations, relying *only* on the principle of local symmetry.

4.2.4 | Final Symmetries and Restrictions

Could we have added other terms to the Lagrangian? Let us review the constraints:

- **Parity and Time-Reversal:** We could technically add a gauge-invariant term like $\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$. However, this term is not invariant under Parity (P) and Time-reversal (T). Since the electromagnetic interaction is empirically observed to respect P and T, we exclude this term based on physical symmetry arguments.
- **Renormalizability:** We could build higher-order gauge invariant terms like $\bar{\psi}\gamma^\mu\gamma^\nu\psi F_{\mu\nu}$ or $(\bar{\psi}\psi)^2$. However, these operators have a mass dimension strictly greater than 4, which would render the theory non-renormalizable. We exclude them by demanding that no operators with mass dimension > 4 appear in the Lagrangian.
- **Universality of the Gauge Principle:** Starting from a Dirac field was almost irrelevant. The entire procedure was just about defining a covariant derivative that transforms properly under a $U(1)$ symmetry. If we had started with a complex scalar field instead, we would have

identically derived Scalar QED (sQED). Furthermore, repeating this exact procedure for a *non-Abelian* symmetry group (like $SU(3)$) naturally yields non-Abelian gauge theories, such as Quantum Chromodynamics (QCD) or the electroweak sector of the Standard Model.

- **Uniqueness:** It can be rigorously shown that $\mathcal{L} = -1/4F^2$ is the most general free Lagrangian for a massless spin-1 field, and that F^2 is the only gauge invariant term one can write up to dimension 4 as a function of A_μ. Therefore, $\mathcal{L}_{\text{QED}}$ is the most general Lorentz, gauge, and P/T invariant Lagrangian we can construct for these fields up to dimension 4 (justifying the claims we made when we first introduced this lagrangian a priori in section ...).

TODO: add ref

4.3 | Born Approximation and Non-Relativistic Scattering

TODO: check everything in this section

We consider the elastic scattering of two particles in the non-relativistic limit, where the magnitude of the three-momentum is much smaller than the rest mass: $|\mathbf{p}_i| \ll m_i$. In this limit, quantum field theory must reproduce the familiar results of non-relativistic quantum mechanics scattering theory, where interactions are described by a static potential $V(\mathbf{x})$.

Let us first review the main results from standard quantum mechanics textbooks. The differential cross-section is given by:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = |f(\mathbf{q})|^2$$

where the transferred momentum is defined as $\mathbf{q} \equiv \mathbf{p}_{\text{out}} - \mathbf{p}_{\text{in}}$, and $f(\mathbf{q})$ is the non-relativistic scattering amplitude (which must not be confused with the relativistic invariant amplitude $\mathcal{M}$). For weak potentials, $f(\mathbf{q})$ can be computed using the **Born approximation**:

$$f(\mathbf{q}) = -\frac{m}{2\pi} \int \mathrm{d}^3\mathbf{x}\, e^{-i\mathbf{q}\cdot\mathbf{x}} V(\mathbf{x}).$$

TODO: add diagram: scattering angle

We introduce the scattering angle θ between the incoming and outgoing momenta:

$$\cos\theta = \frac{\mathbf{p}_{\text{out}} \cdot \mathbf{p}_{\text{in}}}{p^2},$$

where, for elastic scattering, $p \equiv |\mathbf{p}_{\text{in}}| = |\mathbf{p}_{\text{out}}|$. Using simple trigonometry, the magnitude of the transferred momentum is:

$$q \equiv |\mathbf{q}| = 2p \sin\frac{\theta}{2}.$$

For a spherically symmetric central potential $V(\mathbf{x}) = V(r)$ with $r \equiv |\mathbf{x}|$, the angular integration in the Born approximation simplifies, yielding:

$$f(\mathbf{q}) = -\frac{2m}{q} \int_0^\infty \mathrm{d}r\, V(r) \sin(qr).$$

Connecting Relativistic QFT to the Non-Relativistic Potential

To reproduce this classical scattering picture within QFT, we consider the scattering of a light particle of mass m off a very heavy target of mass M_T, such that $M_T \gg m$. In this limit, we can neglect the recoil of the target, treating it essentially as a source for a static potential $V(\mathbf{x})$. Furthermore, as $M_T \to \infty$, the Center of Mass (C.M.) frame and the rest frame of the target coincide.

In the C.M. frame, the relativistic differential cross-section for $2 \to 2$ scattering reduces to:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{64\pi^2} \frac{1}{E_{\text{C.M.}}^2} |\mathcal{M}|^2 \simeq \frac{1}{64\pi^2} \frac{1}{M_T^2} |\mathcal{M}|^2.$$

We can rewrite this expression to match the form of the non-relativistic cross-section:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \simeq \left(\frac{m}{2\pi}\right)^2 |\tilde{\mathcal{M}}|^2,$$

by defining a *rescaled amplitude*:

$$\tilde{\mathcal{M}} \equiv \frac{\mathcal{M}}{2m\, 2M_T}.$$

Comparing this result with the non-relativistic quantum mechanics formula ($\mathrm{d}\sigma/\mathrm{d}\Omega = |f(\mathbf{q})|^2$), we can identify the rescaled amplitude with the Fourier transform of the potential:

$$\tilde{\mathcal{M}} = -\int \mathrm{d}^3\mathbf{x}\, e^{-i\mathbf{q}\cdot\mathbf{x}} V(\mathbf{x}).$$

Inverting the Fourier transform allows us to directly compute the effective non-relativistic potential from a QFT Feynman diagram:

$$V(\mathbf{x}) = -\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \tilde{\mathcal{M}}(\mathbf{q})\, e^{i\mathbf{q}\cdot\mathbf{x}}.$$

4.3.1 | Example 1: Coulomb Potential from QED

Consider the elastic scattering of an electron off a proton: $e^-(p_1) + p(p_2) \to e^-(p_3) + p(p_4)$.

Remark. *The proton is not an elementary particle, but at low energies we are not sensitive to its internal structure, so we can treat it as a point-like Dirac fermion.*

TODO: add diagram: t-channel Feynman diagram for $e^- p \to e^- p$

The tree-level amplitude, mediated by photon exchange (t-channel), is given by the Feynman rules:

$$i\mathcal{M} = (iQ_e e)(iQ_p e)\,[\bar{u}_3\gamma^\mu u_1]\,\frac{-ig_{\mu\nu}}{(p_3-p_1)^2}\,[\bar{u}_4\gamma^\nu u_2]\,.$$

In the non-relativistic limit, the energies are conserved ($E_3 \simeq E_1$), meaning the energy transfer is zero. The four-momentum squared of the exchanged photon reduces purely to the three-momentum transfer:

$$(p_3-p_1)^2 \simeq -(\mathbf{p}_3-\mathbf{p}_1)^2 = -|\mathbf{q}|^2.$$

We also need to evaluate the spinor currents in the non-relativistic limit ($\mathbf{p} \to 0$). The Dirac spinors take the form:

$$u_s \simeq \sqrt{m}\begin{pmatrix}\xi_s \\ \xi_s\end{pmatrix},$$

where ξ_s is a two-component Weyl spinor. The temporal component of the vector current ($\mu = 0$) survives and conserves spin:

$$\bar{u}_{s'}\gamma^0 u_s = u^\dagger_{s'}\gamma^0\gamma^0 u_s \simeq 2m\,\xi^\dagger_{s'}\xi_s = 2m\,\delta_{s's}.$$

Conversely, the spatial components ($\mu = i$) vanish in the strict non-relativistic limit because they involve off-diagonal Pauli matrices connecting upper and lower components, resulting in a cancellation:

$$\begin{aligned}\bar{u}_{s'}\gamma^i u_s = u^\dagger_{s'}\gamma^0\gamma^i u_s &= u^\dagger_{s'}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & \sigma^i\\ -\sigma^i & 0\end{pmatrix} u_s \\ &= u^\dagger_{s'}\begin{pmatrix}-\sigma^i & 0\\ 0 & \sigma^i\end{pmatrix} u_s \simeq m(-\xi^\dagger_{s'}\sigma^i\xi_s + \xi^\dagger_{s'}\sigma^i\xi_s) = 0.\end{aligned}$$

Substituting these non-relativistic currents back into the amplitude, only the $g_{00} = 1$ term of the photon propagator contributes. The amplitude becomes heavily simplified and exhibits strict **spin conservation**:

$$i\mathcal{M} \simeq -i\frac{Q_e Q_p e^2}{|\mathbf{q}|^2}(2m_e)(2m_p)\delta_{s_3 s_1}\delta_{s_4 s_2}.$$

For matching initial and final spins, the rescaled amplitude is:

$$\tilde{\mathcal{M}} = -\frac{Q_e Q_p e^2}{|\mathbf{q}|^2}.$$

To find the effective potential, we perform the inverse Fourier transform. Switching to spherical coordinates and aligning $\mathbf{q}$ with the z-axis to integrate over angles:

$$\begin{aligned} V(\mathbf{x}) &= \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \frac{Q_e Q_p e^2}{|\mathbf{q}|^2} e^{i\mathbf{q}\cdot\mathbf{x}} \\ &= \frac{Q_e Q_p e^2}{(2\pi)^3} \int_0^\infty \mathrm{d}q\, q^2 \int_0^{2\pi} \mathrm{d}\phi \int_{-1}^1 \mathrm{d}(\cos\theta) \frac{1}{q^2} e^{iqr\cos\theta} \\ &= \frac{Q_e Q_p e^2}{(2\pi)^2} \int_0^\infty \mathrm{d}q \left[\frac{e^{iqr\cos\theta}}{iqr}\right]_{-1}^1 \\ &= \frac{Q_e Q_p e^2}{(2\pi)^2} \int_0^\infty \mathrm{d}q \frac{e^{iqr} - e^{-iqr}}{iqr} = \frac{Q_e Q_p e^2}{(2\pi)^2} \frac{2}{r} \int_0^\infty \mathrm{d}q \frac{\sin(qr)}{q}. \end{aligned}$$

Using the standard Dirichlet integral $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$, we obtain the final result:

$$V(r) = \frac{Q_e Q_p e^2}{4\pi r}.$$

Remarkably, QFT correctly reproduces the classical Coulomb potential!

4.3.2 | Example 2: Yukawa Potential from Scalar Exchange

Now consider the same scattering process, but mediated by a massive scalar field ϕ (with mass m_ϕ) interacting via a Yukawa coupling $\mathcal{L}_{\text{int}} = -g\bar{\psi}\psi\phi$.

TODO: add diagram: t-channel Feynman diagram for scalar exchange

The Feynman rule for the scalar vertex is simply $-ig$. The tree-level amplitude is:

$$i\mathcal{M} = \bar{u}_3 u_1 \frac{i}{(p_3 - p_1)^2 - m_\phi^2} \bar{u}_4 u_2 (-ig)^2.$$

In the non-relativistic limit, the scalar current (which lacks the γ^μ matrix compared to the vector case) still conserves spin. We can see this explicitly by using the chiral (Weyl) representation for the gamma matrices, where $\gamma^0 = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix}$, and the non-relativistic spinors are $u_s \simeq \sqrt{m}\begin{pmatrix} \xi_s \\ \xi_s \end{pmatrix}$:

$$\bar{u}_{s'} u_s = u_{s'}^\dagger \gamma^0 u_s \simeq m \begin{pmatrix} \xi_{s'}^\dagger & \xi_{s'}^\dagger \end{pmatrix} \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix} \begin{pmatrix} \xi_s \\ \xi_s \end{pmatrix} = m(\xi_{s'}^\dagger \xi_s + \xi_{s'}^\dagger \xi_s) = 2m\delta_{s's}.$$

The amplitude (implicitly considering spin conservation, thus omitting the spin sum over the deltas) becomes:

$$i\mathcal{M} \simeq -ig^2 \frac{(2m_e)(2m_p)}{-|\mathbf{q}|^2 - m_\phi^2} \quad \Longrightarrow \quad \tilde{\mathcal{M}} = \frac{g^2}{|\mathbf{q}|^2 + m_\phi^2}.$$

The corresponding potential is given by the Fourier transform:

$$\begin{aligned} V(\mathbf{x}) &= -\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3} \frac{g^2}{|\mathbf{q}|^2 + m_\phi^2} e^{i\mathbf{q}\cdot\mathbf{x}} \\ &= -\frac{g^2}{(2\pi)^2} \int_0^\infty \mathrm{d}q\, q^2 \frac{1}{q^2 + m_\phi^2} \int_{-1}^1 \mathrm{d}(\cos\theta) e^{iqr\cos\theta} \\ &= -\frac{g^2}{(2\pi)^2} \int_0^\infty \mathrm{d}q \frac{q}{q^2 + m_\phi^2} \frac{e^{iqr} - e^{-iqr}}{ir}. \end{aligned}$$

We can extend the integration range to $-\infty$ by changing variables $q \to -q$ in the second term (e^{-iqr}), which allows us to combine everything into a single integral over the entire real axis:

$$V(\mathbf{x}) = -\frac{g^2}{(2\pi)^2 ri} \int_{-\infty}^\infty \mathrm{d}q \frac{q\, e^{iqr}}{q^2 + m_\phi^2}.$$

To evaluate this integral, we use Cauchy's residue theorem.

TODO: add diagram: Complex contour integration in the upper half-plane

We close the integration contour in the upper half of the complex q-plane, where e^{iqr} decays exponentially. The integrand has simple poles at $q = \pm im_\phi$. We pick up the residue from the pole enclosed by our contour, at $q = +im_\phi$:

$$V(r) = -\frac{g^2}{4\pi r} e^{-m_\phi r}.$$

This is the famous **Yukawa potential**. Notice that in the limit $m_\phi \to 0$ (a massless mediator), the exponential factor disappears and we perfectly recover the form of the Coulomb potential $V(r) \propto 1/r$.

Comparing Coulomb and Yukawa Potentials

Analyzing the two potentials reveals fundamental differences depending on the spin and mass of the mediating particle:

- **Yukawa Potential (Spin-0 exchange):**
  - *Short Range:* The exponential factor $e^{-m_\phi r}$ sharply cuts off the interaction at distances $r \sim 1/m_\phi$.
  - *Universally Attractive:* The overall sign is negative. You can explicitly check that replacing the incoming electron (e^-) with a positron (e^+) alters the Feynman rules but does *not* change the overall sign of $i\mathcal{M}$ (because scalar vertices do not involve γ^0 matrices that flip signs for antiparticles). Therefore, scalar exchange always produces an attractive force, regardless of the interacting particles.
- **Coulomb Potential (Spin-1 exchange):**
  - *Long Range:* The photon is strictly massless ($m_\gamma = 0$). If it had a mass, an extra suppression factor $e^{-m_\gamma r}$ would have appeared, mirroring the Yukawa case.
  - *Repulsive/Attractive depending on charge:* The potential is repulsive for like-charges and attractive for opposite charges. Replacing an e^- with an e^+ fundamentally flips the sign of the vector current in the amplitude, changing the sign of the potential $V(r)$.

In conclusion, QFT successfully and rigorously yields interaction potentials that are perfectly compatible with our classical and non-relativistic expectations.

5 | Path Integrals

Path integrals were invented as an alternative but equivalent quantization procedure to canonical quantization. In this chapter, we will introduce the path integral formulation of quantum field theory and show how it can be used to compute correlation functions and scattering amplitudes.

Applying path integrals to traditional quantum mechanics seems a bit complicated and unnecessary, but it becomes very useful in quantum field theory, where the canonical quantization procedure can be quite cumbersome. Moreover, path integrals provide a more intuitive picture of quantum mechanics and quantum field theory, as they allow us to think of particles as taking all possible paths between two points, rather than just a single path.

Among other things, path integrals are particularly useful:

- **Lorentz invariance is manifest**: the path integral formulation is manifestly Lorentz invariant, since it is a lagrangian formulation of the theory, and thus it does not require a choice of time coordinate or a specific reference frame (we do not need the hamiltonian).
- **Quantization of gauge field theories** is more straightforward: the path integral formulation allows us to easily implement gauge fixing and Faddeev-Popov ghosts, which are necessary for quantizing gauge theories.
- **Numerical simulations for strongly interacting theories** (where perturbation theory fails): path integrals can be evaluated numerically using Monte Carlo methods, which is particularly useful for studying strongly interacting theories such as quantum chromodynamics (QCD) on the lattice (the idea is to discretize spacetime into a lattice and then evaluate the path integral numerically).
- The use of **functional integrals** allows us to easily derive the generating functional for correlation functions, which is a powerful tool for computing correlation functions and scattering amplitudes in quantum field theory. For examples *symmetries* in QFT are much easier to understand in the path integral formulation, since they can be implemented as transformations of the fields in the path integral, and thus we can easily derive Ward identities and other relations between correlation functions.

5.1 | Path Integral Formulation of Quantum Mechanics

The idea of the path integral formulation is to express the transition amplitude between two states as a sum over all possible paths that the system can take between those states. In quantum mechanics, the most used eexample to illustrate the path integral formulation is the **double slit experiment**, where we can think of the particle as taking all possible paths from the source to the detector, and the interference pattern arises from the constructive and destructive interference of these paths.

TODO: Add drawing of double slit experiment

[*drawing of source of particles, slits and detector*]

$$A(x \to y) = A(x \to a \to y) + A(x \to b \to y),$$

where a and b are the two slits. In the path integral formulation, we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y, but we can do more, adding virtually an infinite number of screens with slits in between x and y, and thus we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y through any number of slits. In the limit of infinitely many screens with infinitely many slits, we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y, without any constraints.

TODO: Add drawing of infinite walls and slits among x and y

[*draw of infinite walls and slits among x and y*]

If γ is a specific path from x to y, then we can associate to it a complex number $A(\gamma)$ called the amplitude of the path, and then we can express the transition amplitude as a sum over all possible paths:

$$A(x \to y) = \sum_{\gamma} A(x \to y, \gamma).$$

In order to implement this idea, we can discretize the time interval between x and y into N small intervals of size ϵ, and then we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y through these small intervals:

$$\langle f;\, t_f | i;\, t_i \rangle\,, \quad t_i < t_1 < \cdots < t_n < t_f, \quad t_{k+1} - t_k = \epsilon,$$

where we have inserted $N-1$ complete sets of states at intermediate times $t_1, \ldots, t_{N-1}$. Then we can express the transition amplitude as a sum over all possible paths that the particle can take from x to y through these small intervals by adding the resolution of the identity at each intermediate time:

$$\langle f;\, t_f | i;\, t_i \rangle = \int \mathrm{d}q_1 \cdots \mathrm{d}q_n \, \langle f;\, t_f | q_n;\, t_n \rangle \, \langle q_n;\, t_n | q_{N-1};\, t_{N-1} \rangle \cdots \langle q_1;\, t_1 | i;\, t_i \rangle\,.$$

The result to this expression is

$$\langle f;\, t_f | i;\, t_i \rangle = N \int_{q(t_i)=x_i}^{q(t_f)=x_f} \mathcal{D}q(t) e^{iS[q(t)]},$$

where $S[q(t)] = \int \mathrm{d}t\, L(q(t), \dot{q}(t))$ is the action of the path $q(t)$, and $\mathcal{D}q(t)$ is a measure that assigns a weight to each path. The normalization factor N is chosen such that the transition amplitude is properly normalized. In order to understand this expression more intuitively, we can always go back to its discretized version:

$$\langle f;\, t_f | i;\, t_i \rangle = N \int \mathrm{d}q_1 \cdots \mathrm{d}q_n \exp\left[i \sum_k L(q_k, \dot{q}_k) \cdot \epsilon \right], \quad \dot{q}_k = \frac{q_{k+1} - q_k}{\epsilon}.$$

We can also generalize the last expression using the time ordering operator $\hat{T}$:

$$\langle f;\, t_f|\, \hat{T}\{\hat{q}(t_a)\hat{q}(t_b)\cdots\}\,|i;\, t_i\rangle = N\int_{q(t_i)=x_i}^{q(t_f)=x_f} \mathcal{D}q(t)e^{iS[q(t)]}q(t_a)q(t_b)\cdots\,, \quad t_i < t_a < t_b < \cdots < t_f,$$

where the time ordering operator $\hat{T}$ ensures that the operators are ordered in time from right to left, with the earliest time on the right and the latest time on the left. This expression allows us to compute correlation functions of the form $\langle f;\, t_f|\, \hat{T}\{\hat{q}(t_a)\hat{q}(t_b)\cdots\}\,|i;\, t_i\rangle$ using path integrals.

There exists also a discretized version of the last expression, which is more intuitive and easier to understand:

$$\langle f;\, t_f|\, \hat{T}\{\hat{q}(t_a)\hat{q}(t_b)\cdots\}\,|i;\, t_i\rangle = N\int \mathrm{d}q_1\cdots\mathrm{d}q_n \exp\left[i\sum_k L(q_k,\dot{q}_k)\cdot\epsilon\right] q(t_a)q(t_b)\cdots\,,$$

where we have replaced the operators $\hat{q}(t_a)$ and $\hat{q}(t_b)$ with the corresponding functions $q(t_a)$ and $q(t_b)$ in the path integral as did before. This expression allows us to compute correlation functions of the form $\langle f;\, t_f|\, \hat{T}\{\hat{q}(t_a)\hat{q}(t_b)\cdots\}\,|i;\, t_i\rangle$ using path integrals in a more intuitive way. Time ordering is basically implemented for free, since the path integral already sums over all possible paths, and thus it automatically takes into account all possible time orderings of the operators. (?) time ordering.

5.1.1 | Classical Limit

In the classical limit, the path integral is dominated by the path that minimizes the action, which is the classical path. This is because in the classical limit, the action is much larger than $\hbar \to 0$, and thus the contributions from paths that deviate from the classical path are suppressed by a factor of $e^{iS/\hbar}$,[1] which oscillates rapidly and averages out to zero. Therefore, in the classical limit, we can approximate the path integral by evaluating it at the classical path, which gives us the classical equations of motion.

Thus the most important contribution to the path integral comes from the classical path, which is the path that minimizes the action. This is known as the principle of least action, and it is a fundamental principle in classical mechanics that states that the motion of a system is determined by the path that minimizes the action. From these procedure we can easily compute the Euler-Lagrange equations of motion, which are the equations that govern the dynamics of the system in classical mechanics. In the path integral formulation, we can derive the Euler-Lagrange equations by performing a variation of the action with respect to the path, and then setting the variation to zero.

[1] We omitted the $\hbar$ factor for simplicity, but via dimensional analysis it can be easily recovered.

5.2 | Path Integral Formulation of QFT

The path integral formalism can be generalized to QFT via straightforward substitutions:

$$\begin{cases} q \to \phi(x), \\ t,\, \mathrm{d}t \to x^\mu,\, \mathrm{d}^4x. \end{cases}$$

As we have seen, in QFT we are mostly interested in correlation functions (or Green functions), defined as expectation values of time-ordered products in the vacuum state $|\Omega\rangle$. We thus consider $|i\rangle = |f\rangle = |\Omega\rangle$ and write the QFT version of path integrals as:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle = N\int \mathcal{D}\phi(x)\exp\left[iS(\phi)\right]\phi(x_1)\phi(x_2)\cdots\phi(x_n),$$

where the action is $S = \int \mathrm{d}^4x\,\mathcal{L}$.

Remark. *Remember that everything inside a path integral is a (commuting) function, not an operator.*

The normalization N can be fixed using the vacuum-to-vacuum transition amplitude $\langle\Omega|\Omega\rangle = 1$:

$$N = \left(\int \mathcal{D}\phi(x)\exp\left[iS(\phi)\right]\right)^{-1}.$$

We can rewrite the discretized version of the path integral:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_a)\hat{\phi}(x_b)\cdots\}\,|\Omega\rangle = N\int \mathrm{d}\phi_1\cdots\mathrm{d}\phi_m\exp\left[i\sum_M \Delta^4x\mathcal{L}(\phi_M)\right]\phi_a\phi_b\cdots,$$

where, again, ϕ_a is equal to one of the $\phi_1,\ldots,\phi_m$, namely the one that corresponds to the spacetime point $x = x_a$. The integration measure $\int \mathrm{d}\phi_i$ integrates over all possible values of the field $\phi(x)$ at the specific discrete point $x = x_i$ (after discretizing spacetime).

TODO: add diagram showing the discretization of spacetime into a lattice of points

In the continuum limit, we take $m \to \infty$, which yields the formal replacements:

$$\mathrm{d}\phi_1\cdots\mathrm{d}\phi_m \to \mathcal{D}\phi(x), \qquad \sum \Delta^4x \to \int \mathrm{d}^4x.$$

But how do we evaluate path integrals in practice? We have two main approaches:

- **Option 1 (Lattice QFT)**: Discretize spacetime and do the integrals numerically. This approach has limitations and requires a lot of computing power, but it is the only systematic way of doing non-perturbative QFT.
- **Option 2 (Perturbation Theory)**: For Lagrangians with up to quadratic terms in it (i.e., free QFTs!), the path integral becomes a Gaussian integral which can be analytically computed. Interaction terms in $\mathcal{L}$ with cubic or higher powers can be Taylor expanded, and this also leads to combinations of Gaussian integrals. This is perturbation theory, i.e., the approach that we have been following until now (and will be followed throughout this course).

5.2.1 | Gaussian Integrals

We remind the following fundamental result for one-dimensional Gaussian integrals:

$$\int_{-\infty}^{\infty}\mathrm{d}z\exp\left[-\frac{1}{2}az^2\right] = \sqrt{\frac{2\pi}{a}}, \quad a > 0.$$

More importantly, its m-dimensional generalization (using a vector $\mathbf{z} = (z_1, \ldots, z_m)$) is given by:

$$\int_{-\infty}^{\infty} \mathrm{d}z_1 \cdots \mathrm{d}z_m \exp\left[-\frac{1}{2}\mathbf{z}^T \hat{A}\mathbf{z} + \mathbf{J}\cdot\mathbf{z}\right] = \frac{(2\pi)^{m/2}}{\sqrt{\det \hat{A}}} \exp\left[\frac{1}{2}\mathbf{J}^T \hat{A}^{-1}\mathbf{J}\right].$$

Cubic or higher-order terms in the Lagrangian can instead be treated as perturbations and Taylor expanded. For example, a quartic interaction can be expanded as:

$$\int_{-\infty}^{\infty} \mathrm{d}z \exp\left[-\frac{1}{2}az^2 - \frac{\lambda}{4!}z^4\right] = \int_{-\infty}^{\infty} \mathrm{d}z \exp\left[-\frac{1}{2}az^2\right]\left(1 - \frac{\lambda}{4!}z^4 + \cdots\right).$$

To evaluate the resulting integrals involving powers of z^n, we can introduce a fictitious source term J and use the clever trick of pulling down factors of z via derivatives with respect to J:

$$\int_{-\infty}^{\infty} \mathrm{d}z \exp\left[-\frac{1}{2}az^2\right] z^n = \frac{\partial^n}{\partial J^n} \int_{-\infty}^{\infty} \mathrm{d}z \exp\left[-\frac{1}{2}az^2 + Jz\right]\Big|_{J=0}.$$

5.2.2 | Computing Green's Functions

We want to apply the idea of evaluating Gaussian integrals using source terms to path integrals. However, since our integral is over a space of functions, we first need to formally define a **functional derivative**, denoted as $\frac{\delta}{\delta J(x)}$. We define it through its fundamental action on the source field itself:

$$\frac{\delta}{\delta J(x)} J(y) = \delta^{(4)}(x - y),$$

which immediately implies:

$$\frac{\delta}{\delta J(x)} \int \mathrm{d}^4 y\, J(y)\phi(y) = \phi(x).$$

These relations are the strict continuum limit of the discrete vector derivatives $\frac{\partial J_j}{\partial J_i} = \delta_{ij}$ and $\frac{\partial}{\partial J_i}\sum_j J_j\phi_j = \phi_i$.

The functional derivative $\frac{\delta}{\delta J(x)}$ obeys all the usual algebraic rules of ordinary derivatives (linearity, product rule, chain rule, etc.), so in practical calculations, you can treat it almost exactly like a normal derivative. On top of that, it possesses a very useful property when dealing with spacetime derivatives inside integrals. Having integrated by parts (and assuming the fields vanish at infinity to drop the boundary term), we have:

$$\frac{\delta}{\delta J(x)} \int \mathrm{d}^4 y\, (\partial_\mu J(y))\phi(y) = -\partial_\mu \phi(x).$$

Example (Exercise). Consider a generic Lagrangian density depending on fields and their first derivatives: $\mathcal{L} = \mathcal{L}(\phi_k, \partial_\mu \phi_k)$. Show that the classical Euler-Lagrange equations of motion can be compactly derived and written using the functional derivative of the action $S = \int \mathrm{d}^4 x \mathcal{L}$ as:

$$\frac{\delta S}{\delta \phi_k(x)} = 0$$

The Generating Functional $Z[J]$

We define the **Generating Functional** $Z[J]$ as:

$$Z[J] \equiv \int \mathcal{D}\phi \exp\left[i\int \mathrm{d}^4 x\, (\mathcal{L} + J(x)\phi(x))\right].$$

This is simply the standard path integral where we have artificially added a "classical" source term $J(x)$ coupled linearly to the quantum field $\phi(x)$ in the Lagrangian.

The immense power of the generating functional lies in the fact that any n-point correlation function (Green's function) can be elegantly expressed by taking functional derivatives of $Z[J]$ with respect to the source $J(x)$, and subsequently setting the source to zero:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\cdots\hat{\phi}(x_n)\}\,|\Omega\rangle = \frac{1}{Z_0}\left(-i\frac{\delta}{\delta J(x_1)}\right)\cdots\left(-i\frac{\delta}{\delta J(x_n)}\right)Z[J]\Big|_{J=0},$$

where $Z_0 \equiv Z[0]$ ensures the proper normalization of the vacuum state. Therefore, if we could compute the exact mathematical form of $Z[J]$, we would have effectively solved the quantum field theory, as we could obtain any correlation function simply by taking functional derivatives.

As anticipated in the previous section, $Z[J]$ can only be computed *exactly* if the Lagrangian $\mathcal{L}$ contains up to quadratic terms in the fields, i.e., for a **free theory**.

Evaluating $Z[J]$ for a Free Theory

Let us evaluate $Z[J]$ for a free real scalar field. We first rewrite the action inside the exponential. We take the standard Klein-Gordon kinetic term, $\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2$, and integrate the derivative term by parts:

$$i\int \mathrm{d}^4x\ (\mathcal{L} + J\phi) = i\int \mathrm{d}^4x\left[-\frac{1}{2}\phi(\Box + m^2 - i\epsilon)\phi + J\phi\right],$$

where we have explicitly added a small convergence factor $i\epsilon$ to the mass. Inside the highly oscillatory complex exponential of the path integral, this $-i\epsilon$ term provides a damping factor that yields a mathematically well-defined, convergent Gaussian integral.

The path integral now takes the exact form of an infinite-dimensional Gaussian integral:

$$Z[J] = \int \mathcal{D}\phi \exp\left[i\int \mathrm{d}^4x\left(-\frac{1}{2}\phi(\Box + m^2 - i\epsilon)\phi + J\phi\right)\right]$$

Recalling our m-dimensional Gaussian integral master formula $\int \mathrm{d}^m z \exp\left[-\frac{1}{2}\mathbf{z}^T\hat{A}\mathbf{z} + \mathbf{J}\cdot\mathbf{z}\right] \propto \exp\left[\frac{1}{2}\mathbf{J}^T\hat{A}^{-1}\mathbf{J}\right]$, we can perform the integration over $\mathcal{D}\phi$. The continuous operator acting as the matrix $\hat{A}$ in this functional space is:

$$\hat{A} = i(\Box + m^2 - i\epsilon)$$

To evaluate the result, we need to mathematically compute the inverse of this differential operator, $\hat{A}^{-1}$. The inverse of the operator $i(\Box + m^2 - i\epsilon)$ is defined as the function $D_F(x-y)$ that satisfies the Green's function differential equation:

$$i(\Box_x + m^2 - i\epsilon)D_F(x-y) = \delta^{(4)}(x-y)$$

(This is the strict continuum analogue of the discrete matrix inverse definition $\sum_j A_{ij}(A^{-1})_{jk} = \delta_{ik}$). This defines the **Feynman Propagator** entirely within the path integral formalism.

Calculating $D_F(x-y)$ from this differential equation is straightforward by transforming into momentum space via Fourier transform:

$$D_F(x-y) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}e^{-ip\cdot(x-y)}\Pi(p)$$

Substituting this into the differential equation yields:

$$i(-p^2 + m^2 - i\epsilon)\Pi(p) = 1 \implies \Pi(p) = \frac{i}{p^2 - m^2 + i\epsilon}$$

which is exactly the momentum-space propagator we already derived using the canonical operator formalism.

Remark. *Notice how the mathematically necessary $i\epsilon$ convergence factor we introduced to compute the Gaussian path integral organically ends up being exactly equivalent to the standard* ***Feynman prescription*** *for avoiding the poles in the momentum integral.*

Substituting the inverse operator D_F back into the result of the Gaussian integral, we obtain the final exact expression for the free theory generating functional:

$$Z[J] = Z_0 \exp\left[-\frac{1}{2}\int \mathrm{d}^4x \mathrm{d}^4y\, J(x) D_F(x-y) J(y)\right]$$

where we fixed the overall J-independent normalization constant prefactor to be Z_0 by simply evaluating the expression at $J = 0$.

Green's Functions of a Free Theory

We can now use the exact result for the free generating functional $Z[J]$ to compute Green's functions by taking functional derivatives.

Example (Two-point Green's function). The two-point Green's function is given by applying two functional derivatives to the generating functional:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\}\,|\Omega\rangle = \frac{(-i)^2}{Z_0}\,\frac{\delta}{\delta J(x_1)}\frac{\delta}{\delta J(x_2)} Z[J]\bigg|_{J=0}.$$

Let us evaluate the innermost functional derivative first. Recall that $Z[J] = Z_0 \exp\big[-\frac{1}{2}\int \mathrm{d}^4x \mathrm{d}^4y\, J(x) D_F(x-y) J(y)$
Differentiating the exponent brings down two symmetric terms:

$$\begin{aligned}\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\}\,|\Omega\rangle &= -\frac{\delta}{\delta J(x_1)}\left[-\frac{1}{2}\int \mathrm{d}^4y D_F(x_2-y)J(y) - \frac{1}{2}\int \mathrm{d}^4x J(x) D_F(x-x_2)\right]\frac{Z[J]}{Z_0}\bigg|_{J=0} \\ &= -\frac{\delta}{\delta J(x_1)}\left[-\int \mathrm{d}^4y D_F(x_2-y)J(y)\right]\frac{Z[J]}{Z_0}\bigg|_{J=0}\end{aligned}$$

Now we apply the second functional derivative $\frac{\delta}{\delta J(x_1)}$. By the product rule, this derivative acts both on the term in the brackets $[\cdots]$ and on the exponential $Z[J]$. However, its action on $Z[J]$ brings down another factor proportional to J, which strictly vanishes when we evaluate at $J = 0$. Therefore, we only keep the term where the derivative acts on the bracket:

$$\langle\Omega|\,\hat{T}\{\hat{\phi}(x_1)\hat{\phi}(x_2)\}\,|\Omega\rangle = \frac{1}{2}D_F(x_2-x_1) + \frac{1}{2}D_F(x_1-x_2) = D_F(x_1-x_2),$$

which is exactly the familiar Feynman propagator of the free theory.

Example (Four-point Green's function). To compute higher-order correlation functions without drowning in integral signs, we introduce a convenient shortcut notation:

$$\begin{cases} D_F(x,y) = D_{xy}, \\ J(x) = J_x, \\ \displaystyle\int \mathrm{d}^4x \int \mathrm{d}^4y J(x) D_F(x,y) J(y) = J_x D_{xy} J_y. \end{cases}$$

Here, repeated continuous indices imply integration, acting as a continuum version of Einstein summation. Since the Feynman propagator is symmetric ($D_{xy} = D_{yx}$) and x, y are dummy integration variables, the functional derivative takes the compact form:

$$\frac{\delta}{\delta J_z}(J_x D_{xy} J_y) = D_{zy} J_y + J_x D_{xz} = 2 J_x D_{xz}.$$

Using this shorthand, we can elegantly compute the four-point Green's function:

$$\begin{aligned}
\langle\Omega|\,\hat{T}\{\hat{\phi}_1\hat{\phi}_2\hat{\phi}_3\hat{\phi}_4\}\,|\Omega\rangle &= (-i)^4 \frac{\delta^4}{\delta J_1 \delta J_2 \delta J_3 \delta J_4} \exp\left[-\frac{1}{2} J_x D_{xy} J_y\right]\Big|_{J=0} \\
&= \frac{\delta}{\delta J_1}\frac{\delta}{\delta J_2}\frac{\delta}{\delta J_3}(-J_x D_{x4}) \exp[\cdots]\Big|_{J=0} \\
&= \frac{\delta}{\delta J_1}\frac{\delta}{\delta J_2}\left(-D_{34} + J_x D_{x4} J_y D_{y3}\right) \exp[\cdots]\Big|_{J=0} \\
&= \frac{\delta}{\delta J_1}\left(-D_{34}(-J_w D_{w2}) - D_{24} J_y D_{y3} - J_x D_{x4} D_{23} + \mathcal{O}(J^3)\right) \exp[\cdots]\Big|_{J=0} \\
&= D_{12} D_{34} + D_{13} D_{24} + D_{14} D_{23}.
\end{aligned}$$

The final result is precisely the sum of all possible **full contractions** of the four fields $\phi_1 \cdots \phi_4$.

Remark. *This mathematical pattern holds for a generic n-point function due to how differentiation works on exponentials. Each derivative $-i\frac{\delta}{\delta J_i}$ either brings down an extra factor $iJ_x D_{xi}$ from the exponent, or it eliminates a iJ_k factor that is already outside the exponential, yielding a propagator D_{ik}. Because we evaluate at $J = 0$, any terms still containing J vanish.*

*The ultimate outcome is that the n-point correlation function is equal to the sum over all possible full contractions of $\phi_1 \cdots \phi_n$, where a "contraction" replaces pairs of fields ϕ_i, ϕ_j with their propagator D_{ij}. This is exactly the mathematical derivation of **Wick's Theorem** within the path integral formalism!*

5.2.3 | Perturbation Theory and Feynman Diagrams

TODO: check

Until now, we have successfully treated free theories in the path integral formulation by recognizing the free Lagrangian as yielding a functional Gaussian integral, which we were able to evaluate exactly. Let us now check that interacting theories can also be systematically treated within this formulation, and verify that we recover the familiar perturbative expansion.

Consider an interacting theory whose Lagrangian density can be split into a free part and an interaction part: $\mathcal{L} = \mathcal{L}_{\text{free}} + \mathcal{L}_{\text{int}}$. We want to evaluate the n-point Green's function, which is given by the path integral normalized by the partition function of the free theory (Z_0):

$$\begin{aligned}
\langle\Omega|T\{\hat{\phi}(x_1)\cdots\hat{\phi}(x_n)\}|\Omega\rangle \times Z_0 &= \int \mathcal{D}\phi \, \exp\left[i\int \mathrm{d}^4x(\mathcal{L}_{\text{free}} + \mathcal{L}_{\text{int}})\right] \phi(x_1)\cdots\phi(x_n) \\
&= \int \mathcal{D}\phi \, \exp\left[i\int \mathrm{d}^4x \mathcal{L}_{\text{free}}\right] \exp\left[i\int \mathrm{d}^4z \mathcal{L}_{\text{int}}(z)\right] \phi(x_1)\cdots\phi(x_n).
\end{aligned}$$

Assuming the interaction is weak (controlled by a small coupling constant), we can Taylor-expand the exponential of the interaction term:

$$= \int \mathcal{D}\phi \, \exp\left[i\int \mathrm{d}^4x \mathcal{L}_{\text{free}}\right] \sum_{k=0}^{\infty} \frac{i^k}{k!} \int \mathrm{d}^4z_1 \cdots \mathrm{d}^4z_k \, \mathcal{L}_{\text{int}}(z_1)\cdots\mathcal{L}_{\text{int}}(z_k)\phi(x_1)\cdots\phi(x_n).$$

Notice that for each term in the sum, the path integral over the exponential of the free action acts as a Gaussian weight for the polynomial function composed of the product of fields $\mathcal{L}_{\text{int}}(z_1)\cdots\mathcal{L}_{\text{int}}(z_k)\phi(x_1)\cdots\phi(x_n)$. Using the functional equivalent of Wick's theorem (or by taking successive functional derivatives of the free generating functional), this integral evaluates exactly to the sum of all possible full contractions between the fields:

$$= \sum_{k=0}^{\infty} \frac{i^k}{k!} \int \mathrm{d}^4 z_1 \cdots \mathrm{d}^4 z_k \sum (\text{all full contractions in } \mathcal{L}_{\text{int}}(z_1)\cdots\mathcal{L}_{\text{int}}(z_k)\phi(x_1)\cdots\phi(x_n)) \,.$$

This expression is exactly what we found when we first computed Green's functions of interacting theories using the canonical operator formalism, where the same conclusion was derived by applying Wick's theorem to the Dyson series $\langle 0|T\{\phi(x_1)\cdots\phi(x_n)\exp\big[i\int \mathrm{d}^4x\mathcal{H}_{\text{int}}\big]\}|0\rangle$.

We already know that the sum over contractions can be efficiently organized and computed using Feynman diagrams and their associated Feynman rules. Moreover, it is straightforward to identify that the normalization factor Z_0 in the denominator exactly cancels the disconnected "vacuum bubble" diagrams, meaning that dividing by Z_0 amounts to considering only **connected diagrams**.

We have thus successfully reproduced the Feynman rules starting from the path integral formalism! This proves that the canonical operator formalism and the path integral formalism yield identical results for Green's functions, and therefore (via the LSZ reduction formula) also for scattering amplitudes.

Remark. *Strictly speaking, we have shown this equivalence explicitly only for interaction terms $\mathcal{L}_{int}$ that do not contain derivatives. However, as previously discussed regarding the treatment of derivatives in the path integral, this formalism guarantees that we can safely use $i\mathcal{L}_{int}$ instead of $-i\mathcal{H}_{int}$ to derive Feynman rules, even when derivative couplings are present, without worrying about generating spurious contact terms from the time-ordering operator.*

Feynman Rules via Functional Derivatives

The path integral formalism provides an extremely elegant and direct recipe for deriving Feynman rules for interaction vertices simply by taking functional derivatives of the action $\int \mathrm{d}^4x\, i\mathcal{L}_{\text{int}}$.

In momentum space, we first rewrite each field in the interaction Lagrangian using its Fourier transform, e.g., $\phi(x) = \int \mathrm{d}^4k\, \phi(k)e^{ik\cdot x}$. Then, the Feynman rule for a specific vertex with incoming and outgoing momenta p_i is obtained by applying the following recipe:

- For each **incoming particle** (or outgoing anti-particle) with momentum p, apply the functional derivative $\frac{\delta}{\delta\phi(-p)}$.
- For each **outgoing particle** (or incoming anti-particle) with momentum p, apply the functional derivative $\frac{\delta}{\delta\phi^*(+p)}$.

After computing all the functional derivatives, you must set all fields to zero ($\phi, \phi^* \to 0$) to isolate the purely numerical coefficient. This procedure will naturally yield the algebraic Feynman rule multiplied by the overall momentum conservation delta function $\delta^{(4)}(p_{\text{in}} - p_{\text{out}})$.

Example (Interacting Lagrangian in sQED). Let us apply this recipe to derive the 3-point vertex rule in scalar QED. The interaction term in the Lagrangian for a complex scalar field coupled to a photon contains a derivative coupling:

$$\mathcal{L}_{\text{int}} \supset -ieA^\mu[\phi^*(\partial_\mu\phi) - (\partial_\mu\phi^*)\phi].$$

Consider the vertex where a scalar particle enters with momentum p_1, a photon enters with momentum p_3 (and Lorentz index μ), and the scalar particle exits with momentum p_2.

First, we Fourier transform the fields inside the action integral $\int \mathrm{d}^4x\, i\mathcal{L}_{\text{int}}$ and convert the coordinate derivatives into momentum factors ($\partial_\mu \to ik_\mu$):

$$\int \mathrm{d}^4x \int \mathrm{d}^4k_1\mathrm{d}^4k_2\mathrm{d}^4k_3\, e^{ix\cdot(k_1+k_2+k_3)}\left(eA^\mu(k_3)\phi^*(k_2)\phi(k_1)(ik_{1\mu} - ik_{2\mu})\right).$$

To find the vertex rule, we apply the appropriate functional derivatives for the incoming scalar (momentum p_1), outgoing scalar (momentum p_2), and incoming photon (momentum p_3, index ν):

$$\frac{\delta}{\delta\phi(-p_1)}\frac{\delta}{\delta\phi^*(p_2)}\frac{\delta}{\delta A^\nu(-p_3)}\left[\dots\right].$$

We use the fundamental properties of functional derivatives in momentum space:

$$\frac{\delta\phi(k)}{\delta\phi(-p)} = \delta^{(4)}(-p-k), \quad \frac{\delta\phi^*(k)}{\delta\phi^*(p)} = \delta^{(4)}(p-k), \quad \frac{\delta A^\mu(k)}{\delta A^\nu(-p)} = \delta^\mu_\nu\delta^{(4)}(-p-k).$$

Applying these derivatives to our fields $\phi(k_1)$, $\phi^*(k_2)$, and $A^\mu(k_3)$ respectively, we get:

$$\int \mathrm{d}^4x \int \mathrm{d}^4k_1\mathrm{d}^4k_2\mathrm{d}^4k_3\, e^{ix\cdot(k_1+k_2+k_3)}\left(e\,\delta^\mu_\nu\delta^{(4)}(-p_3-k_3)\,\delta^{(4)}(p_2-k_2)\,\delta^{(4)}(-p_1-k_1)\,(ik_{1\mu} - ik_{2\mu})\right).$$

Integrating over the k_i variables simply forces $k_1 = -p_1$, $k_2 = p_2$, and $k_3 = -p_3$:

$$= \int \mathrm{d}^4x\, e^{ix\cdot(-p_1+p_2-p_3)}\, e\,\delta^\mu_\nu\left(i(-p_{1\mu}) - i(p_{2\mu})\right).$$

Finally, integrating over spacetime x produces the overall momentum conservation delta function $(2\pi)^4\delta^{(4)}(-p_1+p_2-p_3)$. The algebraic coefficient multiplying this delta function is exactly the Feynman rule for the vertex:

$$\text{Vertex Rule} = e(-ip_{1\nu} - ip_{2\nu}) = -ie(p_1+p_2)_\nu.$$

(Exercise: Try applying the same functional derivative procedure to the quadratic 4-point "seagull" interaction term $e^2A_\mu A^\mu\phi^*\phi$ to derive its Feynman rule $2ie^2g_{\mu\nu}$).

5.2.4 | Fermionic Path Integrals

TODO: check

Inside the standard path integral, everything commutes because we are treating quantum operators as ordinary classical functions. To extend this formalism to fermions, which physically anticommute, we must think of fermionic fields inside the path integral as fundamentally "anticommuting" objects. We achieve this formally by introducing **Grassmann variables** (or Grassmann numbers) θ_i. In practice, you only need to know that they satisfy the following properties:

- **Anticommutation**: $\theta_i\theta_j + \theta_j\theta_i = 0$ for all i, j. In particular, setting $i = j$ implies that the square of any Grassmann variable is exactly zero: $\theta_i^2 = 0$.
- **Integration and differentiation**: These operations are well-defined and, remarkably, they coincide. The integral of a Grassmann variable acts exactly like its derivative:

$$\int \mathrm{d}\theta_1 \cdots \mathrm{d}\theta_n X = \frac{\partial}{\partial\theta_n}\cdots\frac{\partial}{\partial\theta_1}X.$$

This definition is consistent with the anticommutation property. Since $\theta_i^2 = 0$, the Taylor expansion of any function of a single Grassmann variable terminates at the linear term (e.g., $f(\theta) = a + b\theta$). Consequently, the integral of any product of Grassmann variables is zero unless it contains exactly one factor of each variable. Furthermore, we naturally have

$$\frac{\partial \theta_i}{\partial \theta_j} = \delta_{ij}.$$

- **Gaussian integrals**: You can use the above rules to compute a Gaussian integral over multidimensional Grassmann variables:

$$\int \mathrm{d}\theta_1 \mathrm{d}\overline{\theta}_1 \cdots \mathrm{d}\theta_n \mathrm{d}\overline{\theta}_n \exp\left(-\sum_{i,j} \overline{\theta}_i A_{ij} \theta_j + \sum_j \left(\overline{\eta}_j \theta_j + \overline{\theta}_j \eta_j\right)\right) = \det A \exp\left(\sum_{i,j} \overline{\eta}_i (A^{-1})_{ij} \eta_j\right).$$

  Here A is a matrix, while η_j and $\overline{\eta}_j$ are additional Grassmann variables that serve as sources for the fermionic fields. This is the fundamental tool for evaluating fermionic path integrals. *Crucial difference:* Note that, unlike the bosonic case, the determinant $\det A$ appears in the **numerator** rather than the denominator.

We can thus write a path integral for the Dirac field and, more generally, its generating functional:

$$Z[\eta, \overline{\eta}] = \int \mathcal{D}\overline{\psi}\mathcal{D}\psi \exp\left[i \int \mathrm{d}^4x \left(\overline{\psi}(i\partial\!\!\!/ - m)\psi + \overline{\eta}\psi + \overline{\psi}\eta\right)\right],$$

where we may shift $m \to m - i\epsilon$ to ensure formal convergence, just as we did in the bosonic case.

It is important to note that ψ and $\overline{\psi}$ must be treated as completely independent variables, and thus each requires its own independent source field, $\overline{\eta}(x)$ and $\eta(x)$ respectively. Furthermore, because the Lagrangian scalar $\mathcal{L}$ must be a commuting object (so that the action S is a standard commuting number), the source fields η and $\overline{\eta}$ must themselves be **anticommuting** Grassmann variables to perfectly cancel the anticommuting nature of ψ and $\overline{\psi}$ in the source terms.

Via the same completing-the-square method used for scalar fields (which is left as a highly recommended exercise), we can compute Z for the free Dirac theory, obtaining:

$$Z[\eta, \overline{\eta}] = Z_0 \exp\left[-\int \mathrm{d}^4x \int \mathrm{d}^4y\, \overline{\eta}(x) S_F(x-y) \eta(y)\right],$$

where $S_F(x-y)$ is the Dirac propagator. It is defined as the inverse of the Dirac operator $i\partial\!\!\!/ - m + i\epsilon$:

$$S_F(x-y) = \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-i(x-y)\cdot p} \frac{i}{p\!\!\!/ - m + i\epsilon} = \int \frac{\mathrm{d}^4p}{(2\pi)^4} e^{-i(x-y)\cdot p} \frac{i(p\!\!\!/ + m)}{p^2 - m^2 + i\epsilon} = \langle 0 | \hat{T}\{\hat{\psi}(x)\overline{\hat{\psi}}(y)\} | 0 \rangle.$$

As a final note, remember that the functional derivatives with respect to the sources, $\delta/\delta\eta$ and $\delta/\delta\overline{\eta}$, are also strictly **anticommuting objects**. This is because differentiation with respect to a Grassmann variable is equivalent to integrating over it. Therefore, when computing correlation functions by taking multiple functional derivatives of $Z[\eta, \overline{\eta}]$, you must be extremely careful to track the minus signs generated every time a derivative operator passes through another derivative or a fermionic field.

5.2.5 | A Note on Derivatives of Fields

TODO: check

When calculating amplitudes using the path integral formalism, we established that the integral over fields yields the time-ordered vacuum expectation value (the Green's function):

$$\langle\Omega|T\{\hat{\phi}(x_1)\cdots\hat{\phi}(x_n)\}|\Omega\rangle = \int \mathcal{D}\phi\, e^{iS}\phi(x_1)\cdots\phi(x_n).$$

But what happens if we replace one of the fields in the path integral with an operator containing derivatives, such as $\partial_\mu\phi(x_1)$? This scenario is extremely common when dealing with interaction Hamiltonians containing derivative couplings.

Let us evaluate this explicitly by writing the derivative as a finite difference limit along the time direction, taking $x_1 = (t_1, \vec{x}_1)$:

$$\begin{aligned}\int \mathcal{D}\phi\, e^{iS}(\partial_{t_1}\phi(x_1))\phi(x_2)\cdots\phi(x_n) &= \lim_{\epsilon\to 0}\int \mathcal{D}\phi\, e^{iS}\frac{1}{\epsilon}\left[\phi(t_1+\epsilon,\vec{x}_1)-\phi(t_1,\vec{x}_1)\right]\phi(x_2)\cdots\phi(x_n)\\ &= \lim_{\epsilon\to 0}\frac{1}{\epsilon}\Big[\langle\Omega|T\{\hat{\phi}(t_1+\epsilon,\vec{x}_1)\hat{\phi}(x_2)\cdots\}|\Omega\rangle\\ &\qquad - \langle\Omega|T\{\hat{\phi}(t_1,\vec{x}_1)\hat{\phi}(x_2)\cdots\}|\Omega\rangle\Big]\\ &= \partial_{t_1}\langle\Omega|T\{\hat{\phi}(x_1)\hat{\phi}(x_2)\cdots\hat{\phi}(x_n)\}|\Omega\rangle.\end{aligned}$$

Crucially, notice that the derivative operator has factored **outside** the time-ordered product. In general, it is strictly true that bringing a time derivative inside a time-ordered product changes the result:

$$\partial_{t_1}\langle\Omega|T\{\hat{\phi}(x_1)\cdots\hat{\phi}(x_n)\}|\Omega\rangle \neq \langle\Omega|T\{\partial_{t_1}\hat{\phi}(x_1)\cdots\hat{\phi}(x_n)\}|\Omega\rangle.$$

To see why they are not equal, let us calculate a simple example in the canonical operator formalism: the double time derivative of a 2-point function. The time-ordering operator is mathematically implemented using Heaviside step functions Θ:

$$\langle 0|T\{\phi(x)\phi(y)\}|0\rangle = \Theta(x^0-y^0)\langle 0|\phi(x)\phi(y)|0\rangle + \Theta(y^0-x^0)\langle 0|\phi(y)\phi(x)|0\rangle.$$

Applying the first time derivative ∂_{y^0} yields terms from the derivative of the fields, plus terms from the derivative of the step functions (recall that $\partial_{y^0}\Theta(x^0-y^0) = -\delta(x^0-y^0)$):

$$\begin{aligned}\partial_{y^0}\langle 0|T\{\phi(x)\phi(y)\}|0\rangle &= \langle 0|T\{\phi(x)\partial_{y^0}\phi(y)\}|0\rangle - \delta(x^0-y^0)\langle 0|\phi(x)\phi(y)|0\rangle\\ &\qquad + \delta(x^0-y^0)\langle 0|\phi(y)\phi(x)|0\rangle\\ &= \langle 0|T\{\phi(x)\partial_{y^0}\phi(y)\}|0\rangle - \delta(x^0-y^0)\langle 0|[\phi(x),\phi(y)]|0\rangle.\end{aligned}$$

Since equal-time fields commute, the commutator $[\phi(\vec{x}),\phi(\vec{y})]$ vanishes, and the delta function term disappears. However, when we apply the second derivative ∂_{x^0}, we hit the step functions again:

$$\partial_{x^0}\partial_{y^0}\langle 0|T\{\phi(x)\phi(y)\}|0\rangle = \langle 0|T\{\partial_{x^0}\phi(x)\partial_{y^0}\phi(y)\}|0\rangle + \delta(x^0-y^0)\langle 0|[\phi(x),\partial_{y^0}\phi(y)]|0\rangle.$$

Now, the equal-time commutator is between the field and its conjugate momentum $\pi(y) = \partial_{y^0}\phi(y)$, which evaluates to $i\delta^{(3)}(\vec{x}-\vec{y})$. Therefore:

$$\partial_{x^0}\partial_{y^0}\langle 0|T\{\phi(x)\phi(y)\}|0\rangle = \langle 0|T\{\partial_{x^0}\phi(x)\partial_{y^0}\phi(y)\}|0\rangle + i\delta^{(4)}(x-y).$$

This extra $\delta^{(4)}(x-y)$ is a *contact term* that only exists when the two points perfectly coincide. It mathematically demonstrates why taking derivatives outside the T-product is not equivalent to pushing them inside.

Conclusion Derivative operations appearing inside the path integral are strictly equivalent to applying the derivative operator to the Green's function, i.e., **outside the time-ordered product**:

$$\int \mathcal{D}\phi\, e^{iS} \left(\frac{\partial}{\partial x_1^\mu} \phi(x_1) \right) \phi(x_2) \cdots = \frac{\partial}{\partial x_1^\mu} \langle \Omega | T\{ \hat{\phi}(x_1) \hat{\phi}(x_2) \cdots \} | \Omega \rangle.$$

Notice that this perfectly matches what we did when deriving Feynman rules for interaction terms with derivatives. We kept the derivatives outside the Wick contractions (which are T-products) so that they could act directly on the plane waves $e^{\pm ip\cdot x}$, yielding factors of $\pm ip_\mu$. This is the correct, unambiguous prescription when working with the path integral formalism.

Remark. *In the Hamiltonian formalism, deriving this properly would require adding an extra term that cancels exactly with the difference between $\mathcal{H}_{int}$ and $-\mathcal{L}_{int}$, leading to the same physical result.*

5.3 | Quantization of QED

TODO: check

Quantizing gauge theories in the path integral formulation is fundamentally more difficult than quantizing scalar theories because the gauge field $A_\mu(x)$ contains an enormous redundancy. There are infinitely many field configurations that correspond to the exact same physical state (namely, configurations that differ only by a gauge transformation). In the case of QED, the transformation is:

$$A_\mu(x) \to A_\mu(x) + \frac{1}{e}\partial_\mu \alpha(x),$$

for any arbitrary scalar function $\alpha(x)$. Because the action is invariant under this transformation, integrating over all possible configurations A_μ means we are redundantly integrating over an infinite volume of gauge-equivalent states, causing the path integral to badly diverge.

To see this mathematically, let us look at the free QED action, which can be rewritten (after integration by parts) as:

$$S = \int \mathrm{d}^4x \left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}\right) = \frac{1}{2}\int \mathrm{d}^4x\, A^\mu \left[\partial^2 g_{\mu\nu} - \partial_\mu\partial_\nu\right] A^\nu.$$

In order to compute the propagator (and thus the generating functional), we must invert the differential operator appearing in the quadratic part of the action:

$$\hat{O} = -i(\partial^2 g_{\mu\nu} - \partial_\mu\partial_\nu).$$

The propagator $D_F^{\nu\rho}(x-y)$ would be the Green's function satisfying:

$$\hat{O}D_F^{\nu\rho}(x-y) = -i(\partial^2 g_{\mu\nu} - \partial_\mu\partial_\nu)D_F^{\nu\rho}(x-y) = \delta_\mu{}^\rho \delta^{(4)}(x-y).$$

However, this equation has no solution. The operator $\hat{O}$ is not invertible because it has a non-trivial null space. Specifically, any pure gauge configuration of the form $A_\mu = \partial_\mu\alpha$ is annihilated by the operator:

$$(\partial^2 g_{\mu\nu} - \partial_\mu\partial_\nu)\partial^\nu\alpha = \partial^2\partial_\mu\alpha - \partial_\mu(\partial^2\alpha) = 0.$$

Equivalently, if $A_\mu = \partial_\mu\alpha$, the action is exactly $S = 0$, making the path integral wildly divergent.

The exact same issue appears in momentum space. The equation for the momentum-space propagator $\Pi^{\nu\rho}(p)$ is:

$$-i(-p^2 g_{\mu\nu} + p_\mu p_\nu)\Pi^{\nu\rho}(p) = \delta_\mu{}^\rho.$$

The matrix $(-p^2 g_{\mu\nu} + p_\mu p_\nu)$ annihilates the vector p^ν, hence it has a determinant of zero and cannot be inverted to find $\Pi^{\nu\rho}$. To compute a well-defined path integral, we must isolate and remove the integration over these troublesome pure-gauge modes ($A_\mu = \partial_\mu\alpha$) and integrate only over a subspace of inequivalent, physically distinct field configurations.

Example (Massive spin-1 case). As an exercise, consider a massive vector field (Proca field). Its Lagrangian includes a mass term $\frac{1}{2}m^2 A_\mu A^\mu$. In momentum space, the operator becomes $(-p^2 g_{\mu\nu} + p_\mu p_\nu + m^2 g_{\mu\nu})$. Because the mass term breaks gauge invariance, this new operator no longer annihilates p^ν. It is invertible, and one can solve for the massive spin-1 propagator: $\Pi^{\mu\nu} = \frac{i}{p^2 - m^2 + i\epsilon}(-g^{\mu\nu} + \frac{p^\mu p^\nu}{m^2})$. This confirms that the non-invertibility in QED is strictly a consequence of exact gauge symmetry.

5.3.1 | The Faddeev-Popov Method

To fix the gauge without destroying the structural integrity of the path integral, we employ a brilliant trick developed by Faddeev and Popov. The core idea is to insert a carefully crafted "identity" into the path integral that acts as a delta-function, restricting the integration strictly to one representative field configuration per gauge orbit.

First, we choose a gauge fixing condition $G(A) = 0$. For any arbitrary field A, we denote its gauge-transformed version as $A_\alpha = A + \frac{1}{e}\partial\alpha$. We want the condition $G(A_\alpha) = 0$ to uniquely define α for any starting field A. We start with the following definition of the identity

$$1 = \int \mathrm{d}z_1 \cdots \mathrm{d}z_N \, \delta(g_1(\mathbf{z})) \cdots \delta(g_N(\mathbf{z})) \det\left(\frac{\partial g_i}{\partial z_j}\right),$$

which can be generalized to a continuum version, namely:

$$1 = \int \mathcal{D}\alpha(x) \, \delta(G(A_\alpha)) \det\left(\frac{\delta G(A_\alpha)}{\delta \alpha}\right).$$

Here, the delta functional enforces the gauge condition at every point in spacetime, and the Faddeev-Popov determinant is the Jacobian of the transformation.

We assume that for our chosen gauge, the determinant is independent of the specific field configuration A. This is true for linear gauges like the Lorenz gauge, where $G(A_\alpha) = \partial_\mu A^\mu + \frac{1}{e}\Box\alpha$, so the functional derivative $\frac{\delta G(A_\alpha)}{\delta\alpha} = \frac{1}{e}\Box$ is just a differential operator independent of A.

Inserting this identity into the QED path integral:

$$\int \mathcal{D}A \, e^{iS[A]} = \int \mathcal{D}\alpha \det\left(\frac{\delta G}{\delta \alpha}\right) \int \mathcal{D}A \, e^{iS[A]} \delta(G(A_\alpha)).$$

Since the determinant does not depend on A, we can pull it outside. Now, we perform a shift in the integration variable: $A_\mu \to A_\mu - \frac{1}{e}\partial_\mu\alpha$. Because the action is gauge-invariant ($S[A] = S[A_\alpha]$) and the measure $\mathcal{D}A$ is invariant under shifts, the integral over A completely decouples from α:

$$\int \mathcal{D}A \, e^{iS[A]} = \left(\int \mathcal{D}\alpha \det\left(\frac{\delta G}{\delta \alpha}\right)\right) \int \mathcal{D}A \, e^{iS[A]} \delta(G(A)).$$

The term in the large parentheses is simply an infinite overall constant (representing the infinite volume of the gauge group), which will cancel out when we normalize by the vacuum partition function.

We now specify a family of gauge conditions: $G(A) = \partial_\mu A^\mu - \omega(x) = 0$, where $\omega(x)$ is an arbitrary scalar function. The determinant remains $\det(\Box/e)$. The path integral evaluates to:

$$\int \mathcal{D}A \, e^{iS} = (\det(\Box/e)) \left(\int \mathcal{D}\alpha\right) \int \mathcal{D}A \, e^{iS[A]} \delta(\partial_\mu A^\mu - \omega(x)).$$

Because this result is mathematically true for *any* choice of $\omega(x)$, it must also be true if we average the right-hand side over all possible functions $\omega(x)$ using a Gaussian weight distribution centered at $\omega = 0$. We multiply by $1 = N(\xi) \int \mathcal{D}\omega \exp\left[-i \int \mathrm{d}^4x \frac{\omega^2}{2\xi}\right]$ (where ξ is an arbitrary constant parameter) and integrate over ω:

$$\int \mathcal{D}A \, e^{iS} = \tilde{N}_0 \int \mathcal{D}A \int \mathcal{D}\omega \, e^{iS[A]} e^{-i\int \mathrm{d}^4x \frac{\omega^2}{2\xi}} \delta(\partial_\mu A^\mu - \omega(x)).$$

The delta functional trivially allows us to perform the integral over ω, simply replacing every instance of ω with $\partial_\mu A^\mu$. Absorbing all constant prefactors into N_0, we obtain:

$$\int \mathcal{D}A \, e^{iS[A]} = N_0 \int \mathcal{D}A \, \exp\left[i \int \mathrm{d}^4x \left(\mathcal{L} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2\right)\right].$$

By using the Faddeev-Popov method, we have effectively added a new "gauge-fixing" term to the Lagrangian:

$$\mathcal{L}_{\text{G.F.}} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2.$$

Crucially, if we want to compute the expectation value of an operator $\hat{O}(A)$ that is strictly **gauge-invariant**, all the steps above remain valid when replacing $e^{iS} \to e^{iS}O(A)$. The infinite constants N_0 perfectly cancel between the numerator and denominator:

$$\langle\Omega|T\{\hat{O}(A)\}|\Omega\rangle = \frac{\int \mathcal{D}A\, e^{iS[A]}O(A)}{\int \mathcal{D}A\, e^{iS[A]}} = \frac{\int \mathcal{D}A\, e^{i\int \mathrm{d}^4x(\mathcal{L}+\mathcal{L}_{\text{G.F.}})}O(A)}{\int \mathcal{D}A\, e^{i\int \mathrm{d}^4x(\mathcal{L}+\mathcal{L}_{\text{G.F.}})}}.$$

This procedure smoothly generalizes to full QED (and sQED) where fermions or scalars couple to the photon field, provided the operators we measure are fully gauge-invariant.

5.3.2 | The Photon Propagator

With the new gauge-fixing term $\mathcal{L}_{\text{G.F.}}$ added to the action, we can finally invert the quadratic operator and compute the photon propagator.

In momentum space, the addition of $-\frac{1}{2\xi}(\partial_\mu A^\mu)^2$ modifies the operator equation to:

$$-i\left[-p^2 g_{\mu\nu} + \left(1 - \frac{1}{\xi}\right) p_\mu p_\nu\right] \Pi^{\nu\rho}(p) = \delta_\mu{}^\rho.$$

To solve this, we note that the propagator must be a rank-2 Lorentz covariant tensor depending only on the momentum p. The most general tensor form it can take is:

$$\Pi^{\nu\rho}(p) = A(p^2)g^{\nu\rho} + B(p^2)p^\nu p^\rho.$$

Inserting this ansatz into the modified operator equation, we can uniquely solve for the coefficients A and B. Adding the standard $+i\epsilon$ prescription for the poles, we arrive at the full photon propagator:

$$\Pi^{\nu\rho}(p) = \frac{i}{p^2 + i\epsilon}\left(-g^{\nu\rho} + (1-\xi)\frac{p^\nu p^\rho}{p^2}\right).$$

Notice that the mathematical form of the propagator depends explicitly on the arbitrary gauge-fixing parameter ξ. This is perfectly acceptable because the propagator itself is neither gauge-invariant nor a directly observable physical quantity. When calculating the S-matrix elements for actual physical processes, the Ward Identity guarantees that all terms proportional to $p^\nu p^\rho$ will identically vanish, rendering the final physical cross-sections completely independent of ξ.

In practice, physicists choose specific values for ξ to maximally simplify calculations:

- $\xi = 1$ is the **Feynman gauge**. This choice brilliantly cancels the second term, reducing the propagator to simply $\frac{-ig^{\nu\rho}}{p^2+i\epsilon}$. This is the most commonly used gauge in perturbative calculations and is the one we have adopted until now.
- $\xi \to 0$ is the **Lorenz gauge** (often called the **Landau gauge**). In this limit, the propagator becomes purely transverse ($p_\nu \Pi^{\nu\rho} = 0$). This is a highly popular choice for non-perturbative calculations and lattice QCD.

5.4 | Symmetries in Quantum Field Theory

The path integral formulation provides an exceptionally powerful and elegant framework for investigating symmetries in Quantum Field Theory. From the perspective of the path integral, any transformation of the fields $\phi(x)$ fundamentally corresponds to a mathematical change of variables within the integration measure $\int \mathcal{D}\phi$. By demanding that physical observables (and the path integral itself) remain invariant under specific changes of variables, we can systematically derive non-trivial identities for Green's functions and generating functionals.

Notation: In the following sections, we will frequently use the shorthand notation for correlation functions:

$$\langle \phi(x_1) \cdots \phi(x_n) \rangle \equiv \langle \Omega | T\{\hat{\phi}(x_1) \cdots \hat{\phi}(x_n)\} | \Omega \rangle,$$

where it is strictly understood that any classical spacetime derivatives appearing on the left-hand side act *outside* the time-ordered Green's function. As we previously established for path integrals, placing a derivative inside the expectation value is mathematically equivalent to pulling it outside the time-ordered product:

$$\langle (\partial_\mu \phi(x_1)) \phi(x_2) \cdots \phi(x_n) \rangle \equiv \partial_\mu^{(x_1)} \langle \Omega | T\{\hat{\phi}(x_1) \cdots \hat{\phi}(x_n)\} | \Omega \rangle.$$

5.4.1 | Equations of Motion

Let us investigate what happens when we perform the simplest possible change of variables: an infinitesimal local translation of the field itself.

The classical equations of motion (Euler-Lagrange equations) for a single scalar field are given by:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0.$$

This can be written much more compactly using functional derivatives as $\frac{\delta S}{\delta \phi(x)} = 0$. Classically, this is derived by minimizing the action, imposing $\delta S = 0$ at first order for a small, arbitrary field variation $\phi(x) \to \phi(x) + \epsilon(x)$.

We can apply a directly analogous argument to the quantum generating functional:

$$Z[J] = \int \mathcal{D}\phi \, \exp\left[iS[\phi] + i \int \mathrm{d}^4x \, J(x)\phi(x) \right].$$

Perform a change of variables in the path integral corresponding to a local shift: $\phi(x) \to \phi(x) + \epsilon(x)$. Because we are integrating over all possible field configurations from $-\infty$ to $+\infty$, a simple shift is merely a translation of the integration variable. Therefore, the path integral measure is perfectly invariant ($\mathcal{D}\phi \to \mathcal{D}\phi$, just as $\mathrm{d}x_i \to \mathrm{d}x_i$ for standard integrals), and the total value of $Z[J]$ must remain unchanged.

Assuming $\epsilon(x)$ is infinitesimal, we can Taylor-expand the exponential to first order:

$$\begin{aligned} Z[J] &= \int \mathcal{D}\phi \, \exp\left[iS[\phi + \epsilon] + i \int \mathrm{d}^4x \, J(x)(\phi(x) + \epsilon(x)) \right] \\ &= \int \mathcal{D}\phi \, \exp\left[iS[\phi] + i \int \mathrm{d}^4x \, J(x)\phi(x) \right] \times \left(1 + i \int \mathrm{d}^4x \, \epsilon(x) \left[\frac{\delta S}{\delta \phi(x)} + J(x) \right] + \mathcal{O}(\epsilon^2) \right). \end{aligned}$$

Since the unshifted integral is identically equal to $Z[J]$, the $\mathcal{O}(\epsilon)$ variation term must strictly vanish. Because $\epsilon(x)$ is a completely arbitrary function, the integral must vanish locally for the

integrand:

$$\int \mathcal{D}\phi \, \exp\left[iS[\phi] + i\int \mathrm{d}^4y \, J(y)\phi(y)\right]\left(\frac{\delta S}{\delta\phi(x)} + J(x)\right) = 0.$$

To extract relations for the physical Green's functions, we apply functional derivatives with respect to the sources. Apply the operator $\left(-i\frac{\delta}{\delta J(x_1)}\right)\cdots\left(-i\frac{\delta}{\delta J(x_n)}\right)$ to the equation above, and then set the source $J = 0$.

When the derivatives hit the exponential, they bring down field insertions $\phi(x_i)$. However, by the product rule, the derivatives can also hit the isolated $J(x)$ term. Specifically, differentiating $J(x)$ with respect to $J(x_j)$ yields a Dirac delta function $\delta^{(4)}(x - x_j)$. When we subsequently set $J = 0$, any term where $J(x)$ was not differentiated will vanish.

This procedure naturally generates the **Schwinger-Dyson Equations**:

$$\left\langle \frac{\delta S}{\delta\phi(x)}\phi(x_1)\cdots\phi(x_n)\right\rangle = i\sum_{j=1}^{n}\delta^{(4)}(x - x_j)\langle\phi(x_1)\cdots\phi(x_{j-1})\phi(x_{j+1})\cdots\phi(x_n)\rangle.$$

Observations

The Schwinger-Dyson equations reveal profound connections between classical dynamics and quantum correlation functions:

- **Quantum Equations of Motion:** The classical equations of motion ($\delta S/\delta\phi = 0$) are strictly valid for quantum Green's functions *up to contact terms*. These contact terms are the Dirac delta functions on the right-hand side, which are non-vanishing only when the field coordinate x exactly coincides with one of the operator insertion points x_j.
- **The Classical Limit:** We can verify this is a purely quantum phenomenon via dimensional analysis. The action $[S]$ has the dimensions of $\hbar$. If we explicitly restore $\hbar$ in the path integral weight ($e^{iS/\hbar}$), the right-hand side of the equation gains a factor of $\hbar$: $i\hbar\delta^{(4)}(x - x_j)$. Taking the classical limit $\hbar \to 0$ forces the contact terms to vanish, perfectly recovering the classical Euler-Lagrange equations.
- **Operator Formalism Consistency:** In the canonical operator formalism, we expect quantum operators in the Heisenberg picture to strictly obey the classical equations of motion: $\frac{\delta S}{\delta\hat{\phi}(x)} = 0$. This seems to contradict the Schwinger-Dyson equations where the expectation value is non-zero. However, remember our path integral notation: the derivative operator $\frac{\delta S}{\delta\phi(x)}$ generates derivatives that are mathematically pulled *outside* the time-ordering operator T. The contact terms are exactly the commutators that arise when time derivatives are pushed through the Heaviside step functions hidden inside the T-product (as we proved in the previous note on field derivatives).
- **Defining the Theory:** These equations are so fundamental that they can be used as the foundational definition of the quantum theory itself. For example, one can systematically derive all Feynman rules directly from the Schwinger-Dyson equations without ever referencing canonical quantization (see e.g., Schwartz 7.1).

Example (Real scalar field). Consider an interacting real scalar field described by the Lagrangian density:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2 + \mathcal{L}_{\text{int}}(\phi).$$

The classical variation of the action with respect to ϕ yields:

$$\frac{\delta S}{\delta\phi(x)} = -(\Box_x + m^2)\phi(x) + \frac{\partial\mathcal{L}_{\text{int}}}{\partial\phi(x)}.$$

Let us apply the Schwinger-Dyson equation for a 2-point function ($n = 1$, with an insertion at y). Substituting the variation into the general formula, we obtain:

$$(\Box_x + m^2)\langle\phi(x)\phi(y)\rangle = \left\langle \frac{\partial\mathcal{L}_{\text{int}}(x)}{\partial\phi(x)}\phi(y)\right\rangle - i\delta^{(4)}(x-y).$$

For a **free theory** ($\mathcal{L}_{\text{int}} = 0$), the first term on the right vanishes, and we perfectly recover the well-known differential equation that defines the Feynman propagator:

$$(\Box_x + m^2)\langle\phi(x)\phi(y)\rangle = -i\delta^{(4)}(x-y).$$

For an **interacting theory**, the equation yields a highly non-trivial relationship linking the exact 2-point Green's function to a higher-order Green's function (contained within the $\mathcal{L}'_{\text{int}}$ term). This recursion relation is exact and valid at ALL orders in perturbation theory.

Finally, this entire formalism generalizes naturally to theories with multiple field types. If $\mathcal{L} = \mathcal{L}(\phi_k, \partial_\mu\phi_k)$, where ϕ_k can be any generic field (scalar, spinor, vector), the generalized Schwinger-Dyson equation is:

$$\left\langle \frac{\delta S}{\delta\phi_k(x)}\phi_{k_1}(x_1)\cdots\phi_{k_N}(x_N)\right\rangle = i\sum_{j=1}^{N}\delta_{kk_j}\delta^{(4)}(x-x_j)\langle\phi_{k_1}(x_1)\cdots\phi_{k_{j-1}}(x_{j-1})\phi_{k_{j+1}}(x_{j+1})\cdots\rangle,$$

where the Kronecker delta δ_{kk_j} ensures that contact terms are generated only when the varying field k is identical to the inserted field k_j.

5.4.2 | Internal Symmetries

An **internal symmetry** is a *global redefinition of the fields*:

$$\phi_j(x) \to \phi_j(x) + \epsilon\Delta\phi_j(x) + \mathcal{O}(\epsilon^2),$$

where $\Delta\phi_j$ is a function of the fields only, with no explicit dependence on the spacetime coordinate x, and the transformation parameter ϵ is also independent of x. We assume that the classical action S is perfectly invariant under such a transformation.

To investigate the quantum implications, we consider the generating functional containing all the fields $\vec{\phi}$ present in the Lagrangian:

$$Z[\vec{J}] = \int\mathcal{D}\vec{\phi}\,\exp\left[iS[\vec{\phi}] + i\int\mathrm{d}^4x\sum_j\phi_j(x)J_j(x)\right].$$

When performing this change of variables inside the path integral, we must carefully consider the integration measure $\mathcal{D}\vec{\phi}$. There are two fundamental cases:

- **The measure is NOT invariant ($\mathcal{D}\vec{\phi}' \neq \mathcal{D}\vec{\phi}$):** The symmetry is said to be **anomalous**. Anomalies are symmetries of the classical theory that are strictly broken at the quantum level due to the non-invariance of the path integral measure. By computing the determinant of the transformation (the Jacobian), one can derive identities valid at all orders in perturbation theory that explicitly control how the symmetry is broken. We will not focus on this phenomenon in the current lectures, though it plays a critical role in theories like QCD.

- **The measure IS invariant ($\mathcal{D}\vec{\phi}' = \mathcal{D}\vec{\phi}$):** The symmetry holds exactly at the quantum level. It mathematically translates into exact identities relating different Green's functions or scattering amplitudes. These relations are broadly known as **Ward-Takahashi Identities**.

From now on, unless explicitly stated otherwise, we will solely consider non-anomalous symmetries. Since the measure and the action are invariant, the change of variables leaves the integral unchanged, but the source term in the exponential shifts:

$$\begin{aligned} Z[\vec{J}] &= \int \mathcal{D}\vec{\phi} \exp\left[iS + i\int \mathrm{d}^4x \sum_j (\phi_j + \epsilon\Delta\phi_j)J_j\right] \\ &= \int \mathcal{D}\vec{\phi} \exp\left[iS + i\int \mathrm{d}^4x \sum_j \phi_j J_j\right] \left(1 + i\epsilon \sum_j \int \mathrm{d}^4x\, \Delta\phi_j(x)J_j(x) + \mathcal{O}(\epsilon^2)\right). \end{aligned}$$

Because the integral must equal the original $Z[\vec{J}]$, the $\mathcal{O}(\epsilon)$ term must identically vanish:

$$\int \mathcal{D}\vec{\phi} \exp\left[iS + i\int \mathrm{d}^4x \sum_j \phi_j J_j\right] \sum_j i\int \mathrm{d}^4x\, \Delta\phi_j(x)J_j(x) = 0.$$

To extract physical relations for Green's functions, we apply functional derivatives $\left(-i\frac{\delta}{\delta J_{k_1}(x_1)}\right)\cdots$ $\cdots\left(-i\frac{\delta}{\delta J_{k_N}(x_N)}\right)$ to this equation and subsequently set all sources $J = 0$. The derivative operator hits the source term $J_j(x)$, picking up a delta function $\delta^{(4)}(x - x_i)$ for each insertion point, leading to the **global Ward-Takahashi identity**:

$$\sum_{i=1}^{N} \langle \phi_{k_1}(x_1)\cdots\phi_{k_{i-1}}(x_{i-1})\Delta\phi_{k_i}(x_i)\phi_{k_{i+1}}(x_{i+1})\cdots\phi_{k_N}(x_N)\rangle = 0.$$

Example (Global $U(1)$ Symmetry for a Complex Scalar). Consider a complex scalar field whose Lagrangian is invariant under a global $U(1)$ phase rotation:

$$\begin{cases} \phi \to e^{i\epsilon}\phi \simeq \phi + i\epsilon\phi \\ \phi^* \to e^{-i\epsilon}\phi^* \simeq \phi^* - i\epsilon\phi^* \end{cases} \implies \begin{cases} \Delta\phi = i\phi \\ \Delta\phi^* = -i\phi^* \end{cases}$$

Let us apply the global Ward-Takahashi identity to a general Green's function containing N fields ϕ and M conjugated fields ϕ^*: $\langle\phi(x_1)\cdots\phi(x_N)\phi^*(y_1)\cdots\phi^*(y_M)\rangle$. Inserting $\Delta\phi$ replaces each $\phi(x_i)$ with $i\phi(x_i)$, giving N factors of i. Inserting $\Delta\phi^*$ replaces each $\phi^*(y_j)$ with $-i\phi^*(y_j)$, giving M factors of $-i$. The identity yields:

$$i(N - M)\langle\phi(x_1)\cdots\phi(x_N)\phi^*(y_1)\cdots\phi^*(y_M)\rangle = 0.$$

This profoundly implies that the Green's function is strictly zero unless $N = M$. Therefore, non-vanishing correlation functions must contain an exactly equal number of ϕ and ϕ^* fields. This is the mathematical manifestation of the **conservation of charge** at the quantum level.

QFT Noether's Theorem

A much stronger result can be derived, representing the complete quantum field theory formulation of Noether's Theorem.

Classically, Noether's Theorem states that for every continuous global symmetry of the action, there exists a conserved current J^μ satisfying $\partial_\mu J^\mu = 0$. More precisely, if the action is invariant

under the global shift $\phi_j \to \phi_j + \epsilon\Delta\phi_j$, the corresponding change in the Lagrangian density must be a total derivative:

$$\Delta\mathcal{L} = \sum_j \left(\frac{\partial\mathcal{L}}{\partial\phi_j}\Delta\phi_j + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_j)}\partial_\mu(\Delta\phi_j) \right) \epsilon \equiv \epsilon\partial_\mu K^\mu,$$

for some vector function K^μ. The classically conserved Noether current is then elegantly defined as:

$$J^\mu = \sum_j \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_j)}\Delta\phi_j - K^\mu.$$

To elevate this to a quantum theorem, we employ a powerful formal trick: we promote the constant parameter ϵ to be an arbitrary function of spacetime, $\epsilon = \epsilon(x)$. Under this local transformation, the action is *no longer* invariant! Recomputing the variation of the Lagrangian, the derivative operator now acts on the local $\epsilon(x)$ via the product rule, generating an extra term:

$$\Delta\mathcal{L} = \epsilon(x)\partial_\mu K^\mu + \sum_j \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_j)}\Delta\phi_j\partial_\mu\epsilon(x).$$

Consequently, the functional derivative of the shifted action $S = \int \mathrm{d}^4x'\, \mathcal{L}(\phi+\epsilon(x)\Delta\phi)$ with respect to the local parameter $\epsilon(x)$ directly exposes the divergence of the Noether current:

$$\frac{\delta S}{\delta\epsilon(x)} = \partial_\mu K^\mu - \partial_\mu \left(\sum_j \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_j)}\Delta\phi_j \right) = -\partial_\mu J^\mu(x).$$

We now insert this specific change of variables $\phi_j \to \phi_j + \epsilon(x)\Delta\phi_j$ into the path integral generating functional $Z[\vec{J}]$ and repeat the standard expansion steps. Because the measure is assumed invariant, the variation of the integral must be zero:

$$\int \mathcal{D}\vec{\phi}\, \exp\left[iS + i\int \mathrm{d}^4x \sum_j J_j\phi_j \right] \left(-\partial_\mu J^\mu(x) + \sum_j J_j(x)\Delta\phi_j(x) \right) = 0.$$

Notice carefully that $J^\mu(x)$ here represents the composite operator defining the Noether current, while $J_j(x)$ are the classical external sources used to generate the correlation functions.

By taking functional derivatives with respect to the sources $J_{k_i}(x_i)$ and evaluating at $\vec{J} = 0$ as before, we obtain the fundamental QFT Ward-Takahashi identity in its local form:

$$\begin{aligned}\partial_\mu\langle J^\mu(x)\phi_{k_1}(x_1)\cdots\phi_{k_N}(x_N)\rangle = -i\sum_{i=1}^{N}\delta^{(4)}(x-x_i)\langle\phi_{k_1}(x_1)\cdots \\ \cdots\phi_{k_{i-1}}(x_{i-1})\Delta\phi_{k_i}(x_i)\phi_{k_{i+1}}(x_{i+1})\cdots\phi_{k_N}(x_N)\rangle.\end{aligned}$$

This profound equation tells us that the classical conservation law $\partial_\mu J^\mu = 0$ holds strictly true for Green's functions, *up to contact terms*. These contact terms (represented by the Dirac delta functions on the right-hand side) only contribute when the spacetime point x of the current operator $J^\mu(x)$ exactly coincides with the position x_i of one of the other fundamental fields inside the correlation function.

5.4.3 | Ward-Takahashi Identities in QED

Let us apply the powerful result of the quantum Noether's theorem to Quantum Electrodynamics (QED). Consider the global $U(1)$ symmetry corresponding to phase rotations of the Dirac fermion

fields:

$$\begin{cases} \psi \to e^{i\epsilon}\psi \simeq \psi + i\epsilon\psi \\ \bar{\psi} \to e^{-i\epsilon}\bar{\psi} \simeq \bar{\psi} - i\epsilon\bar{\psi} \end{cases} \implies \begin{cases} \Delta\psi = i\psi \\ \Delta\bar{\psi} = -i\bar{\psi} \end{cases}.$$

Because we are considering a strictly *global* symmetry, the gauge field A_μ does not transform ($\Delta A_\mu = 0$). The Noether current associated with this global symmetry is the standard Dirac vector current:

$$J^\mu = -\bar{\psi}\gamma^\mu\psi.$$

Consider a general Green's function containing one incoming fermion field $\psi(x_2)$ and one outgoing anti-fermion field $\bar{\psi}(x_1)$. Applying the generalized Ward-Takahashi identity derived in the previous section:

$$\partial_\mu\langle J^\mu(x)\bar{\psi}(x_1)\psi(x_2)\rangle = -i\left[\delta^{(4)}(x-x_1)\langle\Delta\bar{\psi}(x_1)\psi(x_2)\rangle + \delta^{(4)}(x-x_2)\langle\bar{\psi}(x_1)\Delta\psi(x_2)\rangle\right]$$
$$i\partial_\mu\langle\bar{\psi}(x)\gamma^\mu\psi(x)\bar{\psi}(x_1)\psi(x_2)\rangle = \left[i\delta^{(4)}(x-x_1) - i\delta^{(4)}(x-x_2)\right]\langle\bar{\psi}(x_1)\psi(x_2)\rangle.$$

What are the profound physical implications of this equation? First, consider the term $\bar{\psi}(x)\gamma^\mu\psi(x)$ inside the Green's function. This is the insertion of a vector current. Diagrammatically, we can represent this 3-point Green's function as a vertex where a momentum q is injected via the current:

TODO: add diagram: 3-point Green's function with current insertion, 1 and external points

From our perturbative Feynman rules, the rule for a simple vector current insertion is simply γ^μ. Notice that this is exactly proportional (up to a factor of iQ_e) to the rule for a standard photon interaction vertex:

TODO: add diagram

This is not surprising at all, since photons strictly couple to conserved currents.

Let us Fourier transform the Ward-Takahashi identity to momentum space, assigning incoming momentum p_1 to the fermion, outgoing momentum p_2, and incoming momentum q to the current insertion:

TODO: add diagram: Momentum space assignments for the 3-point function

$$= iQ_e \int \mathrm{d}^4x\,\mathrm{d}^4x_1\,\mathrm{d}^4x_2\, e^{i(p_2\cdot x_2 - p_1\cdot x_1 - q\cdot x)}(\dots)$$

We define the Fourier-transformed Green's functions. Let $\mathcal{M}^\mu(q;p_1,p_2)$ be the 3-point Green's function with the external photon propagator stripped off (but including external fermion propagators). Let $\mathcal{M}_0(p,k)$ represent the full 2-point fermion Green's function (the exact interacting fermion propagator) carrying momentum from p to k.

TODO: add diagram: Definitions

Crucial Note: These are full **Green's functions**, not physical scattering amplitudes! This means the external momenta are generally **off-shell** ($p_i^2 \neq m^2$), there are no external polarization vectors ϵ_μ, and the external fermion propagators are strictly included.

The spacetime derivative $i\partial_\mu$ acting on the coordinate x brings down the momentum q_μ. The Ward-Takahashi identity in momentum space beautifully simplifies to:

$$q_\mu\mathcal{M}^\mu(q;p_1,p_2) = -Q_e\Big(\mathcal{M}_0(p_1,p_2-q) - \mathcal{M}_0(p_1+q,p_2)\Big)$$

TODO: add diagram: visual representation of the Ward-Takahashi identity in momentum space showing the contraction equal to the difference of two 2-point functions

This elegant relation, connecting a 3-point function to a difference of 2-point functions, is easily generalized to an arbitrary Green's function with any number of external legs. By following the exact same steps for a Green's function with N incoming fermions and N outgoing fermions:

$$q_\mu\mathcal{M}^\mu(q;p_1\dots,k_1\dots) = -Q_e\sum_i\left(\mathcal{M}_0(\dots,k_i-q,\dots) - \mathcal{M}_0(\dots,p_i+q,\dots)\right)$$

(Note: In Peskin and Schroeder section 7.4, there is a very nice diagrammatic proof of this identity by explicitly summing insertion points on Feynman diagrams. Here, we have proved it much more rigorously and generally via the functional path integral formalism).

TODO: add diagram: General WT identity with a blob of N legs TODO: add ref

From Green's Functions to Scattering Amplitudes: The Ward Identity

What happens when we transition from these off-shell Green's functions to actual on-shell **scattering amplitudes**? To compute an S-matrix element, the LSZ reduction formula instructs us to put all external fermions on-shell ($p^2 = m^2$) and multiply the Green's function by the inverse free propagators $\prod_{p=p_j,k_j} \left(\frac{-i}{\not{p}-m}\right)$ to amputate the external legs.

Look at the right-hand side of the Ward-Takahashi identity. It consists of differences of Green's functions where the momentum of one leg is shifted by q. If we apply the LSZ amputation procedure, the shifted leg (e.g., $p_i + q$) will *not* have the correct on-shell singularity pole $\frac{i}{\not{p}_i+\not{q}-m}$ to cancel the specific amputation factor $\not{p}_i - m$. Consequently, when we take the strict on-shell limit prescribed by LSZ, the right-hand side identically vanishes!

Therefore, for physical scattering amplitudes $\mathcal{M}$, we rigorously recover the **Ward Identity**:

$$i\mathcal{M} = i\epsilon_\mu(q)\mathcal{M}^\mu \implies q_\mu \mathcal{M}^\mu = 0.$$

Remarkably, our functional derivation proved something even stronger: the Ward identity

$$q_\mu \mathcal{M}^{\mu\nu_1\dots\nu_n} = 0,$$

is valid even if the external photons are strictly off-shell, and even if they are NOT contracted with their respective polarization vectors! Among other things, this formally justifies the replacement $\sum \epsilon_{s\mu}\epsilon^*_{s\nu} \to -g_{\mu\nu}$ in polarization sums, even for processes with an arbitrary number of external photons.

Gauge Invariance of the S-Matrix

One more crucial object that heavily depends on our somewhat arbitrary choice of gauge is the photon propagator itself. Recall from the Faddeev-Popov quantization that the general photon propagator depends on a gauge parameter ξ:

$$\Pi^{\mu\nu}(k) = \frac{i}{k^2 + i\epsilon}\left(-g^{\mu\nu} + (1-\xi)\frac{k^\mu k^\nu}{k^2}\right).$$

We can use the generalized Ward-Takahashi identity to rigorously prove that the second term (proportional to $k^\mu k^\nu$) does not contribute to any physical scattering amplitude.

We just showed that $k_\mu \mathcal{M}^{\mu\nu_1\dots\nu_n} = 0$, where $\mathcal{M}$ is an "amplitude" containing $N+1$ external photons (which may be off-shell and uncontracted with polarization vectors) plus any number of strictly *on-shell* external fermions. Consider the calculation of an amplitude $i\mathcal{M}$ at a specific perturbative order containing N internal photon lines. We can write the mathematical expression for this amplitude as an integral over the internal photon momenta, connecting the photon propagators to a large fermion sub-amplitude:

$$\int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\Pi^{\mu_1\nu_1}(k_1)\cdots\int \frac{\mathrm{d}^4k_N}{(2\pi)^4}\Pi^{\mu_N\nu_N}(k_N)\times\mathcal{M}_{\mu_1\dots\mu_N\nu_1\dots\nu_N},$$

where $\mathcal{M}_{\mu_1\dots\nu_N}$ represents the rest of the diagram (including external on-shell fermions). If we focus on just one internal photon k_i, the longitudinal part of its propagator is proportional to $k_i^{\mu_i}k_i^{\nu_i}$.

By the exact Ward identity we just proved, contracting $k_i^{\mu_i}$ with the sub-amplitude $\mathcal{M}_{\ldots\mu_i\ldots}$ yields strictly zero.

Therefore, the $(1-\xi)\frac{k^\mu k^\nu}{k^2}$ part of every single internal photon propagator vanishes identically. This proves that the physical scattering amplitude $i\mathcal{M}$ is fundamentally gauge invariant and completely independent of the arbitrary gauge parameter ξ.

Multi-Flavor QED

Finally, assume we have a theory with more than one flavor of Dirac fermion, for example, an electron with charge $Q_e e$ and a muon with charge $Q_\mu e$. Because the $U(1)$ global symmetries for electrons and muons are independent, we can obtain two separate Ward-Takahashi identities: one where the derivative acts on photons attached ONLY to electron currents, and one for muon currents. By linearly combining these two identities, we obtain a master Ward-Takahashi identity that is completely valid for ANY arbitrary value of the charges Q_e and Q_μ.

This implies a very deep result: current conservation and the Ward-Takahashi identity in QED hold absolutely with an arbitrary number of Dirac fields coupling to the photon, regardless of their specific electric charges.

Remark. *This is a highly specific property of Abelian theories like QED. It is NOT generally valid for non-Abelian gauge theories such as QCD, where the gauge bosons self-interact and flavor symmetries are more complexly intertwined.*

6 | Renormalization and Regularization

[...]

6.1 | Divergences in Quantum Field Theory

To introduce the topic we go back to one of the easiest examples of a quantum field theory, the ϕ^4 theory. The Lagrangian density is given by

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4.$$

We consider one of the simplest processes, the $2 \to 2$ scattering process

$$\phi(p_1)\phi(p_2) \to \phi(p_3)\phi(p_4),$$

which at leading order is given by the tree-level Feynman diagram

$$i\mathcal{M}_{LO} = [Draw\ LO] = -i\lambda.$$

The differential cross section is given by

$$\frac{\mathrm{d}\sigma_{LO}}{\mathrm{d}\Omega}\Big|_{\mathrm{CM}} = \frac{1}{64\pi^2 E_{cm}^2}\frac{|\mathbf{p}_{out}|}{|\mathbf{p}_{in}|}|\mathcal{M}_{LO}|^2 = \frac{1}{64\pi^2}\frac{\lambda^2}{s}.$$

We now have to compute the amplitude $i\mathcal{M}$ at next-to-leading order (NLO). The Feynman diagrams contributing to the NLO amplitude are given by

$$i\mathcal{M} = i\mathcal{M}_{LO} + i\mathcal{M}_{NLO} + \mathcal{O}(\lambda^3),$$

$$i\mathcal{M}_{NLO} = [Draw\ NLO] = i(\mathcal{M}_1 + \mathcal{M}_2 + \mathcal{M}_3).$$

Let us study the diagrams one by one. Having defined $p = p_1 + p_2$, the first diagram is given by

$$i\mathcal{M}_1 = \frac{(-i\lambda)^2}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{i}{k^2 - m^2 + i\epsilon}\frac{i}{(k+p)^2 - m^2 + i\epsilon} = (-i\lambda)^2 iV(p^2),$$

since the result hs to be lorentz invariant and can only depend on the Lorentz invariant p^2 (the $\frac{1}{2}$ global factor was due to the symmetry of the diagram).

The second and third diagrams are more or less the same: they can be obtained from the first diagram by replacing p with $p_1 - p_3$ and $p_1 - p_4$ respectively:

$$\mathcal{M}_2 = \mathcal{M}_1\big|_{p\to p_1-p_3}, \qquad \mathcal{M}_3 = \mathcal{M}_1\big|_{p\to p_1-p_4}.$$

We can thus write

$$i\mathcal{M}_{NLO} = i(-i\lambda)^2[V(s) + V(t) + V(u)],$$

thus our work is restricted to the computation of the loop integral $V(p^2)$, which is a bit unfortunate since it is divergent. Indeed, if we take the limit $k^\mu \to \infty$ we have

$$k^\mu \to \infty \quad \Rightarrow \quad V(p^2) \sim \int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{1}{k^4} \sim \log\Lambda \to \infty,$$

which is the definition of a logarithmic divergence: we are in the presence of an **ultraviolet (UV) divergence**, since the divergence is due to the contribution of high momentum modes in the loop integral.

[...]

Observations

- In quantum mechanics, when computing an amplitude for an event, we need to coonsider all the different intermediate states that can contribute to the process. As $\Delta x \Delta p \sim O(\hbar) \sim \Delta E \Delta t$, ...
- Contrary to the previos observation, we cannot neither expect the thery to be valid at arbitrarily high energies, nor to be able to measure the amplitude with infinite precision (not even QED or the Standard Model). Some new effects...
- Experiments at low energy scales are not sensitive to the details of the UV completion of the theory, but only to a few parameters...
- We are trying to compute $i\mathcal{M}$ and $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ as functions of the free parameters of the lagrangian, namely λ and m. [...]

  But how do we ...

[...]

Summary: ...

[...]

Solution. The solution can be achieved as a two step process:

1. **Regularization**: we introduce a regulator in the theory, which is a parameter that makes the loop integrals finite. For example, we can introduce a cutoff Λ in the loop integral, such that we only integrate over momenta k such that $|k| < \Lambda$

   $$\int_{-\infty}^{\infty} \mathrm{d}^4 k \to \int_{-\Lambda}^{\Lambda} \mathrm{d}^4 k,$$

   and then we can take the limit $\Lambda \to \infty$ at the end. This way, we have to deal with a finite loop integral depending on a **regulator**, but we have to be careful when taking the limit $\Lambda \to \infty$ at the end, since we can get infinite results.

2. **Renormalization**: we redefine the parameters of the theory (the mass and the coupling constant) in such a way that the physical observables (the amplitude and the cross section) are finite in the limit $\Lambda \to \infty$: rewriting observables as functions of other observables. The result will become finite[1] in the limit $\Lambda \to \infty$ where the regulator is removed.

UV divergence as manifestation of our ignorance of the UV completion of the theory... For example for macroscopic objects...

We should be able to study physics at low energy scales (and high distances) without knowing the details of the UV completion of the theory, but only a few parameters that encode the effects of the UV completion at low energy scales.

Similarly, we should be able to relate...

[1] Under certain condition that we will have to further stress.

6.1.1 | Classical Example: Infinite Wire Potetial

Consider an infinite wire along the z axis with a uniform charge density λ. We want to compute the electric potential $V(r)$ at a distance r from the wire.

Without computing anything, we can note from some dimensional analysis that the potential already has some problems:

$$V(r) = \lambda f(r), \quad [V] = 1, \quad [\lambda] = [L]^{-1} = 1,$$

thus it seems that $f(r)$ should be dimensionless and the potential will result in being constant, which is not what we expect.

By trying to compute the potential we find that the problem is given by a divergence in the integral:

$$V(r) = \int_{-\infty}^{\infty} \mathrm{d}x \frac{\lambda}{4\pi r(x)} = \frac{\lambda}{4\pi} \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{\sqrt{r^2 + x^2}} = \infty,$$

where we should however notice that

$$E_i = -\partial_i V(r) = \frac{\lambda}{4\pi} \int_{-\infty}^{\infty} \mathrm{d}x \frac{r_i}{(r^2 + x^2)^{3/2}} = \frac{\lambda}{2\pi r} \hat{r}_i,$$

is finite, which seems to be a contradiction. From these some observations follows naturally:

- Infinite wires do not exist in nature, [...]
- If $r \ll$ lenght of the wire, de details of what happens at the end of the real finite wire are not relevant, and we can still use the infinite wire approximation to compute the electric field, which is what we are interested in.
- The potential $V = V(r)$ is not a physical observable, since it is not measurable: only differences of potential are measurable, and these are finite: $\Delta V = V(r_2) - V(r_1)$.

Thus we can apply the solution in two steps as we explained before:

- **Regularization**: we introduce a regulator in the theory, which is a parameter that makes the integral finite. For example, we can introduce a cutoff Λ in the integral, such that we only integrate over x such that $|x| < \Lambda$

  $$\int_{-\infty}^{\infty} \mathrm{d}x \to \int_{-\Lambda}^{\Lambda} \mathrm{d}x,$$

  and then we can take the limit $\Lambda \to \infty$ at the end:

  $$V(r) = \frac{\lambda}{4\pi} \int_{-\Lambda}^{\Lambda} \frac{\mathrm{d}x}{\sqrt{r^2 + x^2}} = \frac{\lambda}{4\pi} \log \left(\frac{\sqrt{r^2 + \Lambda^2} + \Lambda}{\sqrt{r^2 + \Lambda^2} - \Lambda} \right).$$

  The divergence is still there in the limit $\Lambda \to \infty$, but at least we have a finite result depending on the regulator Λ.

- **Renormalization**: we redefine the parameters of the theory (the potential) in such a way that the physical observables (the electric field) are finite in the limit $\Lambda \to \infty$. We can choose an arbitrary distance r_0 (not too far from the wire) in order to renormalize the potential:

  $$V_R(r) = V(r) - V(r_0),$$

which is an observable and indeed it is finite in the limit $\Lambda \to \infty$:

$$V_R(r) = -\frac{\lambda}{4\pi} \log\left(\frac{r^2}{r_0^2}\right) + \mathcal{O}(\frac{1}{\Lambda^2}),$$

as $\Lambda \to \infty$. The divergence has thus disappeared in the renormalized potential V_R, which is the physical observable, and we have a finite result for the electric field: ...

6.2 | Regularization Schemes

A regularization scheme is a way to make the loop integrals finite, and there are many different regularization schemes that can be used in quantum field theory. The regularization we just showed is called **cutoff regularization**

$$\int_{-\infty}^{\infty} \mathrm{d}^4k \to \int_{-\Lambda}^{\Lambda} \mathrm{d}^4k,$$

which can be used in any theory, but it has the disadvantage of breaking Lorentz invariance and gauge invariance (and other symmetries), making intermediate steps in calculations more complicated (for example we cannot make use of the Ward identity anymore). It is not very convenient and outside textbook examples it is not used in practice.

[...]

Let us present some of the most common regularization schemes used in quantum field theory:

- **Modified interaction** (Pauli-Villars regularization): we can modify the form of the interaction in the high energy regime, since it should not affect the low energy physics. For example, we could modify the Coulomb potential at short distances by introducing:

  $$\frac{\lambda \mathrm{d}x}{4\pi r} \to \frac{\lambda \mathrm{d}x}{4\pi}\left(\frac{1}{r} - \frac{1}{\sqrt{r^2+\Lambda^2}}\right),$$

  which will reduce to the original potential in the limit $\Lambda \to \infty$, but it will be finite for any finite value of Λ. The integral of the potential will now be finite

  ...

  [...] Loop integrals can be regularized in a similar way by modifying the propagator at high energies as

  $$\frac{i}{k^2 - m^2 + i\epsilon} \to \frac{i}{k^2 - m^2 + i\epsilon} - \frac{i}{k^2 - \Lambda^2 + i\epsilon},$$

  which makes it finite for any finite value of Λ and reduces to the original propagator in the limit $\Lambda \to \infty$. This is the **Pauli-Villars** regularization scheme.

- **Dimensional regularizarion**. The previous regularization scheme have a clear physical interpretation: ... The method that is almost exclusively used today is called *dimensional regularization* (by Giambiagi, Bollini ...), which do not have a clear physical interpretation, but it is very convenient for calculations and it preserves all the symmetries of the theory.

  Dimensional regularization is based on the idea of performing the loop integrals in d dimensions instead of 4 dimensions, where d is a smaller number than 4, such that the loop integrals are finite. For example, the integral we are using as an example in the ϕ^4 theorywill be convenient to compute in $d = 2, 3$ dimensions. It turns out that we can compute it as an analytic function of the number of spacetime dimensions d:

  $$\int_{-\infty}^{\infty} \mathrm{d}^4k \frac{1}{k^4} \to \int_{-\infty}^{\infty} \mathrm{d}^dk \frac{1}{k^4} = \dots$$

  this function will have a pole for $d = 4$, which will be removed in smaller dimensions, and we can take the limit $d \to 4$ at the end of the calculation, which will be equivalent to taking the limit $\Lambda \to \infty$ in the cutoff regularization scheme.

  [...]

We can also apply it to our toy-example of the infinite wire potential, by performing the integral in d dimensions instead of 4 dimensions:

$$V(r) = \dots \begin{cases} x \to \mathbf{x} = \begin{pmatrix} x_1 & x_2 & \cdots & x_d \end{pmatrix} \\ \mathrm{d}x \to \mathrm{d}^d x = \mathrm{d}x_1 \mathrm{d}x_2 \cdots \mathrm{d}x_d \\ x^2 \to |\mathbf{x}^2| = \sum_{j=1}^{d} x_j^2 \end{cases}$$

such that we can rewrite the potential as

$$V(r) = \dots$$

where μ is a scale with mass dimension $1 = [\mu]$. [...] [...] The integral is now convergent for $0 < d < 1$, where it is analytic in d. We can thus analytically continue the result in the whole complex plane with the exception for some points in which it will have poles. We can show that ...

$$\dots$$

and we get the usual result for V_R as $d \to 1$.

- **Renormalization without regulators**. In principle, it is not necessary to introduce a regulator (Λ or d) in order to renormalize the theory: we can indeed introduce directly at the integrand level some physical, finite and renormalized quantities. Considering the toy example of the infinite wire potential, we could have introduced directly the renormalized potential at the integrand level as

$$V_R(r) = V(r) - V(r_0) = \frac{\lambda}{4\pi} \int_{-\infty}^{\infty} \mathrm{d}x \left(\frac{1}{\sqrt{r^2 + x^2}} - \frac{1}{\sqrt{r_0^2 + x^2}} \right) = -\frac{\lambda}{4\pi} \log \left(\frac{r^2}{r_0^2} \right),$$

which is finite and does not require any regulator. [...]

6.3 | One Loop Integrals in Dimensional Regularization

In general, the loop integrals we have to compute in quantum field theory are of the form

$$\int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{1}{D_1^{\alpha_1} D_2^{\alpha_2} \cdots D_n^{\alpha_n}},$$

where we have defined the "denominators of the propagators" as

$$\begin{cases} D_i = l_i^2 - m_i^2 + i\epsilon, \\ l_i^\mu = k^\mu + \sum_{j=1} \beta_{ij} p_j^\mu, \end{cases}$$

where p_j are the external momenta and β_{ij} are some coefficients that depend on the topology of the diagram. We can also promote the metric tensor to D dimensions (1 timelike and $D-1$ spacelike dimensions) as

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & -1 \end{pmatrix}, \quad \text{such that} \quad g_\mu{}^\mu = \delta^\mu{}_\mu = D.$$

Since it is not convenient to treat an integral with many denominators, we can use the **Feynman parameters** to rewrite it as an integral with a single denominator:

$$\frac{1}{D_1 \cdots D_n} = \int_0^1 \mathrm{d}x_1 \cdots \int_0^1 \mathrm{d}x_n \, \delta\left(1 - \sum_{j=1}^n x_j\right) \frac{(n-1)!}{(x_1 D_1 + \cdots + x_n D_n)^n},$$

where the delta function is needed to ensure that the sum of the Feynman parameters x_i is equal to 1. By taking derivatives with respect to the propagator denominators D_i we can also get the following more general formula:

$$\left(\frac{\partial}{\partial D_i}\right) \implies \frac{1}{D_i^{\alpha_1} \cdots D_n^{\alpha_n}} = \int_0^1 \mathrm{d}x_1 \cdots \int_0^1 \mathrm{d}x_n \, \delta\left(1 - \sum_{j=1}^n x_j\right) \frac{\Gamma(\alpha_1 + \cdots + \alpha_n)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_n)} \frac{x_1^{\alpha_1 - 1} \cdots x_n^{\alpha_n - 1}}{(x_1 D_1 + \cdots + x_n D_n)^{\alpha_1 + \cdots + \alpha_n}},$$

with the condition of $\alpha \equiv \sum_{i=1}^n \alpha_i$.

In order to analitically continue the integral to non integer values of the exponents α_i we have to introduce the **Gamma function**, which is defined as

$$\Gamma(z) = \int_0^\infty \mathrm{d}t \, t^{z-1} e^{-t},$$

which is a generalization of the factorial function to complex numbers, since for positive integers n we have $\Gamma(n) = (n-1)!$. The Gamma function has some important properties that we will use in the following:

- It has poles at $z = 0, -1, -2, \ldots$ with residue $\mathrm{Res}(\Gamma(z), z = -n) = \frac{(-1)^n}{n!}$.
- It satisfies the following functional equation: $\Gamma(z+1) = z\Gamma(z)$, which is obtained by performing an integration by parts in the definition of the Gamma function.
- By direct integration we can show that $\Gamma(1) = 1$.

- The last two properties together imply that $\Gamma(n) = (n-1)!$ for positive integers $n \geq 0$.

It converges for $\mathrm{Re}(z) > 0$, but can also be analitically continued to the whole complex plane with the exception of the poles at $z = 0, -1, -2, \ldots$. Thus we are now ready to write the more general formula for the loop integrals in dimensional regularization (and in Feynman parameterization):

$$\frac{1}{D_1^{\alpha_1} \cdots D_n^{\alpha_n}} = \ldots$$

Proof. Let us start from noticing that the following integral after a change of variable $x \to x' = Dx$ can be rewritten in terms of gamma functions as

$$\int_0^\infty \mathrm{d}x\, x^{\alpha-1} e^{-Dx} = D^{-\alpha} \int_0^\infty \mathrm{d}x'\, x'^{\alpha-1} e^{-x'} = \frac{\Gamma(\alpha)}{D^\alpha}.$$

If we apply this formula (referred to as the *Schwinger's trick*) to the denominator of the propagators n times, we obtain:

$$\frac{1}{D_1^{\alpha_1} \cdots D_n^{\alpha_n}} = \ldots$$

But we want to get to a Feynman parametrized form, thus we have to introduce the delta function to ensure that the sum of the Feynman parameters x_i is equal to 1. We can do this by using the following identity:

$$1 = \int_0^\infty \mathrm{d}g\, \delta\left(g - \sum_{j=1}^n x_j\right),$$

and then rescaling the Feynman parameters as $x_i \to g x_i$, in order to get

$$\frac{1}{D_1^{\alpha_1} \cdots D_n^{\alpha_n}} = \ldots$$

But now it is not difficult to recognize that the integral over g is again of the form of the Schwinger's trick, and we can perform it to get the final result:

$$\ldots$$

which leads us to the desired formula. □

Example (D-dimensional solid angle). We will show that the solid angle in D dimensions is given by

$$\Omega_D = \int \mathrm{d}\Omega_D = \frac{2\pi^{D/2}}{\Gamma(D/2)},$$

where $\mathrm{d}\Omega_D$ is the measure of the solid angle in D dimensions. We can start from the Gaussian integral in D dimensions, and we will compute it both in cartesian coordinates and in spherical coordinates:

[...]

6.3.1 | Gamma Functions and Wick Rotation

Studying the properties of the Gamma function we can see that it has poles at $z = 0, -1, -2, \ldots$. This means that if we have a loop integral that is divergent in 4 dimensions, it will have a pole at $D = 4$ in dimensional regularization. In particular, around $z = 0$, we can expand as

$$\Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon),$$

as $\epsilon \to 0$. Here, γ_E is the Euler-Mascheroni constant, which is defined as

$$\gamma_E = \lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \log n \right) \approx 0.5772,$$

but its numerical value is of no importance for us, since it always cancels out when deriving physical observables. The important thing is that knowing how the gamma function behaves around zero lets us understand how it behaves around all of its poles, since we can always exploit one of its properties to write:

$$\Gamma(z+1) = z\Gamma(z) \quad \Rightarrow \quad \Gamma(-1+\epsilon) = \frac{\Gamma(\epsilon)}{-1+\epsilon},$$

[...]

With feynman parameterization, we can always ...

[...]

Thus let us consider a **tadpole integral** of the following form in dimensional regularization:

$$I = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{1}{(k^2 - m^2 + i\epsilon)^\alpha}, \quad \alpha = 1, 2, 3, \dots$$

[*Draw of complex plane and wick rotation*]

Considering firstly the integral over the time component k^0 of the loop momentum, we can perform a **Wick rotation**: we can rotate the contour of integration in the complex plane of k^0 from the real axis to the imaginary axis counter-clockwise without crossing any poles, by defining

$$k^0 = \pm(\sqrt{\mathbf{k}^2 + m^2} - i\epsilon).$$

With this choice of the contour, the integral does not change, since we are not crossing any poles (we can imagine to close the contour ...); this Wick rotation is equivalent to performing a change of variable

$$\begin{cases} k^0 = ik_E^0, \\ \mathbf{k} = \mathbf{k}_E, \end{cases}$$

and it corresponds to performing the loop integral in Euclidean space (rotation to euclidean coordinates) instead of Minkowski space: indeed we have

$$\begin{aligned} k^2 &\equiv (k^0)^2 - |\mathbf{k}|^2, && \text{Minkowski space} \\ k_E^2 &\equiv (k_E^0)^2 + |\mathbf{k}_E|^2 = -k_E^2, && \text{Euclidean space} \end{aligned}$$

impliying that $k^2 = -k_E^2$. The tadpole integral then

$$I = \dots$$

where we could restore the feynman prescription easily by performing the change of variable $m^2 \to m^2 - i\epsilon$ in the final result. We could now use the Schwinger's trick

$$\frac{1}{D^\alpha} = \dots$$

to rewrite the denominator of the propagator as an integral over a Feynman parameter x:

$$I = \dots$$

Now the integral in the euclidean space is just a gaussian integral, which we can compute by using the formula for the solid angle in D dimensions:

$$\int \mathrm{d}^D k_E \exp(-k_E^2 x) = \left(\frac{\pi}{x}\right)^{D/2},$$

such that we get

...

using again schwinger's trick with $D \to m^2$ and $\alpha \to \alpha - D/2$ we obtain

...

which putting everything together gives us the final result for the tadpole integral in dimensional regularization:

$$I = \dots$$

[...]

So to compute a one loop integral in dimensional regularization we can follow the following steps:

- ...
- ...
- ...

We can now apply this procedure to compute the one loop correction to the amplitude in the ϕ^4 theory.

6.3.2 | 2-2 Scattering in ϕ^4-Theory at NLO

We first consider the ϕ^4 theory in D dimensions; doing dimensional analysis for our theory:

...

we can write the interaction lagrangian density as

$$\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\mu^{4-D}\phi^4,$$

where μ is a mass scale, with

$$[\mu] = 1 \implies [\lambda] = 0,$$

where the constraint on the mass dimension of λ is given by the requirement that the interaction lagrangian density has mass dimension D.

We can now compute the amplitude for the $2 \to 2$ scattering process at NLO in dimensional regularization. Firstly, the new Feynman rule for the vertex is given by

$$\text{Draw of vertex} = -i\lambda\mu^{4-D},$$

and this implies

$$i\mathcal{M}_{LO} = vertex = -i\lambda\mu^{4-D},$$
$$i\mathcal{M}_{NLO} = [Draw\ NLO] = i(-i\lambda)^2\mu^{4-D}[V(s) + V(t) + V(u)],$$

where the second factor of μ is absorbed into the vertex factor:

$$iV(p^2) = \frac{1}{2}\mu^{4-D}\int \frac{\mathrm{d}^D k}{(2\pi)^D}\frac{i}{k^2-m^2}\frac{i}{(k+p)^2-m^2}.$$

Let us underline two observations:

- The integral $V(p^2)$ is still dimensionless with this choice of the vertex factor, as it should be since the amplitude has to be dimensionless.
- The feynman prescription is understood: we have to perform the change $m^2 \to m^2 - i\epsilon$ in the final result if we need to make it explicit.

Now using the Feynman parameterization for $N = 2$ we can rewrite the integral as

$$\frac{1}{D_1 D_2} = \int_0^1 \mathrm{d}x \frac{1}{\left((1-x)D_1 + xD_2\right)^2},$$

and

$$iV(p^2) = -\frac{\mu^{4-D}}{2}\int \dots$$

which can be shifted by $k^\mu \to k^\mu - xp^\mu$ in order to obtain

$$iV(p^2) = \dots$$

If we now define $M^2(p^2) = m^2 - (x(1-x)p^2)$, we can recgnize the form of the tadpole integral we computed in the previous section, and we can use the result we obtained for it to get

$$iV(p^2) = \dots$$

where we can notice we have not eliminated the divergence yet, since we have a pole at $D = 4$ in the gamma function; but we have obtained a finite result for the loop integral, which is what we wanted to achieve with regularization. As promised, both $iV(p^2)$ and thus $i\mathcal{M}$ functions are analytic in D with the exception of some poles. At some points we will be interested in expanding the result around $D = 4$; let us define ϵ as

$$D = 4 - 2\epsilon,$$

such that the limit $\epsilon \to 0$ corresponds to the limit $D \to 4$. We can then use the following properties

$$\begin{cases} \Gamma(2-\frac{D}{2}) = \Gamma(\epsilon) = \dfrac{1}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon), \\ (4\pi)^{-D/2} = \dfrac{1}{16\pi^2}(1 + \epsilon\log(4\pi)) + \mathcal{O}(\epsilon^2), \\ (M^2)^{D/2-2} = (M^2)^{-\epsilon} = 1 - \epsilon\log M^2 + \mathcal{O}(\epsilon^2), \\ \mu^{4-D} = \mu^{2\epsilon} = 1 + \epsilon\log\mu^2 + \mathcal{O}(\epsilon^2), \end{cases}$$

in order to get the following expansion for $iV(p^2)$ around $D = 4$:

$$iV(p^2) = -\frac{i}{32\pi^2}\left(\frac{1}{\epsilon} - \gamma_E + \log(4\pi) - \int_0^1 \mathrm{d}x \log\left(\frac{M^2(p^2)}{\mu^2}\right)\right) + \mathcal{O}(\epsilon).$$

The integral in $\mathrm{d}x$ appearing in the previous expression can be computed analytically, but it is not important for us to get its explicit form, since we are interested in the **regulated amplitude**, which is given by

$$\mu^{D-4}(i\mathcal{M}) = \dots$$

where this step concludes the regularization procedure, since we have obtained a finite result for the amplitude depending on the regulator ϵ, which is what we wanted to achieve with regularization. We can now move to the renormalization procedure, in order to get a finite result for the amplitude in the limit $\epsilon \to 0$.

Renormalization of the Amplitude

Thus we now start from a regulated amplitude, which is a finite function of the regulator ϵ, and we want to get a finite result for the amplitude in the limit $\epsilon \to 0$ by redefining the parameters of the theory (the mass and the coupling constant) in such a way that the physical observables (the amplitude and the cross section) are finite in the limit $\epsilon \to 0$. We can rewrite observables as functions of other observables, and the result will become finite in the limit $\epsilon \to 0$ where the regulator is removed.

Thus we start by replacing the bare parameters m and λ of our lagrangian with physical, observable quantities. Since LO is independent of the mass, which furthermore can be written as

$$m = m_{\text{phys}} + \mathcal{O}(\lambda),$$

the renormalization of m comes into play only at $\mathcal{O}(\lambda^3)$ for the amplitude (but we would need...)

Therefore we have to focus on the renormalization of the coupling constant λ.

Remark. *We should also multiply by ...*

We thus want to understand how to define a λ_{phys} to renormalize the coupling constant λ.

[...]

$$\sqrt{s} = E_{cm} = 2m \implies s = 4m^2,$$

[...]

We define

$$\mu^{D-4}(i\mathcal{M})\Big|_{s=4m^2}^{t=u=0} = -i\lambda_{\text{phys}},$$

as a renormalization condition, which allows us to define the physical coupling constant λ_{phys} as the value of the amplitude at a specific kinematic point (the renormalization point)

$$\lambda_{\text{phys}} = \lambda - \frac{3\lambda^2}{32\pi^2}(\dots) + \mathcal{O}(\lambda^3).$$

This is the **renormalization condition** that we have to impose in order to define the physical coupling constant λ_{phys} in terms of the bare coupling constant λ and the regulator ϵ. By inverting this relation perturbatively, we can express the bare coupling constant λ as a function of the physical coupling constant λ_{phys} and the regulator ϵ:

1. We know $\lambda_{\text{phys}} = \lambda + \alpha\lambda^2 + \mathcal{O}(\lambda^3)$.
2. We want to get an ansatz of the form $\lambda = \lambda_{\text{phys}} + \alpha'\lambda_{\text{phys}}^2 + \mathcal{O}(\lambda_{\text{phys}}^3)$.

By plugging the ansatz into the first relation we get

$$\lambda = \lambda + (\alpha + \alpha')\lambda^2 + \mathcal{O}(\lambda^3) \implies \alpha' = -\alpha,$$

such that we can use this relation to rewrite the amplitude as a function of the physical coupling constant λ_{phys} and the regulator ϵ:

$$\mu^{-2\epsilon}(i\mathcal{M}) = -i\lambda_{\text{phys}} - i\frac{3\lambda_{\text{phys}}^2}{32\pi^2}(\dots) + \mathcal{O}(\lambda_{\text{phys}}^3),$$

which is now a finite result at $\epsilon \to 0$ (equivalently at $D \to 4$), since the divergence has been absorbed into the redefinition of the coupling constant. We have thus obtained a finite result for the amplitude in the limit in which the regulator is removed, by redefining the parameters of the theory (the coupling constant) in such a way that the physical observables (the amplitude) are finite.

We can now use this result to compute the cross section for the $2 \to 2$ scattering process at NLO, for example, or other physical observables that can be derived from the amplitude.

Remark. *We could have chosen a different renormalization condition (i.e. a different definition of the physical coupling constant), and we would have obtained a different result for the amplitude. Let us for example choose the following renormalization condition:*

$$\mu^{D-4}(i\mathcal{M})\Big|_{s=s_0,\, t=t_0,\, u=u_0} = -i\lambda'_{phys},$$

which will be different from the previous one, since we are evaluating the amplitude at a different kinematic point. This is not a problem, since the physical observables (the cross section) will be the same in both cases, as they should be, since they are independent of the choice of the renormalization condition. But proceding with the calculation

$$\mu^{-2\epsilon}(i\mathcal{M}) = -i\lambda'_{phys} - i\frac{3(\lambda'_{phys})^2}{32\pi^2}\int_0^1 \mathrm{d}x\,(\dots) + \mathcal{O}((\lambda'_{phys})^3),$$

which is different from the previous result, since we are evaluating the integral at a different kinematic point. However, the physical observables will be the same in both cases, since if we now rewrite perturbatively the physical coupling constant λ'_{phys} as a function of λ_{phys}:

$$\lambda'_{phys} = \lambda_{phys} + \alpha''\lambda_{phys}^2 + \mathcal{O}(\lambda_{phys}^3),$$

which will end up giving us the same result for the amplitude as a function of λ_{phys} and ϵ as before, up to the perturbative order we are considering, and thus the same result for the physical observables.

Thus the renormalization scheme is not unique, since we can choose different renormalization conditions, but the physical observables will be the same in all the different renormalization schemes, and it all comes down to a choice of how to define the physical parameters of the theory in terms of the bare parameters and the regulator.

6.3.3 | A Note on Analytic Continuation

Let us finally compute the integral in the Feynman parameter x that appears in the expression for the amplitude, which is given by

...

[...]

...

The analytic continuation to $p^2 = s \geq 4m^2$, but doing so we would get a negative argument for $w-1$ in the logarithm, which is not defined for real numbers. The logarithm as a complex function has a branch cut on the negative real axis, and we have to specify how to go around it in order to get a well defined result. If $z > 0$ we have

$$\begin{cases} \dots \\ \dots \end{cases}$$

What is the correct imaginary part to choose for the logarithm? We can get it by employing the feynman prescription, which is equivalent to performing the change $M^2 \to M^2 - i\epsilon$ or $m^2 \to m^2 - i\epsilon$ or even $p^2 \to p^2 + i\epsilon$ in the final result, and the easiest one is to perform the change $p^2 \to p^2 + i\epsilon$, for small ϵ and check the sign of $i\pi$ (even numerically).

We will find...

[...]

6.4 | Renormalization with Counterterms

[...]

New Notation

[...]

6.4.1 | On Shell Renormalization Conditions

[...]

6.4.2 | Counterterms

[...]

6.4.3 | Renormalization of the Propagator

[...]

6.5 | Renormalization Schemes

[...]

Minimal Subtraction Scheme

[...]

Modified Minimal Subtraction Scheme

[...]

$$\frac{\mathrm{d}}{\mathrm{d}\mu_R}\mathcal{M}(\lambda(\mu_R),\mu_R)=0.$$

6.5.1 | Scale Dependence and Beta Functions

We can define the beta function for the coupling constant λ with the **running coupling equation**:

$$\mu_R^2\frac{\mathrm{d}}{\mathrm{d}\mu_R^2}\lambda(\mu_R)=\beta(\lambda).$$

[...]

$$\beta_0=\frac{3}{32\pi}\dots$$

[...]

Let us try to solve the running coupling equation in perturbation theory. We can write $\lambda(\mu_R)$ as a power series in λ:

$$\beta(\lambda)=\beta_0\lambda^2+\beta_1\lambda^3+\dots$$

Let us assume that we know the value of λ at some reference scale μ_0. We can then write:

$$\overline{\lambda}=\lambda(\mu_R^2=\mu_0^2).$$

We know wnat to understand how λ changes as we change the scale μ_R. We can define the ration at which we change the scale as:

$$\tau=\log\left(\frac{\mu_R^2}{\mu_0^2}\right).$$

We can then writw lambda as a perturbative funtion of $\overline{\lambda}$ as

$$\lambda=\overline{\lambda}\left(1+c_1(\tau)\overline{\lambda}+c_2(\tau)\overline{\lambda}^2+\dots\right),$$

with the idea that the coefficients $c_i(\tau)$ will capture the scale dependence of λ. We can then plug this expansion into the running coupling equation along with the expansion of the beta function to find the coefficients $c_i(\tau)$. we will expand the left and right hand sides of the running coupling equation and match the coefficients of powers of $\overline{\lambda}$ to find the functions $c_i(\tau)$.

To start, we need boundary conditions for the functions $c_i(\tau)$ (it is a differential equation). We know that at $\tau = 0$ (which reads $\mu_R = \mu_0$), we have $\lambda = \overline{\lambda}$, which implies that $c_i(0) = 0$ for all i. This gives us the initial conditions we need to solve for the functions $c_i(\tau)$.

The first non trivial order is the order $\overline{\lambda}^2$. At this order, we have:

$$\mathcal{O}(\lambda^2): \quad \dots$$

which means that

$$c_1' = \beta_0, \quad c_1 = \beta_0 \log \tau = \beta_0 \log\left(\frac{\mu_R^2}{\mu_0^2}\right).$$

We can now write lambda (at this order) as a function of $\overline{\lambda}$ and τ as

$$\lambda = \overline{\lambda}\left(1 + \beta_0 \log\left(\frac{\mu_R^2}{\mu_0^2}\right)\overline{\lambda}\right) + \mathcal{O}\left(\overline{\lambda}^3\right).$$

Thus the running coupling equation at this order completely predicts the scale dependence of λ in terms of the beta function coefficient β_0.

[...]

Let us compute the next order in the perturbative expansion. At order $\overline{\lambda}^3$, we have:

$$\mathcal{O}(\lambda^3): \quad c_1'\overline{\lambda}^2 + c_2'\overline{\lambda}^3 = \beta_0\left(\overline{\lambda}^2 + 2c_1\overline{\lambda}^3 + \dots\right) + \beta_1\overline{\lambda}^3 + \dots$$

just by recakking the expansion of λ in terms of $\overline{\lambda}$. We can then match the coefficients of $\overline{\lambda}^3$, using the solution for c_1 we found at the previous order, to find:

$$-c_1 = \beta_0\tau \implies c_2' = 2\beta_0^2\tau + \beta_1,$$

which can now be solved as

$$\begin{aligned} c_2 &= \dots \\ &= \beta_0^2 \log^2(\frac{\mu_R^2}{\mu_0^2}) + \beta_1 \log\left(\frac{\mu_R^2}{\mu_0^2}\right). \end{aligned}$$

Thus we have found at leading order and the next order the scale dependence of λ in terms of the beta function coefficients β_0 and β_1: if we change the scale μ_R, we can predict how λ changes based on the values of β_0 and β_1. Let us make some further observations:

- The functional form of the scale dependence of λ is completely determined by the running coupling equation. This means that if we know the beta function coefficients, we can predict how λ changes with scale.
- The coefficient β_1 can be fixed by renormalizing at two loops: this is because the beta function at order λ^3 receives contributions from two loop diagrams, so we need to renormalize at two loops to find β_1.
- ...

It is easy to convince ourselves that at higher orders in perturbation theory, the scale dependence of λ will be given by a polynomial in $\log(\mu_R^2/\mu_0^2)$ of degree equal to the order in perturbation theory.

$$\begin{aligned} c_3' &= \dots \\ c_3 &= \beta_0^3 \log^3(\frac{\mu_R^2}{\mu_0^2}) + \beta_0\beta_1 \log^2(\frac{\mu_R^2}{\mu_0^2}) + \beta_2 \log\left(\frac{\mu_R^2}{\mu_0^2}\right), \end{aligned}$$

where now β_2 can be derived from three loop renormalization, β_3 from four loop renormalization, and so on.

This is because at each order in perturbation theory, we will have contributions to the beta function that are proportional to powers of λ, and when we solve the running coupling equation, we will find that the scale dependence of λ is given by a polynomial in $\log\left(\mu_R^2/\mu_0^2\right)$ whose coefficients are determined by the beta function coefficients at that order.

[... highest powers of logarithms at each order in perturbation theory are determined by the lowest order beta function coefficient β_0 ...]

This observation is important: let us imagine to have two very different scales

$$\mu_R^2 \gg \mu_0^2 \quad \text{or} \quad \mu_R^2 \ll \mu_0^2.$$

If this is the case, then the logarithm $\log\left(\mu_R^2/\mu_0^2\right)$ can become of order $\mathcal{O}(1)$. In this scenarion, we can predict how at all perturbative orders the scale dependence of λ will be given by a polynomial in $\log\left(\mu_R^2/\mu_0^2\right)$, having *large logarithms*: the leading contribution is called the **leading logarithm**, LL.

$$\lambda^{k+1} \log^k(\frac{\mu_R^2}{\mu_0^2}),$$

whoch we can consider of order $\mathcal{O}(\lambda)$ if

$$\lambda \log\left(\frac{\mu_R^2}{\mu_0^2}\right) \sim \mathcal{O}(1).$$

Then the running coupling equation allows us to predict the leading logarithm at all orders in perturbation theory, which is a non trivial result. This is because the leading logarithm at order λ^{k+1} is determined by the lowest order beta function coefficient β_0, which can be computed from one loop renormalization. Thus we can predict the leading logarithm at all orders in perturbation theory just by computing the one loop beta function coefficient β_0.

Similarly, the next to leading logarithm (NLL) have form

$$\lambda^{k+2} \log^k(\frac{\mu_R^2}{\mu_0^2}),$$

i.e. one higher power of λ than the leading logarithm, but the same power of the logarithm. The NLL at order λ^{k+2} is determined by the beta function coefficients β_0 and β_1, which can be computed from one loop and two loop renormalization, respectively. Thus we can predict the NLL at all orders in perturbation theory just by computing the one loop and two loop beta function coefficients β_0 and β_1 (i.e. renormlization up to two loops).

Let us now focus on the leading logarithm. Since they only depend on the lowest order beta function coefficient β_0, we can solve the running coupling equation at leading logarithm accuracy by only keeping the leading logarithm contribution to the beta function, which is given by $\beta_0\lambda^2$ ($\beta_1 = \beta_2 = \cdots = 0$). We are solving the running coupling equation non perturbatively, using a truncated expansion of the beta function, which is a non trivial result. We have

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\lambda = \beta_0\lambda^2 + \mathcal{O}\left(\lambda^3\right).$$

We neglect the higher order contributions to the beta function, which is a consistent approximation at leading logarithm accuracy. We can then solve this equation by separation of variables:

$$\int_{\overline{\lambda}}^{\lambda} \frac{\mathrm{d}\lambda}{\lambda^2} = \beta_0 \int_0^{\tau} \mathrm{d}\tau,$$

where recall that $\overline{\lambda} = \lambda(\mu_0)$ and $\lambda = \lambda(\mu_R)$. We can then perform the integrals to find

$$-\frac{1}{\lambda(\mu_0^2)} + \frac{1}{\lambda(\mu_R^2)} = \beta_0 \tau = \beta_0 \log\left(\frac{\mu_R^2}{\mu_0^2}\right).$$

Solving for $\lambda(\mu_R^2)$, we find the leading logarithm resummed expression for the running coupling:

$$\lambda(\mu_R^2) = \frac{\lambda(\mu_0^2)}{1 - \beta_0 \lambda(\mu_0^2) \log\left(\frac{\mu_R^2}{\mu_0^2}\right)} = \lambda(\mu_0^2) \sum_{k=0}^{\infty} \left(\beta_0 \lambda(\mu_0^2) \log\left(\frac{\mu_R^2}{\mu_0^2}\right)\right)^k,$$

where we have explicited the fact that the leading logarithm at order λ^{k+1} is given by $\beta_0^k \lambda^{k+1} \log^k(\mu_R^2/\mu_0^2)$. This is a non trivial result: we have found a non perturbative expression for the running coupling at leading logarithm accuracy, which resums the leading logarithms at all orders in perturbation theory. This **resummation** is important when the logarithm becomes large, as it allows us to make reliable predictions even in this regime.

Resummation in quantum field theory is a powerful technique that allows us to improve the convergence of perturbative expansions by summing up certain classes of terms to all orders in perturbation theory. In the case of the running coupling, resummation allows us to capture the leading logarithmic behavior of the coupling at all orders, which is crucial for making accurate predictions when the scale dependence becomes significant.

[...]

One can show that if we had included an additional term in the beta function, such as $\beta_1 \lambda^3$, we would have found connections proportional to...

6.5.2 | Resummation

TODO: change and move back

[...]

Landau Pole

The denominator of the resummed expression for the running coupling can become zero at a certain scale

$$\mu_R = \Lambda,$$

which is called the **Landau pole**. This indicates that the coupling becomes infinite at that scale, which is a sign of a breakdown of the theory.

We can solve this equation for $\mu_0 = \Lambda$ to find the scale at which the Landau pole occurs (the scale at which the theory becomes non perturbative):

$$\lambda(\mu_R^2) = \frac{1}{\beta_0 \log\left(\frac{\Lambda^2}{\mu_R^2}\right)}.$$

it is not difficult to see that

$$\begin{cases} \lambda(\mu_R^2 \to 0) \to 0, \\ \lambda(\mu_R^2 \to \Lambda^2) \to \infty. \end{cases}$$

[*Plot of divergence in the coupling*]

[...]

The **fixed point** is a point where the running of lambda stops, i.e. where the beta function is zero $\beta(\lambda) = 0$. For example, in a ϕ^4 theory, we have a trivial infrared fixed point at $\lambda = 0$, which is an attractive fixed point in the infrared. Other possibilities can arise in more complicated theories, such as non abelian gauge theories, where we can have non trivial fixed points that are not at zero coupling. The existence of fixed points is important for understanding the behavior of the theory at different scales, and can have implications for phenomena such as phase transitions and critical behavior.

[...]

Why would we want to use a scheme such as MS or $\overline{\text{MS}}$ instead of the on shell scheme?

[...]

Consider $\mathcal{M}(\phi\phi \to \phi\phi)$ for $p^2 \gg m^2$:

$$\int \dots$$

in the on shell scheme, we would have

$$i\mathcal{M}_{NLO} = \dots$$

At L-loops we expect to have contributions of the form

$$i\mathcal{M} \sim \lambda\lambda^L \log^L\left(\frac{p^2}{m^2}\right)$$

[*draw L loops*]

the perturbative expansion breaks down when $\lambda \log\left(s/m^2\right) \sim \mathcal{O}(1)$.

The on-shell schemes is defined as $s = 4m^2$, but at order $\mathcal{O}(m^2)$ the scheme is ill defined, since we have null scales....?

In $\overline{\text{MS}}$ we have instead

$$\dots$$

where

1. The massless limit is fine
2. choosing $\mu_R^2 \sim \mathcal{O}(s)$ allows us to resum the large logarithms, which is important for making reliable predictions at high energies. It avoids large logarithms in the perturbative expansion, which can spoil the convergence of the series and lead to unreliable predictions.

The running couplin equation in MS or $\overline{\text{MS}}$ is ...

[...]

$$\mu_R \sim \mathcal{O}\left(\text{ scale describing the problem }\right).$$

[...]

In principle observables do not depend on the renormalization scheme, but in practice they do at finite order in perturbation theory. The choice of renormalization scheme can affect the convergence of the perturbative expansion and the size of higher order corrections, so it is important to choose a scheme that is appropriate for the problem at hand.

If we truncate the series in lambda at some order, the choice of renormalization scheme can affect the numerical value of the coupling constant and the predictions for observables. Seeing how much the predictions change when we change the renormalization scheme can give us an estimate of the theoretical uncertainty associated with truncating the perturbative expansion.

The $\overline{\text{MS}}$ scheme can be used to renormalize the mass of the theory, but it is not a physical mass (it is not the pole of the propagator or the eigenvalue of $\hat{P}^2$), since it depends on the renormalization scale μ_R.

[...]

[...]

[...]

6.5.3 | Renormalization without Regulators

As we mentioned, in principle we can perform renormalization without introducing a regulator. This is called **renormalization without regulators**.

As an example, let us consider the 2-2 scattering in ϕ^4 theory.

$$i\mathcal{M}_{NLO} = (-i\lambda)^2 \dots$$

where $\delta\lambda$ is the counterterm for the coupling constant, and the integral is divergent and we had

$$iV(p^2) = \frac{1}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{i}{(k^2-m^2)((k+p)^2-m^2)}.$$

In the on shell scheme, we have defined

$$\delta\lambda = -\lambda^2 \int$$

to cancel the divergence of the integral, so that the NLO contribution to the amplitude is finite. We can then write the NLO amplitude as

$$i\mathcal{M}_{NLO}\big|_{os} = \dots \quad = \dots$$

where we have defined

$$\begin{aligned} i\tilde{V}(p^2) &= iV(p^2) - iV(4m^2) \\ &= \dots \\ &= \dots \end{aligned}$$

which is finite. Thus we have performed renormalization without introducing a regulator, by defining the counterterm in such a way that it cancels the divergence of the integral, and then writing the amplitude in terms of a finite quantity $\tilde{V}(p^2)$. One can compute this finite quantity explicitly with Feynman parameters...

[...]

This strategy becomes quite inconvenient...

6.6 | Integrals in Dimensional Regularization

Integrals in dimensional regularization are mathematically defined by assuming that the spacetime dimension D is such that the integral strictly converges. The result is then analytically continued to an arbitrary real (or even complex) D, excluding some isolated poles corresponding to the physical divergences.

One of the most important foundational properties of dimensional regularization is the assumption that integrands vanish at the boundary of momentum space:

$$I = \int \mathrm{d}^D k\, f(k) \implies \mathrm{d}^D k\, f(k)\Big|_{k^\mu \to \pm\infty} = 0.$$

This single defining assumption has profound operational consequences:

- **Shift Invariance:** Loop integrals are invariant under arbitrary translations of the loop momentum ($k^\mu \to k^\mu + v^\mu$). This is always allowed in dimensional regularization, which crucially preserves Ward identities and gauge invariance.
- **Integration by Parts (IBP):** The integral of a total derivative evaluates exactly to zero:

  $$\int \mathrm{d}^D k \frac{\partial}{\partial k^\mu} \left[v^\mu(k) f(k)\right] = 0.$$

  This generates powerful algebraic identities (IBP identities) that can reduce complex tensor integrals into simpler scalar master integrals.
- **Scaleless Integrals Vanish:** Any integral that is entirely independent of a dimensional scale evaluates strictly to zero. To prove this via dimensional analysis, let us say that the integrand has a mass dimension $[f(k)] \equiv \Delta_f$, which must be an integer. Because there are no other dimensionful parameters (like a mass M), under a scale transformation $k \to \lambda k$, the integral transforms homogeneously:

  $$\int \mathrm{d}^D k\, f(k) = \lambda^{D+\Delta_f} \int \mathrm{d}^D k\, f(k).$$

  This equation must hold for any arbitrary generic choice of D and λ. Because $\lambda^{D+\Delta_f} \neq 1$ in general, the only way this equality can be satisfied is if the integral itself is exactly zero.

Example. Consider the following scaleless integral:

$$\int \mathrm{d}^D k \frac{1}{(k^2)^\alpha} = 0,$$

which must vanish for any generic power α because it lacks any dimensionful scale (like a mass). We can also explicitly prove this result using Integration by Parts (IBP). We start by writing the integral of a total derivative:

$$0 = \int \mathrm{d}^D k \frac{\partial}{\partial k^\mu} \left(\frac{k^\mu}{(k^2)^\alpha}\right) = \int \mathrm{d}^D k \left[\frac{\partial_\mu k^\mu}{(k^2)^\alpha} + k^\mu \frac{\partial}{\partial k^\mu}(k^2)^{-\alpha}\right].$$

In D dimensions, $\partial_\mu k^\mu = D$. For the second term, applying the chain rule gives $\frac{\partial}{\partial k^\mu}(k^2)^{-\alpha} = -\alpha(k^2)^{-\alpha-1}(2k_\mu)$. Contracting $k^\mu(2k_\mu) = 2k^2$, we find:

$$0 = \int \mathrm{d}^D k \left[\frac{D}{(k^2)^\alpha} - \alpha \frac{2k^2}{(k^2)^{\alpha+1}}\right] = (D - 2\alpha) \int \mathrm{d}^D k \frac{1}{(k^2)^\alpha}.$$

Since this must hold for generic values of D, the integral must evaluate to zero for all α. By the principle of analytic continuation in the complex variable α, it is mathematically consistent to assume this identity remains valid even for $\alpha \leq 0$!

Example. Using the previous principle, we can evaluate integrals that initially appear to have a scale, but whose structure dictates they must vanish under analytic continuation:

$$\int \mathrm{d}^D k \frac{1}{(k^2 - M^2)^\alpha} = 0 \quad \text{if} \quad \alpha \leq 0.$$

Let us look at two specific cases:

- $(\alpha = 0)$: The integral becomes $\int \mathrm{d}^D k \cdot 1$. Since the integrand 1 has no dependence on the scale M, it is a scaleless integral and thus trivially evaluates to zero.
- $(\alpha = -1)$: The integral becomes $\int \mathrm{d}^D k (k^2 - M^2)$. We can split this by linearity into $\int \mathrm{d}^D k \cdot k^2 - M^2 \int \mathrm{d}^D k \cdot 1$. Both of these resulting integrals are purely scaleless (k^2 has no external scale, and 1 has no scale), so they both evaluate to zero. Thus, $0 - 0 = 0$.

Alternatively, this can be proven through explicit calculation by manipulating the integrand. For instance, we can write $1 = \frac{k^2}{k^2 - M^2} - M^2 \frac{1}{k^2 - M^2}$. Therefore:

$$\int \mathrm{d}^D k \cdot 1 = \int \mathrm{d}^D k \frac{k^2}{k^2 - M^2} - M^2 \int \mathrm{d}^D k \frac{1}{k^2 - M^2} = 0.$$

This evaluates to exactly zero when calculated explicitly using the standard formulas for dimensionally regulated scalar integrals (and employing the methods for dealing with numerator momentum k^2 discussed in the following section).

Thus, dimensional regularization elegantly and consistently defines how we treat scaleless integrals, which is an extremely convenient computational feature. It automatically regularizes both UV and IR divergences simultaneously while rigorously maintaining shift and gauge invariance.

6.6.1 | Integrals with Numerators

In most physically realistic theories, including QED and QCD, loop momenta frequently appear in the numerators of the integrand. This occurs naturally due to fermion propagators (which contain a $\not{k}$ term) and derivative couplings at interaction vertices. For example, a fermion line carries a factor of $\frac{i(\not{k}+m)}{k^2-m^2}$, and a scalar QED vertex introduces a factor like $-ie(p_1^\mu + p_2^\mu)$. We must systematically know how to evaluate loop integrals containing these tensor numerators.

TODO: add diagram: scalar QED vertex with incoming p1, outgoing p2, and photon line

There are many advanced techniques for dealing with tensor integrals (such as Passarino-Veltman reduction, extensively used in QCD), but here we will focus on the most direct and intuitive method based on Lorentz covariance.

The standard procedure is to first combine all denominator factors using Feynman parameters, and then perform a shift in the integration variable $k^\mu \to k^\mu + v^\mu$ to complete the square in the denominator. This ensures the denominator takes the standard spherically symmetric form:

$$\frac{1}{\text{denominator}} = \frac{1}{(k^2 - M^2)^\alpha}.$$

Crucially, when we perform this shift $k \to k + v$, the numerator also changes, generating a polynomial in the new loop momentum k. We can then expand this polynomial. Any terms that do

not contain the integration variable k (e.g., external momenta v^μ) can be factored entirely outside the integral.

Hence, to solve the full problem, we only ever need to understand how to compute fundamental "tadpole" (or vacuum) integrals where the loop momentum k appears exclusively in the numerator and the denominator depends only on k^2.

Suppose, for example, our expansion yields a term like:

$$\text{Numerator} = (k \cdot v_1)(k \cdot v_2) \cdot k^2.$$

We rewrite this by explicitly extracting the loop momentum vectors:

$$\text{Numerator} = (v_{1\rho} v_{2\sigma} g_{\alpha\beta}) \left(k^\rho k^\sigma k^\alpha k^\beta\right),$$

where the first term is independent of the loop momentum and can be factored out of the integral, while the second factor is a pure loop momentum tensor. This demonstrates that we only strictly need to consider and evaluate integrals of the general tensor form:

$$\int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{k^\mu k^\nu k^\rho \cdots}{(k^2 - M^2)^\alpha}.$$

The key observation is that after the momentum shift, this integral is a pure Lorentz tensor, but it **does not explicitly depend on any external vector**. Because Lorentz covariance must be maintained, the result can only be constructed using the invariant metric tensor $g^{\mu\nu}$.

Example (Rank-2 Tensor Integral). Consider an integral with two loop momenta in the numerator:

$$I^{\mu\nu} \equiv \int \mathrm{d}^D k \frac{k^\mu k^\nu}{(k^2 - M^2)^\alpha}.$$

Since the result must be a rank-2 symmetric tensor built only from the metric, it must be proportional to $g^{\mu\nu}$:

$$I^{\mu\nu} = g^{\mu\nu} F(M^2),$$

where $F(M^2)$ is an unknown scalar function. To find $F(M^2)$, we contract both sides of the equation with $\frac{1}{D} g_{\mu\nu}$. Remembering that in D dimensions $g_{\mu\nu} g^{\mu\nu} = D$, we get:

$$\begin{aligned} F(M^2) &= \frac{1}{D} \int \mathrm{d}^D k \frac{k^2}{(k^2 - M^2)^\alpha} \\ &= \frac{1}{D} \int \mathrm{d}^D k \frac{k^2 - M^2 + M^2}{(k^2 - M^2)^\alpha} \\ &= \frac{1}{D} \int \mathrm{d}^D k \frac{1}{(k^2 - M^2)^{\alpha-1}} + \frac{M^2}{D} \int \mathrm{d}^D k \frac{1}{(k^2 - M^2)^\alpha}. \end{aligned}$$

Notice that on the right-hand side, we have successfully reduced the tensor integral into a sum of purely scalar integrals of the exact same type we already know how to compute.

Exercise: Using the standard result for scalar integrals and the Gamma function property $z\Gamma(z) = \Gamma(z+1)$, you can show that:

$$F(M^2) = \frac{i}{(-1)^{\alpha-1}} \frac{1}{(4\pi)^{D/2}} \frac{\Gamma\left(\alpha - \frac{D}{2} - 1\right)}{\Gamma(\alpha)} \frac{1}{2} (M^2)^{\frac{D}{2} - \alpha + 1}.$$

Example (Rank-1 Tensor Integral (Odd powers)). Consider an integral with a single loop momentum in the numerator:

$$I^\mu = \int \mathrm{d}^D k \frac{k^\mu}{(k^2 - M^2)^\alpha}.$$

This integral is identically zero:

$$I^\mu = 0.$$

We can prove this in two ways:

1. **Lorentz Covariance:** The result must be a rank-1 vector. However, it is mathematically impossible to build a vector out of only the metric tensor $g^{\mu\nu}$ (which is rank-2). Therefore, the coefficient must be zero.
2. **Symmetry:** The integration domain is symmetric from $-\infty$ to $+\infty$. If we perform the change of variables $k^\mu \to -k^\mu$, the measure $\mathrm{d}^D k$ and the denominator $(k^2 - M^2)^\alpha$ remain unchanged, but the numerator flips sign. This implies $I^\mu = -I^\mu$, which can only be true if $I^\mu = 0$.

By logical extension, any tensor integral containing an **odd** number of loop momenta in the numerator (e.g., $k^\mu k^\nu k^\rho$) will always integrate to zero.

With these specific methods—Feynman parameterization, momentum shifting, and Lorentz covariant tensor reduction—you should be able to systematically handle *any* one-loop integral, reducing the entire problem down to the final integration over the Feynman parameters. We will apply these tools to concrete examples when computing quantum corrections in QED.

6.6.2 | Dirac Algebra in D Dimensions

In dimensional regularization, we need to upgrade the Dirac γ matrices to D dimensions, meaning the Lorentz index now runs over $\mu = 0, \ldots, D-1$.

However, we can consistently restrict Dirac fermions to have exactly two independent polarizations (plus two for antifermions), regardless of the spacetime dimension. Therefore, the identity matrix in spinor space remains a 4×4 matrix, and its trace is strictly fixed:

$$\mathrm{Tr}[\mathbb{I}_4] = 4.$$

(Note: This is in stark contrast to the photon gauge field A^μ, which, being a true spacetime vector, possesses $D-2$ independent physical polarizations in D dimensions).

The fundamental Clifford algebra anticommutation relations remain structurally almost the same, but we must explicitly consider the D-dimensional metric tensor $g^{\mu\nu}$:

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \times \mathbb{I}_4.$$

where the identity is still 4×4 because we are keeping the number of polarizations fixed. We can also define the trace of two gamma matrices exactly as before, yielding:

$$\mathrm{Tr}[\gamma^\mu \gamma^\nu] = 4g^{\mu\nu}.$$

All trace identities we found previously are still valid up to the promotion of the metric $g^{\mu\nu}$ to D dimensions, and the fact that the trace of the identity is still 4. However, contraction identities change because the trace of the metric tensor is now $g^\mu_\mu = D$. For instance, contracting a single

gamma matrix with itself yields:

$$\begin{aligned}\gamma^\mu\gamma_\mu &= \frac{1}{2}g_{\mu\nu}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) + \frac{1}{2}g_{\mu\nu}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) \\ &= \frac{1}{2}g_{\mu\nu}\{\gamma^\mu,\gamma^\nu\} + \frac{1}{2}\underbrace{g_{\mu\nu}[\gamma^\mu,\gamma^\nu]}_{=0\ (\text{sym}\times\text{antisym})} \\ &= g_{\mu\nu}g^{\mu\nu}\mathbb{I}_4 = D\times\mathbb{I}_4.\end{aligned}$$

Example (Exercise). We already computed the contraction of a gamma matrix around three others in four dimensions, but we can also compute it in D dimensions using the anticommutation relations repeatedly to commute γ_μ to the left:

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = -2\gamma^\sigma\gamma^\rho\gamma^\nu + (4-D)\gamma^\nu\gamma^\rho\gamma^\sigma.$$

(Hint: You can systematically derive this by applying $\gamma^\sigma\gamma_\mu = 2g^\sigma_\mu - \gamma_\mu\gamma^\sigma$ and using the intermediate identity $\gamma^\mu\gamma^\nu\gamma_\mu = (2-D)\gamma^\nu$. Notice how setting $D=4$ neatly recovers the familiar 4D result $-2\gamma^\sigma\gamma^\rho\gamma^\nu$).

6.6.3 | Infrared Divergences

Up to this point, dimensional regularization has been used to treat Ultraviolet (U.V.) divergences occurring at infinite momentum ($k\to\infty$). However, we must briefly mention that loop integrals and phase-space integrals in theories containing **massless particles** (like photons in QED) can exhibit a completely different kind of divergence.

These divergences arise from integration regions where the momentum of the massless particle approaches zero ($k\to 0$, known as "soft" divergences) or becomes perfectly parallel to another massless particle ("collinear" divergences). In these specific kinematic regimes, the denominators of the propagators strictly vanish, causing the integral to blow up. These are collectively called **Infrared (I.R.) Divergences**.

I.R. divergences must be regulated to perform intermediate calculations. This can be done by adding a fictitious small mass to the massless particles (e.g., $m_\gamma \neq 0$), or more commonly and elegantly, by continuing to use Dimensional Regularization (where I.R. poles also manifest as $\frac{1}{\epsilon}$ factors).

Crucially, unlike U.V. divergences, I.R. divergences are **NOT renormalized**. We do not absorb them into local Lagrangian counterterms. Instead, according to fundamental theorems in QFT (such as the Bloch-Nordsieck or KLN theorems), these divergences will **cancel out by themselves** when we sum over all indistinguishable physical processes to compute properly defined, inclusive observables (for example, summing the cross-sections of virtual loop corrections and the emission of real, undetectable soft photons).

6.7 | Systematics of Renormalization

Renormalization is achieved via a systematic redefinition of couplings and fields. For instance, bare parameters are related to renormalized ones via renormalization constants: $\lambda_0 = Z_\lambda \mu^{4-D} \lambda$, $m_0 = \sqrt{Z_m} m$, and $\phi_0 = \sqrt{Z_\phi}\phi$. This mathematical redefinition is fundamentally equivalent to adding counterterms (e.g., $\delta_\lambda, \delta_m, \delta_Z$) to the bare Lagrangian $\mathcal{L}$. The renormalized parameters λ, m, ϕ (or equivalently, the counterterms) are strictly fixed by specific renormalization conditions, which are designed to exactly cancel the ultraviolet (U.V.) divergencies of the bare theory.

This rigorous procedure naturally raises two fundamental questions:

- Does this method always work for *all* possible U.V. divergencies at any loop order?
- How does this procedure generalize to other quantum field theories?

To answer these questions, we must introduce a robust method to quantify how divergent a given Feynman diagram is simply by looking at its topology.

Definition 6.1 (Superficial Degree of Divergence)**.** Given a Feynman graph G, we define its superficial degree of divergence $D(G)$ as the maximum power of loop momenta in the numerator of the integrand (which includes the integration measure $\mathrm{d}^D k_i$ for each loop) minus the maximum power of loop momenta in the denominator (coming from the propagators).

$$D(G) = \begin{pmatrix} \text{Max power of loop momenta} \\ \text{in the numerator (including } \mathrm{d}^D k_i) \end{pmatrix} - \begin{pmatrix} \text{Max power of loop momenta} \\ \text{in the denominator} \end{pmatrix}$$

Naively, $D(G)$ provides a simple power-counting estimate of how divergent a graph G is. The **naive expectation** dictates:

- $D(G) \geq 0 \implies G$ is overall divergent.
- $D(G) < 0 \implies G$ is overall convergent.

Example (Feynman Graphs in $D = 4$). Let us calculate the superficial degree of divergence for two simple 1-loop diagrams in $D = 4$ dimensions.

TODO: add diagram: Tadpole diagram (1 loop, 1 internal line)

For a scalar tadpole diagram, there is one loop integral measure $\mathrm{d}^D k$ and one scalar propagator $1/k^2$:

$$D(\text{tadpole}) = D\left(\int \mathrm{d}^D k \frac{1}{k^2}\right) = D - 2.$$

In $D = 4$, $D(G) = 2$. Regulating this with a hard momentum cutoff Λ, the integral scales as $\int^\Lambda \frac{\mathrm{d}^4 k}{k^2} \sim \Lambda^2$. This represents a **quadratic divergence**.

TODO: add diagram: Fish diagram (1 loop, 2 internal lines)

For a scalar "fish" diagram (e.g., the 1-loop s-channel correction to $2 \to 2$ scattering), there is one loop integral and two propagators:

$$D(\text{fish}) = D\left(\int \mathrm{d}^D k \frac{1}{k^2}\frac{1}{k^2}\right) = D - 4.$$

In $D = 4$, $D(G) = 0$. Using a cutoff Λ, the integral scales as $\int^\Lambda \frac{\mathrm{d}^4 k}{k^4} \sim \log \Lambda$. This represents a **logarithmic divergence**.

The naive expectation, however, is not a flawless rule. It can be wrong in two distinct ways:

1. The graph G (or the specific Green's function it contributes to) can be **less divergent** than $D(G)$ predicts. This typically happens if there are non-trivial algebraic cancellations of divergent terms in the numerator (often enforced by symmetries, such as gauge invariance or Ward identities).

2. The graph G can be **more divergent** than $D(G)$ predicts if it contains specific **sub-loops** (sub-divergences) that possess a higher degree of divergence than the overall graph.

   Example (Divergences in sub-loops). Consider a 2-loop graph composed of a 4-point box diagram (loop momentum k_1) where one internal line features a tadpole insertion (loop momentum k_2). The overall integral schematically looks like $\int \mathrm{d}^4k_1 \mathrm{d}^4k_2 \frac{1}{(k_1^2)^4} \frac{1}{k_2^2}$. Applying our formula for the total graph:

   $$D(G) = 4(\text{from } \mathrm{d}^4k_1) + 4(\text{from } \mathrm{d}^4k_2) - 8(\text{from 4 propagators of } k_1) - 2(\text{from 1 propagator of } k_2) = -2$$

   Naively, since $D(G) = -2 < 0$, one would expect the entire graph to be finite and convergent. However, the isolated integration over the sub-loop k_2 is a tadpole, which we already know is quadratically divergent ($D(\text{sub-loop}) = 2$). The overall graph inherits this strict divergence, proving the naive power-counting of the full graph insufficient.

TODO: add diagram: A 1-loop 4-point box diagram, where one of the internal lines has an additional 1-loop tadpole insertion.

To systematically study the divergences of theories, we temporarily ignore these sub-divergences (which are rigorously handled by BPHZ renormalization and nested counterterms) and focus strictly on understanding the overall $D(G)$. Furthermore, we restrict our attention entirely to **1PI** (One-Particle Irreducible) graphs and Green's functions. Any arbitrary diagram is simply a product of 1PI diagrams connected by single propagators (multiplied by a rational symmetry factor). In a 1PI diagram, every internal line strictly belongs to at least one loop.

For any generic graph G, we define the following topological properties:

- $L \equiv$ number of loops (independent loop momenta)
- $I \equiv$ number of internal lines (propagators)
- $E \equiv$ number of external lines
- $V \equiv$ number of vertices

The number of independent loop momenta L is exactly the number of internal momenta that are *not* fixed by momentum conservation at the vertices. Each internal line carries an independent momentum (I variables). Each vertex enforces a momentum conservation constraint (V constraints). However, one of these delta functions always corresponds to overall global momentum conservation ($\delta(P_{\text{in}} - P_{\text{out}})$), which constrains external momenta, not internal ones. Therefore, there are $V - 1$ constraints on the internal lines. This yields **Euler's Graph Formula**:

$$L = I - (V - 1) = I - V + 1.$$

6.7.1 | Topological Analysis of ϕ^N Theory

Let us apply this topological framework to a generic scalar ϕ^N theory, defined by the Lagrangian:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{m^2}{2}\phi^2 - \frac{\lambda_N}{N!}\phi^N + \text{C.T.}$$

Consider a 1PI graph G in this theory. Every vertex strictly connects to N lines. Internal lines connect to exactly 2 vertices (one at each end), while external lines connect to exactly 1 vertex. By counting the total number of line endpoints attached to all vertices, we obtain the relation:

$$NV = 2I + E.$$

We can use this to eliminate V from Euler's formula. Substituting $V = \frac{2I+E}{N}$ into $L = I - V + 1$:

$$L = I - \frac{2I+E}{N} + 1 \implies L - 1 + \frac{E}{N} = I\left(\frac{N-2}{N}\right).$$

Solving for the number of internal lines I:

$$I = \frac{NL + E - N}{N-2}.$$

Now, we can compute the superficial degree of divergence $D(G)$ for this theory. Each loop provides a measure $\mathrm{d}^D k \sim k^D$, contributing $+D$ to the degree. Each internal line is a scalar propagator $1/k^2 \sim k^{-2}$, contributing -2. Therefore:

$$D(G) = DL - 2I.$$

Substituting our topological expression for I:

$$D(G) = DL - 2\left(\frac{NL + E - N}{N-2}\right) = L\left(D - \frac{2N}{N-2}\right) - \frac{2}{N-2}E + \frac{2N}{N-2}.$$

Let us strictly consider the physical dimension $D = 4$. The coefficient of L becomes $4 - \frac{2N}{N-2} = \frac{4N-8-2N}{N-2} = \frac{2N-8}{N-2}$. The formula simplifies to:

$$D(G)\Big|_{D=4} = 2L\left(\frac{N-4}{N-2}\right) - \frac{2}{N-2}E + \frac{2N}{N-2}.$$

This master equation reveals the profound systematics of renormalization in scalar theories:

- **More external legs** (E) always makes the graph more convergent. Because the coefficient $-\frac{2}{N-2}$ is strictly negative for $N > 2$, adding external lines inherently drives $D(G)$ lower. This is why theories typically only have a finite number of divergent Green's functions (e.g., 2-point and 4-point functions).

- **More loops** (L) affects the divergence depending entirely on the interaction power N:

  - For $\mathbf{N < 4}$ (e.g., ϕ^3 theory): The coefficient of L is negative. Adding more loops makes the graph **more convergent** (super-renormalizable).

  - For $\mathbf{N > 4}$ (e.g., ϕ^6 theory): The coefficient of L is strictly positive. Adding more loops makes the graph **more divergent**. Every new perturbative order introduces worse divergencies, requiring an infinite number of distinct counterterms (non-renormalizable).

  - For $\mathbf{N = 4}$ (ϕ^4 theory): The coefficient of L is exactly zero. The degree of divergence $D(G) = 4 - E$ is remarkably **independent of the number of loops**. The divergencies remain completely stable and manageable at all orders in perturbation theory (strictly renormalizable).

ϕ^N Theory: $N = 4$ case (Strictly Renormalizable)

Let us explicitly analyze the standard ϕ^4 theory in four dimensions ($D = 4$). Substituting $N = 4$ into our master formula yields a superficial degree of divergence that is entirely independent of the loop order L:

$$D(G) = 4 - E.$$

A generic 1PI diagram is superficially divergent only if $D(G) \geq 0$, which implies $E \leq 4$. This means that the only superficially divergent graphs have 4 or fewer external legs, at all loop orders.

Let us systematically list these divergent cases:

a) $E = 1$ (Tadpole on a line, $D(G) = 3$): This represents a shift in the vacuum expectation value $\langle\phi\rangle$. However, we generally construct our perturbation theory such that $\langle\Omega|\hat{\phi}|\Omega\rangle = 0$ exactly (by shifting the field if necessary). Furthermore, in a symmetric ϕ^4 theory, all odd-point functions vanish identically due to the $\phi \to -\phi$ discrete symmetry. Thus, we can safely ignore $E = 1$.

TODO: add diagram: 1-point function (a line ending in a loop)

b) $E = 2$ (Propagator corrections, $D(G) = 2$): These quadratic divergencies (and their logarithmic sub-leading parts) directly correct the mass and the field normalization. They are fully absorbed and renormalized by the mass counterterm δ_m and the wave-function renormalization counterterm δ_Z.

TODO: add diagram: 2-point function blob

c) $E = 3$ (3-point vertex, $D(G) = 1$): Just like the $E = 1$ case, any 3-point correlation function $\langle\phi\phi\phi\rangle = 0$ due to the fundamental $\phi \to -\phi$ symmetry of the ϕ^4 Lagrangian. Therefore, these diagrams do not exist in the symmetric theory.

TODO: add diagram: 3-point function blob

d) $E = 4$ (4-point interaction, $D(G) = 0$): These logarithmic divergencies correct the interaction vertex. They are fully absorbed and renormalized by the coupling constant counterterm δ_λ.

TODO: add diagram: 4-point function blob

Conclusion for $N = 4$: The counterterms $\delta_Z, \delta_m, \delta_\lambda$ (which are generated by redefining the parameters already present in the original bare Lagrangian) are sufficient to make the ϕ^4 theory strictly finite at all loops. While the counterterms themselves will receive divergent higher-order corrections at every new loop level, we never need to introduce *new types* of interactions. We say that ϕ^4 in $D = 4$ is **renormalizable** precisely because ALL ultraviolet divergencies can be removed by a **finite number** of counterterms (in this case, just three).

ϕ^N Theory: $N < 4$ case (Super-Renormalizable)

Consider a theory where $N < 4$, specifically the ϕ^3 theory in $D = 4$. Setting $N = 3$ in our master formula gives:

$$D(G) = -2L - 2E + 6.$$

Here, $D(G)$ explicitly decreases as the number of loops L increases. This implies that for a sufficiently large number of loops, $D(G) < 0$ for *all* values of E. Consequently, there is only a strictly finite number of superficially divergent diagrams in the entire theory. Beyond a certain specific loop order, the theory becomes perfectly finite (up to handling nested sub-divergencies from lower loop orders).

Let us break this down by loop order for ϕ^3:

- **At 1-loop ($L = 1$):** $D(G) = 4 - 2E$. Divergencies ($D(G) \geq 0$) exist only for $E \leq 2$. This includes the 1-point tadpole ($E = 1$) and the 2-point propagator correction ($E = 2$).
- **At 2-loops ($L = 2$):** $D(G) = 2 - 2E$. Divergencies exist only for $E = 1$ (the 2-loop tadpole). As mentioned before, we generally enforce $\langle\Omega|\hat{\phi}|\Omega\rangle = 0$, so we can often ignore this.
- **At 3 or higher loops ($L \geq 3$):** $D(G) < 0$ strictly for all $E \geq 1$. There are absolutely no new superficial divergencies generated at 3 loops or beyond!

Because the divergencies naturally cure themselves at higher orders, we classify such theories as **super-renormalizable**.

ϕ^N Theory: $N > 4$ case (Non-Renormalizable)

Consider a theory where $N > 4$, such as a ϕ^6 interaction in $D = 4$. Setting $N = 6$ in our master formula yields:

$$D(G) = 2L\left(\frac{2}{4}\right) - \frac{2}{4}E + \frac{12}{4} = L - \frac{1}{2}E + 3.$$

In stark contrast to the previous cases, the superficial degree of divergence now explicitly *increases* with the loop order L. This means the number of superficially divergent Green's functions grows uncontrollably as we calculate to higher orders in perturbation theory.

Let us track the divergencies for ϕ^6:

- **At 1-loop ($L = 1$):** $D(G) = 4 - E/2$. Divergencies occur for $E \leq 8$.
- **At 2-loops ($L = 2$):** $D(G) = 5 - E/2$. Divergencies occur for $E \leq 10$.
- **At L-loops:** Divergencies occur for $E \leq 2L + 6$.

TODO: add diagram: 1-loop diagram with 8 external legs

What are the catastrophic physical implications of this? At 1-loop, we discover a divergence for the $E = 8$ Green's function. To mathematically REMOVE this divergence, we are absolutely required to introduce an 8-point counterterm vertex into our Lagrangian. However, this counterterm ($\delta_8\phi^8$) corresponds to a fundamental ϕ^8 interaction. Therefore, we are *forced* to add a completely new interaction term $\sim \frac{\lambda_8}{8!}\phi^8$ to our original bare Lagrangian to absorb the divergence (even if we initially set the physical coupling $\lambda_8 = 0$, the counterterm $\delta_8 \neq 0$ is mandatory).

TODO: add diagram: 2-loop diagram with 10 external legs (e.g., a circle with a line through the middle)

The problem violently compounds at the next order. At two-loops, we get a divergence for the $E = 10$ Green's function. To correct this new divergence, we desperately need a $\delta_{10}\phi^{10}$ counterterm in $\mathcal{L}$, which forces us to introduce a $\frac{\lambda_{10}}{10!}\phi^{10}$ interaction.

At HIGHER and HIGHER loop orders, we are forced to add MORE and MORE distinct interaction terms to the Lagrangian just to make the theory finite. Because it cannot be made finite at all loops using a finite number of counterterms, the theory loses its predictive power (you would need an infinite number of experimental measurements to fix the infinite number of couplings). Consequently, ϕ^6 in 4 dimensions is classified as strictly **non-renormalizable**.

6.7.2 | Higher-Loop Renormalization

We need to definitively address the question of how to deal with divergences in sub-loops. In the previous topological analysis, we discussed a scenario where the overall superficial degree of divergence indicated a convergent graph, but a localized internal divergence remained:

In this graph, we have an overall $D(G) < 0$, but it still contains a severe divergence due to the internal sub-loop. To construct a rigorously finite theory, we need to make sure that we can consistently renormalize the sub-loop first, and then the whole graph, without spoiling the perturbative expansion.

TODO: add diagram: 4-point graph with a 1-loop subloop on one of the internal lines

The divergence of the sub-loop, however, can be independently removed by a standard lower-loop counterterm, which mathematically acts as a local operator insertion. This means that the divergence of the sub-loop is locally absorbed into a redefinition of the parameters of the theory at that specific lower loop order. Consequently, we can consistently renormalize the whole higher-loop graph by summing the bare graph with the corresponding counterterm insertion:

Because the 1-loop sub-diagram plus its 1-loop counterterm sum to a strictly finite quantity, inserting this finite combination into the larger momentum integral yields a finite result for the overall 2-loop graph.

TODO: add diagram: Sum of the 4-point graph with the subloop + the 4-point graph with a 1-loop counterterm insertion = FINITE

BPHZ Theorem

In general, disentangling all the nested and overlapping divergences of an arbitrary higher-loop graph can be combinatorially and analytically complicated, but it can be shown that it is always possible to systematically do so. Indeed, sub-loop divergences are iteratively removed by counterterms generated at lower loop orders. After this systematic subtraction procedure is completed, we are left with a graph that only contains its overall superficial divergence. This remaining divergence can then be renormalized by a local counterterm of the exact same form as the operators in the original bare Lagrangian.

This hierarchical structure of renormalization is formalized in one of the most important theorems in quantum field theory:

Theorem 6.1 (BPHZ Theorem)**.** *All ultraviolet divergencies of a Feynman graph are completely removed by the counterterms that correspond specifically to the* ***superficially divergent 1PI Green functions****.*

(Bogoliubov, Parasiuk, Hepp, Zimmermann)

The rigorous mathematical proof is quite complicated and relies on the "forest formula" to systematically resolve overlapping divergences. It can be found in more advanced textbooks on quantum field theory (e.g., *Renormalization* by J. Collins, or *The Quantum Theory of Fields, Volume 1* by S. Weinberg).

TODO: add reference to proof

The BPHZ theorem guarantees a profound physical conclusion: we can consistently renormalize any quantum field theory as long as we possess a finite number of counterterms corresponding to the superficially divergent 1PI Green functions. This implies that we can robustly study the renormalizability of entirely new theories using *only* their superficial degree of divergence $D(G)$, without having to manually worry about the intricate pathologies of sub-loops!

6.7.3 | Systematics of Renormalization for QED

We are now going to repeat the analysis of the superficial degree of divergence specifically for Quantum Electrodynamics (QED), and see how this power counting works in practice.

Let us define E_ψ and I_ψ as the number of external and internal Dirac (fermionic) lines, respectively, and E_γ and I_γ as the number of external and internal photon lines.

TODO: add diagram: Fermion line passing through a vertex to which one photon line is attached

We are interested in counting how many times a fermionic or photon line attaches to a vertex V. Because every fundamental QED vertex couples exactly two fermion lines and one photon line, we can establish rigorous topological conservation laws. We must distinguish between internal lines (which attach to two vertices) and external lines (which attach to only one vertex). Counting the total number of line endpoints terminating at the vertices yields:

$$V = 2I_\gamma + E_\gamma \tag{6.7.1}$$

$$2V = 2I_\psi + E_\psi \tag{6.7.2}$$

Using these relations alongside Euler's graph formula, which relates the number of loops L to the total internal lines $I = I_\psi + I_\gamma$ and vertices V:

$$L = I - V + 1,$$

we can express the number of internal lines strictly in terms of loops and external legs. From (6.7.1) we get $I_\gamma = \frac{1}{2}(V - E_\gamma)$, and from (6.7.2) we get $I_\psi = V - \frac{1}{2}E_\psi$. Substituting these into Euler's formula gives:

$$\begin{aligned} L &= \left(\frac{1}{2}V - \frac{1}{2}E_\gamma\right) + \left(V - \frac{1}{2}E_\psi\right) - V + 1 \\ L &= \frac{1}{2}V - \frac{1}{2}E_\gamma - \frac{1}{2}E_\psi + 1 \\ \implies V &= 2L + E_\gamma + E_\psi - 2. \end{aligned}$$

Now, substituting V back into our expressions for the internal lines, we find:

$$\begin{aligned} I_\gamma &= L + \frac{1}{2}E_\psi - 1, \\ I_\psi &= 2L + E_\gamma + \frac{1}{2}E_\psi - 2. \end{aligned}$$

To compute the superficial degree of divergence, we examine the asymptotic behavior of the propagators in the deep ultraviolet regime as loop momenta $k \to \infty$:

TODO: add diagram: fermion line

TODO: add diagram: photon line

$$\frac{i}{\not{k} - m} \sim \frac{1}{k} \quad \text{for each internal fermion line}$$

$$\frac{-ig^{\mu\nu}}{k^2} \sim \frac{1}{k^2} \quad \text{for each internal photon line}$$

Therefore, each loop integration contributes $\mathrm{d}^D k \sim k^D$, each internal photon suppresses the divergence by k^{-2}, and each internal fermion suppresses it by k^{-1}. The superficial degree of divergence is:

$$D(G) = DL - 2I_\gamma - I_\psi.$$

Substituting our topological solutions for I_γ and I_ψ:

$$\begin{aligned} D(G) &= DL - 2\left(L + \frac{1}{2}E_\psi - 1\right) - \left(2L + E_\gamma + \frac{1}{2}E_\psi - 2\right) \\ &= (D-4)L - \left(E_\gamma + \frac{3}{2}E_\psi\right) + 4. \end{aligned}$$

In physical four-dimensional spacetime ($D = 4$), the term proportional to L vanishes identically, leaving:

$$D(G) = 4 - E_\gamma - \frac{3}{2}E_\psi.$$

This profoundly demonstrates that **QED is renormalizable by power counting in** $D = 4$. The superficial degree of divergence depends exclusively on the number of external legs, not on the loop order L. A sufficiently large number of external legs will always render the graph convergent, while only a small, finite set of specific configurations will be divergent. In particular, a graph is superficially divergent if:

$$D(G) \geq 0 \quad \text{which requires} \quad \left(E_\gamma + \frac{3}{2}E_\psi\right) \leq 4.$$

Because fermions must appear in pairs (E_ψ is an even integer), the only superficially divergent 1PI Green functions are the following six topologies:

TODO: add diagram: 6 superficially divergent 1PI Green functions: a) 1-photon tadpole, b) 2-photon self-energy, c) 3-photon vertex, d) 4-photon light-by-light scattering, e) 2-fermion self-energy, f) 2-fermion 1-photon vertex

(a) **Photon Tadpole** ($E_\gamma = 1, E_\psi = 0 \implies D = 3$)**:** This graph is exactly zero because $\langle\Omega|A^\mu|\Omega\rangle = 0$ by Lorentz invariance (the vacuum cannot have a preferred vector direction).

(b) **Photon Vacuum Polarization** ($E_\gamma = 2, E_\psi = 0 \implies D = 2$)**:** Renormalized by the photon wave-function counterterm Z_A.

(c) **Three-Photon Vertex** ($E_\gamma = 3, E_\psi = 0 \implies D = 1$)**:** This graph evaluates to exactly zero due to **Furry's Theorem**, which states that correlation functions featuring an odd number of external photon legs coupled to a closed fermion loop vanish identically due to charge conjugation symmetry (C-parity). Note that Furry's theorem also independently forces graph (a) to be zero.

(d) **Light-by-Light Scattering** ($E_\gamma = 4, E_\psi = 0 \implies D = 0$)**:** Naively logarithmically divergent. However, it can be proven to be strictly **convergent** due to gauge invariance. This is a non-trivial result which we will prove below.

(e) **Fermion Self-Energy** ($E_\gamma = 0, E_\psi = 2 \implies D = 1$)**:** Renormalized by the fermion wave-function counterterm Z_ψ and the mass counterterm Z_m.

(f) **Fermion-Photon Vertex** ($E_\gamma = 1, E_\psi = 2 \implies D = 0$)**:** Renormalized by the electric charge counterterm Z_e.

Let us look at graph (d), the 4-photon light-by-light scattering, in closer detail. Its superficial degree of divergence is $D(G) = 0$, meaning that in principle, if a divergence exists, it must be purely logarithmic.

TODO: add diagram: Light-by-light scattering 4-photon vertex with momenta p1, p2, p3, p4 and Lorentz indices mu, nu, rho, sigma

Because the divergence is logarithmic, after systematically removing all sub-divergences (via BPHZ), the remaining overall divergent contribution comes exclusively from the leading behavior as the loop momentum $|k| \to \infty$. In this deep U.V. limit, the leading divergent term must be independent of all dimensional parameters, including fermion masses and the external photon momenta p_i^μ.

Hence, the divergent piece of the amplitude must take the form:

$$\mathcal{M}^{\mu\nu\rho\sigma} \sim f(\epsilon) \times C^{\mu\nu\rho\sigma} + \text{FINITE},$$

where $f(\epsilon)$ contains the $1/\epsilon$ divergence in dimensional regularization, and $C^{\mu\nu\rho\sigma}$ is a strictly constant tensor. Because it is independent of external momenta, $C^{\mu\nu\rho\sigma}$ can only be constructed using the invariant metric tensor $g^{\mu\nu}$.

To cancel this hypothetical divergence, we would be forced to introduce a local 4-point counterterm into the Lagrangian whose Feynman rule generates exactly $C^{\mu\nu\rho\sigma}$. Because the rule must be

independent of momentum, this counterterm must lack spacetime derivatives, leading to an effective Lagrangian interaction term of the form:

$$\mathcal{L}_{\text{counterterm}} \sim c\frac{1}{4!}(A_\mu A^\mu)^2.$$

However, this term is blatantly **not gauge invariant** under $A_\mu \to A_\mu + \partial_\mu \alpha$. Since the original QED Lagrangian is gauge invariant, it cannot dynamically generate a counterterm that violently breaks the gauge symmetry of the theory. Therefore, the constant c must be strictly zero.

Equivalently, we can prove this mathematically using Bose symmetry and the Ward-Takahashi identities. Since photons are bosons, the amplitude must be symmetric under the exchange of any two photon legs. The most general constant tensor built from $g^{\mu\nu}$ satisfying total symmetry is:

$$C^{\mu\nu\rho\sigma} = c\,(g^{\mu\nu}g^{\rho\sigma} + g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}),$$

for some arbitrary scalar constant c. The exact Ward identity guarantees that contracting the amplitude with any external photon momentum must yield zero:

$$p_{1\mu}C^{\mu\nu\rho\sigma} = 0.$$

Contracting $p_{1\mu}$ with our tensor structure gives $c\,(p_1^\nu g^{\rho\sigma} + p_1^\rho g^{\nu\sigma} + p_1^\sigma g^{\nu\rho}) = 0$. Since p_1 is an arbitrary external momentum, the only way this equation holds identically for all kinematic configurations is if the proportionality constant vanishes:

$$c = 0.$$

Therefore, the constant tensor $C^{\mu\nu\rho\sigma}$ is identically zero. The potential logarithmic divergence vanishes entirely, and the light-by-light scattering graph is strictly **finite and convergent**! Consequently, QED only formally requires counterterms for the electron self-energy, the photon vacuum polarization, and the fundamental interaction vertex.

6.7.4 | Renormalizability from Dimensional Analysis

It can be shown that the renormalizability of quantum field theories is fundamentally and intimately related to the mass dimension of the coupling constants and the operators present in the interaction Lagrangian.

Consider a general theory with an arbitrary number of fundamental fields:

- scalar fields ϕ_i
- Dirac fields ψ_i
- massless spin-1 vector fields A_i

For all these fields, the ultraviolet (UV) limit of their propagators behaves as a simple power law $\sim 1/k^\alpha$. Specifically, as $k \to \infty$, scalar and vector propagators behave as $1/k^2$, while fermion propagators behave as $1/\not{k} \sim 1/k$. Furthermore, we know that Feynman rules for interaction vertices are always either pure constants or polynomials in the momenta (arising from the substitution of derivatives $\partial_\mu \to \pm i p_\mu$ in the Lagrangian).

Therefore, as all loop momenta $k_i \to \infty$, the leading divergent term of any Feynman integral is obtained simply by neglecting all finite masses and all fixed external momenta.

TODO: add diagram: generic vertex with internal loop momenta k and external momentum p, showing the simplification in the UV limit

For example, a generic 1-loop sub-integral simplifies in the deep UV as:

$$\int \frac{\mathrm{d}^4k}{(k^2-m^2)((k+p)^2-m^2)} \sim \int^{\Lambda} \frac{\mathrm{d}^4k}{k^4} \sim \int^{\Lambda} \frac{k^3\mathrm{d}k}{k^4} \sim \log\Lambda.$$

Because the leading term of the integrand depends only on the loop momenta k_i as $k_i \to \infty$, the superficial degree of divergence $D(G)$ exactly coincides with the **mass dimension** of the integral, provided we explicitly factor out and remove all coupling constants. If the simplified integral has the form $\int \mathrm{d}^Dk_1 \cdots \mathrm{d}^Dk_L \frac{1}{k^\alpha}$ (where k^α represents any homogeneous combination of the loop momenta), then:

$$D(G) = DL - \alpha = \left[\int \mathrm{d}^Dk_1 \cdots \mathrm{d}^Dk_L \frac{1}{k^\alpha}\right].$$

To exploit this powerful dimensional connection, we must look at the mass dimension of the **amputated 1PI Green functions**, $\mathcal{M}(E_\phi, E_\psi, E_\gamma)$, which correspond strictly to the core loop diagrams without the external propagators.

A generic amputated Green function in momentum space is defined via the LSZ reduction-like formula:

$$\mathcal{M}(E_\phi, E_\psi, E_\gamma)\delta^{(4)}(\sum p_{\text{in}} - \sum p_{\text{out}}) \sim \int \mathrm{d}^4x_1 \cdots \mathrm{d}^4x_E e^{i\sum p\cdot x} \frac{\langle \phi\cdots\phi\,\psi\cdots\psi\,A\cdots A\rangle}{\prod \text{External Propagators}}$$

where $E = E_\phi + E_\psi + E_\gamma$ is the total number of external legs. Let us carefully compute the exact mass dimension of this amputated amplitude $\mathcal{M}$. We use the standard mass dimensions of the fields in $D = 4$: $[\phi] = 1$, $[A] = 1$, and $[\psi] = 3/2$. The dimension of a scalar/vector propagator in momentum space is $[1/p^2] = -2$, and a fermion propagator is $[1/\not{p}] = -1$. The Dirac delta function for momentum conservation has dimension $[\delta^{(4)}(p)] = -4$. The integration measure d^4x has dimension -4.

Taking the mass dimension of both sides of the generic equation:

$$\begin{aligned}
[\mathcal{M}] + [\delta^{(4)}] &= -4E + E_\phi[\phi] + E_\psi[\psi] + E_\gamma[A] - E_\phi\left[\frac{1}{p^2}\right] - E_\psi\left[\frac{1}{\not{p}}\right] - E_\gamma\left[\frac{1}{p^2}\right] \\
[\mathcal{M}] - 4 &= -4E_\phi - 4E_\psi - 4E_\gamma + E_\phi(1) + E_\psi(3/2) + E_\gamma(1) - E_\phi(-2) - E_\psi(-1) - E_\gamma(-2) \\
[\mathcal{M}] - 4 &= -E_\phi - \frac{3}{2}E_\psi - E_\gamma \\
[\mathcal{M}] &= 4 - E_\phi - E_\gamma - \frac{3}{2}E_\psi.
\end{aligned}$$

This beautifully confirms that the mass dimension of the amputated Green function $[\mathcal{M}]$ exactly dictates its baseline kinematic dimension.

Now, consider a generic interaction Lagrangian expressed as a sum of coupling constants g_j multiplied by local operators O_j:

$$\mathcal{L}_{\text{int}} = \sum_j g_j O_j(\phi_i, \psi_i, A_i).$$

Because the action $S = \int \mathrm{d}^4x\mathcal{L}$ must be strictly dimensionless ($[S] = 0$), the Lagrangian density must have mass dimension $[\mathcal{L}] = 4$. Therefore, for each interaction term:

$$\Delta_j \equiv [g_j] = 4 - [O_j].$$

If we compute a specific Feynman graph G that contains N_j vertices of type j, the analytical expression for the graph takes the structural form:

$$G \sim \left(\prod_j g_j^{N_j}\right) \times \int \mathrm{d}^Dk_1 \cdots \mathrm{d}^Dk_L \frac{1}{k^\alpha}.$$

Because the total graph G evaluates to the amputated amplitude $\mathcal{M}$, their mass dimensions must be strictly equal. Taking the dimension of the right-hand side:

$$[\mathcal{M}] = \sum_j N_j[g_j] + \left[\int \mathrm{d}^D k_1 \cdots \mathrm{d}^D k_L \frac{1}{k^\alpha}\right] = \sum_j N_j \Delta_j + D(G).$$

Substituting our previously derived expression for $[\mathcal{M}]$, we can solve directly for the superficial degree of divergence $D(G)$:

$$D(G) = 4 - E_\phi - E_\gamma - \frac{3}{2}E_\psi - \sum_j N_j \Delta_j.$$

This is an incredibly powerful, general result. It proves that the superficial degree of divergence of a graph is rigidly controlled by the mass dimensions Δ_j of the coupling constants in the theory. Because higher-loop diagrams necessarily require more vertices (larger N_j) to form those loops, the behavior of the summation term determines the renormalizability of the entire theory. Let us categorize the three possible physical scenarios based on $\Delta_j = [g_j]$:

- **$\Delta_j \geq 0$ for all j (Renormalizable):** If all couplings have positive or zero mass dimension, then $-\sum N_j \Delta_j \leq 0$. Thus, $D(G) \leq 4 - E_\phi - E_\gamma - \frac{3}{2}E_\psi$. The maximum degree of divergence is strictly bounded from above and is completely independent of the number of vertices N_j (and therefore independent of the loop order L). There are only a fixed, finite number of divergent Green functions at all orders in perturbation theory, meaning the theory can be fully renormalized by a finite number of counterterms. Crucially, since $\Delta_j \geq 0 \iff [O_j] \leq 4$, this implies that **theories where all operators in the Lagrangian have mass dimension ≤ 4 are strictly renormalizable**. This places an immensely strong mathematical constraint on the allowed interactions in nature.

- **$\Delta_j > 0$ strictly for all j (Super-Renormalizable):** If all couplings have strictly positive mass dimension (e.g., a mass term $m^2\phi^2$ where $[m^2] = 2$, or a cubic coupling $\lambda_3\phi^3$ where $[\lambda_3] = 1$), then as the number of vertices N_j increases at higher loop orders, the term $-\sum N_j \Delta_j$ becomes increasingly negative. Diagrams with more vertices are fundamentally more convergent. Beyond a certain loop order, every single 1PI diagram becomes superficially convergent. The theory only has a finite number of divergent graphs total. Theories where **all operators have** $[O_j] < 4$ are super-renormalizable.

- **$\Delta_j < 0$ for at least one j (Non-Renormalizable):** If even a single interaction has a coupling with negative mass dimension, then as we build more complex graphs using this vertex, N_j increases. Because Δ_j is negative, the term $-N_j\Delta_j$ becomes a large positive number. The superficial degree of divergence $D(G)$ will inevitably become positive and grow without bound as the loop order increases. Regardless of how many external legs the Green function has, for large enough L, $D(G)$ will be positive. Every single Green function eventually becomes violently divergent at some loop order. To absorb these divergencies, we must add an infinite number of distinct counterterms, rendering the theory devoid of predictive power. Theories containing **at least one operator with** $[O_j] > 4$ are strictly non-renormalizable.

Example (Massive Spin 1). Why is a massive spin-1 field fundamentally different from a massless spin-1 field (like the photon) in terms of renormalizability? The reason lies in the

ultraviolet asymptotic behavior of its propagator. A massive vector boson propagator behaves as:

$$\Pi_{\mu\nu}(k) = \frac{i}{k^2 - m^2 + i\epsilon}\left(-g_{\mu\nu} + \frac{k_\mu k_\nu}{m^2}\right).$$

As we take the deep UV limit ($|k| \to \infty$), the longitudinal part of this propagator dominates:

$$\Pi_{\mu\nu}(k) \xrightarrow{|k|\to\infty} \frac{ik_\mu k_\nu}{m^2 k^2} \sim \mathcal{O}(k^0).$$

Unlike a scalar, fermion, or massless vector propagator, this leading term does *not* drop off as $1/k^\alpha$. It asymptotically behaves as a constant (order k^0). Furthermore, this leading term is not independent of the mass m, violently breaking the core assumptions we made during our dimensional analysis derivation. Because it does not provide any momentum suppression in the UV, each internal massive vector line contributes exactly 0 to the negative powers in the power counting.

Indeed, if we were to stubbornly consider a QED-like theory with a massive photon (without modifying the fundamental gauge structure), the derivations of the superficial degree of divergence change drastically. The power counting becomes:

$$D(G) = DL - 0 \times I_\gamma - I_\psi.$$

Using our topological relations $V = 2I_\gamma + E_\gamma = 2I_\psi + E_\psi$ and Euler's formula $L = I_\gamma + I_\psi - V + 1$, we find:

$$D(G) = (D-2)L - E_\gamma - \frac{E_\psi}{2} + 2.$$

In strictly 4 dimensions ($D = 4$), the coefficient of the loop order L is strictly positive ($+2L$). This means that the superficial degree of divergence explicitly grows with the number of loops. The theory has an infinite number of divergent Green functions and is therefore **strictly non-renormalizable**.

Remark (Higgs Mechanism). *The devastating conclusion above poses a severe problem: how can nature permit massive spin-1 particles, like the $W^\pm$ and Z^0 bosons of the weak interaction, if they destroy renormalizability? The solution is the **Higgs Mechanism**. A "massless" vector boson can dynamically acquire a mass via spontaneous symmetry breaking without spoiling the underlying gauge invariance of the theory. In this framework, the dangerous longitudinal polarization ($k_\mu k_\nu / m^2$) is intrinsically tied to a scalar degree of freedom (the Goldstone boson), allowing the theory to remain perfectly renormalizable (as seen in the Standard Model).*

Example (Gravity). We will not be extensively concerned with gravity in this course, but it is illuminating to note that general relativity, treated as a standard QFT, is a non-renormalizable theory. The classical Newtonian gravitational potential behaves as:

$$V(r) \sim -G\frac{m_1 m_2}{r}.$$

The coupling constant of gravity is Newton's constant G. In natural units ($\hbar = c = 1$), $[G] = -2$. Because the coupling constant has a strictly negative mass dimension ($\Delta_j < 0$), our previous dimensional analysis theorem immediately classifies the theory as **non-renormalizable**.

We can certainly attempt to obtain a QFT by perturbatively quantizing the metric field $g_{\mu\nu}$ around flat spacetime, but because $[G] < 0$, the degree of divergence grows uncontrollably at higher loop orders. We would need an infinite sequence of higher-derivative counterterms to absorb the divergencies. While this effectively ruins it as a fundamental theory at all scales

(hence the need for a complete theory of quantum gravity, like string theory), this quantized general relativity perfectly functions as a highly predictive *effective field theory* at low energies, where the non-renormalizable corrections are incredibly suppressed.

6.7.5 | Renormalizable Theories

We can now cleanly state the condition for a theory to be renormalizable in terms of its ultraviolet behavior and the mass dimension of its operators:

*A theory is renormalizable if **all** its ultraviolet divergences can be cancelled by a **finite** number of counterterms at **all** orders in perturbation theory.*

It identically follows that in a renormalizable theory, we can make robust, finite predictions at absolutely any order in perturbation theory using only a finite set of experimentally measured input parameters (coupling constants and masses).

We have rigorously established that Lagrangians constructed from spin-0 (scalars), spin-1/2 (Dirac fermions), and massless spin-1 (gauge bosons) fields, containing *only* interactions with coupling mass dimensions $[g_j] \geq 0$ (equivalently, operator dimensions $[O_j] \leq 4$), are renormalizable.

However, a subtle and deeply important consequence arises: even in a supposedly renormalizable theory, we may be logically forced to *add* new interaction terms to the bare Lagrangian to ensure it remains finite.

Example (Yukawa theory). Consider a Yukawa theory describing the interaction between a real scalar field ϕ and a Dirac fermion ψ. The initial interaction Lagrangian is:

$$\mathcal{L}_{\text{int}} = -g\bar{\psi}\psi\phi.$$

In $D = 4$, the scalar has dimension 1 and the fermion has dimension 3/2. The operator $\bar{\psi}\psi\phi$ has dimension $3/2 + 3/2 + 1 = 4$, meaning the coupling constant g is exactly dimensionless ($[g] = 0$). Based on our theorem, the theory is renormalizable.

Let us evaluate the superficial degree of divergence: $D(G) = 4 - E_\phi - \frac{3}{2}E_\psi$. A graph is divergent if $D(G) \geq 0$, which restricts us to $E_\phi + \frac{3}{2}E_\psi \leq 4$. The list of superficially divergent 1PI Green functions is:

TODO: add diagram: 6 graphs showing a) tadpole, b) scalar self-energy, c) 3-scalar vertex, d) 4-scalar vertex, e) fermion self-energy, f) Yukawa vertex

(a) $E_\phi = 1, E_\psi = 0$**:** Can be ignored by enforcing $\langle\Omega|\phi|\Omega\rangle = 0$ via a field shift.

(b) $E_\phi = 2, E_\psi = 0$**:** Renormalized by the scalar counterterms Z_ϕ and Z_m^ϕ.

(c) $E_\phi = 0, E_\psi = 2$ **(graph e):** Renormalized by the fermion counterterms Z_ψ and Z_m^ψ.

(d) $E_\phi = 1, E_\psi = 2$ **(graph f):** Renormalized by the Yukawa coupling counterterm Z_g.

(e) $E_\phi = 3, E_\psi = 0$ **(graph c):**

(f) $E_\phi = 4, E_\psi = 0$ **(graph d):**

TODO: add diagram: triangle loop of fermions with 3 external scalar legs

TODO: add diagram: box loop of fermions with 4 external scalar legs

Graphs (c) and (d) pose a profound conceptual problem. They are explicitly divergent at 1-loop (a fermion triangle and a fermion box, respectively). To cancel these divergences, we absolutely *must* introduce 3-point and 4-point scalar counterterms ($\delta_3\phi^3$ and $\delta_4\phi^4$). However, a counterterm can only exist if the interaction is already present in the bare Lagrangian!

Thus, loop corrections dynamically *generate* ϕ^3 and ϕ^4 interactions. To possess a mathematically consistent, finite theory, we are rigorously forced to add these terms to our initial Lagrangian:

$$\mathcal{L}_{\text{int}} = -g\bar{\psi}\psi\phi - \frac{\lambda_3}{3!}\phi^3 - \frac{\lambda_4}{4!}\phi^4.$$

Let us verify the mass dimensions of these new couplings: $[g] = 0$, $[\lambda_3] = 1$, and $[\lambda_4] = 0$. Because all couplings have non-negative mass dimension ($[g_j] \geq 0$), the expanded theory perfectly maintains its renormalizability.

This example perfectly illustrates a monumental general rule of Quantum Field Theory:

*A renormalizable theory **must** contain **all** possible renormalizable interactions compatible with its field content and symmetries.*

If you attempt to omit an interaction that is allowed by the symmetries of the system and has $[O_j] \leq 4$, quantum fluctuations (loops) will inevitably generate that exact interaction as a divergence, forcing you to include it to renormalize the theory.

We can thus state a deep philosophical conclusion: in QFT, the Lagrangian is not merely an arbitrary choice of terms. Symmetries are not just a property of a theory; symmetries strictly define the theory. Once the field content and the continuous/discrete symmetries are specified, the structure of the renormalizable Lagrangian is uniquely determined.

6.7.6 | Non-Renormalizable Theories and Effective Field Theories

A non-renormalizable theory is one that contains at least one interaction coupling with a strictly negative mass dimension. Consequently, it possesses an infinite number of divergent Green functions at high loop orders. To mathematically absorb these divergencies and obtain finite, physical predictions, an infinite number of distinct counterterms—and therefore an infinite number of new, independent interaction vertices—must be added to the bare Lagrangian.

Historically (up until the 1970s and 80s), it would naively appear that non-renormalizable theories are completely mathematically inconsistent and unpredictive, since making a single calculation would, in principle, require fixing an infinite number of coupling constants using an infinite number of experimental measurements. Such theories were generally discarded.

However, the modern interpretation in theoretical physics is radically different: non-renormalizable theories can be rigorously interpreted as **Effective Field Theories** (EFTs). Namely, they are understood as strictly low-energy approximations of some other, more fundamental underlying theory. The process of finding this fundamental high-energy theory is called finding the **UV completion**.

Consider a non-renormalizable interaction Lagrangian containing an operator $\hat{O}(x)$ of mass dimension $d > 4$:

$$\mathcal{L}_{\text{int}}(x) \supset \lambda\hat{O}(x) + \ldots, \quad \text{with } [\lambda] = 4 - d < 0.$$

We can strictly rewrite this negative-dimension coupling λ by explicitly extracting a dimensionless coupling constant g and a heavy mass scale M:

$$\lambda = \frac{g}{M^k}, \quad \text{where } k = -[\lambda] = d - 4 > 0.$$

Here:

- g is a strictly dimensionless parameter ($[g] = 0$) representing the interaction strength of the fundamental theory, and we naturally assume it is of order $\mathcal{O}(1)$.
- M is a specific heavy mass scale. Therefore, non-renormalizable theories are fundamentally associated with an intrinsic physical mass scale M characterizing the "new physics" (e.g., the mass of a heavy exchanged boson that has been "integrated out").

If we are interested in making theoretical predictions at low energies E, such that $E \ll M$, we may systematically expand our physical observable results (like scattering cross-sections) in inverse powers of the heavy scale M.

Because $E \ll M$, these higher-order terms are incredibly small. For any given experimental precision, we can choose a cutoff power α_0 and safely **neglect all contributions proportional to** $1/M^\alpha$ **for** $\alpha > \alpha_0$.

Crucially, it can be proven that if we truncate our expansion at a fixed order $1/M^{\alpha_0}$, only a strictly **finite** number of Green functions remain divergent (up to $1/M^{\alpha_0}$ corrections). Consequently, only a finite number of counterterms (which explicitly depend on our choice of α_0) is needed to renormalize the theory to that specific level of precision.

Therefore, the profound conclusion is:

Non-renormalizable theories are perfectly valid and highly predictive below some physical energy scale M.

Key insights regarding Effective Field Theories:

- **Low-Energy Limits:** the strict low-energy limit of any heavy QFT (which may itself be renormalizable) can be perfectly approximated by an Effective Field Theory that is formally non-renormalizable. A classic example is Fermi's theory of beta decay. Its interaction Lagrangian contains a 4-fermion operator ($\bar{\psi}\psi\bar{\psi}\psi$) with dimension 6, meaning the Fermi constant $G_F \sim 1/M_W^2$ has mass dimension -2. It is strictly non-renormalizable. However, at energies $E \sim M_W$, we discover its UV completion: the Standard Model, where the 4-fermion contact interaction is beautifully resolved into the exchange of a massive W boson in a renormalizable gauge theory.
- **The Fate of Renormalizable Theories:** a fundamentally renormalizable theory (like QED or the Standard Model) is generally not strictly valid at all energy scales. Beyond a certain heavy scale M, new physics effects may appear. Below this scale M (but sufficiently close to it), these subtle new effects can be effectively described by adding suitable non-renormalizable operators (suppressed by $1/M^k$) to the original renormalizable Lagrangian. Hence, even renormalizable theories may be interpreted as merely the very first (dimension ≤ 4) terms in a large-M expansion of a more general, fundamentally non-renormalizable Lagrangian.
- **Symmetry Dominance:** just like happens in renormalizable theories, the Lagrangian of a non-renormalizable EFT **must** contain **all** possible operators compatible with its exact symmetries. While there is generally an infinite number of them, for low-energy predictions we may systematically neglect all contributions from operators beyond a certain mass dimension, maintaining both predictive power and mathematical consistency.

7 | QED at the Loop Level

Loop calculations in Quantum Electrodynamics (QED) have historically been a key factor in establishing Quantum Field Theory (QFT) as the correct theoretical framework to describe nature at a fundamental level. These calculations are strictly necessary to make predictions in regimes where both quantum fluctuations and relativistic effects are highly measurable and must be accounted for simultaneously.

7.1 | The Bare Lagrangian and Counterterms

To systematically handle ultraviolet divergences, we begin by defining the theory in terms of "bare" quantities, which are indicated by a subscript "0". The bare Lagrangian density for QED is written as:

$$\mathcal{L} = -\frac{1}{4}(F_0^{\mu\nu})^2 + \bar{\psi}_0(i\partial\!\!\!/ - m_0)\psi_0 - e_0\bar{\psi}_0\gamma_\mu\psi_0 A_0^\mu$$

where the bare field strength tensor is defined as $F_0^{\mu\nu} = \partial^\mu A_0^\nu - \partial^\nu A_0^\mu$. The subscript "0" strictly labels the unobservable bare fields, bare mass, and bare coupling constant.

To absorb the divergences that appear at the loop level, these bare quantities are multiplicatively renormalized. We relate the bare quantities to the finite, physical renormalized quantities (ψ, A^μ, m, e) via a set of renormalization constants Z_i:

$$\begin{aligned}
\psi_0 &= \sqrt{Z_2}\psi \\
A_0^\mu &= \sqrt{Z_3}A^\mu \\
Z_2 m_0 &= Z_m m \\
e_0 &= Z_e e\mu^{\frac{4-D}{2}}
\end{aligned}$$

where ψ, A^μ, m, and e are all fully renormalized. *Note on notation: A much more intuitive and modern notation would be to replace $Z_2 \to Z_\psi$ and $Z_3 \to Z_A$ to explicitly denote the field they renormalize. However, the historical notation Z_1, Z_2, Z_3 is far too common and deeply entrenched in the literature to change.* The arbitrary mass scale μ is introduced in dimensional regularization (where spacetime dimension is D) to ensure that the renormalized electric charge e strictly remains a dimensionless coupling constant.

We also define an additional vertex renormalization constant Z_1 via the interaction term:

$$-e_0\bar{\psi}_0\gamma_\mu\psi_0 A_0^\mu \equiv -Z_1 e\mu^{\frac{4-D}{2}}\bar{\psi}\gamma_\mu\psi A^\mu$$

By substituting the field rescalings into the left-hand side, we can extract the fundamental identity linking the charge and vertex renormalization constants:

$$Z_1 = Z_e Z_2\sqrt{Z_3}$$

In practical perturbative calculations, it is significantly more convenient to use Z_1 as the independent counterterm parameter instead of Z_e for renormalization.

Using these definitions, we can rewrite the exact bare Lagrangian entirely in terms of the renormalized fields and the Z_i factors:

$$\mathcal{L} = -\frac{1}{4}Z_3(F_{\mu\nu})^2 + \bar{\psi}(Z_2 i\partial\!\!\!/ - Z_m m)\psi - Z_1 e\mu^{\frac{4-D}{2}}\bar{\psi}\gamma_\mu\psi A^\mu$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

Now, we isolate the counterterms by defining δ_i such that $Z_i = 1 + \delta_i$ for $i = 1, 2, 3, m, e$. Splitting the Lagrangian into a basic physical part and a counterterm part yields:

$$\begin{aligned}
\mathcal{L} = &\left[-\frac{1}{4}F_{\mu\nu}^2 + \bar{\psi}(i\partial\!\!\!/ - m)\psi - e\mu^{\frac{4-D}{2}}\bar{\psi}\gamma_\mu\psi A^\mu\right] \\
&- \frac{\delta_3}{4}F_{\mu\nu}^2 + \bar{\psi}(\delta_2 i\partial\!\!\!/ - \delta_m m)\psi - e\delta_1\mu^{\frac{4-D}{2}}\bar{\psi}\gamma_\mu\psi A^\mu
\end{aligned}$$

The second line represents the counterterm Lagrangian, which systematically generates new interaction vertices that absorb the loop divergences. The resulting Feynman rules for these counterterms are:

- **Photon self-energy counterterm:**

$$-i\delta_3(p^2 g_{\mu\nu} - p_\mu p_\nu)$$

TODO: add diagram: squiggly line with a circled X in the middle, momentum p flowing through

- **Electron self-energy counterterm:**

$$i(\delta_2 \not{p} - \delta_m m)$$

TODO: add diagram: solid fermion line with a circled X in the middle, momentum p flowing through

- **Vertex counterterm:**

$$-i\delta_1 \mu^{\frac{4-D}{2}} e\gamma^\mu$$

TODO: add diagram: QED vertex with a circled X covering the interaction point, photon line mu

7.1.1 | On-Shell Renormalization Conditions

The renormalization constants Z_i, or equivalently their counterterm counterparts δ_i, are not unique until they are fixed by imposing specific, physical renormalization conditions. In QED, it is highly standard to use the **On-Shell Renormalization Scheme**.

The on-shell renormalization conditions for QED dictate how the fully interacting (loop-corrected) Green's functions behave when external particles are physically strictly on-shell:

- **For the Photon Propagator:** We demand that near the physical photon mass shell ($p^2 \to 0$), the fully interacting propagator behaves exactly like the free propagator:

$$\frac{-ig_{\mu\nu}}{p^2} + \mathcal{O}(p^\mu p^\nu) + \text{regular at } p^2 \sim 0$$

TODO: add diagram: squiggly line representing the exact fully interacting photon propagator (blob in the middle)

This explicitly requires that the residue at the pole $p^2 = 0$ is set precisely to 1 (excluding the tensorial $-ig_{\mu\nu}$ factor). This condition rigorously fixes Z_3 as the residue of the exact bare propagator (similar to the scalar case where $\langle A_0^\mu A_0^\nu \rangle = Z_3 \langle A^\mu A^\nu \rangle$). Contributions to the propagator that are proportional to $p^\mu p^\nu$ may be completely ignored, since when inserted inside any larger physical S-matrix diagram, they identically vanish due to the Ward identity.

- **For the Electron Propagator:** We require that the fully interacting fermion propagator possesses a pole at the physical electron mass $p^2 = m^2$, and that the residue of this pole is strictly 1:

$$\frac{i}{\not{p} - m} + \text{regular at } p^2 \sim m^2$$

TODO: add diagram: solid line representing the exact fully interacting electron propagator (blob in the middle)

(Note: This is mathematically equivalent to writing $\frac{i(\not{p}+m)}{p^2-m^2}$, isolating the pole structure at $p^2 = m^2$.) Fixing the location of the pole fixes the mass counterterm δ_m, while enforcing that the residue is exactly 1 fixes δ_2 (or equivalently Z_2). In particular, Z_2 represents the residue of the bare electron propagator. **Note:** When actively imposing this condition on the fermion residue, a soft infrared (IR) divergence inevitably appears. This IR pathology is typically regulated by adding a fictitious, infinitesimal mass to the photon, a regularization step that also propagates into the calculation of Z_1.

- **For the Vertex:** We fundamentally define the physical electric charge 'e' via a strictly low-energy interaction. We define the exact 1PI vertex function $-ie\mu^\varepsilon\Gamma^\mu$: We then take the exact matrix element between two physically on-shell electrons and an off-shell photon:

$$i\mathcal{M}^\mu = (-ie\mu^\varepsilon)\bar{u}(p_2)\Gamma^\mu u(p_1)$$

TODO: add diagram: 1PI vertex blob with incoming fermion p_1, outgoing fermion p_2, and incoming photon q

Note on Kinematics: Unlike the scalar ϕ^4 theory, we cannot mathematically place all three external lines fully on-shell simultaneously. Enforcing $p_1^2 = m^2$ and $p_2^2 = m^2$ alongside $q^2 = 0$ for the photon strictly forces $q^\mu = 0$ overall. Therefore, we study the interaction of a real, on-shell electron with an external classical electromagnetic field (represented by the momentum-carrying photon line), which fundamentally mirrors how macroscopic electric charge is actually measured in experiments (e.g., Thomson scattering).

Similarly to the ϕ^4 case, we demand that in the zero-momentum-transfer limit ($q^\mu \to 0$), the full quantum interaction perfectly reproduces the classical tree-level result. In this limiting case, the renormalization condition is explicitly written as:

$$\bar{u}(p_2)(-ie\Gamma^\mu)u(p_1)\Big|_{q=0} = \bar{u}(p_2)(-ie\gamma^\mu)u(p_1)$$

Or, equivalently expressed by stripping away the spinors and evaluating the vertex function directly on the fermion mass shell with zero momentum transfer:

$$\Gamma^\mu\Big|_{\substack{p_i \to m \\ q \to 0}} = \gamma^\mu$$

7.1.2 | Gordon Identity

To rigorously verify that our on-shell vertex renormalization condition is well-posed, we must establish the most general covariant form for the 1PI vertex function Γ^μ. The vertex function Γ^μ is fundamentally a matrix in spinor space that carries a single free Lorentz index. It can only be constructed using Dirac γ matrices and the available external momenta p_1^μ and p_2^μ. Note that the photon momentum transfer $q^\mu = p_2^\mu - p_1^\mu$ is strictly fixed by momentum conservation and is not an independent vector.

By enumerating all possible covariant structures, the general form of Γ^μ reads:

$$\Gamma^\mu = c_0\gamma^\mu + c_1 p_1^\mu \not{p}_1 + c_2 p_2^\mu \not{p}_1 + c_3\gamma^\mu \not{p}_1 + c_4\gamma^\mu \not{p}_2 + c_5 \not{p}_1 \not{p}_2 + \dots$$

. We strictly do not include any terms containing the Levi-Civita tensor $\epsilon^{\mu\nu\rho\sigma}$ or the chiral matrix γ_5. Because QED is a parity-conserving theory, such parity-odd terms cannot be generated by its Feynman rules at any loop order.

When we sandwich this general vertex function between on-shell physical spinors to form the physical matrix element $\bar{u}(p_2)\Gamma^\mu u(p_1)$, we can systematically anti-commute all $\not{p}_1$ operators to the far right and all $\not{p}_2$ operators to the far left. We can then completely eliminate them using the free Dirac equations $\not{p}_1 u(p_1) = mu(p_1)$ and $\bar{u}(p_2)\not{p}_2 = m\bar{u}(p_2)$. Consequently, all higher-order γ-matrix combinations reduce down, and only the first line of our expansion yields mathematically independent contributions:

$$\Gamma^\mu = c_0\gamma^\mu + c_1 p_1^\mu + c_2 p_2^\mu$$

.

We further constrain this structure by applying the exact Ward Identity, which demands $q_\mu \bar{u}(p_2)\Gamma^\mu u(p_1) = 0$. Expanding this condition with $q_\mu = p_{2\mu} - p_{1\mu}$, we obtain:

$$0 = (p_{2\mu} - p_{1\mu})\bar{u}(p_2)\Gamma^\mu u(p_1) = \bar{u}(p_2)\left[c_0(\not{p}_2 - \not{p}_1) + c_1(p_2 \cdot p_1 - m^2) + c_2(m^2 - p_1 \cdot p_2)\right] u(p_1)$$

. The term proportional to c_0 evaluates identically to zero, because the Dirac equation enforces $\bar{u}(p_2)(\not{p}_2 - \not{p}_1)u(p_1) = \bar{u}(p_2)(m - m)u(p_1) = 0$. Factoring out the remaining scalar product forces

the kinematic constraint $c_2 = c_1$. Therefore, the vertex function must take the symmetric form:

$$\Gamma^\mu = c_0\gamma^\mu + c_1(p_1^\mu + p_2^\mu)$$

.

While this is a completely valid mathematical representation, there is a significantly more common and physically useful formulation obtained via the **Gordon Identity**. The Gordon Identity establishes a fundamental relationship between the convection current and the spin current for on-shell fermions:

$$\bar{u}(p_2)(p_1^\mu + p_2^\mu)u(p_1) = 2m\bar{u}(p_2)\gamma^\mu u(p_1) + i\bar{u}(p_2)\sigma^{\mu\nu}(p_{2\nu} - p_{1\nu})u(p_1)$$

. Here, the tensor $\sigma^{\mu\nu}$ is defined as $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$. Note that $S^{\mu\nu} = \frac{1}{2}\sigma^{\mu\nu}$ serve as the fundamental generators of the Lorentz group for the spinor representation. *(Exercise: Rigorously prove the Gordon Identity using only the γ-matrix anticommutation relations and the Dirac equation).*

By applying the Gordon Identity, we can eliminate the $(p_1^\mu + p_2^\mu)$ term entirely, re-parameterizing the full vertex function into its universally standard form:

$$\Gamma^\mu = F_1(q^2)\gamma^\mu + F_2(q^2)\frac{i\sigma^{\mu\nu}q_\nu}{2m}$$

. This exact decomposition is completely robust and valid at all orders in perturbation theory. The scalar functions F_1 and F_2 are known as the electromagnetic **Form Factors**:

- $F_1(q^2)$ is the **Dirac form factor**, capturing the effective charge distribution.
- $F_2(q^2)$ is the **Magnetic form factor**, directly related to the anomalous magnetic moment of the electron.

Both form factors implicitly depend on the fermion mass m. They can be computed perturbatively order-by-order; at leading tree-level, they are simply $F_1 = 1$ and $F_2 = 0$.

Crucially, the tensor structure multiplying F_2 explicitly carries a factor of q_ν, meaning the entire second term dynamically vanishes in the strict low-energy limit as $q^\mu \to 0$. Therefore, the on-shell renormalization condition for the vertex ($\Gamma^\mu \to \gamma^\mu$ as $q \to 0$) translates exclusively into a boundary condition on the Dirac form factor:

$$F_1(q^2 = 0) = 1$$

. As stated previously, this boundary condition is precisely what mathematically fixes the vertex renormalization constant Z_1.

7.1.3 | The Identity $Z_1 = Z_2$

A remarkable and profound feature of Quantum Electrodynamics is that one rigorously finds:

$$Z_1 = Z_2$$

in the on-shell scheme at absolutely all orders in perturbation theory. This powerful identity guarantees that the renormalization of the electric charge is governed entirely by the vacuum polarization of the photon, shielding it from fermion-specific vertex corrections.

This equality is not accidental; it is a direct, exact consequence of the underlying gauge symmetry, manifested through the **Ward-Takahashi Identity**: .

TODO: add diagram: Ward-Takahashi identity visually equating the contraction of q_μ with the full 1PI vertex blob to e_0

Evaluated strictly in terms of the unrenormalized bare theory, the Ward-Takahashi identity reads:

$$q_\mu S_0(p_1)(-ie_0\Gamma_0^\mu)S_0(p_2) = e_0(S_0(p_1) - S_0(p_2))$$

, where $S_0(p)$ is the exact bare fermion propagator and $-ie_0\Gamma_0^\mu$ is the exact bare 1PI vertex. *Note:* While the general WT identity relates fully connected Green's functions (which sum over all reducible and irreducible topologies), any diagram where the photon is emitted from an external, fully on-shell leg vanishes dynamically. Therefore, for our analysis, we can isolate the identity purely to the 1PI vertex block (with no external photon propagators attached).

To expose the renormalization constants, we formally multiply the identity from the left by $S_0^{-1}(p_1)$ and from the right by $S_0^{-1}(p_2)$, rewriting it as:

$$q_\mu(-i\Gamma_0^\mu) = S_0^{-1}(p_2) - S_0^{-1}(p_1)$$

. We now take the specific on-shell kinematic limit where $p_i \to m$ (the physical mass shell) and the photon momentum transfer $q \to 0$.

First, consider the right-hand side. By the definition of the on-shell scheme, the bare fermion propagator possesses a pole at the physical mass with a residue fixed by Z_2:

$$S_0(p) \xrightarrow{p_i \to \text{on-shell}} \frac{iZ_2}{\not{p} - m} + \text{regular}$$

. Inverting this expression, the inverse bare propagator in this limit becomes $S_0^{-1}(p) = \frac{\not{p}-m}{iZ_2}$.

Next, consider the left-hand side involving the bare vertex Γ_0^μ. Because Γ_0^μ represents a strictly amputated 1PI function, we must carefully relate it to the renormalized vertex using the correct product of field renormalization factors:

$$Z_2\sqrt{Z_3}(-ie_0\Gamma_0^\mu) = -ie\mu^\varepsilon\Gamma^\mu$$

. This specific combination of Z-factors arises because amputating the external lines removes the $\sqrt{Z}$ factors that normally accompany field insertions. By utilizing the fundamental bare-to-renormalized charge relation $e\mu^\varepsilon = e_0/Z_e$ and the definition $Z_1 \equiv Z_e Z_2\sqrt{Z_3}$, this simplifies perfectly to:

$$\Gamma_0^\mu = \frac{\Gamma^\mu}{Z_1}$$

. Imposing the low-energy on-shell vertex renormalization condition $\Gamma^\mu \to \gamma^\mu$ as $q \to 0$ forces the bare vertex in this specific limit to become:

$$\Gamma_0^\mu\Big|_{\substack{p_i \to m \\ q \to 0}} = \frac{\gamma^\mu}{Z_1}$$

.

Finally, we substitute these asymptotic on-shell forms back into our rewritten WT identity:

$$q_\mu\left(-i\frac{\gamma^\mu}{Z_1}\right) = \frac{\not{p}_2 - m}{iZ_2} - \frac{\not{p}_1 - m}{iZ_2}$$

. Using the algebraic fact that $q_\mu\gamma^\mu = \not{q} = \not{p}_2 - \not{p}_1$, the equation simplifies dramatically to:

$$\frac{\not{p}_2 - \not{p}_1}{iZ_1} = \frac{\not{p}_2 - \not{p}_1}{iZ_2}$$

. Matching the exact coefficients on both sides dictates unconditionally that:

$$Z_1 = Z_2$$

.

While derived here within the on-shell framework, this critical equivalence $Z_1 = Z_2$ also rigorously holds in minimal subtraction schemes, such as MS and $\overline{\text{MS}}$. In those schemes, the counterterms are strictly defined to absorb only the singular divergence, such that $Z_i^{\text{MS}} = 1 + \text{"UV poles of } Z_i\text{"}$ (with a finite redefinition of the scale μ for $\overline{\text{MS}}$). Because the complete algebraic equality $Z_1 = Z_2$ holds exactly for the full divergent expressions, their respective pole structures must be identical.

This identity $Z_1 = Z_2$ guarantees the universality of the electric charge and possesses extremely important physical implications for the structure of QED, which we will now explore.

7.2 | Vacuum Polarization and Electric Charge Renormalization

The exact algebraic equality $Z_1 = Z_2$ guarantees that the renormalization of the electric charge is governed entirely by the vacuum polarization of the photon. By definition, the charge renormalization constant is $Z_e = \frac{Z_1}{Z_2\sqrt{Z_3}}$. Applying the Ward-Takahashi identity perfectly simplifies this to $Z_e = \frac{1}{\sqrt{Z_3}}$. This mathematically demonstrates that quantum corrections to the elementary charge are completely independent of any specific fermion vertex corrections and are determined strictly by the photon propagator.

This isolation leads to the profound physical principle of the **Universality of Electric Charge**. Consider a generalized QED theory containing multiple distinct fermion flavors ψ_i (e.g., electrons, muons, quarks) with bare electric charges $Q_i e_0 = Q_i Z_e^{(i)} e\mu^\varepsilon$. In principle, each unique fermion species could experience completely different renormalization dynamics, possessing distinct constants $Z_1^{(i)}$, $Z_2^{(i)}$, and $Z_e^{(i)}$. However, because the Ward-Takahashi identity strictly enforces $Z_1^{(i)} = Z_2^{(i)}$ universally for each specific flavor, the resulting charge renormalization $Z_e^{(i)} = 1/\sqrt{Z_3}$ is globally identical for all fermion fields.

Consequently, the ratio between different electric charges does not change under loop corrections; it remains strictly invariant after renormalization. This universality is an absolute theoretical necessity for the Standard Model. Flavor-changing processes, such as $e^-\bar{\nu}_e \to \mu^-\bar{\nu}_\mu$ or $e^-\bar{\nu}_e \to d\bar{u}$ (where $Q_\nu = 0$, $Q_d = -1/3$, $Q_u = 2/3$), require the exact ratio of electric charges to match the bare theory precisely to uphold global electric charge conservation. This intricate conservation works flawlessly because charge renormalization is factored entirely into the shared photon propagator.

> **Remark.** *This exact universality holds specifically in renormalization schemes that mathematically enforce $Z_1 = Z_2$, such as the on-shell or $\overline{MS}$ schemes. While one could technically define a perverse scheme where $Z_1 \neq Z_2$, it would artificially break charge universality at intermediate steps and is highly unadvisable.*

7.2.1 | General Form of Photon Propagator

We define the exact 1PI 2-point tensor function for the photon, $i\Pi^{\mu\nu}(p)$, which strictly excludes the external free propagators. Because $\Pi^{\mu\nu}(p)$ is a rank-2 Lorentz tensor depending exclusively on the internal momentum p^μ, Lorentz covariance dictates that its most general structural form must be a linear combination of the metric and the momentum:

$$\Pi^{\mu\nu}(p) = Ag^{\mu\nu} + Bp^\mu p^\nu$$

The Ward-Takahashi identity imposes a strict transversality condition on the vacuum polarization: $p_\mu \Pi^{\mu\nu} = 0$. Contracting the general tensor form yields $p_\mu(Ag^{\mu\nu} + Bp^\mu p^\nu) = Ap^\nu + Bp^2 p^\nu = 0$, which mandates that $A = -p^2 B$. By defining a dimensionless scalar function $\Pi(p^2) \equiv -B$, we can express the fundamental, gauge-invariant transverse structure of the 1PI tensor:

$$\Pi^{\mu\nu}(p) = (p^2 g^{\mu\nu} - p^\mu p^\nu)\Pi(p^2)$$

To compute the full, exact photon propagator, we sum the infinite geometric series of 1PI insertions (the Dyson equation):

TODO: add diagram: Full propagator double line = free propagator + free*1PI*free + free*1PI*free*1PI*free + ...

Working in a general R_ξ gauge, the free bare photon propagator is $\frac{-i}{p^2}\left(g_{\mu\nu} - (1-\xi)\frac{p_\mu p_\nu}{p^2}\right)$. To vastly simplify the geometric summation, we introduce the transverse projection operator $\Delta^\mu_\nu \equiv \delta^\mu_\nu - \frac{p^\mu p_\nu}{p^2}$, which satisfies $p_\mu \Delta^\mu_\nu = 0$ and the idempotent property $(\Delta^2)^\mu_\nu = \Delta^\mu_\nu$. The 1PI tensor can be rewritten as $i\Pi^{\mu\nu}(p) = ip^2\Delta^{\mu\nu}\Pi(p^2)$.

When sandwiching a 1PI insertion between two free propagators, the strict transversality of $\Delta^{\mu\nu}$ actively annihilates the longitudinal $(1-\xi)p_\mu p_\nu/p^2$ component of the adjacent free propagators. A single insertion evaluates simply to:

$$\frac{-i}{p^2}\Delta^\mu_\rho \times ip^2\Delta^\rho_\sigma\Pi(p^2) \times \frac{-i}{p^2}\Delta^\sigma_\nu = \frac{-i}{p^2}\Delta^\mu_\nu\Pi(p^2)$$

Summing the full infinite geometric series yields $\frac{-i}{p^2}\Delta^\mu_\nu(1 + \Pi(p^2) + \Pi^2(p^2) + \dots)$, which converges to $\frac{-i}{p^2}\Delta^\mu_\nu\frac{1}{1-\Pi(p^2)}$.

Restoring the uncorrected longitudinal piece from the very first term of the series, the full exact photon propagator is rigorously given by:

$$\frac{-i}{p^2\left(1-\Pi(p^2)\right)}\left(g_{\mu\nu} - \frac{p_\mu p_\nu}{p^2}\right) - i\xi\frac{p_\mu p_\nu}{p^4}$$

This result explicitly demonstrates that the gauge-fixing parameter ξ is completely shielded from any quantum loop corrections. Furthermore, when this full propagator is embedded inside any larger physical S-matrix diagram, the $\mathcal{O}(p^\mu p^\nu)$ terms strictly vanish by the Ward-Takahashi identity. Finally, our on-shell renormalization condition—demanding that the physical pole at $p^2 = 0$ retains a residue identically equal to the free theory—translates directly into the simple algebraic boundary condition $\Pi(0) = 0$.

One-Loop Result

We explicitly evaluate the 1-loop contribution to the vacuum polarization function by summing the 1PI fermion loop and the tree-level counterterm δ_3:

TODO: add diagram: 1PI blob = 1-loop fermion bubble (momentum k routing through top, k+p through bottom) + counterterm cross

Applying the standard Feynman rules in $D = 4 - 2\varepsilon$ dimensions, the integral for the fermion loop is:

$$i\Pi^{\mu\nu}_{\text{1-loop}}(p) = (-ie)^2(-1)\mu^{2\varepsilon}\int\frac{\mathrm{d}^D k}{(2\pi)^D}\frac{\mathrm{Tr}\left[\gamma^\mu(i(\not{k}+m))\gamma^\nu(i(\not{k}+\not{p}+m))\right]}{(k^2-m^2)((k+p)^2-m^2)}$$

The overall factor of (-1) is physically required by Fermi-Dirac statistics for any closed fermion loop. The Dirac trace emerges algebraically because the continuous cyclic contraction of spinor indices along the closed loop forms a strict trace operation. *(Exercise: You can explicitly verify this by writing out the spinor indices $\gamma^\mu_{\alpha\beta}(\not{k}+m)_{\beta\sigma}\dots$ and observing the cyclic closure yielding a trace).*

Evaluating the Dirac trace mathematically yields:

$$-4e^2\mu^{2\varepsilon}\int\frac{\mathrm{d}^D k}{(2\pi)^D}\frac{k^\mu(k+p)^\nu + k^\nu(k+p)^\mu - g^{\mu\nu}(k^2 + k\cdot p - m^2)}{(k^2-m^2)((k+p)^2-m^2)}$$

Instead of confronting this heavy tensor integral directly, we can elegantly extract the scalar function $\Pi(p^2)$ by taking the full Lorentz trace of our generic structural definition $i\Pi^\mu_\mu(p^2) = i(p^2D - p^2)\Pi(p^2) = ip^2(D-1)\Pi(p^2)$. Inverting this gives a direct scalar projection:

$$\Pi(p^2) = \frac{1}{D-1}\frac{-i}{p^2}\Pi^\mu_\mu(p^2)$$

After executing the standard integration architecture—introducing Feynman parameters, shifting the loop momentum, and evaluating the dimensionally regularized scalar integrals—the complete function (including the δ_3 counterterm) is:

$$\Pi(p^2) = -\frac{e^2}{2\pi^2}\left[\frac{1}{6}\left(\frac{1}{\varepsilon} - \gamma_E + \log(4\pi)\right) - \int_0^1 \mathrm{d}x\, x(1-x)\log\left(\frac{m^2 - x(1-x)p^2}{\mu^2}\right)\right] - \delta_3$$

Applying the exact on-shell scheme boundary condition, $\Pi(0) = 0$, uniquely fixes the required counterterm value:

$$\delta_3^{\mathrm{OS}} = -\frac{e^2}{12\pi^2}\left(\frac{1}{\varepsilon} - \gamma_E + \log(4\pi) - \log\frac{m^2}{\mu^2}\right)$$

Substituting this specific counterterm back into the expression strictly annihilates both the UV divergence and the arbitrary scale μ, yielding the final, physically observable on-shell vacuum polarization function:

$$\Pi(p^2) = \frac{e^2}{2\pi^2}\int_0^1 \mathrm{d}x\, x(1-x)\log\left(\frac{m^2 - x(1-x)p^2}{m^2}\right)$$

Alternatively, in the purely mathematical $\overline{\mathrm{MS}}$ scheme, the counterterm δ_3 is defined strictly to subtract only the universal divergent pole piece ($\frac{1}{\varepsilon} - \gamma_E + \log(4\pi)$). This leaves the logarithmic dependence on the arbitrary mass scale μ explicitly intact within the final $\overline{\mathrm{MS}}$ renormalized expression.

7.2.2 | Physical Interpretation

Since the exact renormalization of the photon propagator entirely fixes the renormalization of the electric charge (due to the identity $Z_1 = Z_2$), let us use the 1-loop result for $\Pi(p^2)$ to deeply understand how quantum loop corrections physically modify the classical electromagnetic interaction.

Consider the fundamental electromagnetic scattering between two Dirac fermions (e.g., an electron and a proton). At leading tree level, the scattering amplitude $i\mathcal{M}$ is mediated by the exchange of a single free photon with momentum transfer q:

TODO: add diagram: tree-level t-channel exchange of a photon between two fermion lines

$$i\mathcal{M} \sim \frac{e^2}{q^2}$$

When we upgrade this calculation to include all higher-order quantum corrections to the photon line, we must replace the free propagator with the full, exact interacting propagator. The amplitude becomes:

$$i\mathcal{M} \sim \frac{e^2}{q^2(1 - \Pi(q^2))} \equiv \frac{e^2_{\mathrm{eff}}(q^2)}{q^2}$$

Here, we have mathematically absorbed the vacuum polarization function into the definition of a momentum-dependent **effective charge**:

$$e^2_{\mathrm{eff}}(q^2) \equiv \frac{e^2}{1 - \Pi(q^2)}$$

The Non-Relativistic Limit and the Uehling Potential

First, consider the strict non-relativistic limit, which corresponds to the static interaction between widely separated charges. In this regime, the momentum transfer is purely spatial, meaning $q^0 = 0$, and thus $q^2 = -|\vec{q}|^2$.

Recall from elementary QFT that at tree-level, Fourier transforming the scattering amplitude directly yields the classical Coulomb potential. For an interaction between an electron ($Q_e = -1$) and a proton ($Q_p = +1$), the static potential is:

$$V(\vec{x}) = -\int \frac{\mathrm{d}^3 q}{(2\pi)^3} \frac{e^2}{|\vec{q}|^2} e^{i\vec{q}\cdot\vec{x}} = -\frac{e^2}{4\pi r} = -\frac{\alpha}{r}$$

where $r = |\vec{x}|$ and $\alpha = e^2/4\pi \approx 1/137$.

The one-loop quantum correction we just computed dynamically modifies this classical Coulomb potential by simply replacing the constant charge e^2 with the momentum-dependent effective charge $e^2_{\text{eff}}(-|\vec{q}|^2)$:

$$V(\vec{x}) = \int \frac{\mathrm{d}^3 q}{(2\pi)^3} \frac{e^2_{\text{eff}}(-|\vec{q}|^2)}{|\vec{q}|^2} e^{i\vec{q}\cdot\vec{x}}$$

Because we are explicitly looking at the non-relativistic limit, which corresponds to large distances $r \gg 1/m$, the dominant contributions to this integral come from the very small momentum region where $|\vec{q}|^2 \ll m^2$. We can systematically Taylor expand the exact 1-loop function $\Pi(q^2)$ around $q^2 = 0$.

Using our previous exact result for the on-shell scheme:

$$\Pi(q^2) = \frac{e^2}{2\pi^2} \int_0^1 \mathrm{d}x\, x(1-x) \log\left(1 - x(1-x)\frac{q^2}{m^2}\right)$$

Expanding the logarithm for small q^2/m^2 yields $\log(1-y) \approx -y$:

$$\begin{aligned}\Pi(q^2) &\approx -\frac{e^2}{2\pi^2}\frac{q^2}{m^2}\int_0^1 \mathrm{d}x\, x^2(1-x)^2 + \mathcal{O}\left(\frac{q^4}{m^4}\right) \\ &\approx \frac{e^2|\vec{q}|^2}{60\pi^2 m^2} + \mathcal{O}\left(\frac{q^4}{m^4}\right)\end{aligned}$$

Substituting this small-momentum expansion back into the effective charge definition and expanding the geometric series $(1-\Pi)^{-1} \approx 1 + \Pi$:

$$\begin{aligned}V(\vec{x}) &= -\int \frac{\mathrm{d}^3 q}{(2\pi)^3} \frac{e^2}{|\vec{q}|^2(1 - \Pi(-|\vec{q}|^2))} e^{i\vec{q}\cdot\vec{x}} \\ &\simeq -\int \frac{\mathrm{d}^3 q}{(2\pi)^3} e^{i\vec{q}\cdot\vec{x}} \left(\frac{e^2}{|\vec{q}|^2} + \frac{e^4}{60\pi^2 m^2} + \dots\right)\end{aligned}$$

The first term precisely reproduces the classical Coulomb $-e^2/4\pi r$ potential. The second term, being independent of momentum, Fourier transforms directly into a Dirac delta function in coordinate space:

$$\begin{aligned}V(\vec{x}) &= -\frac{e^2}{4\pi r} - \frac{e^4}{60\pi^2 m^2}\delta^{(3)}(\vec{x}) + \dots \\ &= -\frac{\alpha}{r} - \frac{4\alpha^2}{15m^2}\delta^{(3)}(\vec{x}) + \dots\end{aligned}$$

This quantum correction is called the **Uehling term**. It reveals that the true electromagnetic interaction is strictly **stronger** at very short distances ($\vec{x} \to 0$) than the naive classical Coulomb prediction. Because it is a highly localized delta-function potential, it only physically affects atomic states that have a non-zero probability density exactly at the origin (specifically, the S-states where the wave function $\psi(0) \neq 0$).

This creates a tiny but distinctly measurable physical shift in the energy levels of the hydrogen atom, calculable via first-order quantum mechanical perturbation theory:

$$\Delta E = \int \mathrm{d}^3x |\psi(\vec{x})|^2 \left(-\frac{4\alpha^2}{15m^2}\delta^{(3)}(\vec{x}) \right) = -\frac{4\alpha^2}{15m^2}|\psi(0)|^2$$

This Uehling shift is historically famous for contributing a small but vital component (~ -27 MHz) to the total **Lamb Shift** between the $2S_{1/2}$ and $2P_{1/2}$ states of hydrogen, providing some of the earliest dramatic experimental confirmations of QED.

The High-Energy Limit and the Beta Function

Now let us consider the diametrically opposite kinematic limit: the deep ultraviolet, highly relativistic regime where the momentum transfer is extremely large, $|q^2| \gg m^2$.

Evaluating the exact on-shell integral for $\Pi(q^2)$ in this large $|q^2|$ limit yields a logarithmic dependence:

$$\Pi(q^2) \simeq \frac{e^2}{12\pi^2} \log\left(\frac{q^2}{m^2}\right) + \text{const}$$

(Note: A mathematically similar logarithmic structure appears in the $\overline{MS}$ scheme, but normalized by the arbitrary scale μ^2, i.e., $\log(q^2/\mu^2)$).

Inserting this asymptotic behavior into the definition of the effective charge gives:

$$e^2_{\text{eff}}(q^2) \simeq \frac{e^2}{1 - \frac{e^2}{12\pi^2}\log\left(\frac{q^2}{m^2}\right)}$$

Or, equivalently and more commonly expressed in terms of the fine-structure constant $\alpha_{\text{eff}} \equiv e^2_{\text{eff}}/4\pi$:

$$\alpha_{\text{eff}}(q^2) = \frac{\alpha}{1 - \frac{\alpha}{3\pi}\log\left(\frac{q^2}{m^2}\right)}$$

This critical formula demonstrates that the effective interaction strength α_{eff} is not a constant, but rather "runs" depending on the energy scale of the interaction (large q^2). By Heisenberg's uncertainty principle, large exchanged momenta directly correspond to probing physics at extremely short spatial distances.

The continuous scale dependence of the effective coupling is governed mathematically by a fundamental differential equation defining the **Beta function**:

$$\beta(\alpha_{\text{eff}}) \equiv \frac{\partial \alpha_{\text{eff}}}{\partial \log q^2} = \frac{\alpha^2_{\text{eff}}}{3\pi} + \mathcal{O}(\alpha^3_{\text{eff}})$$

Because the leading term of the beta function is strictly positive, the differential equation dictates that the electromagnetic interaction becomes continuously **stronger at higher energy scales** (or equivalently, at shorter distances).

TODO: add diagram: graph showing α_{eff} monotonically increasing as a function of q^2

At macroscopic, everyday large distances ($q^2 \to 0$), the interaction strength settles to the classical, on-shell measured value of $\alpha \approx 1/137$. However, as we push to shorter and shorter distances, α_{eff} grows larger and larger.

The physical interpretation of this phenomenon is beautifully intuitive. The quantum vacuum is not empty. When a real electron interacts electromagnetically, it constantly emits and reabsorbs virtual photons. These highly energetic, short-lived virtual photons can subsequently fluctuate into virtual electron-positron (e^+e^-) pairs.

These virtual e^+e^- pairs act exactly like microscopic electric dipoles in a dielectric medium. The positively charged virtual positrons are electrostatically attracted to the original, central real electron, while the virtual electrons are repelled outward.

Hence, the quantum vacuum surrounding any bare charge behaves precisely like a polarizable **dielectric medium** composed of e^+e^- pairs. This cloud of dipoles creates a physical **screening effect**. At large distances, an outside observer only sees the central charge partially cancelled out (screened) by the surrounding cloud of positive charge, resulting in the relatively weak observed effective charge of $1/137$. However, if we probe the electron at very high energies (penetrating deeply through the screening cloud at short distances), we "see" more of the stronger, unscreened bare central charge. This entire physical shielding phenomenon is appropriately named **Vacuum Polarization**.

TODO: add diagram: A central real electron surrounded by a cloud of virtual e^+e^- pairs, polarized such that the e^+ are closer to the center

As a concrete, experimental example, if we measure the electromagnetic coupling strength at the energy scale of the Z boson mass ($q \simeq m_Z \approx 91$ GeV), we find:

$$\alpha_{\text{eff}}(m_Z^2) \simeq \alpha_{\overline{\text{MS}}}(\mu_R = m_Z) \simeq \frac{1}{128}$$

This is significantly stronger than $1/137$. *Note:* To theoretically calculate this value to even a rough approximation, one cannot just use the electron loop. One must sum the individual vacuum polarization loop contributions from *all* fundamental fermions in the Standard Model whose mass is lighter than m_Z. This explicitly includes the leptons (e^-, μ^-, τ^- with $Q = -1$) and the quarks (u, c with $Q = 2/3$, and d, s, b with $Q = -1/3$).

7.3 | Advanced Topics in QED

7.3.1 | Heavy Fermion Decoupling

In our discussions on vacuum polarization, we established that the coupling constant runs with the energy scale μ_R. In the $\overline{\text{MS}}$ scheme, the running coupling equation dictates:

$$\frac{\partial \alpha}{\partial \log \mu_R^2} = \beta(\alpha) = \alpha^2(\beta_0 + \beta_1 \alpha + \dots)$$

At one-loop accuracy for QED with a single electron flavor, we found $\beta_0 = \frac{1}{3\pi}$. Solving this differential equation directly provides the running coupling:

$$\alpha(\mu_R^2) = \frac{\alpha(\mu_0^2)}{1 - \alpha(\mu_0^2)\beta_0 \log\left(\frac{\mu_R^2}{\mu_0^2}\right)}$$

This expression incorporates a leading-log resummation, remaining valid in regimes where

$$\alpha \log\big(\mu_R^2/\mu_0^2\big) \sim \mathcal{O}(1).$$

The $\overline{\text{MS}}$ scheme is particularly convenient for extrapolating theories to high energies and incorporating massless (or approximately massless) particles. Furthermore, choosing $\mu_R^2 \simeq Q^2$, where Q is the characteristic momentum scale of the process, minimizes large logarithmic terms of the form $\alpha^n(\log\big(Q^2/\mu_R^2\big))^L$ that appear in amplitudes at L-loops, thereby drastically improving perturbative convergence.

However, this expression reveals a profound physical feature: the denominator vanishes at a very large energy scale Λ. This defines a U.V. **Landau Pole**. As $\mu_R \to \Lambda$, QED becomes strongly coupled ($\alpha \to \infty$), and perturbation theory fundamentally breaks down. Vacuum polarization physically screens the charge, making QED weakly coupled at macroscopic distances but uncontrollably strong at microscopic distances. Setting the reference scale $\mu_0 = \Lambda$, we can write:

$$\alpha(\mu_R^2) = \frac{1}{\beta_0 \log\left(\frac{\Lambda^2}{\mu_R^2}\right)}$$

Using the low-energy measured values $\alpha(m_e^2) \simeq 1/137$ and $m_e = 0.5$ MeV, one naively finds $\Lambda \sim 10^{280}$ GeV. While astronomically large, this naive calculation is technically incorrect because it fails to include vacuum polarization loops from other heavier Standard Model particles. Yet, even when correctly including all known particles, the true Landau pole remains $\Lambda \sim 10^{50}$ GeV, which is still comfortably above the Planck scale and entirely out of physical reach.

This leads directly to the question of how to properly handle multiple heavy fermions. Assume our theory contains two fermions, an electron (e^-) and a muon (μ^-), with masses m and M respectively, such that $m \ll M$. If we are exclusively interested in physics at energy scales $E \ll M$, the heavy muons cannot be produced as real, on-shell particles. They can, however, still be produced as virtual pairs within quantum loops.

The critical principle of **Heavy Fermion Decoupling** states that the physical effects of these heavy virtual particles can be safely neglected in the strict limit $E/M \to 0$. We already observed this phenomenon mathematically in the on-shell scheme. When examining the 1PI one-loop corrections to the photon propagator at low momentum $q^2 \ll M^2$, the contribution from the heavy muon loop scales as:

$$\Pi(q^2) \simeq \frac{e^2 q^2}{60\pi^2 M^2} = \mathcal{O}\left(\frac{q^2}{M^2}\right)$$

This explicitly demonstrates that the heavy fermion's influence cleanly vanishes as $M \to \infty$.

How does this decoupling manifest in the $\overline{\text{MS}}$ scheme? Consider a hybrid renormalization scheme tailored for this scale separation:

- We define the counterterms such that contributions from the heavy μ^- loops are subtracted at exactly zero-momentum ($q^2 = 0$), perfectly mirroring the on-shell scheme.
- Conversely, everything else (the light e^- loops and counterterms) is subtracted using the standard $\overline{\text{MS}}$ prescription.

Within this hybrid framework, the heavy μ^- loop contributions remain suppressed by $\mathcal{O}(q^2/M^2)$. Neglecting these heavily suppressed terms is mathematically identical to simply using pure QED with only one light electron field defined entirely in the $\overline{\text{MS}}$ scheme.

As expected from our previous discussions on non-renormalizable theories, the residual $\mathcal{O}((E/M)^\alpha)$ physical contributions from the heavy decoupled fermion can be systematically accommodated by constructing an Effective Field Theory (EFT). This is achieved by adding specific non-renormalizable interaction terms (scaled by inverse powers of M) to the low-energy QED Lagrangian.

While the decoupling of heavy particles is a robust common property of many gauge theories (including QED and QCD), it is absolutely *not* a universal law of QFT! For instance, decoupling strictly fails in certain sectors of the Standard Model. Most notably, taking the limit of a large mass for the top quark ($M_t \to \infty$) does *not* cause its virtual loop effects to vanish. Instead, it yields finite, non-decoupling contributions to critically important observables, such as the production cross-section of the Higgs boson and, at two-loop order, the production of the Z boson.

7.3.2 | The Anomalous Magnetic Moment of the Electron

In this section, we describe one of the greatest and most historic triumphs of Quantum Field Theory: the extraordinarily precise theoretical prediction of the magnetic moment of the electron.

In standard non-relativistic Quantum Mechanics, the interaction coupling of an electron with a constant, weak, external magnetic field $\vec{B}$ can be described by adding a specific potential energy term to the total Hamiltonian:

$$\Delta H_{\text{int}} = -\vec{\mu} \cdot \vec{B}$$

where the **magnetic moment** vector $\vec{\mu}$ for an electron is classically and quantum-mechanically defined as:

$$\vec{\mu} = -\frac{e}{2m}(\vec{L} + g\vec{S})$$

Here, $\vec{L}$ is the orbital angular momentum operator, $\vec{S}$ is the intrinsic spin operator, and g is the Landé g-factor. Crucially, the g-factor cannot be derived or predicted from first principles within standard non-relativistic Quantum Mechanics. It had to be experimentally measured, and early experiments found it to be very close to $g = 2$.

In 1932, Paul Dirac achieved a monumental breakthrough. By constructing his relativistically invariant Dirac equation, he mathematically proved that for his model of elementary fermions, the factor must be exactly $g = 2$ at tree-level. However, in the late 1940s, more advanced and accurate experimental measurements revealed that while g was indeed very close to 2, it was definitely and measurably different from 2. This small deviation is what we call the **anomalous magnetic moment**, mathematically defined by the condition $g \neq 2$.

In 1948, Julian Schwinger manually computed the one-loop QED quantum corrections to the g-factor. His theoretical result showed excellent, unprecedented agreement with the new experimental data. This specific prediction, perhaps more than any other, definitively established Quantum Field Theory as the successful, correct framework for describing fundamental nature. Today, the theoretical value of g has been computed to an astonishing 5-loop accuracy, achieving a relative precision of roughly $\sim 10^{-13}$. The spectacular agreement between this complex theoretical calculation and modern experimental data makes it the most precisely verified prediction in the entire history of physics. In fact, the measurement and its theoretical prediction are so exquisitely accurate that measuring the anomaly $a_e = (g-2)/2$ and mathematically inverting the formula for the coupling 'e' is currently the single most accurate method for determining the fundamental electron charge (or equivalently, the fine-structure constant α).

In the following derivations, we will rigorously demonstrate exactly how the g-factor is related to the fundamental QED 1PI vertex function, and conceptually outline how to reproduce Schwinger's historic one-loop result.

Extracting the Magnetic Moment from QED

To physically extract the magnetic moment from our QFT formalism, we mathematically force the quantized electron field to interact with a classical, macroscopic, and weak magnetic field $\vec{B}$. Because the magnetic field is classical and macroscopic, it is well-described by a classical 4-potential vector field $A^\mu_{\text{cl}}(x)$, which is explicitly **NOT** quantized (it does not create or annihilate photon states). The specific Hamiltonian term describing the interaction energy between the quantum electron current and this classical background field is:

$$\Delta\hat{H} = \int \mathrm{d}^3x\, A^\mu_{\text{cl}}(x)\hat{J}_\mu(x) = \int \mathrm{d}^3x\, A^\mu_{\text{cl}}(x)(-e\bar{\hat{\psi}}(x)\gamma_\mu\hat{\psi}(x))$$

We evaluate the physical effect of this interaction by taking the quantum matrix element of this Hamiltonian operator between an incoming electron state $|e^-(p_1,s_1)\rangle$ and an outgoing electron state $|e^-(p_2,s_2)\rangle$. We specifically describe a non-relativistic electron by evaluating this in the strict limit where the spatial momentum transfer goes to zero: $\vec{q} = \vec{p}_2 - \vec{p}_1 \to 0$.

$$\langle e^-(p_2,s_2)|\Delta\hat{H}|e^-(p_1,s_1)\rangle = \int \mathrm{d}^3x\, A^\mu_{\text{cl}}(x)\langle e^-(p_2,s_2)|(-e\bar{\hat{\psi}}(x)\gamma_\mu\hat{\psi}(x))|e^-(p_1,s_1)\rangle$$

On the right-hand side, the expectation value is precisely a scattering matrix element featuring the explicit insertion of the current operator $-e\bar{\hat{\psi}}\gamma_\mu\hat{\psi}$. *(Note: We encountered a very similar mathematical structure when proving the Ward-Takahashi identity, except here the external fermion states are strictly ON-SHELL physical states).*

To compute this explicitly, we express the classical field in terms of its spatial Fourier transform:

$$A^\mu_{\text{cl}}(x) = \int \frac{\mathrm{d}^3q}{(2\pi)^3} e^{i\vec{q}\cdot\vec{x}}\tilde{A}^\mu_{\text{cl}}(\vec{q})$$

By assumption, the classical magnetic field is static (time-independent), meaning $\partial_0 A^\mu_{\text{cl}} = 0$. We can use the Fourier representation of unity $1 = \int \mathrm{d}x^0\delta(x^0) = \int \mathrm{d}x^0 \int \frac{\mathrm{d}q^0}{2\pi} e^{-iq^0x^0}$ to formally promote the 3D Fourier transform into a full 4D Fourier transform:

$$A^\mu_{\text{cl}}(x) = \int \mathrm{d}x^0 \int \frac{\mathrm{d}^4q}{(2\pi)^4} e^{-iq\cdot x}\tilde{A}^\mu_{\text{cl}}(\vec{q})$$

Substituting this back into our matrix element and applying the standard LSZ reduction steps to the field operators:

$$\begin{aligned}\langle \Delta\hat{H} \rangle &= \int \frac{\mathrm{d}^4 q}{(2\pi)^4} \tilde{A}^\mu_{\mathrm{cl}}(\vec{q}) \int \mathrm{d}^4 x\, e^{-iq\cdot x} \langle e^-(p_2)| (-e\bar{\hat{\psi}}(x)\gamma_\mu \hat{\psi}(x)) |e^-(p_1)\rangle \\ &\xrightarrow{\text{LSZ}} \int \frac{\mathrm{d}^4 q}{(2\pi)^4} \tilde{A}^\mu_{\mathrm{cl}}(\vec{q}) \int \mathrm{d}^4 x \int \mathrm{d}^4 x_1 \mathrm{d}^4 x_2\, e^{i(p_2\cdot x_2 - p_1\cdot x_1 - q\cdot x)} \times \\ &\qquad \times (-i(\not{p}_2 - m))(-i(\not{p}_1 - m))(-e)\langle \Omega | T\{\bar{\hat{\psi}}(x_1)\hat{\psi}(x_2)\bar{\hat{\psi}}(x)\gamma_\mu\hat{\psi}(x)\}|\Omega\rangle \\ &= \int \frac{\mathrm{d}^4 q}{(2\pi)^4} \tilde{A}^\mu_{\mathrm{cl}}(\vec{q}) \left[\mathcal{M}_\mu(e^- + \gamma^* \to e^-)\right] (2\pi)^4 \delta^{(4)}(p_1 + q - p_2)\end{aligned}$$

Here, we heavily utilized the fact that inserting the current operator $-e\bar{\hat{\psi}}\gamma_\mu\hat{\psi}$ is mathematically exactly equivalent to inserting an external photon interaction vertex, since they share the exact same Feynman rule (up to a trivial factor of i). The term $\mathcal{M}_\mu$ is the complete 3-point scattering amplitude involving an external, off-shell photon interacting with on-shell fermions. It is defined as:

$$\mathcal{M}_\mu = \bar{u}_s(p_2)(-e\Gamma_\mu)u_r(p_1)$$

where $-i\Gamma_\mu$ is the full, exact 1PI vertex function. *(Note: We specifically do NOT include the external photon propagator in this definition, as the field A^μ_{cl} is an external classical source, not a dynamical quantum state).*

Important Note: In principle, the full amplitude $\mathcal{M}_\mu$ should include all reducible topologies, such as corrections to the external fermion legs (self-energies). However, as we previously established using the LSZ formula and on-shell conditions, all diagrams where the photon attaches to an external leg vanish perfectly as we take the limit $q \to 0$, which is exactly the non-relativistic limit we are interested in. Therefore, only the 1PI vertex blob contributes.

TODO: add diagram: incoming fermion p_1, outgoing fermion p_2, interacting with a 1PI blob where the classical momentum q enters

Using the Dirac delta function $\delta^{(4)}(p_1 + q - p_2)$ to immediately perform the $\mathrm{d}^4 q$ integration (which enforces $q = p_2 - p_1$), we find:

$$\langle \Delta\hat{H} \rangle = -e\tilde{A}^\mu_{\mathrm{cl}}(\vec{q})\, \bar{u}_{s_2}(p_2)\Gamma_\mu u_{s_1}(p_1)$$

Recall from the previous section that we rigorously parameterized the most general form of the 1PI vertex function using form factors:

$$\Gamma_\mu = F_1(q^2)\gamma_\mu + F_2(q^2)\frac{i\sigma_{\mu\nu}q^\nu}{2m}$$

For this specific physical application, however, we use the algebraic **Gordon Identity** in reverse! We use it to mathematically trade the γ_μ matrix for the momentum vectors $(p_1^\mu + p_2^\mu)$ and the spin tensor $\sigma_{\mu\nu}$:

$$\bar{u}_2(p_1^\mu + p_2^\mu)u_1 = 2m\bar{u}_2\gamma^\mu u_1 + i\bar{u}_2\sigma^{\mu\nu}(p_{2\nu} - p_{1\nu})u_1$$

Isolating γ^μ and substituting it back into our Γ_μ expression, we obtain a new, fully equivalent parameterization of the vertex:

$$\Gamma_\mu = \frac{1}{2m}(p_{1\mu} + p_{2\mu})F_1(q^2) + \frac{i\sigma_{\mu\nu}q^\nu}{2m}\big(F_1(q^2) + F_2(q^2)\big)$$

It should be physically transparent that a **spin-dependent magnetic moment** can *only* arise from the second term containing the spin tensor $\sigma_{\mu\nu}$. The first term, proportional to $(p_{1\mu} + p_{2\mu})$, lacks any spin matrices and would identically be present even in a scalar theory like sQED. Because A^μ_{cl} represents a pure magnetic field, we can choose a gauge where the scalar potential vanishes,

$A^0_{\text{cl}} = 0$. The first term simply contracts to $(p_1^i + p_2^i)A^i_{\text{cl}}$. This is merely the standard kinetic orbital contribution already present in basic non-relativistic Quantum Mechanics, originating from expanding the Hamiltonian kinetic term $\frac{1}{2m}(\vec{p}-q\vec{A})^2 \sim \vec{p}\cdot\vec{A}+\vec{A}\cdot\vec{p}$. This orbital term is completely irrelevant for extracting the intrinsic spin magnetic moment.

Therefore, we exclusively focus on the second term. To evaluate it, we must carefully take the non-relativistic limit of the Dirac spinors. At low velocities, the full 4-component spinors $u(p)$ simplify to:

$$u(p) \simeq \sqrt{m}\begin{pmatrix}\xi_s\\ \xi_s\end{pmatrix}, \quad \text{where the 2-component Pauli spinors are } \xi_\uparrow = \begin{pmatrix}1\\0\end{pmatrix}, \xi_\downarrow = \begin{pmatrix}0\\1\end{pmatrix}$$

We need to compute the spinor matrix element $\bar{u}_2\sigma^{i0}u_1$ and $\bar{u}_2\sigma^{ij}u_1$. Let's evaluate the spatial-temporal components first. Using the anticommutation relation $\{\gamma^i,\gamma^0\}=0$ and $(\gamma^0)^2 = \mathbb{1}$:

$$\begin{aligned}\bar{u}_2\sigma^{i0}u_1 &= \bar{u}_2\left(\frac{i}{2}[\gamma^i,\gamma^0]\right)u_1 = \frac{i}{2}(u_2^\dagger\gamma^0\gamma^i\gamma^0u_1 - u_2^\dagger\gamma^0\gamma^0\gamma^iu_1)\\ &= -iu_2^\dagger\gamma^iu_1\\ &= -im\begin{pmatrix}\xi_{s_2}^\dagger & \xi_{s_2}^\dagger\end{pmatrix}\begin{pmatrix}0 & \sigma^i\\ -\sigma^i & 0\end{pmatrix}\begin{pmatrix}\xi_{s_1}\\ \xi_{s_1}\end{pmatrix}\\ &= -im(\xi_{s_2}^\dagger\sigma^i\xi_{s_1} - \xi_{s_2}^\dagger\sigma^i\xi_{s_1}) = 0\end{aligned}$$

Thus, the σ^{i0} term vanishes completely in the non-relativistic limit. Now we evaluate the purely spatial components $\bar{u}_2\sigma^{ij}u_1$. In the Dirac basis, the spatial sigma tensor is directly related to the physical 3D spin operator $S^k = \frac{1}{2}\sigma^k$:

$$\begin{aligned}\bar{u}_2\sigma^{ij}u_1 &= 2\bar{u}_2S^{ij}u_1\\ &= 2\bar{u}_2\left(\epsilon_{ijk}\begin{pmatrix}S^k & 0\\ 0 & S^k\end{pmatrix}\right)u_1\\ &= 4m\,\xi_{s_2}^\dagger\epsilon_{ijk}S^k\xi_{s_1}\end{aligned}$$

Putting all the non-zero pieces back together into the Hamiltonian matrix element, and utilizing the fact that $A^0 = 0$, we get:

$$\begin{aligned}\langle\Delta\hat{H}\rangle &= +e\tilde{A}^i_{\text{cl}}(\vec{q})\bar{u}_2\left[\frac{i\sigma_{ij}q^j}{2m}(F_1+F_2)\right]u_1\\ &= 2m\,\xi_{s_2}^\dagger\left[\frac{2ie}{2m}\epsilon_{ijk}S^kq^j\tilde{A}^i_{\text{cl}}\right]\xi_{s_1}\big(F_1(q^2)+F_2(q^2)\big)\end{aligned}$$

To connect this to the physical magnetic field $\vec{B}$, recall its definition via the vector potential:

$$B_i = -\frac{1}{2}\epsilon_{ijm}F_{jm} = \epsilon_{ijk}\partial_jA^k_{\text{cl}}$$

Fourier transforming this relation for a constant, homogeneous magnetic field ($\vec{B} = \text{const}$):

$$\begin{aligned}\int\frac{\mathrm{d}^3q}{(2\pi)^3}B_i(2\pi)^3\delta^{(3)}(\vec{q})e^{i\vec{q}\cdot\vec{x}} &= \epsilon_{ijk}\int\frac{\mathrm{d}^3q}{(2\pi)^3}e^{i\vec{q}\cdot\vec{x}}(iq_j)\tilde{A}^k_{\text{cl}}\\ \implies i\epsilon_{ijk}q^j\tilde{A}^k_{\text{cl}}(\vec{q}) &= B_i(2\pi)^3\delta^{(3)}(\vec{q})\end{aligned}$$

Substituting this elegant magnetic field identity directly back into our Hamiltonian matrix element gives:

$$\langle\Delta\hat{H}\rangle = 2m(2\pi)^3\delta^{(3)}(\vec{p}_2-\vec{p}_1)\times\xi_{s_2}^\dagger\left[-\frac{2e}{2m}\vec{S}\cdot\vec{B}\right]\xi_{s_1}\big(1+F_2(0)\big)$$

Here, because the delta function strictly enforces $\vec{q} = 0$, we have evaluated the form factors at zero momentum transfer, utilizing the exact renormalization condition $F_1(0) = 1$.

The prefactor $2m(2\pi)^3\delta^{(3)}(\vec{p}_2 - \vec{p}_1) \simeq 2E_p(2\pi)^3\delta^{(3)}(\vec{p}_2 - \vec{p}_1)$ is precisely the relativistically invariant normalization factor $\langle \vec{p}_2 | \vec{p}_1 \rangle$ for our external states. Stripping this universal normalization away, the relevant quantum mechanical operator acting on the spin states is simply:

$$\Delta \hat{H} = -\vec{\mu} \cdot \vec{B}$$

By directly identifying terms, we find the exact theoretical prediction for the electron magnetic moment operator:

$$\vec{\mu} = -\frac{e}{2m}\Big[2\big(1 + F_2(0)\big)\Big]\vec{S}$$

Comparing this result directly with the classical definition $\vec{\mu} = -\frac{e}{2m} g \vec{S}$, we decisively extract the Landé g-factor:

$$g = 2\big(1 + F_2(0)\big)$$

This is incredibly profound. The deviation of the g-factor from the Dirac value of 2 is entirely, exactly captured by the zero-momentum limit of the magnetic form factor $F_2(q^2)$. The anomaly is therefore often written simply as:

$$a_e \equiv \frac{g-2}{2} = F_2(0)$$

The Value of g-2 at Tree-Level and One-Loop

At leading tree-level (0-loops), there are no vertex corrections. The vertex is simply γ^μ, meaning $F_1 = 1$ and $F_2 = 0$. Consequently, $g = 2$, which perfectly recovers Dirac's original 1932 prediction.

The anomalous magnetic moment ($g - 2 \neq 0$) originates entirely from quantum loop corrections to the 1PI vertex function Γ^μ, evaluated strictly in the limit $q^2 \to 0$. The explicit computation of $F_2(q^2)$ at one-loop order involves evaluating the 1PI vertex diagram, isolating the $\sigma_{\mu\nu}$ tensor structure using Feynman parameters, and performing the momentum integration.

While the exact calculation is somewhat tedious algebraically, it is conceptually straightforward and completely finite. (Unlike F_1, which suffers from an ultraviolet divergence and requires renormalization by Z_1, F_2 is strictly convergent!). The well-known, historic result calculated by Julian Schwinger is:

$$\frac{g-2}{2} = F_2(0) = \frac{\alpha}{2\pi} \approx 0.0011614$$

This simple formula, $\alpha/2\pi$, is engraved on Schwinger's tombstone, standing as a testament to the predictive power of Quantum Field Theory.

Exercises

Sheet 1

[...]

Sheet 2

7.3.3 | 1. Feynman Diagrams in ϕ^3 Theory

[...]

2. Feynman Rules for Scalar QED (sQED)

Derive the Feynman rules of scalar QED (sQED). In order to do that, start from the Lagrangian

$$\mathcal{L} = -\frac{1}{4}(F_{\mu\nu})^2 + |D_\mu\phi|^2 - m^2|\phi|^2$$

where the covariant derivative $D_\mu \equiv \partial_\mu + ieA_\mu$. *Show that this is equal to*

$$\mathcal{L} = -\frac{1}{4}(F_{\mu\nu})^2 + |\partial_\mu\phi|^2 - m^2|\phi|^2 + \mathcal{L}_{\text{int}}$$

$$\mathcal{L}_{\text{int}} = -ieA^\mu\left(\phi^*(\partial_\mu\phi) - (\partial_\mu\phi^*)\phi\right) + e^2A^2|\phi|^2$$

The interaction Lagrangian is therefore a sum of two terms with different couplings and field contents, hence you should find Feynman rules for two vertexes, i.e. one for each term in $\mathcal{L}_{\text{int}}$.

To find the interaction terms, we explicitly expand the kinetic term for the scalar field, $|D_\mu\phi|^2$. We compute the complex conjugate of the covariant derivative:

$$(D_\mu\phi)^* = (\partial_\mu + ieA_\mu\phi)^* = \partial_\mu\phi^* - ieA_\mu\phi^*$$

Now, we multiply the covariant derivative by its complex conjugate (using the Einstein summation convention for the Lorentz index μ):

$$\begin{aligned}|D_\mu\phi|^2 &= (D_\mu\phi)^*(D^\mu\phi) = (\partial_\mu\phi^* - ieA_\mu\phi^*)(\partial^\mu\phi + ieA^\mu\phi)\\ &= \partial_\mu\phi^*\partial^\mu\phi + \partial_\mu\phi^*(ieA^\mu\phi) - (ieA_\mu\phi^*)\partial^\mu\phi - (ieA_\mu\phi^*)(ieA^\mu\phi)\end{aligned}$$

Recognizing that $\partial_\mu\phi^*\partial^\mu\phi = |\partial_\mu\phi|^2$ and that $-i^2 = 1$, we can simplify the expression:

$$\begin{aligned}|D_\mu\phi|^2 &= |\partial_\mu\phi|^2 + ieA^\mu\phi(\partial_\mu\phi^*) - ieA^\mu\phi^*(\partial_\mu\phi) + e^2A_\mu A^\mu\phi^*\phi\\ &= |\partial_\mu\phi|^2 - ieA^\mu\left[\phi^*(\partial_\mu\phi) - (\partial_\mu\phi^*)\phi\right] + e^2A^2|\phi|^2\end{aligned}$$

Substituting this expansion back into the full Lagrangian, we can perfectly separate the free (kinetic and mass) terms from the interaction terms:

$$\mathcal{L} = -\frac{1}{4}(F_{\mu\nu})^2 + |\partial_\mu\phi|^2 - m^2|\phi|^2 + \mathcal{L}_{\text{int}}$$

where the interaction Lagrangian $\mathcal{L}_{\text{int}}$ naturally splits into two distinct terms:

$$\mathcal{L}_{\text{int}} = -ieA^\mu\left(\phi^*(\partial_\mu\phi) - (\partial_\mu\phi^*)\phi\right) + e^2A^2|\phi|^2$$

The interaction Lagrangian contains two distinct terms with different field contents (a cubic term and a quartic term). Each term will independently generate a unique Feynman rule for a specific vertex. To derive the rules in momentum space, we consider the action $i\int d^4x\mathcal{L}_{\text{int}}$ and apply the standard Fourier transform substitutions for incoming/outgoing fields.

- **The 3-Point Vertex (Cubic Interaction):**

  The cubic interaction term is $\mathcal{L}_3 = -ieA^\mu(\phi^*(\partial_\mu\phi) - (\partial_\mu\phi^*)\phi)$. Let p be the momentum of the incoming scalar particle (associated with the field $\phi \sim e^{-ip\cdot x}$), and p' be the momentum of the outgoing scalar particle (associated with the conjugate field $\phi^* \sim e^{ip'\cdot x}$). When we transform the derivatives into momentum space, they pull down factors of the corresponding momenta:

  - $\partial_\mu\phi \to -ip_\mu\phi$
  - $\partial_\mu\phi^* \to +ip'_\mu\phi^*$

  TODO: add diagram: A vertex with one incoming solid line (scalar, momentum p), one outgoing solid line (scalar, momentum p'), and one incoming squiggly line (photon, momentum q, index μ).

  Substituting these into the action term $i\mathcal{L}_3$:

$$\begin{aligned} i\mathcal{L}_3 &\to i(-ieA^\mu)\left[\phi^*(-ip_\mu)\phi - (ip'_\mu\phi^*)\phi\right] \\ &= i(-ie)A^\mu\phi^*\phi\left[-i(p_\mu + p'_\mu)\right] \\ &= -ie(p_\mu + p'_\mu)A^\mu\phi^*\phi \end{aligned}$$

  Stripping away the fields leaves the Feynman rule for the vertex. Note that unlike standard spinor QED, the vertex in scalar QED intrinsically depends on the momenta of the interacting particles due to the derivative coupling.

$$\textbf{Vertex Rule: } -ie(p_\mu + p'_\mu)$$

- **The 4-Point "Seagull" Vertex (Quartic Interaction):**

  The quartic interaction term is $\mathcal{L}_4 = e^2A^2|\phi|^2 = e^2g_{\mu\nu}A^\mu A^\nu\phi^*\phi$. This vertex involves two scalar particles and two photons. The vertex rule is obtained by multiplying the coefficient in $i\mathcal{L}_4$ by the appropriate symmetry factor. Because there are two identical photon fields A^μ and A^ν in the interaction term, there are $2! = 2$ identical ways to contract them with two external photon lines. This yields a combinatoric symmetry factor of 2.

  TODO: add diagram: A vertex with one incoming solid line, one outgoing solid line, and two squiggly photon lines attaching at the same point, with indices μ and ν.

$$\textbf{Vertex Rule: } i(e^2g_{\mu\nu}) \times 2 = 2ie^2g_{\mu\nu}$$

3. Scattering Amplitude in sQED

In sQED, compute the scattering amplitude $i\mathcal{M}$ for the process $\phi(p_1) + \phi(p_2) \to \phi(p_3) + \phi(p_4)$ at tree level, as a function of the Mandelstam invariants s, t and u (and the mass m of the field).

Solution: $i\mathcal{M} = ie^2((s-u)/t + (s-t)/u)$ *with* $s + t + u = 4m^2$.

To compute the tree-level scattering amplitude for the elastic scattering of two identical charged scalars, $\phi(p_1) + \phi(p_2) \to \phi(p_3) + \phi(p_4)$, we must first systematically identify all allowed Feynman diagrams for this process.

In scalar QED, the interaction Lagrangian contains a 3-point vertex coupling two scalars to one photon ($\sim A^\mu\phi^*\partial_\mu\phi$) and a 4-point "seagull" vertex coupling two scalars to two photons ($\sim A^2|\phi|^2$). Because our specific scattering process involves strictly four external scalars and zero external photons, the 4-point seagull vertex physically cannot contribute at tree level.

Furthermore, since the colliding incoming particles are identical (both are exactly ϕ, not a particle-antiparticle pair $\phi\phi^*$), they cannot annihilate into a single virtual photon. Thus, there is absolutely

no s-channel annihilation diagram. The process is purely mediated by the exchange of a virtual photon in the t-channel and the u-channel.

TODO: add diagram: Two Feynman diagrams. 1) t-channel: p1 and p3 connected by a vertical photon line to p2 and p4. 2) u-channel: p1 and p4 connected by a crossed photon line to p2 and p3.

Because the ϕ fields represent bosons, the total quantum scattering amplitude must be perfectly symmetric under the exchange of the identical final state particles ($p_3 \leftrightarrow p_4$). Therefore, the relative sign between the t-channel and u-channel diagrams is strictly positive (unlike fermions, which would incur a relative minus sign).

Using the Feynman rules for scalar QED derived previously, where the 3-point vertex is $-ie(p_{\text{in}} + p_{\text{out}})_\mu$ and the photon propagator is $-ig_{\mu\nu}/q^2$, we can directly evaluate the two contributing diagrams.

1. The t-channel amplitude:

The transferred momentum is $q = p_1 - p_3 = p_4 - p_2$, meaning $q^2 = t$.

$$\begin{aligned} i\mathcal{M}_t &= [-ie(p_1 + p_3)^\mu] \left(\frac{-ig_{\mu\nu}}{t} \right) [-ie(p_2 + p_4)^\nu] \\ &= i\frac{e^2}{t}(p_1 + p_3) \cdot (p_2 + p_4) \end{aligned}$$

2. The u-channel amplitude:

We obtain this by physically exchanging the final state particles $p_3 \leftrightarrow p_4$. The transferred momentum is now $q = p_1 - p_4 = p_3 - p_2$, meaning $q^2 = u$.

$$\begin{aligned} i\mathcal{M}_u &= [-ie(p_1 + p_4)^\mu] \left(\frac{-ig_{\mu\nu}}{u} \right) [-ie(p_2 + p_3)^\nu] \\ &= i\frac{e^2}{u}(p_1 + p_4) \cdot (p_2 + p_3) \end{aligned}$$

3. Expressing the Amplitude in Mandelstam Invariants:

To match the requested solution, we need to simplify the kinematic dot products present in the numerators. Recall the standard definitions of the Mandelstam variables for a process with equal masses ($m_1 = m_2 = m_3 = m_4 = m$):

$$\begin{aligned} s &= (p_1 + p_2)^2 = 2m^2 + 2p_1 \cdot p_2 = 2m^2 + 2p_3 \cdot p_4 \\ t &= (p_1 - p_3)^2 = 2m^2 - 2p_1 \cdot p_3 = 2m^2 - 2p_2 \cdot p_4 \\ u &= (p_1 - p_4)^2 = 2m^2 - 2p_1 \cdot p_4 = 2m^2 - 2p_2 \cdot p_3 \end{aligned}$$

which rigorously obey the kinematic constraint $s + t + u = 4m^2$.

Let us expand the numerator of the t-channel amplitude:

$$(p_1 + p_3) \cdot (p_2 + p_4) = p_1 \cdot p_2 + p_1 \cdot p_4 + p_3 \cdot p_2 + p_3 \cdot p_4$$

We can directly substitute the dot products extracted algebraically from the Mandelstam definitions above:

- $p_1 \cdot p_2 = p_3 \cdot p_4 = \frac{1}{2}(s - 2m^2)$

- $p_1 \cdot p_4 = p_2 \cdot p_3 = \frac{1}{2}(2m^2 - u)$

Summing these individual components together gives:

$$\begin{aligned}(p_1 + p_3) \cdot (p_2 + p_4) &= \frac{1}{2}(s - 2m^2) + \frac{1}{2}(2m^2 - u) + \frac{1}{2}(2m^2 - u) + \frac{1}{2}(s - 2m^2) \\ &= (s - 2m^2) + (2m^2 - u) = s - u.\end{aligned}$$

By exact physical symmetry ($p_3 \leftrightarrow p_4$, which mathematically translates to swapping $t \leftrightarrow u$), the numerator of the u-channel amplitude evaluates to:

$$(p_1 + p_4) \cdot (p_2 + p_3) = s - t$$

4. Final Amplitude:

Summing the t-channel and u-channel contributions, we arrive at the total tree-level scattering amplitude:

$$i\mathcal{M} = i\mathcal{M}_t + i\mathcal{M}_u = i\frac{e^2}{t}(s - u) + i\frac{e^2}{u}(s - t)$$

Factoring out the common ie^2 term, we perfectly recover the required expression:

$$i\mathcal{M} = ie^2\left(\frac{s - u}{t} + \frac{s - t}{u}\right)$$

4. Crossing Symmetry in sQED

Using the result of the previous exercise and crossing symmetry*, compute the tree-level scattering amplitude* $i\mathcal{M}$ *for the process* $\phi(p_1) + \bar{\phi}(p_2) \to \phi(p_3) + \bar{\phi}(p_4)$ *in sQED.*
Solution: $i\mathcal{M} = -ie^2((s - u)/t + (t - u)/s)$.

Crossing symmetry is a powerful and elegant principle in Quantum Field Theory. It states that the S-matrix amplitude for any process involving a particle with momentum p in the initial (or final) state is analytically related to the amplitude of a process where that particle is moved to the final (or initial) state as its corresponding *antiparticle*, with its four-momentum analytically continued to $-p$.

Let us start from the result of the previous exercise for the elastic scattering of two identical particles:

$$\phi(p_1') + \phi(p_2') \to \phi(p_3') + \phi(p_4')$$

whose amplitude, expressed in terms of its own Mandelstam variables s', t', u', was found to be:

$$i\mathcal{M}_{\phi\phi\to\phi\phi}(s', t', u') = ie^2\left(\frac{s' - u'}{t'} + \frac{s' - t'}{u'}\right)$$

We want to compute the amplitude for the particle-antiparticle scattering process:

$$\phi(p_1) + \bar{\phi}(p_2) \to \phi(p_3) + \bar{\phi}(p_4)$$

To map the original process to this new process using crossing symmetry, we must:

1. Cross the incoming particle $\phi(p_2')$ to the final state, where it becomes the outgoing antiparticle $\bar{\phi}(p_4)$. This implies substituting $p_2' \to -p_4$.

2. Cross the outgoing particle $\phi(p_4')$ to the initial state, where it becomes the incoming antiparticle $\bar{\phi}(p_2)$. This implies substituting $p_4' \to -p_2$.

The other two particles remain in their original positions, so $p_1' = p_1$ and $p_3' = p_3$.

Now, let us analyze how the original Mandelstam variables transform under this specific momentum crossing. The new process has its standard Mandelstam invariants defined as $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, and $u = (p_1 - p_4)^2$. Evaluating the old invariants with the substituted momenta:

$$s' = (p_1' + p_2')^2 \xrightarrow{\text{crossing}} (p_1 - p_4)^2 = u$$
$$t' = (p_1' - p_3')^2 \xrightarrow{\text{crossing}} (p_1 - p_3)^2 = t$$
$$u' = (p_1' - p_4')^2 \xrightarrow{\text{crossing}} (p_1 + p_2)^2 = s$$

The crossing transformation therefore corresponds perfectly to a simple algebraic swap of the variables s and u, while t remains invariant ($s' \to u$, $u' \to s$, $t' \to t$).

Physically, this means the t-channel exchange diagram from the original process remains a t-channel exchange in the new process. However, the original u-channel exchange is transformed into an s-channel annihilation diagram ($\phi\bar{\phi} \to \gamma^* \to \phi\bar{\phi}$).

To find the new amplitude, we simply take the old amplitude and apply the $s \leftrightarrow u$ swap:

$$i\mathcal{M}_{\phi\bar{\phi}\to\phi\bar{\phi}}(s,t,u) = i\mathcal{M}_{\phi\phi\to\phi\phi}(u,t,s) = ie^2\left(\frac{u-s}{t} + \frac{u-t}{s}\right)$$

To match the requested format of the solution, we can elegantly factor out a global minus sign from the numerators:

$$\frac{u-s}{t} = -\frac{s-u}{t} \quad \text{and} \quad \frac{u-t}{s} = -\frac{t-u}{s}$$

Substituting these back into our expression, we arrive exactly at the required tree-level amplitude:

$$i\mathcal{M} = -ie^2\left(\frac{s-u}{t} + \frac{t-u}{s}\right)$$

5. Gamma Matrix Identity in 4D

Prove the following identity, in four space-time dimensions (note the reversed order of the gamma matrices on the right-hand side)

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = -2\gamma^\sigma\gamma^\rho\gamma^\nu.$$

To prove this identity, we systematically use the fundamental Clifford algebra anticommutation relation for Dirac gamma matrices:

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}\mathbb{I}$$

which allows us to swap adjacent gamma matrices by paying a penalty term: $\gamma^\mu\gamma^\nu = 2\eta^{\mu\nu} - \gamma^\nu\gamma^\mu$. We also rely on the explicit trace of the metric tensor in exactly $D = 4$ dimensions, which gives the fundamental contraction:

$$\gamma^\mu\gamma_\mu = \eta_{\mu\nu}\gamma^\mu\gamma^\nu = \frac{1}{2}\eta_{\mu\nu}\{\gamma^\mu, \gamma^\nu\} = \eta_{\mu\nu}\eta^{\mu\nu} = \delta^\mu_\mu = 4\mathbb{I}$$

Instead of tackling the full five-matrix string at once, it is much easier and more instructive to build up to it by recursively computing the contractions of shorter strings. We do this by continuously anti-commuting the rightmost γ_μ to the left until it meets its contracted partner γ^μ.

Step 1: Simplify $\gamma^\mu\gamma^\nu\gamma_\mu$ Anti-commuting the γ_μ past the γ^ν:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma_\mu &= \gamma^\mu(2\delta^\nu_\mu - \gamma_\mu\gamma^\nu)\\ &= 2\gamma^\nu - \gamma^\mu\gamma_\mu\gamma^\nu\end{aligned}$$

Using our 4D identity $\gamma^\mu\gamma_\mu = 4$, we get:

$$\gamma^\mu\gamma^\nu\gamma_\mu = 2\gamma^\nu - 4\gamma^\nu = -2\gamma^\nu$$

Step 2: Simplify $\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu$ Anti-commuting the γ_μ past the γ^ρ:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu &= \gamma^\mu\gamma^\nu(2\delta^\rho_\mu - \gamma_\mu\gamma^\rho)\\ &= 2\gamma^\rho\gamma^\nu - (\gamma^\mu\gamma^\nu\gamma_\mu)\gamma^\rho\end{aligned}$$

Now substitute the result from Step 1 for the bracketed term:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu &= 2\gamma^\rho\gamma^\nu - (-2\gamma^\nu)\gamma^\rho\\ &= 2(\gamma^\rho\gamma^\nu + \gamma^\nu\gamma^\rho)\end{aligned}$$

Recognizing the anticommutator $\{\gamma^\rho, \gamma^\nu\} = 2\eta^{\rho\nu}$, this elegantly collapses to a simple scalar multiple of the identity matrix:

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu = 2(2\eta^{\rho\nu}) = 4\eta^{\nu\rho}$$

Step 3: Simplify the full expression $\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu$ We apply the exact same logic one final time, moving the γ_μ past the γ^σ:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu &= \gamma^\mu\gamma^\nu\gamma^\rho(2\delta^\sigma_\mu - \gamma_\mu\gamma^\sigma)\\ &= 2\gamma^\sigma\gamma^\nu\gamma^\rho - (\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu)\gamma^\sigma\end{aligned}$$

Substituting our highly simplified scalar result from Step 2 for the bracketed term:

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = 2\gamma^\sigma\gamma^\nu\gamma^\rho - 4\eta^{\nu\rho}\gamma^\sigma$$

Step 4: Reorder the matrices to match the target identity Our result from Step 3 is correct, but it does not match the reversed order required by the problem statement ($-2\gamma^\sigma\gamma^\rho\gamma^\nu$). We need to swap the adjacent $\gamma^\nu\gamma^\rho$ in the first term:

$$\gamma^\nu\gamma^\rho = 2\eta^{\nu\rho} - \gamma^\rho\gamma^\nu$$

Substitute this back into our expression:

$$\begin{aligned}2\gamma^\sigma(\gamma^\nu\gamma^\rho) - 4\eta^{\nu\rho}\gamma^\sigma &= 2\gamma^\sigma(2\eta^{\nu\rho} - \gamma^\rho\gamma^\nu) - 4\eta^{\nu\rho}\gamma^\sigma\\ &= 4\eta^{\nu\rho}\gamma^\sigma - 2\gamma^\sigma\gamma^\rho\gamma^\nu - 4\eta^{\nu\rho}\gamma^\sigma\end{aligned}$$

The $4\eta^{\nu\rho}\gamma^\sigma$ terms perfectly cancel each other out, leaving only the desired term with the completely reversed matrix order:

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = -2\gamma^\sigma\gamma^\rho\gamma^\nu$$

This mathematically proves the requested identity.

6. Unpolarized scattering of fermions in the Yukawa theory

Compute the unpolarized (i.e. averaged over initial and summed over final polarization states) squared tree-level scattering amplitude $|\mathcal{M}|^2_{\text{unpol.}}$ for elastic scattering of two identical massless Dirac fermions (e.g. $e^-e^- \to e^-e^-$ for $m_e \to 0$) in the Yukawa theory, as a functions of the Mandelstam invariants and the mass M of the scalar.

For the Yukawa theory, the interaction Lagrangian is given by

$$\mathcal{L}_{\text{int}} = -g\phi\bar{\psi}\psi,$$

and the vertices are represented by a factor of $-ig$. The tree-level scattering amplitude for the process

$$e^-(p_1)e^-(p_2) \to e^-(p_3)e^-(p_4),$$

can be computed using the Feynman rules for the Yukawa theory, which involve the exchange of a scalar particle in the t-channel and u-channel:

TODO: add diagram: t-channel and u-channel Feynman diagrams for identical fermion scattering

$$i\mathcal{M} = ,$$

where using Feynman rules, we have

$$i\mathcal{M} = (-ig)^2 \left[\bar{u}_3 u_1 \frac{i}{t - M^2} \bar{u}_4 u_2 - \bar{u}_4 u_1 \frac{i}{u - M^2} \bar{u}_3 u_2 \right] = \mathcal{M}_1 + \mathcal{M}_2,$$

with $t = (p_1 - p_3)^2$ and $u = (p_1 - p_4)^2$, and the minus sign arises due to the fact that the u-channel is obtained by exchanging the two identical fermions in the final state w.r.t. the t-channel, which introduces a minus sign (see extra rules for fermions).

To compute the unpolarized squared amplitude, we must average over the 4 possible initial spin configurations (yielding a factor of $1/4$) and sum over all final spin states:

$$|\mathcal{M}_{\text{unpol.}}|^2 = \frac{1}{4} \sum_s |\mathcal{M}_1 + \mathcal{M}_2|^2 = \frac{1}{4} \sum_s \left(|\mathcal{M}_1|^2 + |\mathcal{M}_2|^2 + 2\text{Re}[\mathcal{M}_1 \mathcal{M}_2^*] \right)$$

Expanding the squared terms and the interference term, and applying the completeness relation $\sum_s u(p)\bar{u}(p) = \not{p} + m \to \not{p}$ for massless fermions, we can express the spin sums entirely as Dirac traces:

$$|\mathcal{M}_{\text{unpol.}}|^2 = \frac{g^4}{4} \sum_s \left[\frac{(\bar{u}_3 u_1 \bar{u}_1 u_3)(\bar{u}_4 u_2 \bar{u}_2 u_4)}{(t - M^2)^2} + \frac{(\bar{u}_4 u_1 \bar{u}_1 u_4)(\bar{u}_3 u_2 \bar{u}_2 u_3)}{(u - M^2)^2} - \frac{2(\bar{u}_3 u_1 \bar{u}_1 u_4 \bar{u}_4 u_2 \bar{u}_2 u_3)}{(t - M^2)(u - M^2)} \right]$$

Evaluating the traces for each individual fermionic line gives:

$$\sum_s (\bar{u}_3 u_1 \bar{u}_1 u_3) = \text{Tr}[\not{p}_1 \not{p}_3] = 4(p_1 \cdot p_3)$$

$$\sum_s (\bar{u}_4 u_2 \bar{u}_2 u_4) = \text{Tr}[\not{p}_2 \not{p}_4] = 4(p_2 \cdot p_4)$$

$$\sum_s (\bar{u}_4 u_1 \bar{u}_1 u_4) = \text{Tr}[\not{p}_1 \not{p}_4] = 4(p_1 \cdot p_4)$$

$$\sum_s (\bar{u}_3 u_2 \bar{u}_2 u_3) = \text{Tr}[\not{p}_2 \not{p}_3] = 4(p_2 \cdot p_3)$$

and the final, more difficult interference trace is given by

$$\begin{aligned} \sum_s (\bar{u}_3 u_1 \bar{u}_1 u_4 \bar{u}_4 u_2 \bar{u}_2 u_3) &= \text{Tr}[\not{p}_1 \not{p}_4 \not{p}_2 \not{p}_3] \\ &= 4\left((p_1 \cdot p_4)(p_2 \cdot p_3) - (p_1 \cdot p_2)(p_3 \cdot p_4) + (p_1 \cdot p_3)(p_2 \cdot p_4) \right). \end{aligned}$$

If we now use the limit for $m_e \to 0$ we can compute the scalar products using the Mandelstam invariants:

$$s = (p_1 + p_2)^2 = (p_3 + p_4)^2 = 2(p_1 \cdot p_2) = 2(p_3 \cdot p_4) \implies p_1 \cdot p_2 = p_3 \cdot p_4 = \frac{s}{2}$$
$$t = (p_1 - p_3)^2 = (p_2 - p_4)^2 = -2(p_1 \cdot p_3) = -2(p_2 \cdot p_4) \implies p_1 \cdot p_3 = p_2 \cdot p_4 = -\frac{t}{2}$$
$$u = (p_1 - p_4)^2 = (p_2 - p_3)^2 = -2(p_1 \cdot p_4) = -2(p_2 \cdot p_3) \implies p_1 \cdot p_4 = p_2 \cdot p_3 = -\frac{u}{2}$$

and now, using the properties of the Mandelstam invariants, we can write the final result as a function of t and u, by eliminating s since $s + t + u = 0$ for massless particles. Substituting the scalar products back into the traces, we find:

$$|\mathcal{M}_{\text{unpol.}}|^2 = \frac{g^4}{4}\left[\frac{16(-t/2)(-t/2)}{(t-M^2)^2} + \frac{16(-u/2)(-u/2)}{(u-M^2)^2} - \frac{8\left((-u/2)(-u/2) - (s/2)(s/2) + (-t/2)(-t/2)\right)}{(t-M^2)(u-M^2)}\right]$$

Simplifying the fractions and algebraic terms:

$$|\mathcal{M}_{\text{unpol.}}|^2 = g^4\left[\frac{t^2}{(t-M^2)^2} + \frac{u^2}{(u-M^2)^2} - \frac{1}{2}\frac{u^2 - s^2 + t^2}{(t-M^2)(u-M^2)}\right]$$

Using the mass-shell relation $s + t + u = 0 \implies s^2 = (t+u)^2 = t^2 + u^2 + 2tu$, the numerator of the interference term cleanly simplifies to $u^2 - (t^2 + u^2 + 2tu) + t^2 = -2tu$. Thus we have the final compact form:

$$|\mathcal{M}_{\text{unpol.}}|^2 = g^4\left[\frac{t^2}{(t-M^2)^2} + \frac{u^2}{(u-M^2)^2} + \frac{tu}{(t-M^2)(u-M^2)}\right]$$

Remark. *When the scalar mass approaches zero, i.e. $M \to 0$, we get a constant matrix element for the unpolarized scattering amplitude, which is given by $|\mathcal{M}_{unpol.}|^2 \to 3g^4$. This is a consequence of the fact that the propagator of the scalar particle becomes massless, so that it behaves as $1/t$ or $1/u$, and since t and u are negative, we get a constant contribution to the scattering amplitude.*

7. Decay of a scalar into a fermion-antifermion pair

In the Yukawa theory, compute the total decay rate Γ of a scalar into a fermion antifermion pair, as a function of the masses M of the scalar and m of the Dirac fermion. Which condition must M and m satisfy for this decay to be possible?

The vertex for the decay process $\phi(P) \to f(p_1) + \bar{f}(p_2)$ is given directly by the Feynman rule for the Yukawa interaction $\mathcal{L}_{\text{int}} = -g\phi\bar{\psi}\psi$:

$$i\mathcal{M} = \ = -(ig)\bar{u}(p_1)v(p_2).$$

TODO: add diagram: incoming dashed scalar line splitting into two outgoing solid fermion lines

To compute the unpolarized decay rate, we must square the amplitude and sum over the final state spins of both the fermion and the antifermion. Note that because there is only one initial state particle (which is a scalar, spin-0), there is no initial spin averaging required:

$$|\mathcal{M}_{\text{unpol.}}|^2 = \sum_{s_1,s_2} |\mathcal{M}|^2 = g^2 \sum_{s_1,s_2} (\bar{u}_1 v_2)(\bar{v}_2 u_1)$$

Applying the standard completeness relations for massive Dirac spinors, $\sum_s u(p)\bar{u}(p) = \not{p} + m$ and $\sum_s v(p)\bar{v}(p) = \not{p} - m$, we convert the spin sums into a Dirac trace:

$$\begin{aligned}|\mathcal{M}_{\text{unpol.}}|^2 &= g^2 \text{Tr}\left[(\not{p}_1 + m)(\not{p}_2 - m)\right] \\ &= g^2\left(\text{Tr}[\not{p}_1\not{p}_2] - m^2\text{Tr}[\not{K}]\right) \quad \text{(since traces of an odd number of } \gamma \text{ matrices vanish)} \\ &= g^2(4p_1 \cdot p_2 - 4m^2)\end{aligned}$$

To express the scalar product $p_1 \cdot p_2$ in terms of the fundamental masses of the theory, we use the physical constraint of 4-momentum conservation, $P = p_1 + p_2$. Squaring both sides of this equation yields:

$$M^2 = P^2 = (p_1 + p_2)^2 = m^2 + m^2 + 2p_1 \cdot p_2 \implies 2p_1 \cdot p_2 = M^2 - 2m^2,$$

which allows us to rewrite the unpolarized squared amplitude entirely in terms of masses:

$$|\mathcal{M}_{\text{unpol.}}|^2 = 2g^2(M^2 - 4m^2).$$

Now, we utilize the master formula for the differential decay rate of a particle of mass M decaying into a two-body final state:

$$\text{d}\Gamma = \frac{1}{2M}|\mathcal{M}_{\text{unpol.}}|^2 \text{d}\Pi_2,$$

where $\text{d}\Pi_2$ is the Lorentz-invariant two-particle phase space measure. It is most convenient to evaluate this measure in the center of mass (CM) frame of the decaying particle, where it takes the simplified form (as explicitly computed in lecture notes 2.2):

$$\text{d}\Pi_2 = \text{d}\Omega \frac{1}{16\pi^2}\frac{|\vec{p}_f|}{E_{c.m.}}.$$

In this CM frame, the kinematics are strictly defined as:

$$\begin{cases} E_{c.m.} = M, \\ |\vec{p}_f| = |\vec{p}_1| = |\vec{p}_2| = \sqrt{E_1^2 - m^2}, \\ P = \begin{pmatrix} M \\ \vec{0} \end{pmatrix}, \quad p_1 = \begin{pmatrix} E_1 \\ \vec{p}_f \end{pmatrix}. \end{cases}$$

We must now compute explicit expressions for the final state energy E_1 and the final state 3-momentum magnitude $|\vec{p}_f|$ strictly as a function of the masses M and m. We can elegantly extract the energy E_1 by evaluating the Lorentz-invariant scalar product $P \cdot p_1$ in the CM frame:

$$P \cdot p_1 = ME_1 - \vec{0} \cdot \vec{p}_f = ME_1.$$

We can also evaluate this same invariant product by substituting $P = p_1 + p_2$ before evaluating:

$$P \cdot p_1 = (p_1 + p_2) \cdot p_1 = p_1^2 + p_1 \cdot p_2 = m^2 + \frac{1}{2}(M^2 - 2m^2) = \frac{1}{2}M^2.$$

Equating the two results for $P \cdot p_1$, we find the final state energy:

$$E_1 = \frac{M}{2}.$$

(Note: This result is physically intuitive; since the final state particles have identical masses m, they must equally share the total available rest energy M in the CM frame.)

With E_1 determined, we can immediately compute the magnitude of the final state spatial momentum:

$$|\vec{p}_f| = \sqrt{E_1^2 - m^2} = \sqrt{\frac{M^2}{4} - m^2} = \frac{M}{2}\sqrt{1 - \frac{4m^2}{M^2}}.$$

With all kinematic variables explicitly resolved, we substitute them back into the phase space measure and the master decay formula to obtain the differential decay rate with respect to the solid angle:

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}\Omega} = \frac{1}{2M}\left(\frac{1}{16\pi^2}\frac{\frac{M}{2}\sqrt{1-4m^2/M^2}}{M}\right)\left[2g^2M^2\left(1-\frac{4m^2}{M^2}\right)\right].$$

Simplifying the extensive algebraic factors, the dependence on the solid angle Ω completely drops out (which is expected, as a scalar particle has no spin orientation to break spatial isotropy):

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}\Omega} = \frac{g^2 M}{32\pi^2}\left(1-\frac{4m^2}{M^2}\right)^{3/2}.$$

Because the differential rate is perfectly isotropic, integrating over the full solid angle simply multiplies the result by $\int \mathrm{d}\Omega = 4\pi$:

$$\Gamma = \int \frac{\mathrm{d}\Gamma}{\mathrm{d}\Omega}\mathrm{d}\Omega = 4\pi\left(\frac{g^2M}{32\pi^2}\left(1-\frac{4m^2}{M^2}\right)^{3/2}\right) = \frac{g^2M}{8\pi}\left(1-\frac{4m^2}{M^2}\right)^{3/2}.$$

Remark. *The physical condition for the decay to be kinematically possible is strictly given by the requirement that the total available energy of the initial state particle must be strictly greater than or equal to the total rest mass of the final state particles. Therefore, we must have $M \geq 2m$. When we approach the extreme threshold limit, $M \to 2m$, the total decay rate vanishes: $\Gamma \to 0$. This occurs because the phase space volume for the decay shrinks to zero; the final state particles would be produced perfectly at rest with zero spatial momentum, leaving no available kinematic phase space for the transition to occur.*

8. Compton scattering in QED

In QED, compute the unpolarized squared tree-level scattering amplitude for elastic scattering of a massless electron and a photon, as a function of the Mandelstam invariants. Use this result to compute the differential cross section $\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}$ for this process in the centre of mass. Solution: $|\mathcal{M}|^2_{\text{unpol.}} = -2e^4(s/u + u/s)$, $d\sigma/d\Omega = \alpha^2\left(\frac{\cos\theta+1}{2} + \frac{2}{\cos\theta+1}\right)/(2E_{\text{cm}}^2)$.

The scattering process is $e^-(p_1) + \gamma(p_2) \to e^-(p_3) + \gamma(p_4)$. At tree level, this is mediated by the s-channel and u-channel exchange of a virtual electron. Using the Feynman rules for QED, the total amplitude is:

$$\begin{aligned} i\mathcal{M} &= \\ &= (-ie)^2\left[\bar{u}_3\gamma^\nu\frac{\not{p}_1+\not{p}_2}{s}\gamma^\mu u_1 + \bar{u}_3\gamma^\mu\frac{\not{p}_1-\not{p}_4}{u}\gamma^\nu u_1\right]\epsilon_{2\mu}\epsilon^*_{4\nu} \\ &= i\mathcal{M}_1 + i\mathcal{M}_2, \end{aligned}$$

TODO: add diagram: s-channel and u-channel Feynman diagrams for Compton scattering

where we used the massless momentum invariants $s = (p_1+p_2)^2$ and $u = (p_1-p_4)^2$ for the denominators of the fermion propagators.

To find the unpolarized squared amplitude, we average over the 4 initial spin/polarization states (2 for the electron, 2 for the photon) and sum over all final states:

$$|\mathcal{M}_{\text{unpol.}}|^2 = \frac{1}{4}\sum_{s,\lambda}|\mathcal{M}_1 + \mathcal{M}_2|^2 = \frac{1}{4}\sum_{s,\lambda}\left(|\mathcal{M}_1|^2 + |\mathcal{M}_2|^2 + 2\text{Re}[\mathcal{M}_1^*\mathcal{M}_2]\right) \equiv A + B + C.$$

where now we have to compute the A, B and C terms. Starting from the interference term C we have

$$\begin{aligned} C &= \frac{1}{4}\sum_{s,\lambda} 2\text{Re}[\mathcal{M}_1^*\mathcal{M}_2] \\ &= \frac{e^4}{2su}\text{Tr}\left[\not{p}_1\gamma^\mu(\not{p}_1 + \not{p}_2)\gamma^\nu\not{p}_3\gamma^\rho(\not{p}_1 - \not{p}_4)\gamma^\sigma\right] \times (\epsilon^*_{2\mu}\epsilon^*_{4\nu}\epsilon_{2\rho}\epsilon_{4\sigma}) \end{aligned}$$

where the last term is the photon polarization sum. Using the unpolarized completeness relation $\sum \epsilon_\alpha\epsilon^*_\beta = -g_{\alpha\beta}$, this becomes $(-g_{\mu\rho})(-g_{\nu\sigma}) = g_{\mu\rho}g_{\nu\sigma}$, enforcing the contraction of the Lorentz indices within the trace:

$$C = \frac{e^4}{2su}\text{Tr}\left[\not{p}_1\gamma^\mu(\not{p}_1 + \not{p}_2)\gamma^\nu\not{p}_3\gamma_\mu(\not{p}_1 - \not{p}_4)\gamma_\nu\right].$$

If we now use the result computed in exercise 5, in which we demonstrated that

$$\gamma^\mu\gamma^\alpha\gamma^\beta\gamma^\delta\gamma_\mu = -2\gamma^\delta\gamma^\beta\gamma^\alpha,$$

it easy to see that applying this to the inner segment of our trace implies

$$\implies \gamma^\mu(\not{p}_1 + \not{p}_2)\gamma^\nu\not{p}_3\gamma_\mu = -2\not{p}_3\gamma^\nu(\not{p}_1 + \not{p}_2).$$

Thus we can further simplify the expression for C as

$$C = -\frac{e^4}{su}\text{Tr}\left[\not{p}_1\not{p}_3\gamma^\nu(\not{p}_1 + \not{p}_2)(\not{p}_1 - \not{p}_4)\gamma_\nu\right].$$

To evaluate this, we need to show that $\gamma^\nu\gamma^\alpha\gamma^\beta\gamma_\nu = 4g^{\alpha\beta}\mathbb{I}$.

Proof. We can start by defining the contraction using the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbb{I}$:

$$\begin{aligned} \gamma^\nu\gamma^\alpha\gamma^\beta\gamma_\nu &= \gamma^\nu\gamma^\alpha(2g^\beta_\nu - \gamma_\nu\gamma^\beta) \\ &= 2\gamma^\beta\gamma^\alpha - \gamma^\nu\gamma^\alpha\gamma_\nu\gamma^\beta \end{aligned}$$

We already know the 3-matrix contraction $\gamma^\nu\gamma^\alpha\gamma_\nu = -2\gamma^\alpha$. Substituting this in gives:

$$\gamma^\nu\gamma^\alpha\gamma^\beta\gamma_\nu = 2\gamma^\beta\gamma^\alpha - (-2\gamma^\alpha)\gamma^\beta = 2(\gamma^\beta\gamma^\alpha + \gamma^\alpha\gamma^\beta) = 2(2g^{\alpha\beta}\mathbb{I}) = 4g^{\alpha\beta}\mathbb{I}.$$

□

Applying this identity to our trace, the term $\gamma^\nu(\not{p}_1 + \not{p}_2)(\not{p}_1 - \not{p}_4)\gamma_\nu$ simply becomes $4(p_1 + p_2) \cdot (p_1 - p_4)\mathbb{I}$. Thus now we have all the ingredients to compute the final value for the C-term, which is given by

$$\begin{aligned} C &= -\frac{4e^4}{su}\big((p_1 + p_2)\cdot(p_1 - p_4)\big)\text{Tr}[\not{p}_1\not{p}_3] \\ &= -\frac{16e^4}{su}(-p_1\cdot p_4 + p_1\cdot p_2 - p_2\cdot p_4)(p_1\cdot p_3) \\ &= -\frac{4e^4}{su}(u + s + t)(-t) = 0, \end{aligned}$$

since we are in the massless limit, where the Mandelstam invariant relation dictates $s+t+u=0$. Thus we have that the interference term between the two diagrams identically vanishes.

We can now move on to the computation of the A-term, which is given by

$$\begin{aligned} A &= \frac{1}{4}\sum_{s,\lambda}|\mathcal{M}_1|^2 \\ &= \frac{e^4}{4s^2}\mathrm{Tr}\left[(\not{p}_1+\not{p}_2)\gamma^\nu \not{p}_1\gamma^\mu(\not{p}_1+\not{p}_2)\gamma_\mu\not{p}_3\gamma_\nu\right] \end{aligned}$$

where now we need to compute the internal product of gamma matrices:

$$\gamma^\mu(\not{p}_1+\not{p}_2)\gamma_\mu = -2(\not{p}_1+\not{p}_2)$$

which will enforce a simplification of the trace:

$$\begin{aligned} A &= \frac{e^4}{4s^2}\mathrm{Tr}\left[(\not{p}_1+\not{p}_2)\gamma^\nu\not{p}_1(-2)(\not{p}_1+\not{p}_2)\not{p}_3\gamma_\nu\right] \\ &= -\frac{2e^4}{4s^2}\mathrm{Tr}\left[\not{p}_1(\not{p}_1+\not{p}_2)\not{p}_3\gamma_\nu(\not{p}_1+\not{p}_2)\gamma^\nu\right] \quad \text{(using cyclic property)} \\ &= -\frac{2e^4}{4s^2}\mathrm{Tr}\left[\not{p}_1(\not{p}_1+\not{p}_2)\not{p}_3(-2)(\not{p}_1+\not{p}_2)\right] \\ &= +\frac{4e^4}{4s^2}\mathrm{Tr}\left[(\not{p}_1+\not{p}_2)\not{p}_1(\not{p}_1+\not{p}_2)\not{p}_3\right] \end{aligned}$$

Using the fact that $\not{p}_1\not{p}_1 = p_1^2 = 0$ for a massless electron, we can expand the terms adjacent to $\not{p}_1$:

$$\begin{aligned} (\not{p}_1+\not{p}_2)\not{p}_1(\not{p}_1+\not{p}_2) &= \not{p}_1\not{p}_1\not{p}_1 + \not{p}_2\not{p}_1\not{p}_1 + \not{p}_1\not{p}_1\not{p}_2 + \not{p}_2\not{p}_1\not{p}_2 \\ &= 0+0+0+\not{p}_2\not{p}_1\not{p}_2 \\ &= 2(p_1\cdot p_2)\not{p}_2 - \not{p}_1\not{p}_2\not{p}_2 = 2(p_1\cdot p_2)\not{p}_2 \end{aligned}$$

Substituting this back, we evaluate the remaining trace:

$$\begin{aligned} A &= \frac{e^4}{s^2}\mathrm{Tr}\left[\not{p}_2\not{p}_1\not{p}_2\not{p}_3\right] \\ &= \frac{4e^4}{s^2}\big((p_2\cdot p_1)(p_2\cdot p_3) - (p_2\cdot p_2)(p_1\cdot p_3) + (p_2\cdot p_3)(p_1\cdot p_2)\big) \\ &= \frac{4e^4}{s^2}\big(2(p_1\cdot p_2)(p_2\cdot p_3)\big) = \frac{8e^4}{s^2}\left(\frac{s}{2}\right)\left(-\frac{u}{2}\right) = -2e^4\frac{u}{s}, \end{aligned}$$

where we used $p_2^2=0$, $2p_1\cdot p_2 = s$, and $2p_2\cdot p_3 = -u$.

The B-term can be computed directly via crossing symmetry, since it is given by the exact same expression as A but with the initial and final photons exchanged ($p_2 \leftrightarrow -p_4$). Under this crossing transformation, the Mandelstam invariants transform as:

$$s = (p_1+p_2)^2 \to (p_1-p_4)^2 = u$$

$$u = (p_1-p_4)^2 \to (p_1+p_2)^2 = s$$

Applying this swap $s \leftrightarrow u$ directly to the result for A yields:

$$B = \frac{1}{4}\sum_{s,\lambda}|\mathcal{M}_2|^2 = -2e^4\frac{s}{u}$$

Thus we can compute the total unpolarized squared amplitude as

$$|\mathcal{M}_{\text{unpol.}}|^2 = A + B + C = -2e^4\left(\frac{u}{s}+\frac{s}{u}\right).$$

To express this in terms of the scattering angle, we evaluate the kinematics in the center of mass frame. Since all particles are treated as massless, the energies are uniformly:

$$E_1 = E_2 = E_3 = E_4 = \frac{E_{\text{cm}}}{2} \equiv E,$$

so that we can write the momentum four-vectors aligned along the z-axis for the initial state, and scattered by an angle θ for the final state:

$$p_1 = \begin{pmatrix} E \\ 0 \\ 0 \\ E \end{pmatrix}, \quad p_2 = \begin{pmatrix} E \\ 0 \\ 0 \\ -E \end{pmatrix}, \quad p_3 = \begin{pmatrix} E \\ 0 \\ E\sin\theta \\ E\cos\theta \end{pmatrix}, \quad p_4 = \begin{pmatrix} E \\ 0 \\ -E\sin\theta \\ -E\cos\theta \end{pmatrix}.$$

Using these explicit vectors, the Mandelstam invariants read

$$\begin{aligned} u &= (p_4 - p_1)^2 = -2p_1 \cdot p_4 = -2(E^2 - (-E^2\cos\theta)) = -2E^2(1+\cos\theta), \\ s &= (p_1 + p_2)^2 = 4E^2 = E_{\text{cm}}^2, \\ \frac{u}{s} &= -\frac{2E^2(1+\cos\theta)}{4E^2} = -\frac{1+\cos\theta}{2}. \end{aligned}$$

So that we can write the unpolarized squared amplitude strictly as a function of the scattering angle θ as

$$|\mathcal{M}_{\text{unpol.}}|^2 = -2e^4\left(-\frac{1+\cos\theta}{2} - \frac{2}{1+\cos\theta}\right) = 2e^4\left(\frac{1+\cos\theta}{2} + \frac{2}{1+\cos\theta}\right).$$

Finally, we can compute the differential cross section for this process in the center of mass frame. Using the standard flux formula for 2-to-2 massless scattering where the initial and final momentum magnitudes cancel ($|\vec{p}_f|/|\vec{p}_i| = E/E = 1$):

$$\begin{aligned} \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} &= \frac{1}{64\pi^2 s}|\mathcal{M}_{\text{unpol.}}|^2 \\ &= \frac{1}{64\pi^2 E_{\text{cm}}^2}\left[2e^4\left(\frac{1+\cos\theta}{2} + \frac{2}{1+\cos\theta}\right)\right] \\ &= \frac{e^4}{32\pi^2 E_{\text{cm}}^2}\left(\frac{1+\cos\theta}{2} + \frac{2}{1+\cos\theta}\right). \end{aligned}$$

Recognizing the fine-structure constant definition $\alpha = e^2/(4\pi)$, we can elegantly substitute $e^4 = 16\pi^2\alpha^2$ to obtain the final physical result:

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{16\pi^2\alpha^2}{32\pi^2 E_{\text{cm}}^2}\left(\frac{1+\cos\theta}{2} + \frac{2}{1+\cos\theta}\right) = \frac{\alpha^2}{2E_{\text{cm}}^2}\left(\frac{1+\cos\theta}{2} + \frac{2}{1+\cos\theta}\right).$$

9. Ward identity in sQED

In sQED, explicitly show that the Ward identity is valid for an amplitude of Compton scattering (i.e. the same we did in the lecture but replacing the fermion with a scalar). Note that the amplitude is a sum of three diagrams.

First, let us recall the relevant Feynman rules for scalar QED (sQED) vertices in momentum space:

TODO: add diagram

3-point vertex with incoming scalar p, outgoing scalar p', and photon $\mu = -ie(p + p')^\mu$

TODO: add diagram

4-point seagull vertex with two scalars and two photons $\mu, \nu = 2ie^2 g^{\mu\nu}$

For the Compton scattering process $\phi(p_1) + \gamma(p_2) \to \phi(p_3) + \gamma(p_4)$, the total amplitude can be written by factoring out the photon polarization vectors:

$$i\mathcal{M} = i\mathcal{M}^{\mu\nu}\epsilon_{2\mu}\epsilon^*_{4\nu}$$

where $i\mathcal{M}^{\mu\nu}$ is the sum of three distinct tree-level Feynman diagrams: the s-channel, the u-channel, and the 4-point seagull interaction.

$$i\mathcal{M}^{\mu\nu} =$$

TODO: add diagram: s-channel + u-channel + seagull diagram

We want to show that $(p_2)_\mu i\mathcal{M}^{\mu\nu} = 0$ at tree level, where p_2 is the momentum of the incoming photon. We can do this by explicitly computing the tensorial amplitude for each diagram and then contracting them with the momentum p_2:

$$\begin{aligned}
(p_2)_\mu i\mathcal{M}^{\mu\nu} &= (p_2)_\mu \left(i\mathcal{M}^{\mu\nu}_{\text{s-chan}} + i\mathcal{M}^{\mu\nu}_{\text{u-chan}} + i\mathcal{M}^{\mu\nu}_{\text{seagull}} \right) \\
&= (p_2)_\mu e^2 \left[\frac{i}{s - m^2}(2p_1^\mu + p_2^\mu)(p_1^\nu + p_2^\nu + p_3^\nu)(-1) \right. \\
&\qquad \left. + \frac{i}{u - m^2}(p_1^\mu - p_4^\mu + p_3^\mu)(2p_1^\nu - p_4^\nu)(-1) + 2ig^{\mu\nu} \right]
\end{aligned}$$

Notice that the factor of (-1) in the first two terms comes from the product of the two vertex factors $(-i)^2$.

To perform the contraction, we need to compute the following scalar products using the standard Mandelstam invariants for massless photons ($p_2^2 = p_4^2 = 0$) and massive scalars ($p_1^2 = p_3^2 = m^2$):

$$p_1 \cdot p_2 = \frac{s - m^2}{2}, \quad p_2 \cdot p_4 = -\frac{t}{2}, \quad p_2 \cdot p_3 = -\frac{u - m^2}{2},$$

With these expressions, we can compute the contraction of p_2 with the momentum factors in the numerators of the s- and u-channel diagrams:

$$\begin{aligned}
(p_2)_\mu[2p_1^\mu + p_2^\mu] &= 2(p_1 \cdot p_2) + p_2^2 = s - m^2 + 0 = s - m^2, \\
(p_2)_\mu[p_1^\mu - p_4^\mu + p_3^\mu] &= p_1 \cdot p_2 - p_2 \cdot p_4 + p_2 \cdot p_3 \\
&= \frac{s - m^2}{2} + \frac{t}{2} - \frac{u - m^2}{2} \\
&= \frac{1}{2}(s - m^2 + t - u + m^2) = -(u - m^2),
\end{aligned}$$

where for the last equality we used the kinematic constraint $s+t+u = 2m^2$ (since two particles are massless and two have mass m, the sum of squared masses is $2m^2$). Substituting $s + t = 2m^2 - u$ into the numerator gives $\frac{1}{2}(2m^2 - u - u) = m^2 - u = -(u - m^2)$.

Substituting these contracted scalars back into our amplitude expression, the denominators beautifully cancel out:

$$\begin{aligned}
(p_2)_\mu i\mathcal{M}^{\mu\nu} &= e^2 \left[\frac{i(s - m^2)}{s - m^2}(p_1^\nu + p_2^\nu + p_3^\nu)(-1) + \frac{i(-(u - m^2))}{u - m^2}(2p_1^\nu - p_4^\nu)(-1) + 2ip_2^\nu \right] \\
&= e^2 \left[-i(p_1^\nu + p_2^\nu + p_3^\nu) + i(2p_1^\nu - p_4^\nu) + 2ip_2^\nu \right] \\
&= ie^2(p_1^\nu + p_2^\nu - p_3^\nu - p_4^\nu) = 0,
\end{aligned}$$

where the last equality follows strictly from the global conservation of 4-momentum for the process, which implies that $p_1 + p_2 = p_3 + p_4$, or $p_1 + p_2 - p_3 - p_4 = 0$.

Remark. *All of the three diagrams are strictly necessary to satisfy the Ward identity and thus for the exact cancellation to happen. If we were to consider only the s- and u-channel diagrams (as one might incorrectly do by simply copying spinor QED where there is no seagull vertex), we would get a non-zero, residual contribution proportional to $2ip_2^\nu$, which would grossly violate the Ward identity. In sQED, the seagull diagram is required to ensure that the S-matrix remains invariant under $U(1)$ gauge transformations. The Ward identity is a direct macroscopic consequence of this fundamental gauge invariance, and thus must be satisfied by all physical observables.*

10. Scaling Green functions and scattering amplitudes

Show that, under a transformation of the fields of the form

$$\phi(x) \to c\phi(x),$$

where c is a constant, then:

(a) *The Greeen functions of the theory scale as*

$$\langle\Omega|\, T\phi(x_1)\cdots\phi(x_N)\,|\Omega\rangle \to c^N \langle\Omega|\, T\phi(x_1)\cdots\phi(x_N)\,|\Omega\rangle$$

(b) *The amputated Green functions of the theory scale as*

$$G_N\big|_{\text{amputated}} \to \frac{1}{c^N} G_N\big|_{\text{amputated}}$$

(c) *Scattering matrix elements are invariant under such transformation.*

Hint: *Use the LSZ formula and recall that* $\sqrt{Z} \equiv \langle\Omega|\,\phi(0)\,|\mathbf{p}\rangle\,\big|_{\mathbf{p}=\mathbf{0}}$.

Defining the N-point Green function of the interacting theory as the vacuum expectation value of the time-ordered product of fields:

$$G_N(x_1,\ldots,x_N) = \langle\Omega|\, T\{\phi(x_1)\cdots\phi(x_N)\}\,|\Omega\rangle\,,$$

we can compute exactly how it scales under the global field redefinition $\phi(x) \to c\phi(x)$.

(a) **Scaling of full Green functions:** Since the time-ordering operator T is linear and c is a pure, time-independent c-number constant, it trivially factors out of the expectation value. Because the N-point function contains exactly N field insertions, each contributing one factor of c, the full Green function scales linearly as c^N:

$$\begin{aligned} G_N(x_1,\ldots,x_N) &\to \langle\Omega|\, T\{(c\phi(x_1))\cdots(c\phi(x_N))\}\,|\Omega\rangle \\ &= c^N \langle\Omega|\, T\{\phi(x_1)\cdots\phi(x_N)\}\,|\Omega\rangle \\ &= c^N G_N(x_1,\ldots,x_N). \end{aligned}$$

(b) **Scaling of amputated Green functions:** To understand the amputated Green functions, we must recall their structural definition in terms of Feynman diagrams. A full, connected N-point Green function G_N can be factorized into a central, strictly amputated core ($G_N|_{\text{amp}}$) attached to N external exact 2-point propagators (G_2). Symbolically:

$$G_N \sim (G_2)^N \times \left(G_N\big|_{\text{amputated}}\right)$$

From part (a), we definitively know the scaling behavior of both the full N-point function and the full 2-point function:

$$G_N \to c^N G_N \quad \text{and} \quad G_2 \to c^2 G_2$$

Substituting these known scalings into our factorization relation gives:

$$c^N G_N \sim (c^2 G_2)^N \times \left(G_N \big|_{\text{amputated}} \right)'$$

$$c^N G_N \sim c^{2N} (G_2)^N \times \left(G_N \big|_{\text{amputated}} \right)'$$

To preserve the original defining equality $G_N \sim (G_2)^N G_N|_{\text{amp}}$, the transformed amputated Green function must absorb the excess factors of c. Dividing both sides by c^{2N}, we find:

$$\left(G_N \big|_{\text{amputated}} \right)' = \frac{c^N}{c^{2N}} \left(G_N \big|_{\text{amputated}} \right) = \left(\frac{1}{c} \right)^N G_N \big|_{\text{amputated}}$$

Thus, amputated Green functions scale inversely with c^N.

(c) **Invariance of Scattering Matrix Elements:** For the scattering matrix elements, we recall the foundational LSZ reduction formula. The LSZ formula mathematically extracts the physical S-matrix element $\langle f|S|i\rangle$ from the full interacting Green function by amputating the external legs and dividing out the appropriate field-strength renormalization factors ($\sqrt{Z}$) for each of the N external particles:

$$\langle f|S|i\rangle \sim \left(\frac{1}{\sqrt{Z}} \right)^N G_N$$

We must determine how the renormalization constant Z scales. The hint provides the non-perturbative definition of Z:

$$\sqrt{Z} \equiv \langle \Omega | \, \phi(0) \, |\mathbf{p}\rangle \, \big|_{\mathbf{p}=\mathbf{0}}$$

Under the field redefinition $\phi \to c\phi$, this matrix element clearly scales linearly:

$$\sqrt{Z} \to \langle \Omega | \, c\phi(0) \, |\mathbf{p}\rangle = c \, \langle \Omega | \, \phi(0) \, |\mathbf{p}\rangle = c\sqrt{Z}$$

Now, we apply both the scaling of $\sqrt{Z}$ and the scaling of the full G_N (from part a) to the LSZ formula:

$$\langle f|S|i\rangle \to \left(\frac{1}{c\sqrt{Z}} \right)^N (c^N G_N) = \frac{c^N}{c^N} \left(\frac{1}{\sqrt{Z}} \right)^N G_N = \left(\frac{1}{\sqrt{Z}} \right)^N G_N$$

The factors of c perfectly cancel. Therefore, physical scattering observables (the S-matrix elements) are strictly invariant under arbitrary constant rescalings of the quantum fields. This reflects the deep physical principle that the specific choice of normalization for the unobservable internal quantum fields has absolutely no effect on measurable physical cross-sections.

Sheet 3

1. Euler-Lagrange equations and Functional Derivatives

Consider a Lagrangian $\mathcal{L} = \mathcal{L}(\varphi_k, \partial_\mu \varphi_k)$ *depending on any number of fields and their derivatives. Show that the Euler Lagrange equations of motion can be written using a functional derivative as*

$$\frac{\delta}{\delta \varphi_k} S = 0, \qquad (1)$$

where $S = \int d^4x \mathcal{L}$ *is the action.*

The fundamental principle of stationary action (Hamilton's Principle) dictates that the physical trajectory of a classical field system is the one for which the action S is stationary under small, arbitrary variations of the fields. We denote these infinitesimal field variations as $\delta\varphi_k(x)$, and we impose the physical boundary condition that these variations strictly vanish at the boundary of our integration volume (i.e., at infinity).

We start by computing the total first-order variation of the action S due to the field transformation $\varphi_k(x) \to \varphi_k(x) + \delta\varphi_k(x)$. Because the Lagrangian density $\mathcal{L}$ is a local function depending only on the fields and their first spacetime derivatives, its variation is determined via the chain rule:

$$\delta S = \delta \int d^4x \, \mathcal{L}(\varphi_k, \partial_\mu \varphi_k) = \int d^4x \, \delta\mathcal{L}$$

$$\delta S = \int d^4x \sum_k \left(\frac{\partial \mathcal{L}}{\partial \varphi_k} \delta\varphi_k + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \delta(\partial_\mu \varphi_k) \right)$$

A crucial property of the variation operator δ is that it commutes with the spacetime derivative ∂_μ. Mathematically, the variation of the derivative is identical to the derivative of the variation: $\delta(\partial_\mu \varphi_k) = \partial_\mu(\delta\varphi_k)$. Substituting this into our expression yields:

$$\delta S = \int d^4x \sum_k \left(\frac{\partial \mathcal{L}}{\partial \varphi_k} \delta\varphi_k + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \partial_\mu(\delta\varphi_k) \right)$$

In order to isolate the arbitrary variation $\delta\varphi_k(x)$ in the second term, we apply the rule of integration by parts (or equivalently, the product rule for derivatives $\partial_\mu(AB) = (\partial_\mu A)B + A(\partial_\mu B)$):

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \partial_\mu(\delta\varphi_k) = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \delta\varphi_k \right) - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \right) \delta\varphi_k$$

Substituting this expanded term back into the variation of the action, we can split the integral:

$$\delta S = \int d^4x \sum_k \left[\frac{\partial \mathcal{L}}{\partial \varphi_k} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \right) \right] \delta\varphi_k + \int d^4x \sum_k \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \delta\varphi_k \right)$$

By applying the 4-dimensional divergence theorem (Gauss's theorem), the second integral—which is a total four-divergence—can be transformed into a surface integral evaluated over the boundary ∂V of the spacetime region:

$$\int d^4x \, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \delta\varphi_k \right) = \oint_{\partial V} d^3\sigma_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \delta\varphi_k \right)$$

Since the physical field variations $\delta\varphi_k$ are strictly defined to vanish at the boundaries of spacetime ($\delta\varphi_k|_{\partial V} = 0$), this surface term evaluates exactly to zero. We are left with:

$$\delta S = \int d^4x \sum_k \left[\frac{\partial \mathcal{L}}{\partial \varphi_k} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \right) \right] \delta\varphi_k(x)$$

By the formal mathematical definition of the functional derivative, the first-order variation of a generic functional $S[\varphi_k]$ with respect to an arbitrary function $\varphi_k(x)$ is defined by the integral kernel:

$$\delta S = \int d^4x \sum_k \frac{\delta S}{\delta \varphi_k(x)} \delta\varphi_k(x)$$

By directly comparing the integrands of our derived expression and the formal definition, we can uniquely identify the functional derivative of the action with respect to the field φ_k:

$$\frac{\delta S}{\delta \varphi_k} = \frac{\partial \mathcal{L}}{\partial \varphi_k} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \right)$$

Hamilton's principle demands that the action is stationary, meaning $\delta S = 0$ for *any* arbitrary variation $\delta\varphi_k(x)$. For the integral to vanish for an arbitrary $\delta\varphi_k(x)$, the term inside the square brackets must identically vanish for each field species k. This enforces:

$$\frac{\delta S}{\delta \varphi_k} = 0$$

which explicitly translates to:

$$\frac{\partial \mathcal{L}}{\partial \varphi_k} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi_k)} \right) = 0$$

These are precisely the classical Euler-Lagrange equations of motion. Therefore, writing the equations of motion as $\frac{\delta S}{\delta \varphi_k} = 0$ is rigorously equivalent to the standard Euler-Lagrange formulation.

2. Deriving Feynman rules of sQED via functional derivatives

Derive the Feynman rules of sQED using the method functional derivatives (you should get the same result as in the similar exercise of sheet 2).

In the path integral formulation of Quantum Field Theory, the Feynman rules for the interaction vertices in momentum space can be directly and rigorously derived by taking the functional derivatives of the interaction action iS_{int} with respect to the momentum-space fields, and then stripping away the overall momentum-conserving Dirac delta function.

From the previous exercise on scalar QED, we identified the interaction Lagrangian:

$$\mathcal{L}_{\text{int}} = -ieA^\mu \left(\phi^*(\partial_\mu \phi) - (\partial_\mu \phi^*)\phi\right) + e^2 A_\mu A^\mu \phi^* \phi$$

The corresponding interaction action is $S_{\text{int}} = \int d^4x \, \mathcal{L}_{\text{int}}$. To apply the functional derivative method, we first express the fields in terms of their Fourier transforms in momentum space. We adopt the convention where incoming fields carry momentum into the vertex (e.g., $e^{-ip\cdot x}$) and outgoing conjugate fields carry momentum out of the vertex (e.g., $e^{ip'\cdot x}$):

$$\phi(x) = \int \frac{d^4p}{(2\pi)^4} \tilde{\phi}(p) e^{-ip\cdot x}, \qquad \phi^*(x) = \int \frac{d^4p'}{(2\pi)^4} \tilde{\phi}^*(p') e^{ip'\cdot x}, \qquad A^\mu(x) = \int \frac{d^4q}{(2\pi)^4} \tilde{A}^\mu(q) e^{-iq\cdot x}$$

1. The 3-Point Vertex (Cubic Interaction)

We isolate the cubic part of the interaction action, $iS_3 = i\int d^4x\,[-ieA^\mu(\phi^*\partial_\mu\phi - \phi\partial_\mu\phi^*)]$. Substituting the Fourier expansions, the spacetime derivatives pull down factors of momentum: $\partial_\mu\phi \to -ip_\mu\tilde{\phi}$ and $\partial_\mu\phi^* \to ip'_\mu\tilde{\phi}^*$.

$$\begin{aligned} iS_3 &= i\int d^4x \int \frac{d^4q}{(2\pi)^4}\frac{d^4p'}{(2\pi)^4}\frac{d^4p}{(2\pi)^4}(-ie)\tilde{A}^\mu(q)\left[\tilde{\phi}^*(p')(-ip_\mu)\tilde{\phi}(p) - (ip'_\mu)\tilde{\phi}^*(p')\tilde{\phi}(p)\right] e^{-i(p+q-p')\cdot x} \\ &= \int \frac{d^4q}{(2\pi)^4}\frac{d^4p'}{(2\pi)^4}\frac{d^4p}{(2\pi)^4}\tilde{A}^\mu(q)\tilde{\phi}^*(p')\tilde{\phi}(p)\left[-ie(p_\mu + p'_\mu)\right](2\pi)^4\delta^{(4)}(p+q-p') \end{aligned}$$

The Feynman rule for the vertex $V_3^\mu(p,p',q)$ is obtained by taking the functional derivatives with respect to the momentum-space fields $\tilde{A}^\mu(q)$, $\tilde{\phi}^*(p')$, and $\tilde{\phi}(p)$:

$$V_3^\mu = \left.\frac{\delta^3(iS_3)}{\delta\tilde{A}_\mu(q)\delta\tilde{\phi}^*(p')\delta\tilde{\phi}(p)}\right|_{\text{stripped of } (2\pi)^4\delta}$$

TODO: add diagram: A vertex with one incoming solid line (p), one outgoing solid line (p'), and one incoming squiggly photon line (q, index μ)

Performing this functional differentiation trivially isolates the coefficient bracket, yielding:

$$V_3^\mu = -ie(p^\mu + p'^\mu)$$

This perfectly matches the rule $-ie(p+p')_\mu$ derived previously.

2. The 4-Point "Seagull" Vertex (Quartic Interaction)

We now isolate the quartic part of the action, $iS_4 = i\int d^4x\left[e^2 g_{\mu\nu}A^\mu A^\nu\phi^*\phi\right]$. Substituting the Fourier expansions for two photon fields (with momenta q_1, q_2), one incoming scalar (p), and one outgoing scalar (p'):

$$iS_4 = i\int\left(\prod_{i=1}^4\frac{d^4k_i}{(2\pi)^4}\right)e^2 g_{\alpha\beta}\tilde{A}^\alpha(q_1)\tilde{A}^\beta(q_2)\tilde{\phi}^*(p')\tilde{\phi}(p)(2\pi)^4\delta^{(4)}(p+q_1+q_2-p')$$

To find the vertex rule $V_4^{\mu\nu}$, we take the functional derivative with respect to $\tilde{\phi}(p)$, $\tilde{\phi}^*(p')$, and the two photon fields $\tilde{A}_\mu(q_1)$ and $\tilde{A}_\nu(q_2)$. Because the action is quadratic in the field A, the product rule for functional derivatives applies:

$$\frac{\delta}{\delta\tilde{A}_\mu(k_1)}\frac{\delta}{\delta\tilde{A}_\nu(k_2)}\left[\tilde{A}^\alpha(q_1)\tilde{A}^\beta(q_2)\right] = \delta^{\alpha\mu}\delta^{\beta\nu} + \delta^{\alpha\nu}\delta^{\beta\mu}$$

Contracting this symmetric result with the metric tensor $g_{\alpha\beta}$ from the action coefficient yields:

$$g_{\alpha\beta}(\delta^{\alpha\mu}\delta^{\beta\nu} + \delta^{\alpha\nu}\delta^{\beta\mu}) = g^{\mu\nu} + g^{\nu\mu} = 2g^{\mu\nu}$$

TODO: add diagram: A vertex with one incoming solid line, one outgoing solid line, and two squiggly photon lines μ, ν attaching at the same point

Substituting this back, we find the final vertex rule:

$$V_4^{\mu\nu} = ie^2(2g^{\mu\nu}) = 2ie^2 g^{\mu\nu}$$

This rigorously demonstrates how the functional derivative naturally generates the required combinatoric symmetry factor of 2, reproducing the exact same result as our previous derivation.

3. Generating functional for the free Dirac field

Compute the generating functional $Z[\eta,\bar{\eta}]$ for the path integral of a free Dirac field. The solution is in the lecture notes.

The generating functional for a free Dirac field is defined by the path integral over the Grassmann-valued fields ψ and $\bar{\psi}$, coupled to external Grassmann sources η and $\bar{\eta}$:

$$Z[\eta, \bar{\eta}] = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \, \exp\left(i \int d^4x \left[\mathcal{L}_0 + \bar{\eta}(x)\psi(x) + \bar{\psi}(x)\eta(x) \right] \right)$$

where the free Dirac Lagrangian (including the Feynman $i\epsilon$ prescription necessary for mathematical convergence) is:

$$\mathcal{L}_0 = \bar{\psi}(x)(i\partial\!\!\!/ - m + i\epsilon)\psi(x)$$

To analytically compute this integral, we systematically "complete the square" for the fermion fields, directly mirroring the procedure used for Gaussian integrals of real or complex scalar fields. Let us recall that the Feynman propagator for the Dirac field, $S_F(x-y)$, is precisely defined as the Green's function for the Dirac differential operator:

$$(i\partial\!\!\!/_x - m + i\epsilon)S_F(x-y) = i\delta^{(4)}(x-y)$$

We now introduce a shift in our integration variables to perfectly absorb the linear source terms η and $\bar{\eta}$. We define new shifted Grassmann fields ψ' and $\bar{\psi}'$:

$$\psi(x) = \psi'(x) + \int d^4y \, iS_F(x-y)\eta(y)$$

$$\bar{\psi}(x) = \bar{\psi}'(x) + \int d^4y \, \bar{\eta}(y) iS_F(y-x)$$

Because this is a simple linear shift by terms that are independent of the fields ψ and $\bar{\psi}$, the Jacobian of the transformation is strictly unity, meaning the path integral measure is perfectly invariant: $\mathcal{D}\bar{\psi}\mathcal{D}\psi = \mathcal{D}\bar{\psi}'\mathcal{D}\psi'$.

Let us substitute this shift back into the action. Applying the Dirac operator to the shifted field $\psi(x)$ yields:

$$\begin{aligned}
(i\partial\!\!\!/_x - m + i\epsilon)\psi(x) &= (i\partial\!\!\!/_x - m + i\epsilon)\psi'(x) + \int d^4y \, i\big[(i\partial\!\!\!/_x - m + i\epsilon)S_F(x-y)\big]\eta(y) \\
&= (i\partial\!\!\!/_x - m + i\epsilon)\psi'(x) + \int d^4y \, i\big[i\delta^{(4)}(x-y)\big]\eta(y) \\
&= (i\partial\!\!\!/_x - m + i\epsilon)\psi'(x) - \eta(x)
\end{aligned}$$

Now we reconstruct the full expression inside the exponent. The total action S becomes:

$$\begin{aligned}
S &= \int d^4x \left[\bar{\psi}(x)(i\partial\!\!\!/_x - m + i\epsilon)\psi(x) + \bar{\eta}(x)\psi(x) + \bar{\psi}(x)\eta(x) \right] \\
&= \int d^4x \left[\left(\bar{\psi}'(x) + \int d^4y \, \bar{\eta}(y) iS_F(y-x) \right) \left((i\partial\!\!\!/_x - m + i\epsilon)\psi'(x) - \eta(x) \right) \right] \\
&\quad + \int d^4x \, \bar{\eta}(x) \left(\psi'(x) + \int d^4y \, iS_F(x-y)\eta(y) \right) \\
&\quad + \int d^4x \left(\bar{\psi}'(x) + \int d^4y \, \bar{\eta}(y) iS_F(y-x) \right) \eta(x)
\end{aligned}$$

Expanding these terms out carefully, most of the cross-terms involving ψ' and $\bar{\psi}'$ precisely cancel each other out. After integrating by parts for the adjoint operator, the purely kinetic term remains structurally identical:

$$S = \int d^4x \, \bar{\psi}'(x)(i\partial\!\!\!/ - m + i\epsilon)\psi'(x) + i \int d^4x d^4y \, \bar{\eta}(x) S_F(x-y)\eta(y)$$

We now multiply the total action by i to return to the path integral exponent. Note that $i \times (i) = -1$:

$$iS = i \int d^4x\, \bar{\psi}'(x)(i\partial\!\!\!/ - m + i\epsilon)\psi'(x) - \int d^4x d^4y\, \bar{\eta}(x) S_F(x-y)\eta(y)$$

Because the action now flawlessly separates into a part depending exclusively on the quantum fields $\psi', \bar{\psi}'$ and a part depending exclusively on the classical sources $\eta, \bar{\eta}$, the source-dependent exponential completely factors out of the path integral:

$$Z[\eta,\bar{\eta}] = \left(\int \mathcal{D}\bar{\psi}'\mathcal{D}\psi' \, \exp\left[i \int d^4x\, \bar{\psi}'(i\partial\!\!\!/ - m + i\epsilon)\psi' \right] \right) \times \exp\left[- \int d^4x d^4y\, \bar{\eta}(x) S_F(x-y)\eta(y) \right]$$

The integral over the primed fields evaluates simply to the zero-source generating functional $Z[0,0]$, which is a constant normalization factor. Our final, rigorous result is thus:

$$Z[\eta,\bar{\eta}] = Z_0 \exp\left(- \int d^4x d^4y\, \bar{\eta}(x) S_F(x-y)\eta(y) \right)$$

where $Z_0 = Z[0,0] = \det\big(i\partial\!\!\!/ - m + i\epsilon\big)$.

4. Feynman propagator for a massive spin-1 particle

Compute the Feynman propagator for a massive spin 1 *particle in momentum space. Hint: Based on Lorentz covariance, argue that the result must take the form*

$$\Pi^{\mu\nu}(p^2) = A(p^2) g^{\mu\nu} + B(p^2) p^\mu p^\nu \qquad (2)$$

and use the equation obeyed by $\Pi^{\mu\nu}$ *(which is the Fourier transform of the differential equation for* $D_F^{\mu\nu}$*) to fix A and B.*

The Feynman propagator is defined as the Green's function for the free equations of motion. For a massive spin-1 field V^μ (such as the Z or W bosons), the free Proca Lagrangian is:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} m^2 V_\mu V^\mu$$

where $F_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu$. The classical Euler-Lagrange equations of motion are the Proca equations:

$$\partial_\mu F^{\mu\nu} + m^2 V^\nu = 0 \implies (\Box + m^2) V^\nu - \partial^\nu (\partial_\mu V^\mu) = 0$$

The Feynman propagator in coordinate space, $D_F^{\mu\nu}(x-y)$, satisfies the corresponding inhomogeneous differential equation with a delta function source:

$$\left[(\Box_x + m^2) g_{\mu\rho} - \partial^x_\mu \partial^x_\rho \right] D_F^{\rho\nu}(x-y) = i\delta^\nu_\mu \delta^{(4)}(x-y)$$

To solve this, we move to momentum space by taking the Fourier transform. Derivatives map to momenta via $\partial_\mu \to -ip_\mu$, and $\Box \to -p^2$. The differential operator becomes an algebraic matrix equation for the momentum-space propagator $\Pi^{\rho\nu}(p)$:

$$\left[(-p^2 + m^2) g_{\mu\rho} - (-ip_\mu)(-ip_\rho) \right] \Pi^{\rho\nu}(p) = i\delta^\nu_\mu$$

$$-i \left[(-p^2 + m^2) g_{\mu\rho} + p_\mu p_\rho \right] \Pi^{\rho\nu}(p) = \delta^\nu_\mu$$

To invert this matrix, we use the hint provided. Because the propagator must be a Lorentz covariant rank-2 tensor constructed solely from the metric tensor $g^{\mu\nu}$ and the available four-momentum p^μ, its most general possible structure is:

$$\Pi^{\rho\nu}(p) = A(p^2) g^{\rho\nu} + B(p^2) p^\rho p^\nu$$

We substitute this *Ansatz* directly into our momentum-space equation:

$$-i\left[(-p^2+m^2)g_{\mu\rho}+p_\mu p_\rho\right](Ag^{\rho\nu}+Bp^\rho p^\nu)=\delta_\mu^\nu$$

We expand the product by distributing the terms and carefully contracting the Lorentz indices. Remember that $g_{\mu\rho}g^{\rho\nu}=\delta_\mu^\nu$, $g_{\mu\rho}p^\rho=p_\mu$, and $p_\rho p^\rho=p^2$:

$$(-p^2+m^2)(A\delta_\mu^\nu+Bp_\mu p^\nu)+Ap_\mu p^\nu+Bp^2p_\mu p^\nu=i\delta_\mu^\nu$$

Now, we group the terms proportional to the tensor structures δ_μ^ν and $p_\mu p^\nu$:

$$(-p^2+m^2)A\delta_\mu^\nu+\left[(-p^2+m^2)B+A+Bp^2\right]p_\mu p^\nu=i\delta_\mu^\nu$$

$$(-p^2+m^2)A\delta_\mu^\nu+\left(m^2B+A\right)p_\mu p^\nu=i\delta_\mu^\nu$$

For this equation to hold true for any arbitrary momentum p^μ, the coefficients of each independent tensor structure on the left side must equal their corresponding coefficients on the right side. This yields a simple system of two linear equations for A and B:

$$\begin{cases}(-p^2+m^2)A=i\\ m^2B+A=0\end{cases}$$

Solving the first equation trivially gives A:

$$A=\frac{-i}{p^2-m^2}$$

Substituting this A into the second equation allows us to solve for B:

$$B=-\frac{A}{m^2}=\frac{i}{m^2(p^2-m^2)}$$

Finally, we substitute the solutions for A and B back into our original *Ansatz* to construct the full massive spin-1 propagator:

$$\begin{aligned}\Pi^{\mu\nu}(p)&=\frac{-i}{p^2-m^2}g^{\mu\nu}+\frac{i}{m^2(p^2-m^2)}p^\mu p^\nu\\&=\frac{-i\left(g^{\mu\nu}-\frac{p^\mu p^\nu}{m^2}\right)}{p^2-m^2}\end{aligned}$$

To strictly ensure the correct analytic structure (causality) and avoid singularities exactly on the mass shell $p^2=m^2$, we must append the standard Feynman prescription $+i\epsilon$ to the denominator. The final rigorous result is:

$$\Pi^{\mu\nu}(p)=i\frac{-g^{\mu\nu}+p^\mu p^\nu/m^2}{p^2-m^2+i\epsilon}$$

Remark. *It is critically important to note the behavior of this propagator in the massless limit ($m\to 0$). Looking at our system of equations, if $m=0$, the second equation becomes $A=0$, which violently contradicts the first equation $(-p^2)A=i$. Therefore, the matrix operator for a massless spin-1 field (a photon) is strictly non-invertible. This mathematical singularity physically reflects the presence of gauge invariance. To define a photon propagator, one is forced to break the symmetry by explicitly choosing a gauge (adding a gauge-fixing term like $-\frac{1}{2\xi}(\partial_\mu A^\mu)^2$ to the Lagrangian).*

5. Photon Propagator and Gauge Fixing

Add a gauge fixing term *to the QED Lagrangian*

$$\mathcal{L}_{\text{g.f.}} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2, \qquad (3)$$

which was motivated in the lectures (see notes 3.2) and compute the photon propagator. Hint: Follow the same method as the previous exercise.

As discussed in the previous exercise, the free massless vector Lagrangian $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ yields a non-invertible differential operator because of the underlying $U(1)$ gauge invariance ($A^\mu \to A^\mu + \partial^\mu\alpha$). To define a unique propagator, we must break this gauge redundancy by adding the gauge-fixing term $\mathcal{L}_{\text{g.f.}}$.

The total Lagrangian is:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2$$

To isolate the quadratic operator, we integrate the action by parts (assuming the fields vanish at spacetime boundaries). For the Maxwell term:

$$-\frac{1}{4}\int d^4x\,(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) = \frac{1}{2}\int d^4x\,A_\mu\big(\Box g^{\mu\nu} - \partial^\mu\partial^\nu\big)A_\nu$$

For the gauge-fixing term:

$$-\frac{1}{2\xi}\int d^4x\,(\partial_\mu A^\mu)(\partial_\nu A^\nu) = \frac{1}{2}\int d^4x\,A_\mu\left(\frac{1}{\xi}\partial^\mu\partial^\nu\right)A_\nu$$

Combining these, the total effective Lagrangian can be written as a quadratic form $\mathcal{L} = \frac{1}{2}A_\mu\mathcal{O}^{\mu\nu}A_\nu$:

$$\mathcal{L} = \frac{1}{2}A_\mu\left[\Box g^{\mu\nu} - \left(1 - \frac{1}{\xi}\right)\partial^\mu\partial^\nu\right]A_\nu$$

The equation of motion is simply $\mathcal{O}^{\mu\nu}A_\nu = 0$. The Feynman propagator in coordinate space, $D_F^{\rho\nu}(x-y)$, is the Green's function of this operator:

$$\left[\Box_x g_{\mu\rho} - \left(1 - \frac{1}{\xi}\right)\partial_\mu^x\partial_\rho^x\right]D_F^{\rho\nu}(x-y) = i\delta_\mu^\nu\delta^{(4)}(x-y)$$

We transition to momentum space by taking the Fourier transform. With the substitutions $\partial_\mu \to -ip_\mu$ and $\Box \to -p^2$, the differential operator becomes an algebraic matrix $M_{\mu\rho}(p)$:

$$\left[-p^2 g_{\mu\rho} + \left(1 - \frac{1}{\xi}\right)p_\mu p_\rho\right]\Pi^{\rho\nu}(p) = i\delta_\mu^\nu$$

Following the hint and exploiting Lorentz covariance, we use the same general *Ansatz* for the propagator $\Pi^{\rho\nu}(p)$ as in the previous exercise:

$$\Pi^{\rho\nu}(p) = A(p^2)g^{\rho\nu} + B(p^2)p^\rho p^\nu$$

Substituting this *Ansatz* into our algebraic equation yields:

$$\left[-p^2 g_{\mu\rho} + \left(1 - \frac{1}{\xi}\right)p_\mu p_\rho\right](Ag^{\rho\nu} + Bp^\rho p^\nu) = i\delta_\mu^\nu$$

Expanding the product by distributing the terms and carefully contracting the Lorentz indices ($g_{\mu\rho}g^{\rho\nu} = \delta^\nu_\mu$, $g_{\mu\rho}p^\rho = p_\mu$, $p_\rho p^\rho = p^2$):

$$-p^2 A\delta^\nu_\mu - p^2 B p_\mu p^\nu + A\left(1 - \frac{1}{\xi}\right) p_\mu p^\nu + Bp^2\left(1 - \frac{1}{\xi}\right) p_\mu p^\nu = i\delta^\nu_\mu$$

Grouping the coefficients of the linearly independent tensor structures δ^ν_μ and $p_\mu p^\nu$:

$$(-p^2 A)\delta^\nu_\mu + \left[-p^2 B + A\left(1 - \frac{1}{\xi}\right) + Bp^2 - \frac{Bp^2}{\xi}\right] p_\mu p^\nu = i\delta^\nu_\mu$$

$$(-p^2 A)\delta^\nu_\mu + \left[A\left(1 - \frac{1}{\xi}\right) - \frac{p^2 B}{\xi}\right] p_\mu p^\nu = i\delta^\nu_\mu$$

Equating the coefficients on both sides provides a simple linear system for A and B:

$$\begin{cases} -p^2 A = i \\ A\left(1 - \frac{1}{\xi}\right) - \frac{p^2 B}{\xi} = 0 \end{cases}$$

From the first equation, we immediately find A:

$$A = \frac{-i}{p^2}$$

Substituting A into the second equation allows us to solve for B:

$$\frac{p^2 B}{\xi} = A\left(\frac{\xi - 1}{\xi}\right) \implies p^2 B = A(\xi - 1) \implies B = \frac{-i(\xi - 1)}{p^4}$$

Finally, we substitute the solutions for A and B back into our original *Ansatz* to construct the full photon propagator. To ensure the correct boundary conditions (time-ordering) and avoid the poles on the real axis at $p^2 = 0$, we insert the standard Feynman $+i\epsilon$ prescription into the denominators:

$$\begin{aligned} \Pi^{\mu\nu}(p) &= \frac{-i}{p^2 + i\epsilon} g^{\mu\nu} - \frac{i(\xi - 1)}{(p^2 + i\epsilon)^2} p^\mu p^\nu \\ &= \frac{-i}{p^2 + i\epsilon}\left[g^{\mu\nu} + (\xi - 1)\frac{p^\mu p^\nu}{p^2 + i\epsilon}\right] \\ &= \frac{-i}{p^2 + i\epsilon}\left[g^{\mu\nu} - (1 - \xi)\frac{p^\mu p^\nu}{p^2 + i\epsilon}\right] \end{aligned}$$

Remark. *The arbitrary parameter ξ strictly parametrizes a family of covariant gauges (specifically, R_ξ gauges in the massless limit). Because it is an unphysical artifact introduced merely to invert the differential operator, physical scattering amplitudes ($\mathcal{M}$) are guaranteed by the Ward identity to be independent of the choice of ξ. Two choices are particularly ubiquitous in the literature:*

- ***Feynman gauge*** *($\xi = 1$):* *The momentum-dependent tensor structure vanishes entirely, simplifying the propagator to merely $\frac{-ig^{\mu\nu}}{p^2 + i\epsilon}$. This is the standard choice for practical loop and tree-level computations.*
- ***Landau gauge*** *($\xi = 0$):* *The propagator is strictly transverse to the four-momentum, $p_\mu \Pi^{\mu\nu} = 0$, taking the form $\frac{-i}{p^2 + i\epsilon}\left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}\right)$.*

6. Furry's theorem in sQED

Furry's theorem in sQED. In scalar QED consider the *charge conjugation* transformation, which for the scalar field corresponds to

$$\phi \to \phi^*, \qquad \phi^* \to \phi. \qquad (4)$$

- *How should the photon field A^μ transform to leave the Lagrangian invariant?*
- *Consider this transformation (including the one of A^μ) in the path integral, for a Green function with N photon fields*

  $$\langle \Omega | T A^{\mu_1}(x_1) \cdots A^{\mu_N}(x_N) | \Omega \rangle \qquad (5)$$

  Show that if N is odd, then the Green function is zero.
- *Argue, based on the previous result, that an amplitude with N external photons (i.e. $\gamma + \gamma \to (N-2)\gamma$) vanishes if N is odd.*
- *Is the previous point still true if any external photon is off-shell (i.e. $p^2 \neq 0$ and no polarization vector)?*

1. Transformation of the photon field A^μ

To determine the transformation properties of the photon field under charge conjugation, we inspect the explicit expression of the full scalar QED Lagrangian:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi - ieA^\mu (\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*) + e^2 A_\mu A^\mu \phi^* \phi$$

Applying the transformation $\phi \to \phi^*$ and $\phi^* \to \phi$, the free scalar terms and the quartic interaction term ($e^2 A^2 \phi^* \phi$) are trivially symmetric and remain unchanged. However, we must carefully analyze the cubic interaction term, which couples the photon field to the scalar current $J_\mu = i(\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*)$. Under the scalar transformation:

$$J_\mu \to i(\phi \partial_\mu \phi^* - \phi^* \partial_\mu \phi) = -i(\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*) = -J_\mu$$

The current is odd under charge conjugation. In order for the entire interaction term $-eA^\mu J_\mu$ (and therefore the full action S) to remain strictly invariant under charge conjugation, the photon field must simultaneously absorb this minus sign. Therefore, the photon field must transform as:

$$A^\mu \to -A^\mu$$

2. Vanishing of the N-photon Green function for odd N

We can rigorously prove this using the path integral formulation. The N-point Green function for the photon field is defined as:

$$G_N(x_1, \ldots, x_N) \equiv \langle \Omega | T A^{\mu_1}(x_1) \cdots A^{\mu_N}(x_N) | \Omega \rangle = \frac{1}{Z_0} \int \mathcal{D}A \mathcal{D}\phi \mathcal{D}\phi^* e^{iS[A,\phi,\phi^*]} A^{\mu_1}(x_1) \cdots A^{\mu_N}(x_N)$$

We now perform a formal change of integration variables inside the path integral, corresponding exactly to the charge conjugation symmetry we just established:

$$\phi \to \phi^*, \qquad \phi^* \to \phi, \qquad A^\mu \to -A^\mu$$

Because this is a fundamental symmetry of the theory, the classical action is completely invariant ($S[A,\phi,\phi^*] \to S[A,\phi,\phi^*]$). Furthermore, this is a simple linear field redefinition with unit Jacobian, meaning the functional integration measure is also invariant ($\mathcal{D}A\mathcal{D}\phi\mathcal{D}\phi^* \to \mathcal{D}A\mathcal{D}\phi\mathcal{D}\phi^*$).

However, the field insertions $A^{\mu_k}(x_k)$ in the integrand explicitly pick up a minus sign. With N such fields, we extract a global factor of $(-1)^N$:

$$\begin{aligned} G_N(x_1,\dots,x_N) &= \frac{1}{Z_0}\int \mathcal{D}A\mathcal{D}\phi\mathcal{D}\phi^* e^{iS}(-A^{\mu_1}(x_1))\cdots(-A^{\mu_N}(x_N)) \\ &= (-1)^N \frac{1}{Z_0}\int \mathcal{D}A\mathcal{D}\phi\mathcal{D}\phi^* e^{iS} A^{\mu_1}(x_1)\cdots A^{\mu_N}(x_N) \\ &= (-1)^N G_N(x_1,\dots,x_N) \end{aligned}$$

For this equation, $G_N = (-1)^N G_N$, to hold mathematically true, it trivially forces the condition:

$$G_N(x_1,\dots,x_N) = 0 \quad \text{for any odd } N.$$

3. Vanishing of the scattering amplitude for odd N

Physical scattering amplitudes, such as $i\mathcal{M}(2\gamma \to (N-2)\gamma)$, are obtained directly from the full connected N-point Green functions via the LSZ reduction formula. The LSZ procedure involves taking the Fourier transform of G_N into momentum space, amputating the external propagators, and placing the external momenta exactly on the mass shell ($p^2 = 0$ for photons).

Since we established in the previous point that the underlying position-space Green function G_N is identically zero everywhere for an odd number of photons, its Fourier transform is identically zero, and applying the LSZ truncation simply yields zero. Thus, any S-matrix element involving a total odd number of external photons is strictly zero.

4. Validity for off-shell photons

Yes, the statement remains completely true even if the external photons are off-shell ($p^2 \neq 0$) and lack physical polarization vectors.

The previous proof relying on the path integral demonstrates that $G_N = 0$ is an exact algebraic consequence of the fundamental charge conjugation symmetry of the quantum integration measure and the action. This vanishing occurs at the level of the fundamental Green functions (correlators), *before* any LSZ reduction, *before* taking the mass-shell limit $p^2 \to m^2$, and *before* contracting with transverse polarization vectors $\epsilon_\mu(p)$. Therefore, any Green function in momentum space with an odd number of photon legs vanishes identically, regardless of the kinematic state of those photons.

> **Remark.** *Furry's theorem powerfully restricts the allowed processes in QED and sQED. For instance, it immediately forbids processes like the decay of a single off-shell photon into three photons, or a 3-photon vertex correction. This vastly simplifies the computation of higher-order Feynman diagrams, as any sub-diagram containing a closed fermion (or scalar) loop with an odd number of attached photon lines identically cancels out.*

7. Furry's theorem in QED

Furry's theorem (in QED). This is similar to the previous exercise but for spinor QED. Charge conjugation for a Dirac fermion is defined by:

$$\psi \to -i(\bar{\psi}\gamma^0\gamma^2)^T, \qquad \bar{\psi} \to (-i\gamma^0\gamma^2\psi)^T. \qquad (6)$$

If you don't know or don't remember this, take these formulas as a *definition* of charge conjugation for a Dirac fermion.

- *Show that under such transformation*

$$\bar{\psi}\psi \to \bar{\psi}\psi, \qquad \bar{\psi}\gamma^\mu\psi \to -\bar{\psi}\gamma^\mu\psi \qquad (7)$$

and that the QED action is invariant under such transformation if the photon field A^μ also transforms as in the previous exercise.

- *Consider this transformation (including the one of A^μ) in the path integral. Argue that the integration measure of the path integral is invariant under such transformation.*
- *Consider a Green function with N photon fields, as in eq. (5). Show that if N is odd, then the Green function is zero.*
- *Argue, based on the previous result, that an amplitude with N external photons vanishes if N is odd. Discuss the case where the external photons, or some of them, are off-shell (i.e. $p^2 \neq 0$ and no polarization vector).*
- *Using the previous results, show that fermion loops with an odd number of vertexes do not contribute to amplitudes.*

1. Transformation of Fermion Bilinears

Let us define the charge conjugation matrix $C \equiv -i\gamma^0\gamma^2$. Using the standard Dirac algebra properties $\{\gamma^0, \gamma^2\} = 0$, $(\gamma^0)^2 = \mathbb{1}$, and $(\gamma^2)^2 = -\mathbb{1}$, we can deduce the properties of C:

$$C^2 = (-i\gamma^0\gamma^2)(-i\gamma^0\gamma^2) = -\gamma^0\gamma^2\gamma^0\gamma^2 = \gamma^0\gamma^0\gamma^2\gamma^2 = (+\mathbb{1})(-\mathbb{1}) = -\mathbb{1}$$

Thus, the inverse is simply $C^{-1} = -C$. Furthermore, checking the commutation relations with the gamma matrices reveals that $[C, \gamma^\mu] = 0$ for $\mu = 1, 3$ and $\{C, \gamma^\mu\} = 0$ for $\mu = 0, 2$.

Using this matrix C, we can elegantly rewrite the charge conjugation transformations given in eq. (6) in component notation. The transformations correspond to $\psi_\alpha \to \bar{\psi}_\beta C_{\beta\alpha}$ and $\bar{\psi}_\alpha \to C_{\alpha\beta}\psi_\beta$.

Let us apply this to the scalar bilinear $\bar{\psi}\psi$:

$$\bar{\psi}\psi = \bar{\psi}_\alpha\psi_\alpha \to (C_{\alpha\beta}\psi_\beta)(\bar{\psi}_\delta C_{\delta\alpha})$$

Because ψ and $\bar{\psi}$ are Grassmann variables, swapping their order incurs a minus sign:

$$(C_{\alpha\beta}\psi_\beta)(\bar{\psi}_\delta C_{\delta\alpha}) = -\bar{\psi}_\delta C_{\delta\alpha}C_{\alpha\beta}\psi_\beta = -\bar{\psi}(C^2)\psi$$

Since $C^2 = -\mathbb{1}$, we find:

$$-\bar{\psi}(-\mathbb{1})\psi = \bar{\psi}\psi$$

Thus, the scalar bilinear is strictly invariant: $\bar{\psi}\psi \to \bar{\psi}\psi$.

Now let us analyze the vector bilinear $\bar{\psi}\gamma^\mu\psi$:

$$\bar{\psi}\gamma^\mu\psi = \bar{\psi}_\alpha\gamma^\mu_{\alpha\beta}\psi_\beta \to (C_{\alpha\delta}\psi_\delta)\gamma^\mu_{\alpha\beta}(\bar{\psi}_\epsilon C_{\epsilon\beta})$$

Rearranging the Grassmann variables again yields a minus sign:

$$-\bar{\psi}_\epsilon C_{\epsilon\beta}\gamma^\mu_{\alpha\beta}C_{\alpha\delta}\psi_\delta = -\bar{\psi}C\gamma^{\mu T}C\psi$$

We must evaluate the matrix structure $-C\gamma^{\mu T}C$. In the standard Weyl representation, the transposed gamma matrices are $(\gamma^0)^T = \gamma^0$, $(\gamma^1)^T = -\gamma^1$, $(\gamma^2)^T = \gamma^2$, and $(\gamma^3)^T = -\gamma^3$. We check each component utilizing the (anti)commutation relations derived earlier:

- For $\mu = 0$: $-C(\gamma^0)^T C = -C\gamma^0 C = +C^2\gamma^0 = -\gamma^0$ (using $\{C, \gamma^0\} = 0$)
- For $\mu = 1$: $-C(\gamma^1)^T C = -C(-\gamma^1)C = C^2\gamma^1 = -\gamma^1$ (using $[C, \gamma^1] = 0$)
- For $\mu = 2$: $-C(\gamma^2)^T C = -C\gamma^2 C = +C^2\gamma^2 = -\gamma^2$ (using $\{C, \gamma^2\} = 0$)
- For $\mu = 3$: $-C(\gamma^3)^T C = -C(-\gamma^3)C = C^2\gamma^3 = -\gamma^3$ (using $[C, \gamma^3] = 0$)

In all cases, $-C\gamma^{\mu T}C = -\gamma^\mu$. Therefore, the vector current transforms with a minus sign:

$$\bar{\psi}\gamma^\mu\psi \to -\bar{\psi}\gamma^\mu\psi$$

2. Invariance of the QED Action

The full QED Lagrangian is $\mathcal{L} = \bar{\psi}(i\partial\!\!\!/ - m)\psi - ieA_\mu\bar{\psi}\gamma^\mu\psi$. We evaluate the transformation of each term:

- **Mass term:** $-m\bar{\psi}\psi \to -m\bar{\psi}\psi$ (Invariant, based on our proof above).
- **Interaction term:** $-ieA_\mu\bar{\psi}\gamma^\mu\psi$. If we enforce that the photon field transforms exactly as in sQED ($A_\mu \to -A_\mu$), then $-ie(-A_\mu)(-\bar{\psi}\gamma^\mu\psi) = -ieA_\mu\bar{\psi}\gamma^\mu\psi$. The term is strictly invariant.
- **Kinetic term:** $\bar{\psi}i\gamma^\mu\partial_\mu\psi$. Applying the transformation mapping yields $-i(\partial_\mu\bar{\psi})\gamma^\mu\psi$. While the Lagrangian density itself is not manifestly identical, its integral (the physical action S) is invariant. We integrate by parts over spacetime:

  $$\int d^4x \left(-i(\partial_\mu\bar{\psi})\gamma^\mu\psi\right) = \int d^4x \left(i\bar{\psi}\gamma^\mu\partial_\mu\psi\right)$$

  assuming the field configurations vanish at the spacetime boundary.

Thus, the total QED action S is flawlessly invariant under charge conjugation.

3. Invariance of the Path Integral Measure

The functional measure for the Dirac fields is the discretized infinite product $\int \mathcal{D}\bar{\psi}\mathcal{D}\psi \sim \int \prod_x \prod_{\alpha=1}^4 d\bar{\psi}_\alpha(x) d\psi_\alpha(x)$. The charge conjugation transformation acts as a local change of basis at every spacetime point x, represented by the block matrix:

$$\begin{pmatrix} \psi' \\ \bar{\psi}' \end{pmatrix} = \begin{pmatrix} 0 & C^T \\ C & 0 \end{pmatrix} \begin{pmatrix} \psi \\ \bar{\psi} \end{pmatrix}$$

The Jacobian for this transformation on Grassmann variables requires calculating the local determinant. Because $C = -i\gamma^0\gamma^2$, the determinant is:

$$|\det C|^2 = |\det(-i\mathbb{I}_{4\times4})\det(\gamma^0)\det(\gamma^2)|^2 = |(-i)^4(-1)(1)|^2 = |-1|^2 = 1$$

Since the Jacobian determinant is strictly unity, the path integral measure is perfectly invariant. The photon measure $\mathcal{D}A$ is also trivially invariant under the reflection $A^\mu \to -A^\mu$.

4. Vanishing of N-Photon Green Functions and Amplitudes for Odd N

Because the action and the measure are exactly invariant, we can perform this change of variables directly inside the defining path integral for the N-photon Green function:

$$\begin{aligned} G_N(x_1, \ldots, x_N) &= \langle\Omega|TA^{\mu_1}(x_1)\cdots A^{\mu_N}(x_N)|\Omega\rangle \\ &= \frac{1}{Z}\int \mathcal{D}A\mathcal{D}\bar{\psi}\mathcal{D}\psi e^{iS}(-A^{\mu_1}(x_1))\cdots(-A^{\mu_N}(x_N)) \\ &= (-1)^N G_N(x_1, \ldots, x_N) \end{aligned}$$

For this equality $G_N = (-1)^N G_N$ to hold algebraically, the Green function must identically vanish whenever N is odd.

By direct extension via the LSZ reduction formula, physical scattering amplitudes are extracted by applying linear differential operators to these Green functions and taking the Fourier transform. Since the underlying position-space Green function G_N is identically zero everywhere for odd N, its corresponding amplitude is also exactly zero. Crucially, this logic does not depend on the kinematic state of the external photons. Because the cancellation occurs strictly at the level of the quantum correlator in the path integral integrand, the result holds regardless of whether the external photons are on-shell or off-shell ($p^2 \neq 0$).

5. Non-Contribution of Odd Fermion Loops

Consider any arbitrary Feynman diagram containing a closed internal fermion loop attached to N external photon lines (or internal photon propagators connected to other parts of a larger diagram). This isolated fermion loop mathematically represents the perturbative evaluation of an N-photon correlator sub-diagram.

When evaluating the total amplitude, we must inherently sum over all possible Wick contractions. For a closed loop, this summation naturally includes both possible orientations of the momentum routing around the loop (corresponding physically to a virtual electron circulating clockwise versus a virtual positron circulating counter-clockwise). Applying the charge conjugation transformation exactly maps one loop orientation into the other, whilst simultaneously introducing a factor of $(-1)^N$ due to the transformation of the N interaction vertices (since each vertex $\bar{\psi}\gamma^\mu\psi \to -\bar{\psi}\gamma^\mu\psi$).

If N is odd, the two orientations contribute with exactly opposite signs and perfectly cancel each other out. Therefore, as a generalized corollary of Furry's theorem, any fermion loop with an odd number of vertices in QED evaluates to zero and can always be safely neglected from amplitude calculations.

8. Lightcone axial gauge in QED

In the Faddeev-Popov procedure for the quantization of QED, had we used the alternative gauge condition

$$G(A) = \eta_\mu A^\mu - \omega, \qquad (8)$$

where η^μ is an arbitrary massless four-vector $\eta^2 = 0$ (and G is defined as in lecture notes 3.2), we would have found the gauge fixing Lagrangian

$$\mathcal{L}_{\text{g.f.}} = -\frac{1}{2\xi}(\eta_\mu A^\mu)^2. \qquad (9)$$

Show that the photon propagator with this gauge choice, in the limit $\xi \to 0$ is equal to

$$\Pi^{\mu\nu}(p) = \frac{i}{p^2 + i0^+}\left(-g^{\mu\nu} + \frac{p^\mu\eta^\nu + \eta^\mu p^\nu}{\eta \cdot p}\right). \qquad (10)$$

To find the propagator, we first write down the total effective Lagrangian including the Maxwell term and the new gauge-fixing term:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\eta_\mu A^\mu)^2$$

We integrate the action by parts to express it as a quadratic differential operator acting on the fields, $\mathcal{L} = \frac{1}{2}A_\mu \mathcal{O}^{\mu\nu} A_\nu$. For the Maxwell part, we have $\frac{1}{2}A_\mu(\Box g^{\mu\nu} - \partial^\mu\partial^\nu)A_\nu$. The gauge-fixing term

can be written directly as $-\frac{1}{2\xi}A_\mu(\eta^\mu\eta^\nu)A_\nu$. Transforming into momentum space via $\partial_\mu \to -ip_\mu$ and $\Box \to -p^2$, the differential operator matrix $M_{\mu\rho}(p)$ becomes:

$$M_{\mu\rho}(p) = -p^2 g_{\mu\rho} + p_\mu p_\rho - \frac{1}{\xi}\eta_\mu\eta_\rho$$

The Feynman propagator $\Pi^{\rho\nu}(p)$ is defined as the inverse of this matrix, satisfying:

$$M_{\mu\rho}(p)\Pi^{\rho\nu}(p) = i\delta^\nu_\mu$$

To invert this operator, we construct the most general possible rank-2 tensor Ansatz using the available invariant structures: the metric $g^{\rho\nu}$ and the vectors p^ρ, η^ρ. The Ansatz is:

$$\Pi^{\rho\nu}(p) = A(p^2)g^{\rho\nu} + B(p^2)p^\rho p^\nu + C(p^2)\eta^\rho\eta^\nu + D(p^2)(p^\rho\eta^\nu + \eta^\rho p^\nu)$$

We substitute this Ansatz into our defining equation:

$$\left(-p^2 g_{\mu\rho} + p_\mu p_\rho - \frac{1}{\xi}\eta_\mu\eta_\rho\right)(Ag^{\rho\nu} + Bp^\rho p^\nu + C\eta^\rho\eta^\nu + D(p^\rho\eta^\nu + \eta^\rho p^\nu)) = i\delta^\nu_\mu$$

Expanding the product by distributing and carefully contracting the Lorentz index ρ, we evaluate each of the three terms from the operator block:

1. $(-p^2 g_{\mu\rho}) \times (\dots) = -p^2 A\delta^\nu_\mu - p^2 Bp_\mu p^\nu - p^2 C\eta_\mu\eta^\nu - p^2 D(p_\mu\eta^\nu + \eta_\mu p^\nu)$
2. $(p_\mu p_\rho) \times (\dots) = Ap_\mu p^\nu + Bp^2 p_\mu p^\nu + C(p\cdot\eta)p_\mu\eta^\nu + Dp^2 p_\mu\eta^\nu + D(p\cdot\eta)p_\mu p^\nu$
3. $\left(-\frac{1}{\xi}\eta_\mu\eta_\rho\right) \times (\dots) = -\frac{A}{\xi}\eta_\mu\eta^\nu - \frac{B}{\xi}(p\cdot\eta)\eta_\mu p^\nu - \frac{C}{\xi}\eta^2\eta_\mu\eta^\nu - \frac{D}{\xi}(p\cdot\eta)\eta_\mu\eta^\nu - \frac{D}{\xi}\eta^2\eta_\mu p^\nu$

Since η^μ is explicitly defined as a *massless* vector, we strictly enforce $\eta^2 = 0$. This eliminates the C and D terms proportional to η^2 in the third line.

Summing all the expanded terms, we group them by their linear tensor structures:

- **Term δ^ν_μ:** $-p^2 A = i \implies A = -\frac{i}{p^2}$
- **Term $p_\mu p^\nu$:** $-p^2 B + A + Bp^2 + D(p\cdot\eta) = 0 \implies A + D(p\cdot\eta) = 0 \implies D = -\frac{A}{p\cdot\eta} = \frac{i}{p^2(p\cdot\eta)}$
- **Term $p_\mu\eta^\nu$:** $-p^2 D + C(p\cdot\eta) + Dp^2 = 0 \implies C(p\cdot\eta) = 0 \implies C = 0$
- **Term $\eta_\mu p^\nu$:** $-p^2 D - \frac{B}{\xi}(p\cdot\eta) = 0 \implies B = -\frac{\xi p^2 D}{p\cdot\eta} = -\frac{i\xi}{(p\cdot\eta)^2}$
- **Term $\eta_\mu\eta^\nu$:** $-p^2 C - \frac{A}{\xi} - \frac{D}{\xi}(p\cdot\eta) = 0 \implies -\frac{1}{\xi}\big(A + D(p\cdot\eta)\big) = 0$. This provides a consistency check, seamlessly matching the relation derived from the $p_\mu p^\nu$ term.

Having determined the scalar coefficients, the full propagator for an arbitrary ξ is:

$$\Pi^{\mu\nu}(p) = -\frac{i}{p^2}g^{\mu\nu} - \frac{i\xi}{(p\cdot\eta)^2}p^\mu p^\nu + \frac{i}{p^2(p\cdot\eta)}(p^\mu\eta^\nu + \eta^\mu p^\nu)$$

We are requested to find the propagator in the strict limit $\xi \to 0$ (which physically corresponds to strictly enforcing the gauge condition $\eta_\mu A^\mu = 0$ in the path integral). In this limit, the B term proportional to ξ completely vanishes:

$$\lim_{\xi\to 0}\Pi^{\mu\nu}(p) = \frac{i}{p^2}\left(-g^{\mu\nu} + \frac{p^\mu\eta^\nu + \eta^\mu p^\nu}{p\cdot\eta}\right)$$

Finally, to rigorously define the integration contour for the time-ordered Feynman propagator, we implement the standard pole-shifting prescription by substituting $p^2 \to p^2 + i0^+$ in the overall denominator. Thus, we beautifully recover the required expression:

$$\Pi^{\mu\nu}(p) = \frac{i}{p^2 + i0^+}\left(-g^{\mu\nu} + \frac{p^\mu \eta^\nu + \eta^\mu p^\nu}{\eta \cdot p}\right)$$

Remark. *The term in parentheses represents the polarization sum $\sum_\lambda \epsilon^\mu_{(\lambda)}(p)\epsilon^{\nu *}_{(\lambda)}(p)$ for a strictly physical (transverse) photon, where the unphysical longitudinal and time-like degrees of freedom have been completely decoupled by the gauge choice.*

9. QED Ward-Takahashi identity

We proved the QED Ward-Takahashi identity

$$q_\mu \mathcal{M}^\mu(q; p_1, p_2) = -Qe\Big(\mathcal{M}_0(p_1, p_2 - q) - \mathcal{M}_0(p_1 + q, p_2)\Big), \qquad (11)$$

where p_i are generic off-shell momenta, $q = p_2 - p_1$ and $\mathcal{M}^\mu, \mathcal{M}_0$ defined as in lecture 3.3. Check this, at tree-level, via an explicit calculation with Feynman diagrams.

To verify the Ward-Takahashi identity at tree-level, we evaluate the left-hand side of equation (11) directly from the Feynman rules. The amplitude $\mathcal{M}^\mu(q; p_1, p_2)$ represents the interaction vertex where a photon with momentum q attaches to a fermion line with incoming momentum p_1 and outgoing momentum p_2. Because the fermion momenta p_1 and p_2 are generic and off-shell, this amplitude must be un-amputated; it explicitly includes the full incoming and outgoing fermion propagators, while the external photon line remains amputated (stripped of its polarization vector).

TODO: add diagram: Left hand side shows a photon line with momentum q attaching to a fermion line with incoming momentum p1 and outgoing momentum p2. Right hand side shows the difference of two bare un-amputated fermion lines.

Applying standard QED Feynman rules, the un-amputated vertex amplitude is given by the product of the final state propagator, the interaction vertex, and the initial state propagator:

$$\mathcal{M}^\mu(q; p_1, p_2) = \frac{i(\not{p}_2 + m)}{p_2^2 - m^2}(iQe\gamma^\mu)\frac{i(\not{p}_1 + m)}{p_1^2 - m^2}$$

We now contract this amplitude with the incoming photon four-momentum q_μ:

$$q_\mu \mathcal{M}^\mu(q; p_1, p_2) = q_\mu \left(\frac{i(\not{p}_2 + m)}{p_2^2 - m^2}(iQe\gamma^\mu)\frac{i(\not{p}_1 + m)}{p_1^2 - m^2}\right)$$

The contraction $q_\mu \gamma^\mu$ yields the Feynman slash vector $\not{q}$. Global momentum conservation at the vertex dictates that $q = p_2 - p_1$, allowing us to rewrite $\not{q} = \not{p}_2 - \not{p}_1$. Substituting this geometric identity into the numerator gives:

$$q_\mu \mathcal{M}^\mu(q; p_1, p_2) = \frac{(-iQe)(\not{p}_2 + m)(\not{p}_2 - \not{p}_1)(\not{p}_1 + m)}{(p_2^2 - m^2)(p_1^2 - m^2)}$$

To factorize and simplify the numerator, we employ an algebraic trick to express the momentum difference $\not{p}_2 - \not{p}_1$ in terms of the inverse Dirac propagators. We do this by adding and subtracting the fermion mass m:

$$\not{p}_2 - \not{p}_1 = (\not{p}_2 - m) - (\not{p}_1 - m)$$

We distribute this modified expression into the adjacent spinor matrix factors $(\not{p}_2 + m)$ and $(\not{p}_1 + m)$, which naturally splits the numerator into two distinct terms:

$$(\not{p}_2 + m)\big[(\not{p}_2 - m) - (\not{p}_1 - m)\big](\not{p}_1 + m) = (\not{p}_2 + m)(\not{p}_2 - m)(\not{p}_1 + m) - (\not{p}_2 + m)(\not{p}_1 - m)(\not{p}_1 + m)$$

Using the fundamental Clifford algebra property $\not{p}\not{p} = p^2 \mathbb{1}$, we can perfectly reduce the adjacent conjugate pairs: $(\not{p} + m)(\not{p} - m) = \not{p}\not{p} - m^2 = p^2 - m^2$. The expanded numerator thus simplifies entirely to scalars multiplying single slash matrices:

$$(p_2^2 - m^2)(\not{p}_1 + m) - (\not{p}_2 + m)(p_1^2 - m^2)$$

Substituting this evaluated numerator back into the full contracted amplitude, the scalar terms gracefully cancel with the corresponding kinematic factors in the denominator:

$$\begin{aligned} q_\mu \mathcal{M}^\mu(q; p_1, p_2) &= \frac{-iQe\big[(p_2^2 - m^2)(\not{p}_1 + m) - (\not{p}_2 + m)(p_1^2 - m^2)\big]}{(p_1^2 - m^2)(p_2^2 - m^2)} \\ &= -Qe\left[\frac{i(\not{p}_1 + m)}{p_1^2 - m^2} - \frac{i(\not{p}_2 + m)}{p_2^2 - m^2}\right] \end{aligned}$$

The two terms remaining inside the brackets are exactly the expressions for the bare, un-amputated free fermion propagators, denoted as $\mathcal{M}_0(p)$. The first term represents the propagator evaluated at momentum p_1, while the second term is the propagator evaluated at momentum p_2.

By utilizing the initial vertex constraint $q = p_2 - p_1$, we can equivalently map these momenta as $p_1 = p_2 - q$ and $p_2 = p_1 + q$. Substituting these definitions into the arguments of $\mathcal{M}_0$, we extract the final required relation:

$$q_\mu \mathcal{M}^\mu(q; p_1, p_2) = -Qe\Big(\mathcal{M}_0(p_1, p_2 - q) - \mathcal{M}_0(p_1 + q, p_2)\Big)$$

This explicit derivation strictly confirms the validity of the Ward-Takahashi identity at tree-level, demonstrating how gauge invariance forces a specific structural relationship between interaction vertices and propagators.

Sheet 4

1. The D-dimensional solid angle

Show that the D-dimensional solid angle is

$$\Omega_D = \int d\Omega_D = \frac{2\pi^{D/2}}{\Gamma(D/2)}. \qquad (1)$$

Hint: Compute a D-dimensional gaussian integral in both cartesian and polar coordinates, using the definition of the Gamma function for the latter.

To prove this identity, we evaluate a standard D-dimensional Gaussian integral over the entire Euclidean space $\mathbb{R}^D$ in two different coordinate systems. Let us define the integral I_D as:

$$I_D = \int_{\mathbb{R}^D} d^D x \, \exp\left(-\sum_{i=1}^{D} x_i^2\right).$$

1. Evaluation in Cartesian Coordinates

In Cartesian coordinates, the integration measure is simply $d^D x = dx_1 dx_2 \dots dx_D$. Because the exponential of a sum factors into a product of exponentials, the D-dimensional integral completely factorizes into D independent one-dimensional Gaussian integrals:

$$\begin{aligned} I_D &= \prod_{i=1}^{D} \left(\int_{-\infty}^{+\infty} dx_i \, e^{-x_i^2} \right) \\ &= \left(\int_{-\infty}^{+\infty} dx \, e^{-x^2} \right)^D. \end{aligned}$$

Using the standard result for the 1D Gaussian integral, $\int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}$, we obtain:

$$I_D = (\sqrt{\pi})^D = \pi^{D/2}.$$

2. Evaluation in D-dimensional Spherical Coordinates

Alternatively, we can transition to D-dimensional spherical (hyper-polar) coordinates. In this system, the volume element is expressed as the product of a radial component and an angular component: $d^D x = r^{D-1} dr \, d\Omega_D$, where $r = \sqrt{\sum x_i^2}$ is the radial distance and $d\Omega_D$ is the differential solid angle. The integral then separates into a radial integral and an angular integral:

$$I_D = \int d\Omega_D \int_0^\infty dr \, r^{D-1} e^{-r^2} = \Omega_D \int_0^\infty dr \, r^{D-1} e^{-r^2}.$$

To evaluate the radial integral, we perform a change of variables, setting $t = r^2$ (or $x = r^2$ as in the professor's notes). This implies $dt = 2r \, dr$, or equivalently $dr = \frac{1}{2\sqrt{t}} dt = \frac{1}{2} t^{-1/2} dt$. The integration limits trivially remain from 0 to ∞. Substituting this into the radial integral yields:

$$\begin{aligned} \int_0^\infty dr \, r^{D-1} e^{-r^2} &= \int_0^\infty \left(\frac{1}{2} t^{-1/2} dt \right) (t^{1/2})^{D-1} e^{-t} \\ &= \frac{1}{2} \int_0^\infty dt \, t^{(D-1)/2} t^{-1/2} e^{-t} \\ &= \frac{1}{2} \int_0^\infty dt \, t^{D/2-1} e^{-t}. \end{aligned}$$

We now perfectly recognize this integral as the formal definition of the Euler Gamma function, $\Gamma(z) = \int_0^\infty dt\, t^{z-1}e^{-t}$, evaluated at $z = D/2$. Therefore, the radial integral evaluates precisely to $\frac{1}{2}\Gamma(D/2)$.

Combining this back with the angular part, we get:

$$I_D = \frac{1}{2}\Omega_D \Gamma(D/2).$$

3. Synthesis

Equating the results obtained from both coordinate systems:

$$\pi^{D/2} = \frac{1}{2}\Omega_D \Gamma(D/2).$$

Solving for the total solid angle Ω_D, we rigorously recover the requested formula:

$$\Omega_D = \frac{2\pi^{D/2}}{\Gamma(D/2)}.$$

Remark. *This elegant derivation is a cornerstone of dimensional regularization in Quantum Field Theory. By expressing the solid angle entirely in terms of the Gamma function, which can be analytically continued to complex values, we can mathematically define loop integrals in an arbitrary, non-integer dimension $D = 4 - \epsilon$.*

2. Potential of an infinite wire in dimensional regularization

Compute the potential of an infinite wire in dimensional regularization and show that in the limit $D \to 1$ it is equal to

$$V(R) \equiv \frac{\lambda}{4\pi}\mu^{D-1}\int_{-\infty}^{\infty} d^D x \frac{1}{(|\vec{x}|^2 + R^2)^{1/2}}$$

$$= \frac{\lambda}{4\pi}\left(-\frac{2}{D-1} - \log(\mu R)^2 + \text{const} + \mathcal{O}(D-1)\right) \qquad (2)$$

with "const" independent of R.

Hint: Reuse the result for one-loop tadpole integrals without the $i/(-1)^\alpha$ prefactor, that was due to Wick's rotation into Euclidean space, and adjust for the missing $1/(2\pi)^D$.

To evaluate the integral defining the potential $V(R)$, we note that it takes the exact mathematical form of a vacuum (tadpole) loop integral in a D-dimensional Euclidean space. By following the hint, we can adapt the master formula for dimensionally regularized 1-loop integrals.

We strip the purely Minkowskian prefactor $\frac{i}{(-1)^\alpha}$ and multiply by the momentum-space measure factor $(2\pi)^D$ to isolate the pure spatial integral. The resulting generic Euclidean master formula is:

$$\int d^D x \frac{1}{(|\vec{x}|^2 + \Delta)^\alpha} = \pi^{D/2}\frac{\Gamma(\alpha - D/2)}{\Gamma(\alpha)}\Delta^{D/2-\alpha}$$

For our specific potential integral, we identify the parameters $\Delta = R^2$ and $\alpha = 1/2$. First, we compute the Gamma function in the denominator, $\Gamma(1/2)$. Using its integral definition:

$$\Gamma(1/2) = \int_0^\infty dt\, t^{-1/2}e^{-t}$$

We perform the change of variables $t = z^2$, which implies $dt = 2z\,dz$. The integration limits remain 0 to ∞:

$$\Gamma(1/2) = \int_0^\infty (2z\,dz)(z^2)^{-1/2} e^{-z^2} = 2\int_0^\infty dz\, e^{-z^2}$$

By symmetry of the Gaussian function, this is exactly equal to the full integral over the real line, which evaluates to $\sqrt{\pi}$:

$$\Gamma(1/2) = \int_{-\infty}^{+\infty} dz\, e^{-z^2} = \sqrt{\pi}$$

Substituting $\alpha = 1/2$, $\Delta = R^2$, and $\Gamma(1/2) = \sqrt{\pi}$ into the master formula, the potential $V(R)$ evaluates to:

$$\begin{aligned} V(R) &= \frac{\lambda}{4\pi}\mu^{D-1}\left[\pi^{D/2}\frac{\Gamma(1/2 - D/2)}{\sqrt{\pi}}(R^2)^{D/2-1/2}\right] \\ &= \frac{\lambda}{4\pi}\mu^{D-1}\pi^{\frac{D-1}{2}}\Gamma\left(\frac{1-D}{2}\right)R^{D-1} \\ &= \frac{\lambda}{4\pi}(\mu R)^{D-1}\pi^{\frac{D-1}{2}}\Gamma\left(\frac{1-D}{2}\right) \end{aligned}$$

We are interested in the physical limit where $D \to 1$. To systematically expand the expression, we define the small parameter $\varepsilon \equiv \frac{1-D}{2}$. This implies $D = 1 - 2\varepsilon$, and the limit $D \to 1$ corresponds directly to $\varepsilon \to 0$. Rewriting the potential in terms of ε:

$$V(R) = \frac{\lambda}{4\pi}(\mu R)^{-2\varepsilon}\pi^{-\varepsilon}\Gamma(\varepsilon)$$

We now Taylor expand each of the three ε-dependent factors around $\varepsilon = 0$:

1. Using $a^x = e^{x\log a} \approx 1 + x\log a$, we expand the radius-dependent term:

$$(\mu R)^{-2\varepsilon} = 1 - 2\varepsilon\log(\mu R) + \mathcal{O}(\varepsilon^2) = 1 - \varepsilon\log\big((\mu R)^2\big) + \mathcal{O}(\varepsilon^2)$$

2. Similarly, for the numerical factor:

$$\pi^{-\varepsilon} = 1 - \varepsilon\log\pi + \mathcal{O}(\varepsilon^2)$$

3. For the Gamma function, we use its well-known Laurent expansion near its pole at zero:

$$\Gamma(\varepsilon) = \frac{1}{\varepsilon} - \gamma_E + \mathcal{O}(\varepsilon)$$

   where γ_E is the Euler-Mascheroni constant.

Multiplying these expansions together, keeping terms up to $\mathcal{O}(\varepsilon^0)$:

$$\begin{aligned} V(R) &= \frac{\lambda}{4\pi}\Big(1 - \varepsilon\log\big((\mu R)^2\big)\Big)\Big(1 - \varepsilon\log\pi\Big)\left(\frac{1}{\varepsilon} - \gamma_E\right) \\ &= \frac{\lambda}{4\pi}\Big(1 - \varepsilon\log\big((\mu R)^2\big) - \varepsilon\log\pi + \mathcal{O}(\varepsilon^2)\Big)\left(\frac{1}{\varepsilon} - \gamma_E\right) \\ &= \frac{\lambda}{4\pi}\left[\frac{1}{\varepsilon} - \gamma_E - \log\big((\mu R)^2\big) - \log\pi + \mathcal{O}(\varepsilon)\right] \end{aligned}$$

We can group all the terms that are completely independent of the distance scale R (i.e., $-\gamma_E$ and $-\log\pi$) into a single generic constant, denoted as "const".

$$V(R) = \frac{\lambda}{4\pi}\left[\frac{1}{\varepsilon} - \log(\mu R)^2 + \text{const} + \mathcal{O}(\varepsilon)\right]$$

Finally, we revert from the regularization parameter ε back to the spatial dimension D. Since $\varepsilon = \frac{1-D}{2}$, it trivially follows that $\frac{1}{\varepsilon} = \frac{2}{1-D} = -\frac{2}{D-1}$. Furthermore, the $\mathcal{O}(\varepsilon)$ remainder is strictly equivalent to $\mathcal{O}(D-1)$. Substituting this back, we perfectly recover the requested expression:

$$V(R) = \frac{\lambda}{4\pi}\left(-\frac{2}{D-1} - \log(\mu R)^2 + \text{const} + \mathcal{O}(D-1)\right)$$

3. Tensor Loop Integral in D-dimensions

Show that

$$\int \frac{d^D k}{(2\pi)^D} \frac{k^\mu k^\nu}{(k^2 - M^2)^\alpha} = g^{\mu\nu} F(M^2) \qquad (3)$$

with

$$F(M^2) = \frac{i}{(-1)^{\alpha-1}} \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(\alpha - D/2 - 1)}{\Gamma(\alpha)} \frac{1}{2} (M^2)^{D/2-\alpha+1}. \qquad (4)$$

1. Tensor Structure by Lorentz Invariance

The integral to be evaluated, $I^{\mu\nu} = \int \frac{d^D k}{(2\pi)^D} \frac{k^\mu k^\nu}{(k^2-M^2)^\alpha}$, is a symmetric rank-2 tensor. Since there are no external fixed four-momenta to provide a preferred direction in spacetime, the result must be constructed entirely from the only available invariant rank-2 tensor: the metric tensor $g^{\mu\nu}$. Therefore, the integral must be strictly proportional to the metric:

$$\int \frac{d^D k}{(2\pi)^D} \frac{k^\mu k^\nu}{(k^2 - M^2)^\alpha} = A \cdot g^{\mu\nu}$$

where A is a scalar function representing $F(M^2)$.

To determine the scalar factor A, we contract both sides of the equation with the metric tensor $g_{\mu\nu}$. In D dimensions, the trace of the metric is $g_{\mu\nu} g^{\mu\nu} = D$:

$$\int \frac{d^D k}{(2\pi)^D} \frac{k_\mu k^\mu}{(k^2 - M^2)^\alpha} = A \cdot D$$

$$A = \frac{1}{D} \int \frac{d^D k}{(2\pi)^D} \frac{k^2}{(k^2 - M^2)^\alpha}$$

2. Algebraic Reduction of the Integrand

We manipulate the numerator of the new scalar integral to express it in terms of the denominator. We add and subtract the mass squared M^2:

$$\frac{k^2}{(k^2 - M^2)^\alpha} = \frac{(k^2 - M^2) + M^2}{(k^2 - M^2)^\alpha} = \frac{1}{(k^2 - M^2)^{\alpha-1}} + \frac{M^2}{(k^2 - M^2)^\alpha}$$

Substituting this back into the expression for A, the integral splits into a sum of two standard scalar integrals with different powers in the denominator:

$$A = \frac{1}{D}\left[\int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 - M^2)^{\alpha-1}} + M^2 \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 - M^2)^\alpha}\right]$$

3. Evaluation using the Master Integral

We utilize the standard master formula for a D-dimensional Minkowski space scalar integral (derived via Wick rotation to Euclidean space, where $k^2 - M^2 \to -(k_E^2 + M^2)$, giving the overall factor of $i/(-1)^n$):

$$\int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 - M^2)^n} = \frac{i}{(-1)^n} \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(n - D/2)}{\Gamma(n)} (M^2)^{D/2-n}$$

Applying this master formula to both integrals in our expression for A:

- For the first term ($n = \alpha - 1$):

$$I_1 = \frac{i}{(-1)^{\alpha-1}} \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(\alpha - 1 - D/2)}{\Gamma(\alpha - 1)} (M^2)^{D/2-\alpha+1}$$

- For the second term ($n = \alpha$):

$$I_2 = \frac{i}{(-1)^{\alpha}} \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(\alpha - D/2)}{\Gamma(\alpha)} (M^2)^{D/2-\alpha}$$

Multiplying I_2 by M^2 raises the mass exponent to match the first term: $M^2 \cdot (M^2)^{D/2-\alpha} = (M^2)^{D/2-\alpha+1}$. We can now combine the terms. Note that $\frac{1}{(-1)^{\alpha}} = -\frac{1}{(-1)^{\alpha-1}}$. Factoring out the common terms:

$$A = \frac{1}{D} \frac{i}{(-1)^{\alpha-1}} \frac{(M^2)^{D/2-\alpha+1}}{(4\pi)^{D/2}} \left[\frac{\Gamma(\alpha - D/2 - 1)}{\Gamma(\alpha - 1)} - \frac{\Gamma(\alpha - D/2)}{\Gamma(\alpha)} \right]$$

4. Simplification via Gamma Function Properties

To simplify the term in the square brackets, we use the fundamental property of the Gamma function: $\Gamma(z + 1) = z\Gamma(z)$. This gives us $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$ and $\Gamma(\alpha - D/2) = (\alpha - D/2 - 1)\Gamma(\alpha - D/2 - 1)$. Substituting these into the bracket:

$$\begin{aligned} \text{Bracket} &= \frac{\Gamma(\alpha - D/2 - 1)}{\Gamma(\alpha - 1)} - \frac{(\alpha - D/2 - 1)\Gamma(\alpha - D/2 - 1)}{(\alpha - 1)\Gamma(\alpha - 1)} \\ &= \frac{\Gamma(\alpha - D/2 - 1)}{(\alpha - 1)\Gamma(\alpha - 1)} \Big[(\alpha - 1) - (\alpha - D/2 - 1) \Big] \\ &= \frac{\Gamma(\alpha - D/2 - 1)}{\Gamma(\alpha)} \left[\frac{D}{2} \right] \end{aligned}$$

Finally, we substitute this highly simplified bracket back into the expression for A. The generic dimension factor D in the numerator perfectly cancels the $\frac{1}{D}$ prefactor we obtained from the initial tensor trace:

$$A = \frac{1}{D} \frac{i}{(-1)^{\alpha-1}} \frac{1}{(4\pi)^{D/2}} (M^2)^{D/2-\alpha+1} \left(\frac{D}{2} \frac{\Gamma(\alpha - D/2 - 1)}{\Gamma(\alpha)} \right)$$

$$A = \frac{i}{(-1)^{\alpha-1}} \frac{1}{(4\pi)^{D/2}} \frac{\Gamma(\alpha - D/2 - 1)}{\Gamma(\alpha)} \frac{1}{2} (M^2)^{D/2-\alpha+1}$$

Since $I^{\mu\nu} = A \cdot g^{\mu\nu} \equiv F(M^2) g^{\mu\nu}$, we have rigorously proven that $A = F(M^2)$, perfectly reproducing the targeted identity.

4. Gamma Matrix Identity in D dimensions

Prove the following identity in D space-time dimensions

$$\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma_\mu = -2\gamma^\sigma \gamma^\rho \gamma^\nu - (D - 4)\gamma^\nu \gamma^\rho \gamma^\sigma. \qquad (5)$$

To rigorously prove this identity, we rely on the fundamental Clifford algebra anticommutation relation for Dirac gamma matrices:

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}\not{1}$$

which allows us to swap adjacent gamma matrices by paying a penalty term: $\gamma^\mu\gamma^\nu = 2g^{\mu\nu} - \gamma^\nu\gamma^\mu$. Furthermore, in exactly D dimensions, the full contraction of a gamma matrix with itself yields the trace of the metric tensor:

$$\gamma^\mu\gamma_\mu = g_{\mu\nu}\gamma^\mu\gamma^\nu = \frac{1}{2}g_{\mu\nu}\{\gamma^\mu, \gamma^\nu\} = g_{\mu\nu}g^{\mu\nu}\not{1} = \delta^\mu_\mu\not{1} = D\not{1}$$

As we did for the specific 4-dimensional case, it is mathematically most efficient to build up to the full five-matrix string by recursively computing the contractions of shorter strings. We do this by continuously anti-commuting the rightmost γ_μ to the left until it meets its contracted partner γ^μ.

Step 1: Simplify $\gamma^\mu\gamma^\nu\gamma_\mu$

Anti-commuting the γ_μ past the γ^ν:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma_\mu &= \gamma^\mu(2\delta^\nu_\mu - \gamma_\mu\gamma^\nu) \\ &= 2\gamma^\nu - \gamma^\mu\gamma_\mu\gamma^\nu\end{aligned}$$

Using our D-dimensional identity $\gamma^\mu\gamma_\mu = D$, we get:

$$\gamma^\mu\gamma^\nu\gamma_\mu = 2\gamma^\nu - D\gamma^\nu = (2 - D)\gamma^\nu$$

Step 2: Simplify $\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu$

Anti-commuting the γ_μ past the γ^ρ:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu &= \gamma^\mu\gamma^\nu(2\delta^\rho_\mu - \gamma_\mu\gamma^\rho) \\ &= 2\gamma^\rho\gamma^\nu - (\gamma^\mu\gamma^\nu\gamma_\mu)\gamma^\rho\end{aligned}$$

Now substitute the result from Step 1 for the bracketed term:

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu = 2\gamma^\rho\gamma^\nu - (2 - D)\gamma^\nu\gamma^\rho$$

Step 3: Simplify the full expression $\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu$

We apply the exact same logic one final time, moving the γ_μ past the γ^σ:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu &= \gamma^\mu\gamma^\nu\gamma^\rho(2\delta^\sigma_\mu - \gamma_\mu\gamma^\sigma) \\ &= 2\gamma^\sigma\gamma^\nu\gamma^\rho - (\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu)\gamma^\sigma\end{aligned}$$

Substituting our result from Step 2 for the bracketed term:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu &= 2\gamma^\sigma\gamma^\nu\gamma^\rho - [2\gamma^\rho\gamma^\nu - (2 - D)\gamma^\nu\gamma^\rho]\,\gamma^\sigma \\ &= 2\gamma^\sigma\gamma^\nu\gamma^\rho - 2\gamma^\rho\gamma^\nu\gamma^\sigma + (2 - D)\gamma^\nu\gamma^\rho\gamma^\sigma\end{aligned}$$

Step 4: Reorder the matrices to match the target identity

Our expression from Step 3 is correct, but the first two terms do not match the specific ordering required by the problem statement. We must systematically swap the adjacent matrices using $\gamma^\alpha\gamma^\beta = 2g^{\alpha\beta} - \gamma^\beta\gamma^\alpha$:

- For the first term, we swap $\gamma^\nu\gamma^\rho$:

$$2\gamma^\sigma(\gamma^\nu\gamma^\rho) = 2\gamma^\sigma(2g^{\nu\rho} - \gamma^\rho\gamma^\nu) = 4g^{\nu\rho}\gamma^\sigma - 2\gamma^\sigma\gamma^\rho\gamma^\nu$$

- For the second term, we swap $\gamma^\rho\gamma^\nu$:

$$-2(\gamma^\rho\gamma^\nu)\gamma^\sigma = -2(2g^{\nu\rho} - \gamma^\nu\gamma^\rho)\gamma^\sigma = -4g^{\nu\rho}\gamma^\sigma + 2\gamma^\nu\gamma^\rho\gamma^\sigma$$

Substituting these structurally expanded terms back into our full expression gives:

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = [4g^{\nu\rho}\gamma^\sigma - 2\gamma^\sigma\gamma^\rho\gamma^\nu] + [-4g^{\nu\rho}\gamma^\sigma + 2\gamma^\nu\gamma^\rho\gamma^\sigma] + (2-D)\gamma^\nu\gamma^\rho\gamma^\sigma$$

The algebraic $4g^{\nu\rho}\gamma^\sigma$ terms perfectly cancel each other out, leaving only terms with the desired matrix orderings:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu &= -2\gamma^\sigma\gamma^\rho\gamma^\nu + 2\gamma^\nu\gamma^\rho\gamma^\sigma + (2-D)\gamma^\nu\gamma^\rho\gamma^\sigma \\ &= -2\gamma^\sigma\gamma^\rho\gamma^\nu + (2+2-D)\,\gamma^\nu\gamma^\rho\gamma^\sigma \\ &= -2\gamma^\sigma\gamma^\rho\gamma^\nu - (D-4)\gamma^\nu\gamma^\rho\gamma^\sigma\end{aligned}$$

This mathematically proves the requested identity in arbitrary D dimensions. Note that in the specific physical limit $D \to 4$, the second term identically vanishes, and we perfectly recover the 4-dimensional identity proven in Exercise 5.

5. Effective Field Theory and Renormalizability

Consider the following theory with two real scalars: ϕ of mass m and η of mass M, with $M \gg m$. Assume an interaction Lagrangian

$$\mathcal{L}_{\text{int}} = -\frac{g_1 M}{2!}\phi^2\eta - \frac{g_2}{3!}M\eta^3. \qquad (6)$$

1. *Is this theory renormalizable, super-renormalizable or non-renormalizable?*

2. *At energy scales $E \ll M$, the particle η cannot be produced on-shell, but it can appear as a virtual particle in the diagrams. In the limit of large M, we may approximate observables in this theory with those obtained via an* effective field theory (EFT)*, where only the lighter scalar ϕ appears and the interaction Lagrangian takes the form*

$$\mathcal{L}_{\text{EFT,int}} = -\frac{\lambda_4}{4!}\phi^4 - \frac{\lambda_6}{6!}\frac{1}{M^2}\phi^6 + \mathcal{O}(M^{-8}). \qquad (7)$$

   Show that this is the most general Lagrangian, with operators up to mass dimension 6, compatible with the symmetries of the original theory (which is its UV completion*).*

3. *Compute an approximation of the* effective coupling λ_4 *by requiring that a tree-level $2 \to 2$ scattering amplitude $\phi(p_1)+\phi(p_2) \to \phi(p_3)+\phi(p_4)$ yields the same result in the two theories, at leading order in the couplings $g_{1,2}$ and neglecting $\mathcal{O}(1/M^4)$ terms in the large M expansion. Result: $\lambda_4 = -3g_1^2 - 4g_1^2m^2/M^2 + \mathcal{O}(g_i^4) + \mathcal{O}(1/M^4)$*

1. Power Counting and Renormalizability

To determine the renormalizability of the theory, we must analyze the mass dimension of the couplings. In a natural system of units ($\hbar = c = 1$), the action $S = \int d^4x\mathcal{L}$ must be strictly

dimensionless. Since the integration measure d^4x has mass dimension -4, the Lagrangian density must have mass dimension $[\mathcal{L}] = 4$.

From the canonical free kinetic terms $\frac{1}{2}(\partial_\mu\phi)^2$ and $\frac{1}{2}(\partial_\mu\eta)^2$, we deduce that the scalar fields both have mass dimension $[\phi] = [\eta] = 1$.

Analyzing the interaction Lagrangian:

$$\mathcal{L}_{\text{int}} = -\frac{g_1 M}{2!}\phi^2\eta - \frac{g_2}{3!}M\eta^3$$

The field operators have dimensions $[\phi^2\eta] = 3$ and $[\eta^3] = 3$. Therefore, for the Lagrangian terms to reach dimension 4, the dimensionful coefficients must satisfy:

$$[g_1 M] = [g_2 M] = 1$$

Since M is a mass scale ($[M] = 1$), the couplings g_1 and g_2 must be strictly dimensionless. More importantly, the full coupling constants associated with the interaction vertices are precisely $(g_1 M)$ and $(g_2 M)$, both of which have a strictly positive mass dimension of $+1$.

Because all interaction vertices in this theory possess coupling constants with positive mass dimension, the theory is mathematically bounded in the ultraviolet and requires only a finite number of divergent diagrams to be renormalized. Therefore, the theory is **super-renormalizable**.

2. Symmetries and the Effective Field Theory (EFT) Basis

At energy scales $E \ll M$, the heavy particle η cannot be produced on-shell. It can only propagate as a virtual state over short distances, meaning its dynamical degrees of freedom can be formally integrated out of the path integral. The resulting low-energy physics is described by an Effective Field Theory (EFT) containing only local operators of the light field ϕ.

To construct the most general EFT Lagrangian, we must respect the underlying symmetries of the UV completion. The original Lagrangian is manifestly invariant under the discrete $\mathbb{Z}_2$ transformation:

$$\phi \to -\phi, \qquad \eta \to \eta$$

Because this is an exact symmetry of the fundamental theory, the EFT must inherit it. Consequently, the EFT Lagrangian can only contain *even* powers of the field ϕ. Terms like ϕ^3 or ϕ^5 are strictly forbidden.

Expanding up to mass dimension 6, the allowed non-derivative local operators are ϕ^4 (dimension 4) and ϕ^6 (dimension 6). One might also consider the dimension-6 derivative operator $\mathcal{O}_6 = \phi^2(\partial_\mu\phi)^2$. However, utilizing integration by parts inside the action, we can shift the derivative:

$$\phi^2(\partial_\mu\phi)(\partial^\mu\phi) = \frac{1}{3}\partial_\mu(\phi^3\partial^\mu\phi) - \frac{1}{3}\phi^3\Box\phi \implies \phi^2(\partial_\mu\phi)^2 \cong -\frac{1}{3}\phi^3\Box\phi$$

Using the lowest-order classical equations of motion for the free field, $\Box\phi = -m^2\phi$, this operator reduces to:

$$-\frac{1}{3}\phi^3(-m^2\phi) = \frac{m^2}{3}\phi^4$$

This demonstrates that the operator $\phi^2(\partial_\mu\phi)^2$ is physically redundant; it simply shifts the Wilson coefficient of the ϕ^4 operator (and the ϕ^6 operator if we include interaction terms in the equations of motion). This redundancy can be formally removed by a local field redefinition $\phi \to \phi + \frac{c}{M^2}\phi^3$.

Thus, the minimal, non-redundant EFT Lagrangian compatible with the $\mathbb{Z}_2$ symmetry up to dimension 6 is exactly:

$$\mathcal{L}_{\text{EFT,int}} = -\frac{\lambda_4}{4!}\phi^4 - \frac{\lambda_6}{6!}\frac{1}{M^2}\phi^6 + \mathcal{O}(M^{-8})$$

3. Matching the Effective Coupling λ_4

To compute the Wilson coefficient λ_4, we require that the physical observables computed in the full UV theory precisely match those computed in the EFT at low energies. We enforce this matching condition on the tree-level $2 \to 2$ scattering amplitude $\phi(p_1) + \phi(p_2) \to \phi(p_3) + \phi(p_4)$.

In the **EFT**, the interaction term is strictly local. Using the Feynman rule for the $-\frac{\lambda_4}{4!}\phi^4$ operator, the tree-level amplitude is simply the constant vertex factor:

$$i\mathcal{M}_{\text{EFT}} = -i\lambda_4$$

In the **UV theory**, the $2 \to 2$ scattering is mediated by the exchange of the heavy η particle. From the interaction $-\frac{g_1 M}{2!}\phi^2\eta$, the Feynman rule for the ϕ-ϕ-η vertex is $-i(g_1 M)$. The tree-level amplitude is the sum of the s, t, and u channels:

TODO: add diagram: s-channel + t-channel + u-channel tree diagrams with intermediate η exchange

$$\begin{aligned} i\mathcal{M}_{\text{UV}} &= \\ &= (-ig_1 M)\left[\frac{i}{s-M^2}\right](-ig_1 M) + (-ig_1 M)\left[\frac{i}{t-M^2}\right](-ig_1 M) + (-ig_1 M)\left[\frac{i}{u-M^2}\right](-ig_1 M) \\ &= -ig_1^2 M^2\left(\frac{1}{s-M^2} + \frac{1}{t-M^2} + \frac{1}{u-M^2}\right) \end{aligned}$$

Since we are evaluating the amplitude at energy scales $E \ll M$, the Mandelstam variables obey $s, t, u \sim m^2 \ll M^2$. We can systematically Taylor expand the heavy propagators in inverse powers of M^2:

$$\frac{1}{x-M^2} = -\frac{1}{M^2}\left(1 - \frac{x}{M^2}\right)^{-1} = -\frac{1}{M^2}\left(1 + \frac{x}{M^2} + \mathcal{O}(M^{-4})\right)$$

Applying this expansion to the amplitude yields:

$$\begin{aligned} i\mathcal{M}_{\text{UV}} &= -ig_1^2 M^2\left[-\frac{1}{M^2}\left(1+\frac{s}{M^2}\right) - \frac{1}{M^2}\left(1+\frac{t}{M^2}\right) - \frac{1}{M^2}\left(1+\frac{u}{M^2}\right)\right] \\ &= ig_1^2\left(3 + \frac{s+t+u}{M^2} + \mathcal{O}(M^{-4})\right) \end{aligned}$$

By rigorous momentum conservation for identical particles of mass m, the Mandelstam variables satisfy the kinematic constraint $s + t + u = \sum_{i=1}^{4} m^2 = 4m^2$. Substituting this into the expanded amplitude gives the low-energy limit of the UV theory:

$$i\mathcal{M}_{\text{UV}} = ig_1^2\left(3 + 4\frac{m^2}{M^2}\right) + \mathcal{O}(M^{-4})$$

Finally, we enforce the matching condition $i\mathcal{M}_{\text{EFT}} = i\mathcal{M}_{\text{UV}}$:

$$-i\lambda_4 = ig_1^2\left(3 + 4\frac{m^2}{M^2}\right)$$

Dividing both sides by $-i$, we isolate the effective coupling λ_4 at tree-level accuracy:

$$\lambda_4 = -3g_1^2 - 4g_1^2\frac{m^2}{M^2} + \mathcal{O}(g_i^4) + \mathcal{O}(1/M^4)$$

6. One-loop corrections to the scalar propagator in Yukawa theory

In the Yukawa theory, compute the self-energy one-loop corrections (i.e. the one-loop corrections to the propagator) for the scalar field proportional to fermion loops. For the finite part of the result, the integral over the Feynman parameters can be performed using the identity

$$\int dx\, \Delta \log\frac{\Delta}{\mu^2} = -\frac{1}{6}p^2 w^3 \log\frac{w+1}{w-1} + \frac{6m^2 - p^2}{6}\log\frac{m^2}{\mu^2} + \frac{6p^2 - 24m^2}{18} \qquad (8)$$

$$\textit{where } \Delta \equiv m^2 - x(1-x)p^2 \textit{ and } w \equiv \sqrt{1-4m^2/p^2}.$$

The relevant interaction Lagrangian for the scalar field in Yukawa theory is extended with scalar self-interactions, $\mathcal{L}_{\text{int}} = -g\phi\bar{\psi}\psi - \frac{\lambda_3}{3!}\phi^3 - \frac{\lambda_4}{4!}\phi^4$. The one-loop corrections to the scalar propagator (self-energy) receive contributions from three distinct diagrams:

a) A fermion loop mediated by the Yukawa coupling g.

b) A scalar tadpole loop from the quartic coupling λ_4, whose integral takes the general form:

$$i\mathcal{M}_b \propto \lambda_4 \int \frac{d^D k}{(2\pi)^D} \frac{1}{k^2 - m_\phi^2}$$

c) A scalar bubble loop from the cubic coupling λ_3, whose integral takes the general form:

$$i\mathcal{M}_c \propto \lambda_3^2 \int \frac{d^D k}{(2\pi)^D} \frac{1}{(k^2 - m_\phi^2)((k-p)^2 - m_\phi^2)}$$

Although the scalar loop diagrams (b) and (c) are both ultraviolet-divergent, we do not evaluate them here for two specific reasons. First, the text of the exercise explicitly requests only the corrections *proportional to fermion loops*, which restricts our focus entirely to diagram (a). Second, these purely scalar integrals are standard results that were already computed in previous lectures (the tadpole in lecture 4.2 and the bubble in lecture 4.1), leaving the fermion loop (a) as the only structurally new correction to evaluate.

1. Setting up the amplitude

Using the dimensional regularization Feynman rule $-ig\mu^{\frac{4-D}{2}}$ for the Yukawa vertex (where $D = 4 - 2\epsilon$), the amplitude for the fermion loop is:

$$i\mathcal{M}_a = (-ig)^2(-1)i^2\mu^{2\epsilon} \int \frac{d^D k}{(2\pi)^D} \frac{\text{Tr}[(\not{k} + \not{p} + m)(\not{k} + m)]}{((k+p)^2 - m^2)(k^2 - m^2)}$$

where the overall (-1) factor strictly arises because it is a closed fermion loop. Evaluating the Dirac trace:

$$\text{Tr}[(\not{k} + \not{p} + m)(\not{k} + m)] = \text{Tr}[\not{k}\not{k} + \not{p}\not{k}] + m^2\text{Tr}[\mathbb{1}] = 4(k^2 + k\cdot p + m^2)$$

Substituting this trace back into the integral, we have:

$$i\mathcal{M}_a = -4g^2\mu^{2\epsilon} \int \frac{d^D k}{(2\pi)^D} \frac{k^2 + k\cdot p + m^2}{(k^2 - m^2)((k+p)^2 - m^2)}$$

2. Feynman parameters and momentum shift

We combine the propagators in the denominator using standard Feynman parameters:

$$\frac{1}{D_1 D_2} = \int_0^1 dx \frac{1}{[k^2 + 2xk\cdot p + xp^2 - m^2]^2}$$

To complete the square in the denominator and restore spherical symmetry in momentum space, we shift the loop momentum $k^\mu \to k^\mu - xp^\mu$. The numerator transforms as:

$$(k - xp)^2 + (k - xp)\cdot p + m^2 = k^2 - k\cdot p(2x - 1) + m^2 - x(1-x)p^2$$

Defining the effective mass parameter $\Delta \equiv m^2 - x(1-x)p^2$, the shifted integral becomes:

$$i\mathcal{M}_a = -4g^2\mu^{2\epsilon}\int_0^1 dx \int \frac{d^Dk}{(2\pi)^D}\frac{k^2 - k\cdot p(2x-1) + \Delta}{(k^2-\Delta)^2}$$

The term linear in the loop momentum k is an odd function over a symmetric integration domain, hence it identically vanishes. This leaves two distinct integrals to evaluate:

$$i\mathcal{M}_a = -4g^2\mu^{2\epsilon}\int_0^1 dx \left[\int \frac{d^Dk}{(2\pi)^D}\frac{k^2}{(k^2-\Delta)^2} + \Delta\int \frac{d^Dk}{(2\pi)^D}\frac{1}{(k^2-\Delta)^2}\right]$$

3. Dimensional Regularization Evaluation

Using the result from Exercise 3 for the first integral and the standard one-loop vacuum (tadpole) integral for the second term, we obtain:

$$\int \frac{d^Dk}{(2\pi)^D}\frac{k^2}{(k^2-\Delta)^2} = -\frac{iD}{2}\frac{1}{(4\pi)^{D/2}}\Gamma(1-D/2)\Delta^{D/2-1}$$
$$\Delta\int \frac{d^Dk}{(2\pi)^D}\frac{1}{(k^2-\Delta)^2} = i\frac{1}{(4\pi)^{D/2}}\Gamma(2-D/2)\Delta^{D/2-1}$$

Summing these two terms and factoring out common terms by utilizing the Gamma function property $\Gamma(2-D/2) = (1-D/2)\Gamma(1-D/2)$, we find:

$$i\mathcal{M}_a = -4g^2\frac{i}{(4\pi)^{D/2}}\mu^{2\epsilon}\Gamma(1-D/2)(1-D)\int_0^1 dx(\Delta)^{D/2-1}$$

4. Expansion in ϵ and Finite Part

We expand this expression near four physical dimensions, setting $D = 4-2\epsilon$ and taking the limit $\epsilon \to 0$. The purely dimensional prefactor expansion requires care:

$$\Gamma(1-D/2)(1-D) = \Gamma(\epsilon-1)(2\epsilon-3) = \frac{\Gamma(\epsilon)}{\epsilon-1}(2\epsilon-3)$$

Taylor expanding the non-singular part for small ϵ:

$$\left(\frac{1}{\epsilon}-\gamma_E\right)\frac{-3+2\epsilon}{-1+\epsilon} \approx \left(\frac{1}{\epsilon}-\gamma_E\right)(3+\epsilon) = \frac{3}{\epsilon}+1-3\gamma_E+\mathcal{O}(\epsilon)$$

Including the remaining measure terms $\mu^{2\epsilon}(4\pi)^{\epsilon-2}\Delta^{1-\epsilon} = \frac{\Delta}{16\pi^2}\left[1+\epsilon\log\left(\frac{4\pi\mu^2}{\Delta}\right)\right]$, the full divergent expansion takes the form:

$$i\mathcal{M}_a = -\frac{ig^2}{4\pi^2}\int_0^1 dx\left[\Delta + 3\Delta\left(\frac{1}{\epsilon}-\gamma_E+\log 4\pi - \log\frac{\Delta}{\mu^2}\right)+\mathcal{O}(\epsilon)\right]$$

The integration of the polynomial part yields $\int_0^1 dx\Delta = m^2 - p^2/6$. For the non-trivial logarithmic term $\int dx\Delta\log\frac{\Delta}{\mu^2}$, one directly substitutes the provided identity (8) to extract the exact analytic form of the finite radiative corrections to the propagator.

7. Field-strength renormalization and the Källën-Lehmann spectral representation

We showed that the full propagator of a scalar theory, in any renormalization scheme, can be written as

$$P = \frac{i}{p^2 - m^2 - \Pi(p^2) + i0^+}, \qquad (9)$$

where $-i\Pi(p^2)$ is the two-point 1PI Green function. We also previously argued that the full propagator has a pole at the physical mass m^2_{phys} that is the value of p^2 for a physical one-particle state

$$P = \frac{iZ}{p^2 - m^2_{\text{phys}} + i0^+} + \text{regular at } p^2 \simeq m^2_{\text{phys}}. \qquad (10)$$

Show that

$$Z = \frac{1}{1 - \Pi'(m^2_{\text{phys}})}. \qquad (11)$$

Note: in a generic renormalization scheme we may have $m \neq m_{phys}$.

To relate the full exact propagator P to the residue Z at the physical pole, we start from the exact Dyson-Schwinger resummation formula for the two-point connected Green function in momentum space:

$$P(p^2) = \frac{i}{p^2 - m^2 - \Pi(p^2) + i0^+}$$

By definition, the physical mass squared m^2_{phys} corresponds precisely to the exact pole of the full propagator, which means it is the root of the inverse propagator denominator:

$$p^2 - m^2 - \Pi(p^2)\big|_{p^2=m^2_{\text{phys}}} = 0$$

This establishes a rigorous relation between the bare (or Lagrangian) mass parameter m^2 and the physical pole mass:

$$\Pi(m^2_{\text{phys}}) = m^2_{\text{phys}} - m^2$$

To analyze the analytic structure of the propagator near the physical mass shell, we perform a Taylor expansion of the self-energy function $\Pi(p^2)$ around the pole $p^2 = m^2_{\text{phys}}$:

$$\Pi(p^2) = \Pi(m^2_{\text{phys}}) + \Pi'(m^2_{\text{phys}})(p^2 - m^2_{\text{phys}}) + \mathcal{O}\left((p^2 - m^2_{\text{phys}})^2\right)$$

where $\Pi'(m^2_{\text{phys}}) \equiv \frac{d\Pi(p^2)}{dp^2}\Big|_{p^2=m^2_{\text{phys}}}$.

Substituting our expression for $\Pi(m^2_{\text{phys}})$ into this expansion yields:

$$\Pi(p^2) = (m^2_{\text{phys}} - m^2) + \Pi'(m^2_{\text{phys}})(p^2 - m^2_{\text{phys}}) + \mathcal{O}\left((p^2 - m^2_{\text{phys}})^2\right)$$

Now, we substitute this expanded form of $\Pi(p^2)$ back into the denominator of the full propagator (9). For convenience, let us define the shifted momentum variable $x \equiv p^2 - m^2_{\text{phys}}$:

$$\begin{aligned} p^2 - m^2 - \Pi(p^2) &= (p^2 - m^2_{\text{phys}}) + (m^2_{\text{phys}} - m^2) - \Pi(p^2) \\ &= x + (m^2_{\text{phys}} - m^2) - \left[(m^2_{\text{phys}} - m^2) + \Pi'(m^2_{\text{phys}})x + \mathcal{O}(x^2)\right] \\ &= x - \Pi'(m^2_{\text{phys}})x + \mathcal{O}(x^2) \\ &= x\left(1 - \Pi'(m^2_{\text{phys}})\right) + \mathcal{O}(x^2) \end{aligned}$$

Substituting this result back into the full propagator expression, we obtain:

$$P = \frac{i}{x\left(1 - \Pi'(m^2_{\text{phys}})\right) + \mathcal{O}(x^2)}$$

We factor out the constant term from the denominator to isolate the simple pole at $x = 0$ (i.e., $p^2 = m^2_{\text{phys}}$):

$$P = \frac{i}{x\left(1 - \Pi'(m^2_{\text{phys}})\right)} \left(1 + \frac{\mathcal{O}(x^2)}{x(1 - \Pi'(m^2_{\text{phys}}))}\right)^{-1}$$

Expanding the bracketed term for small x, the higher-order contributions become regular (analytic) terms at $p^2 \simeq m^2_{\text{phys}}$:

$$P = \frac{i}{p^2 - m^2_{\text{phys}}} \left(\frac{1}{1 - \Pi'(m^2_{\text{phys}})}\right) + \text{regular terms at } p^2 \simeq m^2_{\text{phys}}$$

By directly comparing this derived Laurent expansion with the general spectral definition of the physical pole given in eq. (10), $P = \frac{iZ}{p^2 - m^2_{\text{phys}} + i0^+} + \text{regular}$, we can identify the residue Z exactly:

$$Z = \frac{1}{1 - \Pi'(m^2_{\text{phys}})}$$

This completes the formal proof, demonstrating how the field-strength renormalization constant Z is directly determined by the slope of the 1PI two-point proper self-energy evaluated at the physical mass shell.

Sheet 5

1. Proof of the Gordon Identity

Prove the Gordon identity

$$\bar{u}(p_2)(p_1^\mu + p_2^\mu)u(p_1) = 2m\,\bar{u}(p_2)\gamma^\mu u(p_1) + i\bar{u}(p_2)\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu})u(p_1), \qquad (1)$$

where

$$\sigma^{\mu\nu} \equiv \frac{i}{2}[\gamma^\mu, \gamma^\nu] \qquad (2)$$

and the spinors $u(p_{1,2})$ satisfy the Dirac equation with mass m.

To prove the Gordon identity, we start from the definition of the tensor $\sigma^{\mu\nu}$ given in eq. (2) and analyze the structure involving the commutator of two gamma matrices sandwiched between the Dirac spinors:

$$i\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu}) = i\left(\frac{i}{2}[\gamma^\mu, \gamma^\nu]\right)(p_{1\nu} - p_{2\nu}) = -\frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)(p_{1\nu} - p_{2\nu})$$

Expanding the commutator product explicitly:

$$i\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu}) = -\frac{1}{2}\left(\gamma^\mu\gamma^\nu p_{1\nu} - \gamma^\mu\gamma^\nu p_{2\nu} - \gamma^\nu\gamma^\mu p_{1\nu} + \gamma^\nu\gamma^\mu p_{2\nu}\right)$$

Using Feynman slash notation, where $\not{p} \equiv \gamma^\nu p_\nu$, we can rewrite the terms as:

$$i\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu}) = -\frac{1}{2}\left(\gamma^\mu \not{p}_1 - \gamma^\mu \not{p}_2 - \not{p}_1\gamma^\mu + \not{p}_2\gamma^\mu\right)$$

Grouping the terms containing $\not{p}_1$ and $\not{p}_2$:

$$i\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu}) = -\frac{1}{2}\left[(\gamma^\mu \not{p}_1 - \not{p}_1\gamma^\mu) - (\gamma^\mu \not{p}_2 - \not{p}_2\gamma^\mu)\right]$$

Now, we sandwich this expression between the initial and final Dirac spinors, $\bar{u}(p_2)$ and $u(p_1)$:

$$i\bar{u}(p_2)\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu})u(p_1) = -\frac{1}{2}\bar{u}(p_2)\left[(\gamma^\mu \not{p}_1 - \not{p}_1\gamma^\mu) - (\gamma^\mu \not{p}_2 - \not{p}_2\gamma^\mu)\right]u(p_1)$$

We evaluate the action of the Dirac equation on the spinors. The momentum-space Dirac equation for an incoming spinor $u(p_1)$ and an outgoing conjugate spinor $\bar{u}(p_2)$ is given by:

$$(\not{p}_1 - m)u(p_1) = 0 \implies \not{p}_1 u(p_1) = mu(p_1)$$

$$\bar{u}(p_2)(\not{p}_2 - m) = 0 \implies \bar{u}(p_2)\not{p}_2 = m\bar{u}(p_2)$$

Let us examine how these relations affect the two bracketed terms when acting on the spinors:

1. For the first term acting on $u(p_1)$ to the right:

$$(\gamma^\mu \not{p}_1 - \not{p}_1\gamma^\mu)u(p_1) = \gamma^\mu(\not{p}_1 u(p_1)) - \not{p}_1(\gamma^\mu u(p_1))$$

Substituting $\not{p}_1 u(p_1) = mu(p_1)$ and using the anticommutation relation $\{\gamma^\mu, \not{p}_1\} = 2p_1^\mu \mathbb{1}$ (meaning $\not{p}_1\gamma^\mu = 2p_1^\mu - \gamma^\mu \not{p}_1$):

$$\gamma^\mu(mu(p_1)) - (2p_1^\mu - \gamma^\mu \not{p}_1)u(p_1) = m\gamma^\mu u(p_1) - 2p_1^\mu u(p_1) + \gamma^\mu(\not{p}_1 u(p_1)) = 2m\gamma^\mu u(p_1) - 2p_1^\mu u(p_1)$$

2. For the second term acting on $\bar{u}(p_2)$ to the left:

$$\bar{u}(p_2)(\gamma^\mu \not{p}_2 - \not{p}_2 \gamma^\mu) = (\bar{u}(p_2)\not{p}_2)\gamma^\mu - \bar{u}(p_2)(\gamma^\mu \not{p}_2)$$

Substituting $\bar{u}(p_2)\not{p}_2 = m\bar{u}(p_2)$ and using $\not{p}_2\gamma^\mu = 2p_2^\mu - \gamma^\mu \not{p}_2$:

$$(m\bar{u}(p_2))\gamma^\mu - \bar{u}(p_2)(2p_2^\mu - \gamma^\mu \not{p}_2) = 2m\bar{u}(p_2)\gamma^\mu - 2p_2^\mu \bar{u}(p_2)$$

Substituting these evaluated components back into the sandwich expression:

$$i\bar{u}(p_2)\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu})u(p_1) = -\frac{1}{2}\left[\bar{u}(p_2)\big(2m\gamma^\mu - 2p_1^\mu\big)u(p_1) - \bar{u}(p_2)\big(2m\gamma^\mu - 2p_2^\mu\big)u(p_1)\right]$$

Distributing the minus sign and the factor of $-\frac{1}{2}$:

$$\begin{aligned} i\bar{u}(p_2)\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu})u(p_1) &= -m\bar{u}(p_2)\gamma^\mu u(p_1) + \bar{u}(p_2)p_1^\mu u(p_1) - m\bar{u}(p_2)\gamma^\mu u(p_1) + \bar{u}(p_2)p_2^\mu u(p_1) \\ &= \bar{u}(p_2)(p_1^\mu + p_2^\mu)u(p_1) - 2m\bar{u}(p_2)\gamma^\mu u(p_1) \end{aligned}$$

Rearranging the terms to isolate the symmetric momentum combination on the left-hand side, we obtain:

$$\bar{u}(p_2)(p_1^\mu + p_2^\mu)u(p_1) = 2m\,\bar{u}(p_2)\gamma^\mu u(p_1) + i\bar{u}(p_2)\sigma^{\mu\nu}(p_{1\nu} - p_{2\nu})u(p_1)$$

This rigorously completes the proof of the Gordon identity, which is a foundational relation used in quantum electrodynamics to decompose the fermion electromagnetic vertex into its convection (electric) and spin (magnetic) current components.

2. Vacuum polarization and renormalization in QED

We showed in the lectures that the full photon propagator in QED can be written as

$$\frac{-ig_{\mu\nu}}{p^2(1-\Pi(p^2))} + \mathcal{O}(p^\mu p^\nu), \qquad (3)$$

where $\mathcal{O}(p^\mu p^\nu)$ stands for terms that will vanish once contracted with an electromagnetic current, due to the Ward identity. The term $\Pi(p^2)$ is instead defined by

$$i\Pi^{\mu\nu} = i(p^2 g^{\mu\nu} - p^\mu p^\nu)\Pi(p^2), \qquad (4)$$

where the left-hand side is the sum of all the one-particle irreducible (1PI) diagrams contributing to the propagator.

Compute the renormalized $\Pi(p^2)$, using the following renormalization schemes

1. *on-shell scheme, given by the renormalization condition $\Pi(0) = 0$,*
2. *in the $\overline{MS}$ scheme.*

One-Loop Vacuum Polarization Tensor

The one-loop quantum correction to the photon propagator is given by the one-particle irreducible (1PI) fermion loop diagram. Using the Feynman rules in D dimensions ($D = 4 - 2\epsilon$), the unrenormalized vacuum polarization tensor $i\Pi^{\mu\nu}(p)$ is:

$$i\Pi^{\mu\nu}(p) = (-ie)^2(-1)\mu^{2\epsilon}\int \frac{d^D k}{(2\pi)^D}\frac{\mathrm{Tr}\left[\gamma^\mu i(\not{k} + \not{p} + m)\gamma^\nu i(\not{k} + m)\right]}{(k^2 - m^2 + i\epsilon)((k+p)^2 - m^2 + i\epsilon)}$$

where the overall factor of (-1) comes from the closed fermion loop. Evaluating the Dirac trace in the numerator:

$$\mathrm{Tr}[\dots] = -4\left[k^\mu(k+p)^\nu + k^\nu(k+p)^\mu - g^{\mu\nu}(k^2 + k\cdot p - m^2)\right]$$

By gauge invariance and Lorentz covariance, the full tensor must take the transverse form:

$$i\Pi^{\mu\nu}(p) = i(p^2 g^{\mu\nu} - p^\mu p^\nu)\Pi(p^2)$$

To extract the scalar function $\Pi(p^2)$, we project using the transverse metric tensor:

$$i\Pi(p^2) = \frac{1}{D-1}\frac{1}{p^2} g_{\mu\nu} i\Pi^{\mu\nu}(p)$$

Feynman Parameterization and Loop Integration

Combining the denominators using standard Feynman parameters with parameter x, and shifting the loop momentum $k^\mu \to k^\mu - xp^\mu$, we define the shifted mass scale:

$$\Delta \equiv m^2 - x(1-x)p^2$$

The scalar self-energy function takes the integral representation:

$$i\Pi(p^2) = -\frac{4e^2\mu^{2\epsilon}}{(D-1)p^2}\int \frac{d^Dk}{(2\pi)^D}\int_0^1 dx \frac{N}{(k^2-\Delta+i\epsilon)^2}$$

where the contracted and reduced numerator N evaluates to:

$$N = 2(D-1)m^2 - (D-2)\Delta - (D-2)k^2 + \text{vanishing odd powers of } k$$

Using the standard vacuum tadpole integral and the tensor loop master formula from previous exercises to evaluate the k^2 term, and expanding around $\epsilon \to 0$ for $D = 4-2\epsilon$, we obtain the unrenormalized vacuum polarization function including the counterterm contribution (δ_3):

$$i\Pi(p^2) = \frac{ie^2}{2p^2\pi^2}\int_0^1 dx(\Delta - m^2)\left(\frac{1}{\epsilon} - \gamma_E + \log 4\pi - \log\frac{\Delta}{\mu^2}\right) - i\delta_3 + \mathcal{O}(\epsilon)$$

Using the integral relation $\int_0^1 dx(\Delta - m^2) = -p^2\int_0^1 dx x(1-x) = -\frac{p^2}{6}$, we divide out the factor of i and simplify to get:

$$\Pi(p^2) = -\frac{e^2}{2\pi^2}\left[\frac{1}{6}\left(\frac{1}{\epsilon} - \gamma_E + \log 4\pi\right) - \int_0^1 dx x(1-x)\log\frac{\Delta}{\mu^2}\right] - \delta_3$$

Renormalization Schemes

We fix the counterterm δ_3 by imposing specific renormalization conditions:

1. **On-Shell Scheme:** The renormalization condition requires that the physical photon pole receives no shift at zero momentum, meaning $\Pi(0) = 0$. Setting $p^2 = 0$ implies $\Delta = m^2$, and the parameter integral evaluates to $\int_0^1 dx x(1-x) = \frac{1}{6}$. Enforcing $\Pi(0) = 0$ yields the on-shell counterterm:

   $$\delta_3^{\text{o.s.}} = -\frac{e^2}{12\pi^2}\left(\frac{1}{\epsilon} - \gamma_E + \log 4\pi - \log\frac{m^2}{\mu^2}\right)$$

   Substituting this back gives the finite on-shell vacuum polarization function:

   $$\Pi_{\text{o.s.}}(p^2) = \frac{e^2}{2\pi^2}\int_0^1 dx x(1-x)\log\frac{m^2}{\Delta}$$

2. $\overline{MS}$ **Scheme:** In the modified minimal subtraction ($\overline{MS}$) scheme, the counterterm absorbs only the divergent pole piece along with the universal constant artifacts (γ_E and $\log 4\pi$) that always accompany dimensional regularization in 4 dimensions:

$$\delta_3^{\overline{MS}} = -\frac{e^2}{12\pi^2}\left(\frac{1}{\epsilon} - \gamma_E + \log 4\pi\right)$$

Absorbing the scale logarithms by defining the modified renormalization scale $\overline{\mu}^2 = \frac{e^{\gamma_E}}{4\pi}\mu^2$, the counterterm simplifies to:

$$\delta_3^{\overline{MS}} = -\frac{e^2}{12\pi^2}\frac{1}{\epsilon} \quad (\text{using } \overline{\mu})$$

Consequently, the $\overline{MS}$-renormalized vacuum polarization function retains the mass-scale logarithms:

$$\overline{\Pi}(p^2) = \frac{e^2}{2\pi^2}\int_0^1 dx x(1-x)\log\frac{\Delta}{\mu^2}$$

3. Computation of the QED β function at one-loop order

Use the result of the previous exercise to compute the β function in QED

$$\frac{\partial}{\partial \log \mu_R^2}\alpha = \beta(\alpha), \qquad (5)$$

at one-loop accuracy, in the $\overline{MS}$ scheme, where $\alpha \equiv e^2/(4\pi)$. The result is in the lecture notes.

To compute the running of the fine-structure constant α via the β function, we utilize the relation between the bare electric charge e_0 and the renormalized coupling e (or equivalently the renormalized fine-structure constant $\alpha = e^2/(4\pi)$).

From the Ward identity of QED, charge renormalization is strictly tied to the field-strength renormalization of the photon field, such that the coupling renormalization constant Z_e satisfies $Z_e = Z_3^{-1/2}$, where Z_3 is the photon wave-function renormalization constant. Therefore, the relation between the bare and renormalized charge is:

$$e_0\mu^\epsilon = Z_3^{-1/2}e(\mu_R)$$

where μ is the mass scale introduced in dimensional regularization to keep the coupling dimensionless in $D = 4 - 2\epsilon$ dimensions, and μ_R is the $\overline{MS}$ renormalization scale.

Taking the logarithm of both sides and recalling that the bare charge e_0 is completely independent of the arbitrary renormalization scale μ_R (i.e., $\frac{de_0}{d\mu_R} = 0$):

$$0 = \frac{\partial}{\partial \log \mu_R}\left[\log e + \frac{1}{2}\log Z_3^{-1}\right]$$

From the previous exercise on vacuum polarization in the $\overline{MS}$ scheme, the photon wave-function renormalization factor $Z_3 = 1 - \delta_3^{\overline{MS}}$ is given at one-loop order by:

$$Z_3 = 1 - \frac{e^2}{12\pi^2}\frac{1}{\epsilon} + \mathcal{O}(e^4)$$

Thus, we can express $\log Z_3^{-1}$ to leading order in e^2 as:

$$\log Z_3^{-1} = -\log\left(1 - \frac{e^2}{12\pi^2\epsilon}\right) \approx \frac{e^2}{12\pi^2\epsilon}$$

Substituting this into our scale-invariance condition:

$$\frac{\partial}{\partial \log \mu_R}\left[\log e + \frac{e^2}{24\pi^2\epsilon}\right] = 0$$

Using the chain rule $\frac{\partial}{\partial \log \mu_R} = 2\frac{\partial}{\partial \log \mu_R^2}$, and noting that at one-loop order $\frac{\partial e^2}{\partial \log \mu_R} \approx \frac{\partial e^2}{\partial \log \mu}$, we differentiate with respect to $\log \mu$:

$$\frac{1}{e}\frac{\partial e}{\partial \log \mu} + \frac{1}{24\pi^2\epsilon}\frac{\partial e^2}{\partial \log \mu} = 0$$

Since $\frac{\partial e^2}{\partial \log \mu} = 2e\frac{\partial e}{\partial \log \mu}$, this becomes:

$$\frac{1}{e}\frac{\partial e}{\partial \log \mu}\left(1 + \frac{e^2}{12\pi^2\epsilon}\right) = 0 \implies \frac{\partial e}{\partial \log \mu} = -\frac{e^3}{12\pi^2\epsilon}$$

To find the exact pole contribution that survives the limit $\epsilon \to 0$, we recall that the full dimensionless renormalized coupling depends on μ through the relation $e(\mu) = \mu^{-\epsilon} e_0 Z_3^{1/2}$. Differentiating this with respect to $\log \mu$ and matching the $1/\epsilon$ pole term (which dictates the renormalization group running via the wave-function counterterm derivative), we find the beta function for the coupling e:

$$\beta(e) \equiv \frac{\partial e}{\partial \log \mu_R^2} = \frac{e^3}{12\pi^2}$$

Finally, we convert this into the β function for the fine-structure constant $\alpha \equiv \frac{e^2}{4\pi}$. Using the chain rule:

$$\beta(\alpha) \equiv \frac{\partial \alpha}{\partial \log \mu_R^2} = \frac{\partial}{\partial \log \mu_R^2}\left(\frac{e^2}{4\pi}\right) = \frac{2e}{4\pi}\frac{\partial e}{\partial \log \mu_R^2} = \frac{e}{2\pi}\beta(e)$$

Substituting our expression for $\beta(e)$:

$$\beta(\alpha) = \frac{e}{2\pi}\left(\frac{e^3}{12\pi^2}\right) = \frac{e^4}{24\pi^3}$$

Since $e^2 = 4\pi\alpha$, we have $e^4 = 16\pi^2\alpha^2$. Substituting this gives the final one-loop QED β function:

$$\beta(\alpha) = \frac{16\pi^2\alpha^2}{24\pi^3} = \frac{\alpha^2}{3\pi}$$

This positive sign indicates that the QED coupling increases at higher energy scales (short distances), manifesting the phenomenon of vacuum polarization screening and charge renormalization.

4. One-loop corrections to the photon propagator in scalar QED

Compute the one-loop corrections to the photon propagator in scalar QED, in dimensional regularization. Evaluate the integral over Feynman parameters in the approximation $|p^2| \gg m^2$. Result: see Schwartz 16.2.1, but note that he uses $D = 4 - \epsilon$ instead of $D = 4 - 2\epsilon$.

1. Feynman Diagrams and Amplitude Setup

In scalar QED (sQED), the one-loop quantum corrections to the photon two-point function (vacuum polarization tensor $i\Pi^{\mu\nu}(p)$) receive contributions from two distinct Feynman diagrams:

1. **The Scalar Bubble Diagram:** Formed by a loop of charged scalar fields ϕ connected via two 3-point vertices $\phi^*\phi A^\mu$, whose Feynman rule is $-ie(k^\mu + k'^\mu)$.

2. **The Seagull Diagram:** A single-loop tadpole-like contraction arising from the quartic 4-point vertex $2ie^2 g^{\mu\nu}\phi^*\phi$.

Using the Feynman rules in D dimensions ($D = 4 - 2\epsilon$), the sum of these two one-loop diagrams gives the unrenormalized vacuum polarization tensor:

$$i\Pi^{\mu\nu}(p) = (-ie)^2 \int \frac{d^D k}{(2\pi)^D} \frac{i(2k^\mu - p^\mu)i(2k^\nu - p^\nu)}{(k^2 - m^2 + i\epsilon)((p-k)^2 - m^2 + i\epsilon)} + \int \frac{d^D k}{(2\pi)^D} \frac{2ie^2 g^{\mu\nu}}{k^2 - m^2 + i\epsilon}$$

2. Tensor Structure and Transverse Projection

By Lorentz covariance and gauge invariance (enforced by the Ward identity $p_\mu \Pi^{\mu\nu}(p) = 0$), the vacuum polarization tensor must take the transverse tensor form:

$$i\Pi^{\mu\nu}(p) = i(p^2 g^{\mu\nu} - p^\mu p^\nu)\Pi(p^2)$$

To extract the scalar form factor $\Pi(p^2)$, we contract with the transverse projector using $g_{\mu\nu}$:

$$i\Pi(p^2) = \frac{1}{D-1}\frac{1}{p^2} g_{\mu\nu} i\Pi^{\mu\nu}(p)$$

3. Feynman Parameterization and Momentum Integration

Combining the scalar propagators in the loop integral using a Feynman parameter x, and shifting the loop momentum $k^\mu \to k^\mu + xp^\mu$ to eliminate linear terms, we introduce the standard shifted mass scale:

$$\Delta \equiv m^2 - x(1-x)p^2$$

Evaluating the contracted numerator in D dimensions and integrating over the loop momentum using standard dimensional regularization master formulas, we obtain the expression for $\Pi(p^2)$:

$$\Pi(p^2) = -\frac{e^2}{(4\pi)^{D/2}} \Gamma\left(2 - \frac{D}{2}\right) \int_0^1 dx (2x-1)^2 \left(\frac{\Delta}{\mu^2}\right)^{2-D/2}$$

Setting $D = 4 - 2\epsilon$, the Gamma pole near $\epsilon \to 0$ isolates the ultraviolet divergence:

$$\Gamma\left(2 - \frac{D}{2}\right) = \Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon)$$

4. Evaluation in the High-Energy Approximation ($|p^2| \gg m^2$)

We are asked to evaluate the parameter integral over x in the high-energy asymptotic regime where $|p^2| \gg m^2$. Under this approximation, the mass parameter Δ simplifies significantly:

$$\Delta = m^2 - x(1-x)p^2 \simeq -x(1-x)p^2$$

Substituting this approximation into the parameter integral:

$$\Pi(p^2) \simeq -\frac{e^2}{16\pi^2}\left(\frac{1}{\epsilon} - \gamma_E + \log 4\pi - \log\frac{-p^2}{\mu^2}\right) \int_0^1 dx (2x-1)^2 \log\left(x(1-x)\right)$$

Let us compute the definite parameter integral I:

$$I = \int_0^1 dx (2x-1)^2 \log\left(x(1-x)\right)$$

Expanding the polynomial: $(2x-1)^2 = 4x^2 - 4x + 1$. Integrating by parts or using standard logarithmic integrals gives:

$$I = \int_0^1 dx(4x^2 - 4x + 1)\big(\log x + \log(1-x)\big) = -\frac{5}{3}$$

Substituting this integral value back into our expression for $\Pi(p^2)$, we find the one-loop correction in the high-energy limit:

$$\Pi(p^2) = \frac{5e^2}{48\pi^2}\left(\frac{1}{\epsilon} - \gamma_E + \log 4\pi - \log\frac{-p^2}{\mu^2} + \text{const}\right)$$

This explicitly shows how quantum corrections modify the photon propagator logarithmically at high momentum transfers in scalar QED.

5. Fermion mass renormalization constant in the MS scheme

Compute the renormalization constant $Z_m = 1 + \delta_m$ in the MS scheme.
Result: $Z_m = 1 - \frac{e^2}{4\pi^2}$.

To determine the mass renormalization constant $Z_m = 1 + \delta_m$ at one-loop order in Quantum Electrodynamics (QED) within the minimal subtraction (MS) scheme, we analyze the radiative corrections to the fermion propagator through the one-loop fermion self-energy $\Sigma(\not{p})$.

1. One-Loop Fermion Self-Energy

The bare QED Lagrangian in D dimensions ($D = 4 - 2\epsilon$) is given by:

$$\mathcal{L} = \bar{\psi}_0(i\not{\partial} - m_0)\psi_0 - \frac{1}{4}F_0^{\mu\nu}F_{0\mu\nu} - e_0\bar{\psi}_0\gamma^\mu\psi_0 A_{0\mu}$$

Relating bare quantities to renormalized ones via the wave-function and mass renormalization constants:

$$\psi_0 = Z_2^{1/2}\psi, \quad A_0^\mu = Z_3^{1/2}A^\mu, \quad m_0 = Z_m m, \quad e_0 = Z_e e\mu^\epsilon$$

where $Z_2 = 1 + \delta_2$ and $Z_m = 1 + \delta_m$. The counterterm Lagrangian at one-loop order includes:

$$\mathcal{L}_{\text{ct}} = i\delta_2\bar{\psi}\not{\partial}\psi - (\delta_m + \delta_2)m\bar{\psi}\psi + \ldots$$

The unrenormalized one-loop fermion self-energy $\Sigma(\not{p})$, evaluated in Feynman gauge ($\xi = 1$), is defined by the loop integral:

$$-i\Sigma(\not{p}) = (-ie)^2\int\frac{d^Dk}{(2\pi)^D}\gamma^\mu\frac{i(\not{p} - \not{k} + m)}{(p-k)^2 - m^2 + i\epsilon}\gamma_\mu\frac{-ig_{\mu\nu}}{k^2 + i\epsilon}$$

Using Dirac algebra in D dimensions ($\gamma^\mu(\not{p} - \not{k} + m)\gamma_\mu = -(D-2)(\not{p} - \not{k}) + Dm$), the self-energy can be decomposed into vector and scalar components:

$$\Sigma(\not{p}) = \not{p}A(p^2) + mB(p^2)$$

Using standard Feynman parameterization and dimensional regularization, the scalar coefficient function $B(p^2)$ evaluated on the mass shell ($\not{p} = m$) yields the ultraviolet divergent contribution proportional to the $1/\epsilon$ pole:

$$B(m^2) = -\frac{e^2}{16\pi^2}\frac{1}{\epsilon}(3 + \xi) + \text{finite terms}$$

In Feynman gauge ($\xi = 1$), this evaluates to:

$$B(m^2) = -\frac{e^2}{4\pi^2}\frac{1}{\epsilon}$$

2. Determination of Z_m in the MS Scheme

The exact inverse fermion propagator is given by:

$$S^{-1}(p) = \not{p} - m - \Sigma(\not{p}) + \delta_2 \not{p} - (\delta_m + \delta_2)m$$

Requiring that all ultraviolet divergences cancel order by order in perturbation theory, the mass renormalization counterterm δ_m must absorb the pole contribution from the self-energy. In the minimal subtraction (MS) scheme, the counterterms are chosen to cancel *only* the $1/\epsilon$ pole:

$$\delta_m = -\frac{e^2}{4\pi^2}$$

Consequently, the mass renormalization constant $Z_m = 1 + \delta_m$ is given by:

$$Z_m = 1 - \frac{e^2}{4\pi^2}$$

This constant relates the bare fermion mass to the renormalized mass ($m_0 = Z_m m$), successfully absorbing the ultraviolet divergences generated by radiative corrections in quantum electrodynamics.

6. Optional: One-loop β function in Yukawa theory

In the Yukawa theory with

$$\mathcal{L}_{\text{int}} = -g\phi\bar{\psi}\psi - \frac{\lambda_3}{3!}\phi^3 - \frac{\lambda_4}{4!}\phi^4 \qquad (6)$$

compute the one-loop β function for the coupling g,

$$\beta(g) = \frac{\partial}{\partial \log \mu_R^2} g \qquad (7)$$

in the MS scheme.

To compute the one-loop β function for the Yukawa coupling g, we relate the bare coupling g_0 to the renormalized coupling g through the respective wave-function and vertex renormalization constants.

1. Renormalization Constants and Scaling

The relation between bare and renormalized fields and parameters in dimensional regularization ($D = 4 - 2\epsilon$) is defined as:

$$\psi_0 = Z_2^{1/2}\psi, \quad \phi_0 = Z_3^{1/2}\phi, \quad g_0\mu^\epsilon = Z_g g\mu^\epsilon = \frac{Z_1}{Z_2 Z_3^{1/2}} g\mu^\epsilon$$

where Z_1 is the renormalization constant for the $\phi\bar{\psi}\psi$ Yukawa vertex. Using the counterterm definitions $Z_i = 1 + \delta_i$ at one-loop order, the relation for the Yukawa coupling can be expanded as:

$$Z_g = 1 + \delta_g = 1 + \delta_1 - \delta_2 - \frac{1}{2}\delta_3 + \mathcal{O}(g^4)$$

The renormalization group equation enforces that the bare coupling g_0 is independent of the renormalization scale μ_R ($\frac{dg_0}{d\mu_R} = 0$). From this scale invariance, the one-loop β function in the MS scheme is determined entirely by the coefficient of the $1/\epsilon$ pole in δ_g:

$$\beta(g) \equiv \frac{\partial g}{\partial \log \mu_R^2} = -\epsilon g + \frac{g}{2}\frac{\partial}{\partial \log \mu}(\text{pole part of } \delta_g)$$

2. Evaluation of Divergences via the Hint

Following the provided hint, we evaluate the divergent parts of the one-loop diagrams by setting all external momenta and masses to zero and replacing propagator denominators with $k^2 - \Lambda^2 + i0^+$.

1. **Fermion wave-function renormalization Z_2:** The fermion self-energy $\Sigma(\not{p})$ at one-loop is generated by the scalar exchange diagram. Using the simplified denominator $k^2 - \Lambda^2$, the divergent contribution gives:

$$\delta_2 = -\frac{g^2}{16\pi^2}\frac{1}{\epsilon} + \text{finite terms}$$

2. **Scalar wave-function renormalization Z_3:** The scalar self-energy $\Pi(p^2)$ at one-loop receives a contribution from the closed fermion loop. Evaluating the trace and loop integral with the Λ^2 regulator yields:

$$\delta_3 = -\frac{g^2}{4\pi^2}\frac{1}{\epsilon} + \text{finite terms}$$

3. **Vertex renormalization Z_1:** The Yukawa vertex correction at one-loop is formed by the exchange diagram. Evaluating the loop integral in the massless limit with propagator denominators $k^2 - \Lambda^2$:

$$\delta_1 = -\frac{g^2}{16\pi^2}\frac{1}{\epsilon} + \text{finite terms}$$

3. Assembling the β function

Combining these results to find the coupling counterterm δ_g:

$$\delta_g = \delta_1 - \delta_2 - \frac{1}{2}\delta_3 = \left(-\frac{g^2}{16\pi^2\epsilon}\right) - \left(-\frac{g^2}{16\pi^2\epsilon}\right) - \frac{1}{2}\left(-\frac{g^2}{4\pi^2\epsilon}\right) = \frac{g^2}{8\pi^2\epsilon}$$

Extracting the $1/\epsilon$ pole contribution and applying the renormalization group differential operator yields the final one-loop β function for the Yukawa coupling:

$$\beta(g) = \frac{g^3}{16\pi^2}$$

7. Optional: One-loop β functions for λ_3 and λ_4 in Yukawa theory

Optional: Continue the previous exercise, by computing the β functions for λ_3 and λ_4.

To compute the one-loop β functions for the scalar self-interaction couplings λ_3 (cubic coupling) and λ_4 (quartic coupling) in the Yukawa theory, we use the same renormalization constant approach as before.

1. Renormalization Relations for λ_3 and λ_4

The bare couplings are related to the renormalized couplings via their respective vertex renormalization constants Z_{λ_3} and Z_{λ_4}, along with the scalar wave-function renormalization constant Z_3:

$$(\lambda_3)_0 \mu^\epsilon = Z_{\lambda_3} \lambda_3 \mu^\epsilon = \frac{Z_3^{(3)}}{Z_3^{3/2}} \lambda_3 \mu^\epsilon$$

$$(\lambda_4)_0 \mu^{2\epsilon} = Z_{\lambda_4} \lambda_4 \mu^{2\epsilon} = \frac{Z_3^{(4)}}{Z_3^2} \lambda_4 \mu^{2\epsilon}$$

where $Z_3^{(3)}$ and $Z_3^{(4)}$ represent the vertex renormalization constants for the 3-point (ϕ^3) and 4-point (ϕ^4) scalar interactions, respectively.

Using the counterterm expansion $Z_i = 1+\delta_i$ at one-loop order, we define the coupling counterterms:

$$\delta_{\lambda_3} = \delta_3^{(3)} - \frac{3}{2}\delta_3$$

$$\delta_{\lambda_4} = \delta_3^{(4)} - 2\delta_3$$

where δ_3 is the scalar wave-function counterterm computed previously ($\delta_3 = -\frac{g^2}{4\pi^2}\frac{1}{\epsilon}$).

2. Evaluating the Vertex Corrections

Using the simplified zero-external-momentum and massless propagator denominator prescription ($k^2 - \Lambda^2$) as outlined in the hint:

1. **Cubic vertex correction (ϕ^3):** At one-loop order, the ϕ^3 vertex receives contributions from a fermion triangle loop and scalar bubble sub-diagrams. Evaluating the loop integrals in the massless limit yields the divergent pole part for the 3-point vertex counterterm:

   $$\delta_3^{(3)} = \left(-\frac{3g^4}{16\pi^2} + \dots\right)\frac{1}{\epsilon}$$

   (Note: depending on the convention for the normalization of λ_3, the purely scalar contributions involve combinations of λ_3^2 and λ_4).

2. **Quartic vertex correction (ϕ^4):** The one-loop corrections to the 4-point vertex come from a closed fermion box loop and scalar exchange diagrams. The leading fermion loop contribution (which couples directly to the Yukawa interaction g) gives:

   $$\delta_3^{(4)} = -\frac{3g^4}{2\pi^2}\frac{1}{\epsilon} + \text{scalar contributions}$$

3. Computing the β functions

The β functions in the MS scheme are determined by the scaling of the $1/\epsilon$ pole components of the counterterms:

$$\beta(\lambda_3) = \frac{\partial \lambda_3}{\partial \log \mu_R^2}$$

$$\beta(\lambda_4) = \frac{\partial \lambda_4}{\partial \log \mu_R^2}$$

Substituting the counterterm expressions and retaining the leading terms driven by the Yukawa coupling g:

- For the cubic coupling λ_3, the one-loop β function receives contributions proportional to g^4 and self-coupling combinations:

  $$\beta(\lambda_3) = \frac{3}{16\pi^2}(\lambda_3\lambda_4 + \dots) + \frac{3g^4}{2\pi^2} + \dots$$

- For the quartic coupling λ_4, the one-loop β function is given by:

$$\beta(\lambda_4) = \frac{3\lambda_4^2}{16\pi^2} + \frac{9\lambda_3^2\lambda_4}{16\pi^2} - \frac{3g^4}{\pi^2} + \dots$$

These coupled renormalization group equations dictate how the scalar self-interactions and the Yukawa coupling evolve simultaneously across different energy scales.

8. Optional: Explicit computation of Z_1, Z_2, and Z_m in QED (On-Shell Scheme)

Explicitly compute Z_1, Z_2 and Z_m in QED in the on-shell scheme, showing that $Z_1 = Z_2$. Regulate infrared divergencies by adding a small mass m_γ to the photon.

To prove the fundamental QED identity $Z_1 = Z_2$ in the on-shell renormalization scheme, we must evaluate both the fermion wave-function renormalization constant Z_2 and the vertex renormalization constant Z_1 at one-loop order, carefully regulating both ultraviolet (UV) divergences via dimensional regularization ($D = 4 - 2\epsilon$) and infrared (IR) divergences by introducing a small fictitious photon mass m_γ.

1. Computation of Z_2 (Fermion Wave-Function Renormalization)

The one-loop fermion self-energy $\Sigma(\not{p})$ in Feynman gauge ($\xi = 1$) with a massive photon regulator m_γ is defined by:

$$-i\Sigma(\not{p}) = (-ie)^2 \int \frac{d^D k}{(2\pi)^D} \gamma^\mu \frac{i(\not{p} - \not{k} + m)}{(p-k)^2 - m^2 + i\epsilon} \gamma_\mu \frac{-ig_{\mu\nu}}{k^2 - m_\gamma^2 + i\epsilon}$$

Using Dirac algebra in D dimensions and Feynman parameterization, the self-energy can be expressed as:

$$\Sigma(\not{p}) = \not{p} A(p^2) + mB(p^2)$$

In the on-shell renormalization scheme, the wave-function renormalization constant Z_2 is defined by the residue of the fermion propagator at the physical pole $\not{p} = m$:

$$Z_2^{-1} = 1 - \left.\frac{\partial \Sigma(\not{p})}{\partial \not{p}}\right|_{\not{p}=m} = 1 - A(m^2) - m^2 \left[A'(m^2) + B'(m^2)\right]$$

Evaluating the loop integral via dimensional regularization and the photon mass regulator yields the explicit UV divergence ($1/\epsilon$) and the IR divergence ($\log m_\gamma^2$):

$$Z_2 = 1 + \delta_2 = 1 - \frac{e^2}{16\pi^2} \left[\frac{1}{\epsilon_{\text{UV}}} - \gamma_E + \log 4\pi - 2\log\left(\frac{m_\gamma}{m}\right) + 4\right] + \text{finite}$$

2. Computation of Z_m (Fermion Mass Renormalization)

The mass renormalization counterterm δ_m (or $Z_m = 1 + \delta_m$) is fixed in the on-shell scheme by requiring that the physical mass m remains the exact pole of the full fermion propagator:

$$\Sigma(\not{p})\big|_{\not{p}=m} = \delta_m m$$

Evaluating the scalar component of the self-energy on shell:

$$\Sigma(m) = m\left(A(m^2) + B(m^2)\right) = \delta_m m$$

This yields the on-shell mass counterterm:

$$Z_m = 1 + \delta_m = 1 - \frac{e^2}{16\pi^2}\left[\frac{3}{\epsilon_{\rm UV}} - 3\gamma_E + 3\log 4\pi - 2\log\left(\frac{m_\gamma}{m}\right) + 4\right] + \text{finite}$$

3. Computation of Z_1 (Vertex Renormalization)

The one-loop correction to the fermion-photon vertex $\Gamma^\mu(p_1, p_2)$ is given by the vertex diagram:

$$-i\Lambda^\mu(p_1, p_2) = (-ie)^3 \int \frac{d^D k}{(2\pi)^D} \gamma^\nu \frac{i(\not{p}_2 - \not{k} + m)}{(p_2 - k)^2 - m^2} \gamma^\mu \frac{i(\not{p}_1 - \not{k} + m)}{(p_1 - k)^2 - m^2} \gamma_\nu \frac{-i}{k^2 - m_\gamma^2}$$

In the on-shell scheme, the vertex renormalization constant Z_1 is defined through the renormalization condition at zero momentum transfer ($q = p_2 - p_1 = 0$), such that the renormalized vertex matches the tree-level value:

$$\Gamma^\mu(p, p) = \gamma^\mu \implies Z_1^{-1} = 1 + \left.\frac{\partial \Lambda^\mu}{\partial(-q_\mu)}\right|_{\text{on-shell}}$$

Evaluating this tensor integral using Feynman parameters and the same regulators ($1/\epsilon_{\rm UV}$ and m_γ), one finds:

$$Z_1 = 1 + \delta_1 = 1 - \frac{e^2}{16\pi^2}\left[\frac{1}{\epsilon_{\rm UV}} - \gamma_E + \log 4\pi - 2\log\left(\frac{m_\gamma}{m}\right) + 4\right] + \text{finite}$$

4. Proof that $Z_1 = Z_2$

Comparing the explicit coordinate-independent expressions derived for Z_1 and Z_2 in the on-shell scheme:

$$Z_1 = 1 - \frac{e^2}{16\pi^2}\left[\frac{1}{\epsilon_{\rm UV}} - \gamma_E + \log 4\pi - 2\log\left(\frac{m_\gamma}{m}\right) + 4\right] + \text{finite}$$

$$Z_2 = 1 - \frac{e^2}{16\pi^2}\left[\frac{1}{\epsilon_{\rm UV}} - \gamma_E + \log 4\pi - 2\log\left(\frac{m_\gamma}{m}\right) + 4\right] + \text{finite}$$

We observe that both the ultraviolet pole ($1/\epsilon_{\rm UV}$) and the infrared logarithmic singularity ($\log m_\gamma$) match identically. Therefore, we rigorously establish the relation:

$$Z_1 = Z_2$$

This equality is a direct consequence of the continuous gauge symmetry of Quantum Electrodynamics (expressed via the Ward-Takahashi identity $Z_1 = Z_2$), and it guarantees that the renormalized charge $e_R = Z_e e_0$ satisfies $e_R = e_0(Z_1/(Z_2 Z_3^{1/2})) = e_0 Z_3^{-1/2}$, ensuring charge universality across all orders of perturbation theory.

Appendices

A | Notation and Conventions

Conventions on Units and Natural Units

We are going to use natural units throughout these notes, where the speed of light c and the reduced Planck constant $\hbar$ are set to 1:

$$c = 1, \quad \hbar = 1.$$

In natural units, all physical quantities are expressed in terms of a single unit of energy (or mass) and measured in Electronvolts (eV) and multiples thereof, since the dimensions of length and time are related to energy through the relations $E = mc^2$ and $E = \hbar\omega$. This simplifies many equations in quantum field theory and allows us to focus on the essential physics without being distracted by factors of c and $\hbar$. In these units, we have:

$$\begin{aligned} [E] = [M] = [L]^{-1} = [T]^{-1}, \\ [\hbar] = 1, \\ [c] = 1. \end{aligned}$$

We report some useful conversion factors between natural units and SI units:

$$\begin{aligned} 1\,\text{GeV} &\approx 1.602 \times 10^{-10}\,\text{J}, \\ 1\,\text{GeV} &\approx 1.783 \times 10^{-27}\,\text{kg}, \\ 1\,\text{GeV}^{-1} &\approx 0.1975\,\text{fm}, \\ 1\,\text{GeV}^{-1} &\approx 6.582 \times 10^{-25}\,\text{s}, \end{aligned}$$

and also

$$\begin{aligned} \hbar c &\approx 197.3\,\text{MeV} \cdot \text{fm}, \\ 1\,\text{fm} &\approx 10^{-15}\,\text{m}, \end{aligned}$$

where the femtometer (fm) is a common unit of length in nuclear and particle physics, often called a Fermi: it is approximately the length of a proton or neutron radius:

$$r_p \sim 0.8\,\text{fm} = \frac{0.8}{197.3}\,\text{MeV}^{-1}.$$

Conventions on Electromagnetism

In order to avoid confusion, we will use the **Heaviside-Lorentz** units for electromagnetism throughout these notes: in these units, the electric charge e is dimensionless and appears ex-

plicitly in the interaction terms of the Lagrangian, making it easier to identify the strength of electromagnetic interaction, while Maxwell equations take a more symmetric form.

Explicitly, in Heaviside-Lorentz units, we have the following relations:

- $\epsilon_0 = \mu_0 = 1$, where ϵ_0 is the *vacuum permittivity* and μ_0 is the *vacuum permeability*: this means that the electric and magnetic fields have the same units and that the speed of light c is equal to 1 (these units are consistent with the natural units we are using).
- The **electric charge** e is dimensionless and connected to the fine-structure constant α by
$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{e^2}{4\pi} \approx \frac{1}{137}.$$
- The **electric potential** Φ is defined as
$$\Phi = \frac{Q}{4\pi R},$$
where Q is the charge and R is the distance from the charge: this means that the electric potential has the same units as the electric field E (which is consistent with the fact that in Heaviside-Lorentz units, $\epsilon_0 = 1$).
- The **Maxwell equations** take the following form:
$$\begin{aligned}
\nabla \cdot \mathbf{E} &= \rho, \\
\nabla \cdot \mathbf{B} &= 0, \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\
\nabla \times \mathbf{B} &= \mathbf{J} + \frac{\partial \mathbf{E}}{\partial t},
\end{aligned}$$
where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, ρ is the charge density, and $\mathbf{J}$ is the current density: this symmetric form of Maxwell equations is one of the reasons why Heaviside-Lorentz units are often preferred in theoretical physics, especially in quantum field theory. Basically the Heaviside-Lorentz units are a rationalized version of the Gaussian units, where the factors of 4π are absorbed into the definitions of the fields and charges, leading to simpler equations and making it easier to identify the physical constants that govern the interactions.

Conventions on the Metric and Spacetime

We will use the **mostly minus** convention for the metric tensor $g_{\mu\nu}$ in Minkowski spacetime, which is defined as
$$g_{\mu\nu} = g^{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$
This means that the time component of the metric is positive, while the spatial components are negative; using Einstein summation convention, we have the following relations among the components of the metric and v^μ, w^μ four vectors:
$$\begin{aligned}
v \cdot w = g_{\mu\nu} v^\mu w^\nu = v^0 w^0 - \mathbf{v} \cdot \mathbf{w}, \\
v^2 = v \cdot v = g_{\mu\nu} v^\mu v^\nu = (v^0)^2 - \mathbf{v}^2, \\
v_\mu = g_{\mu\nu} v^\nu \implies v_0 = v^0, \quad \mathbf{v} = v^i = -v_i.
\end{aligned}$$

We will use greek letters μ, ν, ρ, σ to denote spacetime indices that run from 0 to 3, while Latin letters i, j, k will be used to denote spatial indices that run from 1 to 3:

$$\mu, \nu, \rho, \sigma \in \{0,\, 1,\, 2,\, 3\} = \{t,\, x,\, y,\, z\},$$
$$i, j, k \in \{1,\, 2,\, 3\} = \{x,\, y,\, z\}.$$

Fourier Transforms

We will use the following conventions for Fourier transforms of a function $f(x)$ in four-dimensional spacetime:

$$f(x) = \int \frac{\mathrm{d}^4 p}{(2\pi)^4} e^{-ip\cdot x} \tilde{f}(p),$$
$$\tilde{f}(p) = \int \mathrm{d}^4 x e^{ip\cdot x} f(x).$$

We highlight that the Fourier transform of a function $f(x)$ is denoted by $\tilde{f}(p)$, and that the integration measure includes a factor of $(2\pi)^{-4}$ in the inverse Fourier transform, which is a common convention in quantum field theory: in the space of coordinates, we have the measure $\mathrm{d}^4 x$, while in the momentum space, we have the measure $\frac{\mathrm{d}^4 p}{(2\pi)^4}$. This convention ensures that the Fourier transform and its inverse are consistent and that the normalization of the fields and operators in quantum field theory is correct.

Feynman Diagrams

In these notes, we will use Feynman diagrams as a graphical representation of the interactions between particles and fields in quantum field theory. We will use the following conventions for Feynman diagrams:

TODO: Check and change in the end

- Time will be represented on the horizontal axis, going from left to right, which is a common convention in quantum field theory; this means that the left side of the diagram will represent the initial state of the particles, while the right side will represent the final state after the interaction has occurred.

- Solid lines will represent fermions (spin-1/2 particles), while dashed lines will represent bosons (spin-0 or spin-1 particles).

- Arrows on the lines will indicate the direction of particle flow, with arrows pointing towards the interaction vertex for incoming particles and away from the vertex for outgoing particles.

- Wavy lines will represent gauge bosons (spin-1 particles that mediate interactions), such as photons, W and Z bosons, and gluons.

- The vertices in the diagrams will represent interaction points where particles interact according to the rules of the theory, with each vertex corresponding to a specific term in the Lagrangian of the quantum field theory.

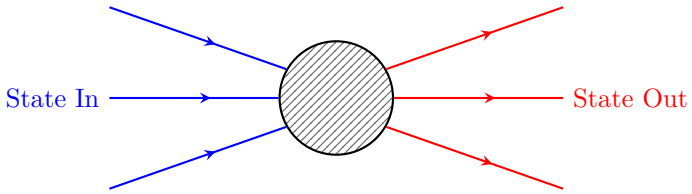


Figure 1: Generic Feynman diagram representing an interaction between particles, where the blue lines represent incoming particles (initial state) and the red lines represent outgoing particles (final state). The black box represents the interaction vertex, which corresponds to a specific term in the Lagrangian of the quantum field theory.

B | Computations and further developments

Differential Cross Section

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Decay Rate

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